Deriving the Particle Zoo from Observer Consistency

B. Müller Mario Poneder Alexander Osika Kai Xue Daniel Ari Friedman

October 25, 2026

Paper release: `r2053` **Released:** October 25, 2026

Author affiliations: Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks; Daniel Ari Friedman, Active Inference Institute and FractiAI.

Abstract

Observer-Patch Holography describes physical measurements through finite observer records and repair. Complete reversible response and endogenous transport constrain the local Standard Model gauge Lie algebra. A declared spin lift of its finite response gives a fifteen-state matter representation; charge and Casimir operators recover one generation's five internal multiplets without a field-name table. Exact invariant tensors admit the three charged-fermion interaction channels with one Higgs doublet, and an optional dimension-five neutrino operator. Their coefficients, vacuum and physical mass interpretation are separate inputs.

Conditional field constructions share a Lorentzian principal cone and observer-clock conversion. A finite CAR field shares the scalar spatial operator; a full cochain control removes edge zero modes caused by truncation. Neither construction identifies the observed chiral spectrum. The source family band's unbroken invariant intertwiners give degenerate family masses, exposing the required symmetry-breaking attachment.

The paper also distinguishes calibrated mass fits, conditional hierarchy coordinates and rejected formulas. No nonzero physical pole mass is claimed as a source-only prediction. A Standard Model identification requires the selected interacting fields and their spectral response on the same observer history, with controlled physical normalization and refinement.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-

history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Because nothing lies outside that structure, recovered structure needs no a-priori justification, nothing unrecovered may be assumed, and every change of description is implemented by the structure's own operations. Agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with its endogeneity clause; and least-informative state selection. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

What This Paper Contributes

The Standard Model gauge paper proves that complete reversible port response and endogenous observer transport force the local Standard Model gauge Lie algebra. Its conditional matrix and matter realization supplies the hypercharge lattice, a three-color carrier, the common $\mathbb{Z}_6$ kernel, and the maximal faithful matter image $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$. It also supplies the distinct conditional compact-gauge/Tannaka reconstruction route, a conditional generation window $3 \leq N_g \leq 5$, and the connection and metric carrier roles. Under the single-complete-object, faithful-action, and operational-cost premises, the exact screen-band theorem selects rank three inside that window. The declared finite simulator recovers the same projector at its lowest positive generator frequency. Tensoring that response band with the declared generation table gives a conditional complex rank-45 candidate. Chirality and the diagonal $\mathbb{Z}_6$ action are properties of the table. A separate finite source-domain receipt checks the declared operator $D_\sigma \otimes I_{45}$ and conditional inheritance of a positive dimensionless gap. The source does not select this matter action or bridge the twelve-port Spin packet to the local operator domain. Identification with physical matter poles and continuum fields is a separate premise. Classical massless modes require explicit action, background, and phase premises; quantum poles require a stronger particle certificate. This paper studies the downstream particle readout after the local closure coordinate is supplied.

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/docs/BASIS.md

The displayed $\mathbb{Z}_6$ quotient is the maximal faithful image of the declared representation. It is not a source-selected physical global form. Two structural routes feed that downstream calculation. Incidence expresses J as a polynomial in adjacency, while target-blind port readback derives the signed response. For the finite-current theorem, write $\mathfrak{g} = D(\mathbb{R}^{12}) \subset \mathfrak{u}(H)$. Faithfulness gives dimension twelve, and commutator closure makes $\mathfrak{g}$ compact reductive. Transitivity gives one fixed line for the alternating group on five letters, A_5. Endogenous proper-carrier transport makes this action inner. It therefore fixes the centre pointwise. The centreless alternative is $\mathfrak{su}(2)^4$, whose factorwise fixed dimensions are zero or three and cannot sum to one. The centre has dimension one, and compact-simple classification splits the eleven-dimensional semisimple part as $3+8$. Thus

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

This argument assumes no ambient continuous gauge group. The declared matrix current and matter chain are conditional. Transportable sectors plus Tannaka reconstruction give a separate conditional compact group. The target-free matter reduct does not select the generation count inside the window. Under separate complete-band and cost-order premises, the screen-band theorem selects rank three. Tensoring that response band with the declared generation table gives a conditional rank-45 candidate. A separate local receipt checks a declared tensor-identity operator and conditional gap inheritance. It neither selects the matter action nor transports the twelve-port Spin packet to the local domain. Matter-pole identification, continuum Spin/locality, the physical seam choice, the third persistence leg, and laboratory identification are premises of any physical reading. Completeness for extra light sectors is a separate attachment premise. The intertwiner between the two current objects is likewise a premise.

A single screen-cell coordinate $P_\star$ feeds the electromagnetic, electroweak, and mass branches. The exterior matter witness fixes four weak doublets. Conditional on the shared physical load carrier, the selected branch emits $v/E_\star$. Separately, its declared comparison maps make the defined commuting-square defect $\epsilon_H^{\mathrm{decl}} = 0$ identically. Those maps are not emitted by the source antecedents, so the identity is not a Higgs naturalness result; a weak scale in GeV also requires an independently closed $E_\star$. Each declared pixel map has a machine-certified interval proof of a unique fixed point. On the target-anchored empirical-closure branch, the measured charged-lepton triple lies inside every outward-rounded arithmetic enclosure. These are diagnostic intervals rather than prospective prediction intervals. The branch inverts exactly at the witness: one anchor-gap value closes all three masses on the measured triple at once, inside the certified band. The scheme bridge is therefore a declared confirm-or-refute target. Absolute charged-lepton and first-principles hadron masses lie outside the theorem, and empirical endpoint data are identified wherever a transport bridge uses them.

1 Introduction

Observer-Patch Holography (OPH) asks a concrete follow-up question after the gauge structure is fixed. Can one local screen constant organize the particle spectrum, or are the masses and mixings a collection of unrelated inputs?

The Standard Model gauge paper separates the local Lie algebra from its matrix and matter realizations. Complete reversible response and endogenous transport force

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

Conditional on the separately declared matrix current, rank-15 matter table, and descent packet,

the maximal faithful matter image is

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3.$$

The exact hypercharge lattice and color count belong to this conditional realization. Separate intrinsic-CP and weak-sector premises give the window $3 \le N_g \le 5$, which does not by itself select the generation count. The graph carrying the alternating group on five letters, A_5, together with anomaly cancellation and the target-free matter reduct does not select a count. A strengthened screen interface does: among single complete faithful bands in the window, the exact Laplacian costs $5-\sqrt{5} < 6 < 5+\sqrt{5}$ select the rank-three band uniquely. The declared unitary simulator recovers the same residue at its lowest positive generator frequency. Tensoring this mode with the declared generation table gives a conditional rank-45 candidate. Chirality and the diagonal $\mathbb{Z}_6$ action come from that table. A separate local-domain receipt checks a declared tensor-identity operator and conditional gap inheritance. The source does not select the matter action or bridge the twelve-port Spin packet to that domain. Physical family interpretation requires matter-pole, continuum, seam-selection, persistence, and laboratory receipts. The paper also supplies the exact hypercharge lattice and a conditional exterior representation witness. If a trace-balanced block carrier $V = C \oplus W$, with dimensions $3+2$ and hypercharges $(-1/3, 1/2)$, is physically selected, then $\Lambda^2 V \oplus \Lambda^4 V$ branches exactly as $Q \oplus u^c \oplus e^c \oplus d^c \oplus L$. It has the three one-Higgs invariant lines, cancels all five gauge and mixed anomalies, and contains four weak doublets per generation. The determinant balance, Spin lift, non-vacuum exterior package, and center/deck descent are receipt-checked under the response and matter contracts. The scalar scan fixes compatible charges and Yukawa channels, not $H = W$ or scalar multiplicity. The exterior census uses left Weyl fields Q, u^c, d^c, L, e^c. Four-component Yukawa terms instead use the charge-conjugated singlets u_R, d_R, e_R: $\bar{Q}\tilde{H}u_R$, $\bar{Q}Hd_R$, and $\bar{L}He_R$. In $q = 6Y$ units their charges are respectively $-1-3+4 = 0$, $-1+3-2 = 0$, and $3+3-6 = 0$. These identities and the singlet conjugation dictionary are kernel-checked against the same exterior table. They fix notation and charge compatibility; they do not determine Yukawa coefficients or a spacetime fermion action. Laboratory identification of the current and flux sectors, exclusion of extra light sectors, attachment of the A_5 face representation to physical families, scalar attachment/dynamics, and continuum QFT are separate premises. The presentation-invariant normal-form framework of Ref. [2] isolates the same-source confluence, cross-source boundary-identification, liveness, and refinement criteria used upstream. It quotients hidden coordinates, labels, worker layout, and ancillas when they leave observer-visible records unchanged. Visible port incidence, topology, response maps, and repair laws are physical branch data. The framework supplies no particle-sector selector or numerical particle output; every such claim below is conditional on the specialist gauge branch and the particle-specific receipts.

The particle branch is a one-fixed-point forward reconstruction problem with an explicit closure matrix. It propagates one dimensionless pixel ratio P through the spectrum. The construction does not emit photon, gluon, or graviton particle masses from symmetry labels. It records their conditional classical carrier modes together with the quantum-particle certificates those modes consume as premises, while the Higgs candidate is on its declared quantitative surface. The dimensionless hierarchy implication is conditional, and the companion zero-defect square is a declared-map identity rather than a source-derived naturality result. The restricted quark source-spread non-identifiability theorem, the common-scale rejection of its reciprocal-ray candidate, and the weighted-cycle neutrino branch are visible as obstruction or comparison surfaces. Target-anchored quark mass textures, the mixed-convention formula diagnostics, and compare-only absolute neutrino attachments are withheld from public prediction tables. The P-closure root, the electroweak W/Z rows, charged leptons, and hadrons obey the sector-specific boundaries stated below.

1.1 Structural particle content and its attachment boundary

On the declared branch, particle carrier roles are constrained in three separate layers:

1. overlap consistency on the observer-patch network;
2. the A1/A2 fixed-space theorem for the local Standard Model gauge Lie algebra, followed by the conditional matrix current, rank-15 matter table, anomaly balance, and tensor descent;
3. the conditional rank-three screen selection, its abstract tensor product with the generation table, and the separate declared local operator context.

The first layer says what kinds of local data can be compared consistently across neighboring patches. The second forces the local Lie algebra. Its declared matrix and matter realization gives the exact charge lattice, color carrier, common $\mathbb{Z}_6$ kernel, and maximal faithful image conditionally. The third separates the finite carrier theorem from its declared physical completion. The result is an exact contract-conditional finite carrier and one-generation matter packet. The selected response band and generation table give a conditional rank-45 candidate. The separate local-domain branch verifies the declared tensor-identity operator without selecting it from the source. A physical particle interpretation requires source binding, laboratory current identification, matter-pole and continuum attachment, interaction, scalar attachment/dynamics, and QFT receipts.

The transportable-sector/Tannaka route is a separate conditional compact-group reconstruction. Physical source binding of the finite current and equality with the Tannaka current are premises of any physical identification. The conditional rank-45 candidate does not become a physical family without source-selected local action, cross-domain Spin transport, matter-pole, continuum, seam-selection, persistence, and symmetry-descent receipts.

Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and Deriving Standard Model Gauge Structure from Observer Overlap Consistency [9, 10] is therefore central to this paper. Complete reversible response and endogenous transport force its local gauge Lie algebra

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

The separately declared matrix and matter realization gives the maximal faithful image and color count

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3.$$

The same conditional packet contains one scalar-doublet channel and a rank-15 internal matter witness. The charge lattice and color count use the stated matrix and matter premises. The quotient is the maximal faithful image of that representation, while selection of the physical global form is a separate premise. Separate intrinsic-CP and weak-sector premises give $3 \leq N_g \leq 5$. The target-free matter reduct leaves the three-generation value unselected. On the strengthened screen interface, exact cost minimization selects the canonical rank-three band and the declared simulator realizes its response residue. Its tensor product with the declared generation table is a conditional rank-45 candidate. A separate finite local-domain receipt checks a declared tensor-identity operator. It does not select that action or identify the local domain with the Spin packet. Physical matter-pole and continuum attachments require separate derivations.

On that selected one-Higgs branch the scalar carrier has a canonical local screen-chart model. Let $C_{\mathrm{EW}} \cong \mathbb{CP}^1$ be the support-visible electroweak chart and fix the positive Hopf line-bundle convention by the neutral component condition $Q(\phi^0) = 0$. Borel–Weil gives

$$H_{\mathrm{OPH}} = H^0(C_{\mathrm{EW}}, \mathcal{O}(1)) \cong \mathbb{C}^2,$$

the first nontrivial holomorphic section space. With OPH's hypercharge and $\mathbb{Z}_6$ normalization this is exactly the $(1,2)_{1/2}$ Higgs carrier. Projectivization classifies a nonzero section direction as a point of $\mathbb{P}(H_{\mathrm{OPH}}) \cong \mathbb{CP}^1$, but it forgets the scalar hypercharge phase. For the lower-component vacuum vector

$$\phi_0 = \frac{v_{\mathrm{SM}}}{\sqrt{2}}\begin{pmatrix}0\\1\end{pmatrix}, \qquad v_{\mathrm{SM}} \neq 0,$$

one has

$$e^{i\alpha T_3}e^{i\beta Y}\phi_0 = e^{i(\beta-\alpha)/2}\phi_0.$$

Consequently $[\phi_0]$ has the projective two-torus stabilizer $\mathrm{U}(1)_{T_3}\times\mathrm{U}(1)_Y$, modulo the inherited finite center, whereas the vector ϕ_0 is fixed only when $\beta = \alpha$, locally. Its connected stabilizer is therefore the electromagnetic diagonal $\mathrm{U}(1)_Q$, generated by $Q = T_3 + Y$. Thus the projective geometry explains the carrier-ray classification, while the chosen nonzero vector supplies the symmetry-breaking statement. This explains the representation, charge, and symmetry-breaking geometry of the one-Higgs slot. It does not derive m_H, the quartic, v_{SM}, or Coleman–Weinberg dynamics; the weak scale and Higgs/top quantitative surface are separate parts of the OPH quantitative branch. The stored $\epsilon_H^{\mathrm{decl}} = 0$ row is only a declared-map commuting-square identity and does not close Higgs naturalness.

The symmetry-breaking geometry of that slot carries a machine-checked composition (`Lean/Screen/ElectroweakBreakingComposition.lean`). One typed premise bundle extends the committed structure premises by five declared finite constants: a quartic coupling $\lambda > 0$, a model-shell radius $r > 0$, two coupling constants $g, g' > 0$, and one Yukawa coefficient y. On the doublet carrier whose complex dimension and $q = 6Y$ charge are the committed table entries $(2,3)$, one theorem on that bundle proves five exact statements. The global minimizer set of the declared potential $\lambda(|h|^2 - r^2)^2$ is exactly the nonzero shell $|h|^2 = r^2$. The stabilizer of the chosen minimum $(0, r)$ inside the four-parameter electroweak direction space is exactly one unbroken line, the (g', g) mix of the third weak and hypercharge directions. The potential along every ray from the minimum is an exact polynomial identity whose quadratic coefficient is the second variation; that coefficient is positive semidefinite, and its kernel equals the set of infinitesimal gauge images at the minimum, with exactly three real parameters. The induced mass form vanishes exactly on the unbroken line and takes the normalized values $g^2r^2/4$ and $(g^2 + g'^2)r^2/4$ on the charged and neutral broken directions, so their tree ratio is $g^2/(g^2 + g'^2)$ in the declared constants. The epsilon-paired Yukawa line produces at the minimum a bilinear with exact coefficient yr, vanishing exactly when $y = 0$ or $r = 0$, and either committed parity sector supplies a colorless doublet/singlet row pair whose committed charges balance against the scalar row charge or its conjugate. The five constants are declared bundle parameters: no committed theorem selects λ, r, g'/g, or y, the committed invariant-form receipts leave the relative coupling normalization free, and the composition supplies no source action, physical vacuum, measured mass, or family structure. The quantitative electroweak and Higgs/top surfaces are the separate branches described in their own sections. If the finite minimum is identified with the conventional vector displayed above, then $r = v_{\mathrm{SM}}/\sqrt{2}$. Translating the finite coefficients into physical mass formulae additionally requires the scalar kinetic and generator normalizations and a physical vacuum attachment, neither of which is supplied here.

The neutral sector of that composition carries a machine-checked coupling dictionary (`Lean/Screen/NeutralCurrentDictionary.lean`). A weak slot is a row of the committed component table together with a weak component index: its third isospin is $\pm\frac{1}{2}$ on the two components of a doublet row and 0 on a singlet row, its hypercharge is the committed $q = 6Y$ charge column, and its electric charge is the sum $Q = T_3 + Y$ in the committed integer units. The

neutral coupling of a slot along a direction is the diagonal action $g a_3 T_3 + g' b\, Y$, and on every neutral direction the committed infinitesimal action is exactly diagonal with the two slot couplings of the scalar row, so the normalization is the committed one, with no freedom left. The unbroken direction of the breaking composition, normalized by the common direction length $g_Z = \sqrt{g^2 + g'^2}$, couples every slot of either committed parity sector at exactly $e\,Q$ with $e = gg'/g_Z$, and the neutral broken direction, normalized the same way, couples it at exactly $g_Z(T_3 - Q \sin^2\theta)$, where $\sin^2\theta = g'^2/(g^2 + g'^2)$ is a defined abbreviation in the declared constants and never a measured value. The Z direction is exactly the orthogonal complement of the unbroken line in the neutral plane, two committed rows with different electric charges have provably different photon couplings, and in either parity sector the zero-charge component of the colorless doublet row has photon coupling exactly zero and Z coupling exactly $\pm(g^2 + g'^2)/2$. The dictionary is exact tree-level algebra on the committed grammar: it supplies no physical current operator, source action, mixing angle, or decay amplitude, and the chirality and source-action attachments are not supplied.

The charged sector of the same composition carries the companion machine-checked dictionary (`Lean/Screen/ChargedCurrentDictionary.lean`). The two charged directions are orthogonal to the photon and Z directions and to each other under the declared pairing, span exactly the orthogonal complement of the neutral plane, and both carry the broken mass value $g^2 v^2/4$. Their two complex combinations are, in exact closed form, the strict shift operators that move the lower weak slot of a doublet row to the upper one and back with coefficient g; normalizing by the derived direction length $\sqrt{2}$ of the charged pair gives the coupling $g/\sqrt{2}$, a constant derived from the committed pairing rather than inserted, pinned to the committed infinitesimal action on the scalar row, and any drifted coefficient breaks the alignment identity. At positive g a nonzero charged coupling exists exactly on the committed partner pairs, the two weak components of a doublet row, and committed singlets are annihilated; every nonzero coupling raises the committed electric charge by exactly one unit of the $q = 6Y$ normalization, from the committed charge columns alone. The dictionaries compose: the commutator of the photon-direction action with the raising and lowering shifts is exactly $\pm i\, g_Z e$ times the shift, so the charged directions carry exactly one declared electromagnetic unit per committed photon normalization; the commutator of the two charged actions lands on the photon and Z actions with exact coefficients $-gg'/(g^2 + g'^2)$ and $-g^2/(g^2 + g'^2)$; and the raising/lowering commutator is $2gi$ times the third-weak action. These joins are exact operator-commutator identities of the committed action on the scalar-row carrier; no abstract bracket on the direction space is committed. The dictionary supplies no physical current operator, propagator, decay amplitude, or flavor content, and the same chirality and source-action attachments are not supplied.

A kinematic internal-clock packet is machine-checked on the same declared Hermitian Lorentz module that carries the mass-shell receipts (`Lean/Geometry/InternalClockRestFrequency.lean`). For one declared positive mass parameter and one declared plane-wave phase form, the rest phase along a frame worldline advances at exactly the mass parameter in every oriented chart, the declared internal clock obeys the exact rotation law $i\,\partial_\tau z = m\, z$, and its period set is exactly the $2\pi/m$ integer lattice, the frequency-rigidity identity for the declared clock. The spatial phase gradient reads off the momentum coordinate on the de Broglie orientation of the phase, and the temporal rate reads off the energy coordinate on the opposite invariant orientation, tied by the committed shell identity, and the zero-mass branch degenerates exactly: the clock is constant, every real number is a period, and no oriented chart brings a future null vector to rest. The oscillation is a declared plane-wave form on a declared module: no localized particle state, self-starting oscillation, source-produced state, laboratory frequency unit, or Compton value is derived, and the clock-calibration and physical-spacetime attachments are not supplied.

The declared plane-wave functions have pointwise amplitude-modulus invariance under their

declared free evolution (`Lean/Geometry/FreeEvolutionPersistence.lean`). The free evolution is the translation flow along a declared frame worldline, acting on wave functions by precomposition, and it is an exact one-parameter group. Under this flow a plane-wave function changes by a unit-modulus global phase only, so its mass-shell membership, its momentum and energy readouts, and its unit modulus are exact constants of the flow, and two functions with distinct positive mass or momentum labels stay distinct at every parameter value. Along the wave's own frame the global phase is exactly the internal clock, and this identity is a clause of the composed machine-checked receipt. On the zero-mass branch the massive parameterization degenerates to the constant function and the frame label is not recoverable, while a future null vector carries the shell, global-phase-covariance, and pointwise-modulus clauses of the packet on the zero-mass shell. This is pointwise algebra for a declared kinematic wave under a declared flow, separate from the record-repair stability results: no physical particle, decay rate, localized wave packet, Hilbert-space normalization, survival probability, or laboratory persistence time is claimed, and the clock calibration, spacetime attachment, and source production premises are not supplied.

The gyromagnetic ratio of the spin sector has an exact algebraic origin (`Lean/EventAlgebra/PauliMinimalCoupling.lean`). Over any complex algebra, for a declared triple X with commutators $[X_i, X_j] = i\,q\,\epsilon_{ijk}B_k$, the square of the Pauli contraction is exactly

$$(\sigma\cdot X)^2 = X\cdot X - q\,(\sigma\cdot B), \qquad X\cdot X := \sum_i X_i X_i.$$

This ordered algebraic sum is not an adjoint norm and carries no positivity claim. With a separately supplied Hermitian kinetic-momentum realization, the spin sector inherits the full coupling q where the orbital convention assigns $q/(2m)$ per unit angular momentum: gyromagnetic ratio two. The commutation data, the coupling, and the carrier are declared. A commuting triple has identically vanishing spin term, the Pauli matrices themselves inhabit the hypothesis class with $q = 2$ and a nonzero spin term, and the coefficient is pinned on that inhabitant. No gauge potential, Dirac equation, Spin structure, laboratory calibration, or spacetime attachment is constructed, and no anomalous-moment content is claimed.

This also delimits the phrase "and no others." The finite OPH claim concerns the low-energy packet implied by the stated response, matter, and descent contracts. Additional connected gauge generators, additional light Higgs multiplets, extra light chiral families, or low-energy supersymmetric partners are absent from the declared completion because that completion declares them absent. This is a declaration, not source-derived exhaustion of physical sectors. Propagating gauge-particle existence requires the complete primitive data, action, phase, and quantum-pole premises below.

The individual families then emerge for different reasons. The realized electromagnetic, color, and dynamical-metric branches identify connection and metric carrier roles. Explicit Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic actions then yield classical transverse or TT massless modes on their stated backgrounds and phases. Quantum photon, gluon, or graviton states require the separate physical-Hilbert-space, pole-residue, and asymptotic/phase receipt. The weak bosons arise when the electroweak gauge sector is propagated through its quantitative closure branch. Quarks and leptons arise from the conditional chiral matter packet together with the rank-three screen selection. Their physical three-family interpretation requires the conditional rank-45 tensor candidate to be identified with physical matter-pole residues and continuum fields, with the local matter action and cross-domain Spin transport selected. Their family splittings are then read from the deeper overlap-transport and excitation machinery, not from a second arbitrary postulate that says "copy the family three times and assign masses by hand." Once color is realized and confined, stable hadrons are composite readout channels of the quark/gluon sector.

The role of the pixel constant P is also important to state clearly. Here, P does not decide whether an electromagnetic particle pole exists, whether there are three colors, or whether the candidate family band is physically attached. The electromagnetic representation role, the color carrier, and the declared generation completion belong to the finite packet before the quantitative closure step. What P does is set the shared quantitative scale on which receipt-certified content is read out numerically. So the logic of the paper is:

$$\begin{aligned}\text{observer consistency} &\to \text{realized gauge/matter branch}\\ &\to \text{representation roles} \to \text{conditional action-level carrier modes}\\ &\to \text{common scale } P \to \text{receipt-gated quantitative spectrum.}\end{aligned}$$

That is how the derivation explains why a universe with the realized branch and the shared pixel scale exhibits this particle zoo.

1.2 Logical basis and reading rule

The particle derivation uses the same premises A1–A3 as the SM/GR derivation in Ref. [9, 10]. They are the oriented observer-patch federation and spherical support, observer agreement, and conditional maximum randomness. On those premises, the SM/GR derivation in Ref. [9, 10] supplies the structural chain used here: the local Standard Model gauge Lie algebra forced by complete response and endogenous transport; the conditional maximal faithful matter image, hypercharge lattice, and color triplet $N_c = 3$ under explicit matrix and matter contracts; a separate receipt-conditional Tannaka classification; and the conditional generation window. Under the named complete-band and operational-cost premises, an exact screen theorem selects the rank-three response band, and the declared unitary response places its residue at the lowest positive generator frequency. Extra-light-sector completeness, laboratory current identification, and physical attachment of that finite band to three matter families are premises the chain does not supply. Refs. [5, 6] supply the patch-net, repair, measurement, and regulated screen language for flavor transport and observer-facing readout.

The quantitative particle side adds one local closure variable. The common pixel ratio P, reported by the incomplete outer/inner declared map of the flagship synthesis [3], feeds the forward electroweak map and its descendants. The public empirical comparison branch displays

$$\alpha^{-1}(0) = 137.035999177(21), \qquad \alpha(0) \simeq 0.00729735256433, \qquad P_C \simeq 1.6309682094.$$

The comparison pixel P_C is defined from the measured endpoint; it is not a source-root coordinate. The computation has a fixed source order: golden-ratio entropy balance gives φ, boundary Gaussian normalization supplies the $\sqrt{\pi}$ width, a trial P feeds the source map through unification, running, and electroweak anchoring, and Ward-projected electromagnetic transport gives the Thomson endpoint used by the outer/inner pixel fixed point. The declared numerical map fixes a unique root on its certified interval. The map is incomplete because no derivation identifies its numerical endpoint with the physical electromagnetic readout. The first-principles diagnostic trunk records the certified coordinates

$$\begin{aligned}P_{\mathrm{fwd}} &= 1.630972095858897\ldots,\\ \alpha^{-1}_{\mathrm{root}} &= 136.994835177413\ldots.\end{aligned}$$

The root is the interval-certified unique fixed point of the declared numerical map. The approximately 0.041 inverse-alpha difference to the public Thomson endpoint is a residual of that

incomplete map. It neither derives a QCD/hadronic contribution nor identifies a physical endpoint relation. The QCD-free hierarchy witness is the cleaner first-principles stress test. Combining the source/root value with the finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel, $\alpha_U(P_C)$, gives the mixed-provenance no-hadron diagnostic $A^{\mathrm{fp}}_{\alpha_U} = 137.0359595136\ldots$, below the Thomson endpoint. It is mixed-provenance comparison bookkeeping rather than a source-only fine-structure prediction. The certified self-consistent gauge-width fixed point is $\alpha^{-1} = 137.035660136946577\ldots$; the mixed diagnostic combines the inner value at P_{fwd} with α_U at the CODATA-derived comparison pixel and is excluded from the physical result. The detailed endpoint table appears in Section 4. The separate proposed cosmic record-closure target $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, with $N_{\mathrm{CRC}} = \log D_\star$, defined through the correctable code of the reachable public record atoms under globally coupled checkpoint kernels, belongs to the cosmological-capacity branch and supplies no particle theorem. The Newton coupling uses the separate selected no-G scale certificate $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$, so the particle branch does not derive G by back-solving the pixel ratio. A fixed-$D = 24$ public checkpoint packet, carrier representation, whole-fiber scalarization, and finite extension and refinement receipts are exact. A bounded all-rung counterfamily has incompatible zero sets under shared base, positivity, carrier, and executable finite controls. The exact arithmetic establishes nonidentifiability for that bounded completion class. Universal all-rung membership in the complete A1–A3 capacity-source contract and an executable-to-Lean bridge are absent. Direct N is not evaluable on the incomplete source antecedent, and the stronger source-class verdict does not follow. A positive direct result requires completion of that antecedent, one physical slack zero, and a universe-level carrier attachment. Its identification with the electroweak bridge is conditional on a positive, unital, refinement-natural identification of the screen load with the electroweak load; the bridge value $N_0 = 3.5321315434\ldots \times 10^{122}$ on the source-forward branch is about 6.6 percent above the weighted Planck base-ΛCDM capacity $N_\Lambda = 3.3126\ldots \times 10^{122}$. Under the separately declared one-class finite-presence attachment, this becomes $3.2920978773\ldots \times 10^{122}$; the distinct Poisson carrier gives $3.3000722254\ldots \times 10^{122}$. Their respective residuals are -0.63 and -0.39 percent. Exact positive compositional completions of the same local datum select neither global action nor a blocked-event semantics. The named-law branch is therefore not evaluable on this source class. Both comparisons are retrospective. If the missing physical identifications are supplied, the pair $(P_\star, N_{\mathrm{CRC}})$ determines dimensionless curvature products such as $\Lambda_\star a_{\mathrm{cell}} = 3\pi P_\star/N_{\mathrm{CRC}}$, not the SI scale product $\Lambda_\star N_{\mathrm{CRC}}$ by itself. The rounded $N_\Lambda \simeq 3.313 \times 10^{122}$ display is a cosmological central-value label, not the high-precision input for G. Structural carriers, quantitative outputs, conditional calculations, and continuation calculations therefore inherit the declared theorem premises, with particle-facing quantitative burden carried by the local declared-map root $P_\star$. The public fine-structure display row is a comparison endpoint conditioned on the measured Thomson value. The first-principles diagnostic row and empirical hadron closure rows are separate support records. Section 4 records that boundary in full.

2 P-Closure and the Reverse-Engineering Strategy

The reverse-engineering claim is easiest to state plainly. In ordinary phenomenological use, the Standard Model does not derive the observed particle masses and mixings from one common microscopic constant. It is usually presented with 19 free parameters in the minimal massless-neutrino theory, and with at least 26 once neutrino masses and mixing are included, depending on neutrino-sector conventions. OPH organizes that situation around one universal pixel ratio P from the outer/inner closure relation described in the flagship synthesis. Whether any parameter compression is realized follows from counting the declared settings against the independently selected basins. The same P

must drive all downstream bosonic, quark, lepton, neutrino, and hadronic branches.

One clarification matters. The broader self-closure formulation uses the universe-level equation

$$N = \log M_0(\mathfrak{U}_N),$$

where M_0 is the multiplicative size of the largest correctable public record code. Its selector-free finite form is stable global screen capacity

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star,$$

for a universe supplied with logarithmic capacity N. The observers paper [4] specifies the finite readback directly by $M_0(q) = \alpha(G_q)$, the independence number of the compound confusability graph of the reachable public record atoms under the globally coupled checkpoint family. Under whole-fiber scalarization and a faithful capacity carrier it is deflationary; under confusability-reflecting capacity extension it is monotone, so top-down iteration reaches the greatest fixed point on a declared finite chain. The fixed-$D = 24$ packet is exact inside its declared finite source category. A bounded all-rung counterfamily has incompatible fixed sets under shared base agreement, positivity, the carrier bound, and executable finite controls. This establishes nonidentifiability for the bounded completion class. It does not establish universal all-rung membership in the complete A1–A3 capacity-source contract or an executable-to-Lean bridge. Direct N is not evaluable on the incomplete source antecedent, and the stronger source-class verdict does not follow. A positive result requires a complete source antecedent, one physical slack zero, and a universe-level carrier attachment. The electroweak identification additionally requires the common screen/electroweak load-carrier hypothesis. The conditional bridge value $N_0 = 3.5321315434\ldots \times 10^{122}$ is about 6.6 percent above the weighted Planck base-ΛCDM capacity $N_\Lambda = 3.3126\ldots \times 10^{122}$, a propagated 2.4 to 2.5 one-dimensional sigma. The conditional finite-presence and Poisson branches instead lie 0.63 and 0.39 percent below that coordinate, respectively. Exact positive composition countermodels select neither global action from the finite source. Their physical carriers are distinct and both comparisons are retrospective. Read through $\Lambda_\star \ell_\star^2 = 3\pi/N$, the same bridge is the zero-dial relation $\Lambda \ell_P^2 = 3e^{-6\pi/(P\alpha_U(P))}$ only after the independent capacity, horizon, and common-load receipts. Neither reserve factor is part of the common-load theorem itself. The same exponent runs the weak hierarchy of this paper. The particle-spectrum derivation studied here uses the local declared-map root $P_\star$ in place of a long particle-by-particle parameter list.

Observation is allowed to supply a branch hint. That is a consequence of the OPH picture, where the universe is a closed fixed structure and internal observers can read approximate coordinates from experiments. The proof standard is stricter: the exact $P_\star$ used by the particle branch is computed by the self-referential numerical pixel equation, and the root is unique on the declared interval. A further derivation is required before that root is a physical endpoint. The pixel equation and its uniqueness statement are carried by the shipped interval contraction certificate: the closure map is a self-map with a derivative bound certified by interval arithmetic on the declared interval. The domain-global statement is discharged as well: the companion domain-global certificate bounds $\sup|g'| < 1$ on every piece of a 256-piece subdivision of the declared numerical domain ($\alpha^{-1} \in [100, 200]$, both readout maps, empty exceptional set), so each declared readout map has exactly one fixed point on that numerical domain. This certificate does not provide the missing physical endpoint relation.

2.1 Why a single P matters

The structural theorems determine an exact contract-conditional finite representation packet, but not *what physical particle world* realizes it or what absolute scale in GeV it occupies. The overlap,

modular, gauge, anomaly, and admissibility arguments give the declared Standard Model packet, the exact hypercharge pattern, and a three-color carrier. The generation window is conditional; separate band premises select rank three. Tensoring the selected response band with the declared generation table gives a conditional rank-45 candidate. A distinct local-domain receipt checks a declared tensor-identity operator and conditional gap inheritance without source-selecting the action or transporting the Spin packet. Physical matter-pole and continuum interpretation is a separate premise. The carrier-mode theorem is an additional action-level result and does not emit zero-GeV quantum particles. These structural results do not by themselves tell us the numerical values of the W boson mass, the Higgs boson mass, or the up-quark mass.

The role of the pixel closure is to supply one common quantitative scale variable for the entire downstream spectrum. In the flagship synthesis, the fine-structure branch asks for the nonzero detuning of a holographic screen cell such that the cell's outer geometric displacement from perfect self-similar equilibrium equals the electromagnetic observation scale emitted by the encoded branch under the common physical carrier identification. That identification is a separate receipt. The outer side of the closure is

$$P = \varphi + \alpha_{\mathrm{in}}(P)\sqrt{\pi}.$$

The first-principles computation is a five-step source chain. First, the golden-ratio entropy balance of the local screen cell supplies $\varphi = (1+\sqrt{5})/2$. Second, boundary maximum-entropy normalization fixes the width of the leading electromagnetic detuning to $\sqrt{\pi}$. Third, a trial P is sent through the source map

$$M_U(P) = E_P e^{-2\pi} P^{1/6}, \qquad E_{\mathrm{cell}}(P) = \frac{E_P}{\sqrt{P}},$$

followed by the heat-kernel closure

$$\bar{\ell}_{\mathrm{SU}(2)}(t_2(P)) + \bar{\ell}_{\mathrm{SU}(3)}(t_3(P)) = \frac{P}{4},$$

which selects $\alpha_U(P)$ and the running family $\alpha_i(m_Z; P)$. Fourth, electroweak mixing gives the source anchor

$$a_0(P) = \alpha_2^{-1}(m_Z; P) + \frac{5}{3}\alpha_1^{-1}(m_Z; P).$$

Fifth, Ward-projected $\mathrm{U}(1)_Q$ transport gives the Thomson endpoint

$$A_T(P) = T_Q(a_0(P), F_{\mathrm{src}}(P)) = \alpha_{\mathrm{em}}^{-1}(0; P),$$

and the cell closes when

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}.$$

The numerical solve is therefore a root problem for

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)}.$$

The branch interval is localized from the observer-facing data, the solver evaluates $A_T(P)$ from the declared source map at each candidate point, and the accepted value is the unique zero of H on that interval. The measured endpoint can locate the interval. It cannot replace the root condition. This proves uniqueness only for the declared numerical map. The derivation connecting its root to the observer-supporting physical electromagnetic endpoint is missing. The CODATA-conditioned comparison readout $\alpha^{-1}(0) = 137.035999177(21)$ gives

$$P \simeq 1.6309682094.$$

The same equation is the local ruler for the downstream particle rows. The five-layer OPH diagnostic trunk emits the certified source-side diagnostic point

$$P_{\mathrm{fwd}} = 1.630972095858897\ldots,$$
$$\alpha_{\mathrm{root}}^{-1} = 136.994835177413\ldots.$$

The root is an incomplete declared-map output, not a physical fine-structure prediction. The detailed table records how the diagnostic trunk, endpoint residual, and source-spectral payload fit together. A separate hardware note reports an optical-cavity check of the same declared geometry. It is a non-discriminating engineering check and supplies no physical endpoint. Once the diagnostic map coordinate is set, the particle question becomes whether all downstream readouts can be written as

$$X_j = G_j(P)$$

with no new sector-specific constants inserted by hand. This paper studies exactly that downstream map.

2.2 How the electroweak branch works

In the implementation used here, the quantitative electroweak branch is the first major readout from P. Its algebraic output family is

$$\{M_W^{\mathrm{chart}},\ M_Z^{\mathrm{chart}},\ \alpha_{\mathrm{em}}^{-1}(q^2),\ \sin^2\theta_W(q^2),\ v\},$$

evaluated from one declared input P on the printed running/matching/threshold/scheme package. The superscript chart marks a mass-chart coordinate, not a certified complex pole. The construction is straightforward in spirit: start from P, build the electroweak running family, select the physical carrier point on that family, and read off the W boson and Z boson pair from that selected point. The Higgs/top critical stage inherits that same electroweak core and does not introduce a new free-input sector. On the declared Ward-projected transport branch, the low-energy electromagnetic row is represented by the Thomson endpoint

$$\alpha_{\mathrm{Th}}^{-1}(P) = \lim_{q^2\to 0} \alpha_{\mathrm{em}}^{-1}(q^2; P)$$

of that same electromagnetic transport family. The forward logical order on this branch is:

$$P \longmapsto \big(M_U(P), E_{\mathrm{cell}}(P)\big) \longmapsto \alpha_U(P) \longmapsto \big(t_U(P), t_{\mathrm{tr}}(P)\big)$$
$$\longmapsto \big(t_2(P), t_3(P), v_{\mathrm{chart}}(P)\big) \longmapsto \alpha_i(\mu_*; P).$$

Here $v_{\mathrm{chart}}(P)$ is the mass-chart coordinate. It is not a renormalized vacuum expectation value in any fixed scheme, and no receipt identifies it with one. The forward transmutation certificate records that the same runtime basis reconstructs the unified diffusion parameter $t_U(P) = 4\pi^2\alpha_U(P)$ and transmutation exponent $t_{\mathrm{tr}}(P) = 2\pi/((N_c+1)\alpha_U(P))$ as the pixel-closure solve itself. The runtime subgraph reads measured electroweak data only for validation. No frozen provenance receipt excludes target dependence or selects one repair law.

The electroweak branch is the place where the named pixel coordinate organizes the weak-sector rows. The reference-fitted inverse adapter displays

$$M_W = 80.3625\ \mathrm{GeV}, \qquad M_Z = 91.1879\ \mathrm{GeV}.$$

The selected-carrier chart, the runtime-target-separated formula candidate, and the reference-fitted coherent repair diagnostic are recorded separately in the support table below.

2.3 What the electroweak branch fixes

Under the declared electroweak running, matching, threshold, and scheme premises, the pixel variable P supplies the source basis $(\alpha_U, \alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\text{source}}, v)$ used by the candidate chart. The displayed value law is an exact implication of the five-part quotient-path certificate stated below, but the finite carrier does not emit that certificate. The candidate law evaluates the mass-side pair (W, Z) together with the running-family anchor (a_0, s_0, v). The electromagnetic row is physically read only after Ward projection to the unbroken $\mathrm{U}(1)_Q$ channel; its low-energy value is the $q^2 \to 0$ endpoint of the same transport family. The derivation also includes two explicit comparison calculations with distinct provenance: the exact selected-carrier chart and the reference-fitted coherent repair diagnostic. The same support logic applies to the flavor branches, whose premises differ. The electroweak quantitative branch states its source payload, same-scheme remainder, and interval certificate explicitly.

For empirical comparison, a documented piecewise $e^+e^- \to$ hadrons compilation, built from resonance parameters and perturbative QCD, evaluates the subtracted dispersion integral to $\Delta\alpha^{(5)}_{\text{had}}(m_Z) = 0.027609 \pm 0.000112$. Inserting it into the endpoint map with the frozen source anchor and lepton transport packet gives $\alpha^{-1} = 136.3827548175$ on $[136.3670480603, 136.3984651934]$, at $P = 1.6310415204$, on the empirical-closure row class. The comparison to the measured endpoint sits in an explicitly compare-only block: the measured value lies outside the interval, and the certified same-scheme anchor gap is $[0.6198609041, 0.6505569679]$ inverse-alpha units. The anchor $a_0(P) = \alpha^{-1}_{\text{em}}(m_Z^2; P)$ is the one-loop renormalization-group value run from the OPH unification scale to m_Z; the physical five-flavor on-shell value is 128.939, and the difference from the OPH anchor 128.308 is 0.631, at the lower edge of the certified gap. The gap records transport missing from the one-loop anchor; it is not evidence by itself that a source chain closes. A source-only endpoint needs the OPH hadronic spectral measure, a target-independent choice among the admitted maps, and a typed construction identifying the two closure readings as one quantity. The empirical endpoint is a comparison calculation and supplies none of those objects. The exact exterior weak multiplicity is four. Its use as the transmutation factor $\beta_{\text{EW}} = 4$ is conditional on the common screen/electroweak load-carrier identification; overloaded β-ratios appear only on benchmark readouts and are not part of the theorem contract.

3 Results at a Glance

The particle derivation carries the local pixel scale into the directly comparable rows below. Detailed scope statements are collected in the support sections that follow.

Chain	OPH output	Role
P-closure and fine structure	certified incomplete-map root 136.994835177413 . . . (certified source-root row); mixed diagnostic 137.0359595136 . . .; certified gauge-width fixed point 137.035660136946577 . . .; empirical endpoint $136.3827548175 \in [136.3670480603, 136.3984651934]$; measured 137.035999177(21)	distinct support classes: incomplete declared-map output, mixed diagnostic excluded from physical output, empirical hadron closure with anchor gap $[0.6198609041, 0.6505569679]$, and measurement
Classical carrier modes	two transverse Maxwell modes; $2 \dim G$ perturbative pure-Yang–Mills modes; two pure-Einstein TT modes	conditional quadratic-action results; quantum particles require separate Hilbert-space, pole, and phase premises

Chain	OPH output	Role
Electroweak W/Z	target-free running-chart coordinates (80.330, 91.119) GeV; strict one-loop consumer and separate scalar analytic receipts; candidate value-law coordinates (80.3770000154, 91.1879780779) GeV	the analytic receipts exclude zeros on declared principal-sheet boxes and isolate, for each of W and Z, one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. They identify neither chart with the physical resonance sheet and prove no source selection, unique continuation, sign bridge, full-matrix Laurent residue, current amplitude, independent numerical replay, or W/Z mass comparison
Electroweak hierarchy and declared-map square	$v/E_\star$ on each named pixel branch $N_{\mathrm{EW}}(P_C) \simeq 3.53235 \times 10^{122}$ $N_0(P_{\mathrm{fwd}}) \simeq 3.53213 \times 10^{122}$ $\mathcal{B}_{\mathrm{EW}} = 0,\ \epsilon_H^{\mathrm{decl}} = 0$	selected dimensionless bridge; $\epsilon_H^{\mathrm{decl}} = 0$ follows from the declared comparison maps and is not a source-derived naturalness result. Identifying either bridge coordinate with cosmic capacity requires a source-derived direct public-record closure with one selected physical capacity and the typed same-load bridge. Source-derived comparison maps followed by an independent residual evaluation are likewise required before assigning a physical naturality defect. The source-forward uncorrected value is 6.6 percent above the weighted Planck base-ΛCDM coordinate. The separately premised finite-presence and Poisson candidates lie 0.63 and 0.39 percent below it. Exact compositional countermodels select neither action. Both reserve comparisons are retrospective; the horizon–record identification and $E_\star$ are separate premises

Chain	OPH output	Role
Higgs/top relation	double-criticality branch $(m_H, m_t) = (125.77, 172.63)$ GeV at the declared boundary-scale candidate $E_\star e^{-\pi} P^{-1/6}$ (two loops), with $m_H = 125.72$ GeV on the fit-free curve at the measured top; target-anchored declared-surface fit (125.1995304097, 172.3523553288) GeV kept separate; no source-only physical mass emitted	the criticality family has zero continuous parameters over the boundary scale. Its scale selection assumes that the boundary record has exactly two equal-depth anchor registers, one geometric and one transmutational, together with the declared Gaussian/affine record and port-additive repair premises. The target-anchored fit carries no predictive force. This branch does not supply the source root, physical scale, rigidity, provenance, uncertainty enclosure, or complex pole
Charged leptons	target-anchored witness values withheld; finite eight-path digital carrier and empirical transport intervals retained as diagnostics	the model proves schema satisfiability and a central record dilation, not physical source selection; the intervals use target-anchored ratios, measured α, and empirical transport
Selected-frame quarks	source-only predictions withheld; conditional six-mass replay in Section 9.6; reciprocal-ray product 0.835323 and held-out error 21.56% at M_Z	common-scale data reject the reciprocal-ray candidate; the generic interface has six scalars and requires a source-derived flavor-orbit selector
Neutrino absolute attachment	compare-only absolute masses withheld; scale-free and mixing comparison rows retained	weighted-cycle comparison branch
Hadrons	first-principles strong-binding descent plus empirical $e^+e^- \to$ hadrons closure surface	OPH hadron construction for first-principles masses; measured hadron data for empirical closure rows
Affine event-record stitching	certified affine event supports with separately typed conditioned-frame balls, followed by cross-boundary token continuation or AMBIGUOUS	conditional observer-record theorem; not a species, mass, or coupling input

Sector	Theorem scope	Derivation	Output(s)	Caveat / theorem boundary
Carrier roles and modes	structural group/content theorem plus conditional action theorem	explicit transverse/TT quadratic kernels with no derived particle mass	no derived particle mass	action, background, and phase are hypotheses; quantization, positive-residue pole, and asymptotic/deconfinement conditions are separate
Electroweak bosons	running-chart calculation, strict one-loop consumer, scalar analytic receipts, conditional quotient transport, and comparison adapter	declared zero-selector map; selected carrier → two-coordinate chart; complete renormalized packet → strict scalar consumer; finite quotient-path certificate → conditional value law	target-free running-chart coordinates (80.330, 91.119) GeV and candidate value-law coordinates (80.3770000154, 91.1879780779) GeV are chart outputs; the source does not select the map, and the analytic receipts concern an external fixture that is not composed with either chart	principal-sheet zero exclusion and one simple channel-specific lower-half algebraic-chart zero for each of W and Z do not identify a physical resonance sheet or supply a unique continuation, sign bridge, full-matrix Laurent residue, current amplitude, independent replay, or W/Z mass comparison

Sector	Theorem scope	Derivation	Output(s)	Caveat / theorem boundary
Electroweak hierarchy and declared-map square	conditional dimensionless bridge plus packaging identity	local $P \to \alpha_U \to v/E_\star$ hierarchy branch $\to$ exterior weak multiplicity $3+1=4$ $\to$ mathematical capacity bridge; separately declared RG/Higgs comparison maps $\to$ commuting-square identity; conditional screen branch: 12 ports and an oriented 24-slot register	$N_{\mathrm{EW}}(P_C) = 3.5323546226929906511\ldots \times 10^{122}$, $N_0(P_{\mathrm{fwd}}) = 3.532131543418936\ldots \times 10^{122}$, $\mathcal{B}_{\mathrm{EW}} = 0$, $\epsilon_H^{\mathrm{decl}} = 0$	Supports the dimensionless hierarchy implication under its named premises. The zero defect is forced only by the declared comparison maps; identical source antecedents admit noncommuting completions, so no source-derived naturality value is emitted. The 24-slot register does not produce four weak loads. Equality with cosmic capacity follows from self-reference only after a source-derived direct public-record closure supplies the cosmic reading and a typed common-load bridge identifies it with the screen and electroweak readings. A physical naturality test requires source-derived maps and an independent evaluation of both residuals. The finite-presence and Poisson factors require distinct extra premises and give retrospective comparison coordinates. An independent physical $E_\star$, the public Thomson endpoint, theorem-level W/Z, and the other listed mass rows are separate surfaces.
Higgs/top stage	double-criticality family from the gauge sector plus a target-anchored fit	criticality law $\lambda = 0,\ \beta_\lambda = 0$ at one source scale $\to$ fully constrained (m_H, m_t) family $\to$ declared boundary-scale candidate	no source-only mass; double-criticality branch $(m_H, m_t) =$ (125.77, 172.63) GeV, with $m_H = 125.72$ GeV on the fit-free curve at the measured top; target-anchored fit (125.1995304097, 172.3523553288) GeV	A physical mass requires the declared Gaussian/affine record and port-additive repair premises, the two equal-depth boundary-record anchors, a source root and physical scale, finite quotient-path selection, same-scheme running, target-independent rigidity, provenance, uncertainty enclosure, and a complex pole. The companion top coordinate is not a separate public top-mass prediction.
Quark family	source constraints, conditional constructions and common-scale rejection	one-scheme dimensionless Yukawas $\to$ reciprocal-ray test $\to$ six-scalar generic interface and flavor-selector boundary	source-only predictions withheld; conditional numerical rows in Sections 9.6 and 9.7; at M_Z, $\rho_u \rho_d = 0.835323$ and the endpoint-granted held-out error is 21.56%	The reciprocal-ray candidate is physically rejected across the tested running scales. The $(\mathbb{R}_{>0})^2$ theorem is a restricted lower-bound obstruction. Mixed-convention template and target-conditioned cumulant residuals are diagnostics. The full-entropy selector is excluded. Conditional renewal and exterior Higgs tensors are supplied; physical family-sensitive interaction and source-selected transport remain open
Charged leptons	conditional construction plus exact target-anchored witness	shared excitation dictionary $\to$ ordered charged carrier $\to$ exact centered readback $\to$ determinant-line lift on theorem-level physical charged data	exact centered readback; same-family target-anchored charged triple withheld from public prediction tables	public charged masses are not emitted from P. The theorem branch does not emit a theorem-level sector-isolated charged determinant exponent vector. It does not attach a source-side determinant character to the physical charged determinant line. The determinant-line lift and algebraic mass readout apply only on theorem-level physical charged data
Neutrinos	target-informed conditional candidate plus auxiliary checks	template family transport $\to$ same-label scalar certificate $\to$ weighted-cycle candidate $\to$ compare-only bridge and absolute attachment $\to$ shared-basis identity	declared scale-free candidate and PMNS/Majorana comparison rows; absolute mass attachments withheld from public prediction tables	exact linear algebra conditional on the declared template and selectors. The candidate fails the NuFIT 6.1 correlated $(\sin^2\theta_{23}, \delta_{\mathrm{CP}})$ profile, and neither the scale-free point nor the bridge invariant is a prediction or theorem [25]
Hadrons	strong-binding construction absent; empirical comparison contract	source-derived hadronic spectral backend contract, including quotient ensemble, source QCD parameter map, Ward current accounting, spectral exports, and empirical $e^+e^- \to$ hadrons payload schema	no first-principles hadron masses; empirical comparison rows are separate from first-principles OPH results	First-principles hadron prediction requires a working OPH hadron construction, such as GLORB/Echosahedron, with the Ward-projected two-current spectral measure, higher-point/transition spectral sectors, same-scheme remainder, and systematics. The empirical comparison uses separate $e^+e^- \to$ hadrons input and cannot establish the first-principles theorem

Sector	Theorem scope	Derivation	Output(s)	Caveat / theorem boundary
Affine event-record stitching	conditional certificate theorem	physically faithful source-causal continuum with affine charts and observer clock atlas → affine event support for each descended token and clock slice → separately typed conditioned-frame ball where required → positive event-location gap → overlap descent → real transverse interface-crossing germs → common-chart sector/gauge transport → ID-independent one-to-one assignment gap → coarse/fine contraction check	nonbranching event-record paths, rays, or isolated records when the location and stitch certificates pass; otherwise AMBIGUOUS, REJECTED, or INTERACTION REQUIRED	This conditional theorem concerns observer-visible records after the source-causal continuum certificate has supplied physical event charts. It does not derive particle species, masses, gauge charges, scattering amplitudes, or geodesic motion. Repeated implementation IDs, nearest-neighbor fits, frame coordinates used as event positions, and file-boundary coincidences are inadmissible evidence.

The theorem rows and target-anchored rows describe different objects. The theorem table carries a two-modulus quark non-identifiability result and a source-domain charged no-go boundary. The target-anchored rows contain same-family witnesses and diagnostic calculations. The electromagnetic, color, and metric carrier roles have conditional classical-mode receipts but no derived quantum mass rows; the W/Z numbers are convention-dependent chart and adapter coordinates; and the Higgs boson sits on the declared Higgs/top critical surface.

The following sections use the premises and caveats stated in the table.

Non-hadron results. The non-hadron results split by branch as follows:

The values and benchmark checks are

$$\alpha^{-1}(0) = 137.035999177(21), \qquad P \simeq 1.6309682094,$$

$$N_{\mathrm{EW}}(P_C) = 3.5323546226929906511\ldots \times 10^{122},$$
$$N_0(P_{\mathrm{fwd}}) = 3.532131543418936\ldots \times 10^{122}, \qquad \mathcal{B}_{\mathrm{EW}} = 0, \qquad \epsilon_H^{\mathrm{decl}} = 0.$$

The source-forward bridge value is about 6.6 percent above the weighted Planck base-ΛCDM capacity $N_\Lambda = 3.31292709806038\ldots \times 10^{122}$. The conditional finite-presence and Poisson carriers give $3.292097877326465\ldots \times 10^{122}$ and $3.300072225377652\ldots \times 10^{122}$, respectively. Their comparisons are retrospective. Exact countermodels obey the declared positive composition and cut-count regrouping laws while producing different global effects, so the finite source selects neither branch. Physical equality requires an executed public-record fixed point plus the typed common screen/electroweak load-carrier map. No source-only nonzero particle mass follows. The selected-carrier, value-law, inverse-adapter, and conditional Higgs coordinates appear only in the technical comparison sections below. Target-anchored charged-lepton and quark witness values, including the mixed-convention quark target packet, appear only in the exact-fit comparison calculations. Compare-only absolute neutrino attachments are withheld from public prediction tables. The neutrino comparison surface keeps representative splitting checks such as

$$\Delta m^2_{21} = 7.488059465106851 \times 10^{-5}\,\mathrm{eV}^2,$$
$$\Delta m^2_{31} = 2.5123118727618473 \times 10^{-3}\,\mathrm{eV}^2,$$
$$\Delta m^2_{32} = 2.4374312781107786 \times 10^{-3}\,\mathrm{eV}^2.$$

The table states each derivation chain and the caveat that prevents a stronger theorem claim.

How to read mismatches and exact hits. Exact numerical agreement is not by itself a proof of a blind prediction. The provenance record classifies the W/Z row as target-used frozen-reference reproduction, the charged-lepton triple as an empirically anchored current-family witness, and the mixed-convention quark packet as a target-anchored selected-frame comparison rather than a public source-only prediction. The same record classifies the hierarchy formula as conditional and the stored $\epsilon_H^{\mathrm{decl}} = 0$ formula as a declared-map packaging identity. The source antecedents do not select those maps and admit nonzero-defect completions; the strict source-root, source-derived comparison-map, and physical-scale certificates are separate premises. Where a row does not match, or where it matches only under constrained premises, the explanations are as follows. The source/root certificate gives $\alpha_{\mathrm{root}}^{-1} = 136.994835177413\ldots$ and $P_{\mathrm{fwd}} = 1.630972095858897\ldots$, but the declared map is incomplete. The certified gauge-width map gives $\alpha^{-1} = 137.035660136946577\ldots$, with gauge-width residual 2.5×10^{-6} relative to the measured 137.035999177(21). Its absolute residual 3.3904×10^{-4} exceeds the experimental standard uncertainty 2.1×10^{-8}; this numerical proximity does not establish agreement at measured precision. The no-hadron packet that combines different pixel coordinates is no fixed point and is excluded from physical output. Source-only closure requires Ward-projected same-scheme transport, target-independent map selection, and the typed same-quantity bridge. The separate endpoint-accounting diagnostic is

$$\Delta_{\mathrm{H,cal}}^{\mathrm{fp}} = \alpha_U(P_C)\, C_{24,Q}, \qquad C_{24,Q} = 1.000964785973232625384951114070247 5\ldots.$$

The $C_{24,Q}$ factor is endpoint accounting rather than a source-emitted theorem. At the public endpoint pixel value

$$P \simeq 1.6309682094$$

the endpoint residual package requires

$$\Delta_{\mathrm{source}}(P) = 0.041465861005223389053448715357314044\ldots.$$

That scalar belongs to the source-side hadronic spectral transport and scheme-remainder map. The declared empirical route would use a separately labeled $e^+e^- \to$ hadrons spectral input. No same-scheme integrated endpoint payload is emitted here. The first-principles transport row excludes an inserted comparison endpoint. The electroweak support record names the residual map and the RG/matching/threshold/scheme packet. An observer inside the branch measures the dressed Thomson coupling, not the undressed source diagnostic $1/136.994835\ldots$, because a zero-momentum electromagnetic measurement sees the Ward-projected $\mathrm{U}(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme finite endpoint matching have been included. The electroweak W/Z row is therefore a chart and prescription comparison, not a physical mass benchmark. Charged leptons carry an underived bridge: the corpus does not derive the determinant trace-lift attachment $3\mu(r) = S_M(r)$ from the electroweak descendants of P to physical charged data, the bridge artifact is a current-corpus gap certificate, and a future P-driven charged-mass law stays admissible. The auxiliary direct-top PDG row differs from the theorem coordinate by 0.50 GeV, or 0.76 combined standard deviations, because Q007TP and Q007TP4 are distinct extraction codomains; the direct-top response kernel carries a fixed-ray point statement, the reference central pair 0.7 sigma off the one-scalar ray on the declared linear surface. Neutrino absolute masses are not directly measured, and PMNS-angle residuals are visible comparison tension outside the theorem branch. First-principles hadron masses have no emitted prediction because the required Ward-projected hadronic spectral measure must come from a working OPH hadron construction. Empirical hadron closure rows carry a separate support class.

Local unification surface. The bosonic rows carry one cross-branch statement. The same pixel input P, fixed on the outer/inner closure relation of the flagship synthesis, fixes the electroweak and criticality trunk

$$P \longmapsto \alpha_U(P) \longmapsto \big(t_U(P), t_{\mathrm{tr}}(P)\big) \longmapsto v_{\mathrm{chart}}(P) \longmapsto (M_W, M_Z)\,.$$

$$\sigma_{\mathrm{crit}} = \alpha_U(P)\cos(2\theta_{W0})/\sqrt{\pi} \longmapsto (m_H, m_t)\,.$$

Here $\alpha_U(P)$ is the branch value selected by the same forward pixel-closure solve, so the bosonic trunk contains no inverse electroweak readback of the internal transmutation data. On the companion gravity side, the same pixel law packages

$$\bar{\ell}_{\mathrm{SU}(2)}(t_{2,\mathrm{run}}) \;+\; \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\mathrm{run}}) = P/4.$$

The gravity normalization itself is emitted by the selected scale certificate $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$. With $a_{\mathrm{cell}} = P\ell_\star^2$, the Newton area law gives $G_{\mathrm{geom}} = \ell_\star^2$, so the pixel cancels. The local unification surface separates one explicit familiar-unit readout package from three mathematical surfaces: the W/Z comparison, the conditional declared-surface Higgs/top coordinate, and the gravity-side readout with its strict classical-regime clause. On the gravity side, the stated local extension surface uses the lifted product presentation of the realized quotient branch and identifies

$$\bar{\ell}_{\mathrm{shared}} = \bar{\ell}_{\mathrm{SU}(2)}(t_{2,\mathrm{run}}) \;+\; \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\mathrm{run}}).$$

On that same surface the pixel law fixes $\bar{\ell}_{\mathrm{shared}} = P/4$, and the local SI readout is

$$G_{\mathrm{SI}} = \frac{c^3\ell_\star^2}{\hbar}$$

from the selected scale certificate. The χ_ν protected-reserve branch uses the same public endpoint convention as

$$P_\chi = P_{\mathrm{C}} = 1.630968209403959\ldots, \qquad \epsilon_{\mathrm{res}} = \frac{P_\chi}{24},$$

after the same-collar shared-edge budget and one-class $\mathbb{Z}_6$ trace receipts pass. This does not make $P/4$ a primitive Hilbert-space dimension: $P/4$ is the local screen-cell entropy budget in the particle and hierarchy branches, while $P_\chi/24$ is the shared scalar-reserve density used by the χ_ν collar branch. On that declared extension surface the same familiar-unit package reads

$$L_{\mathrm{loc}} = \sqrt{a_{\mathrm{cell}}}\,\widehat{L}(P), \qquad t_{\mathrm{loc}} = \frac{\sqrt{a_{\mathrm{cell}}}}{c}\,\widehat{T}(P),$$

$$E_{\mathrm{loc}} = \frac{\hbar c}{\sqrt{a_{\mathrm{cell}}}}\,\widehat{E}(P), \qquad \Theta_{\mathrm{loc}} = \frac{\hbar c}{k_B\sqrt{a_{\mathrm{cell}}}}\,\widehat{\Theta}(P),$$

with dimensionless $\widehat{L}, \widehat{T}, \widehat{E}, \widehat{\Theta}$. Thus, at fixed P, the local ruler is $\sqrt{a_{\mathrm{cell}}}$; seconds are that ruler divided by the structural Lorentz output c; and GeV and Kelvin are downstream familiar-unit displays of the inverse local ruler through $\hbar$ and k_B. On that declared extension surface the local scale-readout bundle is

$$c = 299792458\,\mathrm{m/s},$$
$$G = 6.674299995910528 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}},$$

No nonzero particle-mass coordinate is a source-only prediction. The W/Z inverse adapter is a compare-only quantity. Conditional six-mass and source-VEV matching calculations are displayed

in their technical sections; the supplied flavor law, finite vacuum attachment and external calibrations remain explicit. The structural Lorentz output is the common invariant null cone and speed $c_\star$. The decimal 299792458 m/s is exact by the SI definition of the metre, not a predicted magnitude. The paper keeps the G row as an exact scale-readback value, with no literal zero-difference identity against the rounded benchmark 6.6743×10^{-11}.

The detailed inventory below records the source-derived theorems, conditional continuations, and target-anchored comparisons sector by sector. Conditional numerical entries use the declared external calibrations and attachments in Sections 9.6 and 9.7; they are distinct from source-only predictions.

3.1 Detailed particle inventory

Family	Particle	Theorem scope	OPH value or strongest derived benchmark	Premise boundary
Conditional carrier modes	electromagnetic	action/phase receipt	two transverse classical $k^2 = 0$ modes; $\mu^2_{Q,\mathrm{quad}} = 0$ only in the displayed Maxwell action	physical Hilbert space, positive-residue pole, and stable asymptotic photon receipt absent
Conditional carrier modes	color	perturbative/pre-confinement action receipt	$2 \dim G$ transverse classical $k^2 = 0$ modes; $\mu^2_{\mathrm{YM},\mathrm{quad}} = 0$ only in the displayed pure-Yang–Mills expansion	positive-residue pole and deconfined asymptotic colored sector absent; no free gluon is claimed on the confining branch
Conditional carrier modes	tensor	action/background receipt	two Einstein transverse-traceless classical $k^2 = 0$ modes; $\mu^2_{\mathrm{EH},\mathrm{TT}} = 0$ only on the pure-Einstein branch	metric quantization, positive physical Hilbert space, positive-residue pole, and asymptotic/EFT particle interpretation absent
Electroweak branch	W boson	no source-only physical mass	conditional source-VEV matching: $M_{W,\mathrm{BW}} = 79.8622$ GeV; separate strict scalar-chart receipt	Landau-VEV and gauge attachment assumed; lower than the reference by 0.5070 GeV; strict scalar chart is not identified with the physical resonance sheet
Electroweak branch	Z boson	no source-only physical mass	conditional source-VEV matching: $M_{Z,\mathrm{BW}} = 90.5905$ GeV; separate strict scalar-chart receipt	same finite-attachment boundary as W; lower than the reference by 0.5975 GeV
Higgs/top critical stage	Higgs boson	conditional finite matching	complex-pole coordinate 124.8803 GeV in Section 9.7; no source-only prediction	Landau-VEV attachment and equal-capacity criticality assumed; physical source selection and the full error budget are unproved
Quark family	top quark	conditional finite matching and separate retrospective replay	matched complex-pole coordinate 170.6255 GeV; separate native-chart replay 172.0645 GeV	finite VEV attachment assumed; replay formula target-informed; these are separate constructions and neither is a source-only prediction
Quark family	up quark	conditional retrospective replay; reciprocal-ray candidate rejected	native-chart coordinate 2.15896 MeV; source-only prediction withheld	target-informed flavor law and light-MSbar chart at 2 GeV assumed; physical family-sensitive interaction unselected
Quark family	down quark	same conditional replay and common-scale rejection	native-chart coordinate 4.71104 MeV; source-only prediction withheld	same target-informed law and light-MSbar chart attachment
Quark family	strange quark	conditional replay; held-out reciprocal-ray failure	native-chart coordinate 93.0701 MeV; source-only prediction withheld	same law and chart attachment; $m_s/m_{ud} = 27.094659$ lies outside the quoted PDG interval
Quark family	charm quark	conditional replay; held-out reciprocal-ray failure	native-chart coordinate 1.272260 GeV; source-only prediction withheld	target-informed law and MSbar self-scale chart assumed; no common-scale physical Yukawa matrix emitted
Quark family	bottom quark	conditional replay; reciprocal-ray candidate rejected	native-chart coordinate 4.189458 GeV; source-only prediction withheld	same law and self-scale chart attachment; physical Higgs normalization and family interaction are unselected
Charged leptons	electron	continuation gap	n/a; exact same-carrier centered readback exists once a charged source pair is emitted, but the absolute scale is blocked by the closed common-shift no-go; benchmark target $g_e^\star = 0.0457789$, equivalently $\Delta_e^{\mathrm{abs},\star} = 3.00398633$	close branch-generator splitting, then emit the lift whose descended scalar is $\mu_{\mathrm{phys}}(Y_e)$; within that lift the exact smaller forcing object is the physical identity-mode equalizer, after which $\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\mathrm{phys}}(Y_e)\,\mathbf{1}$, $A_{\mathrm{ch}}(Y_e) = \mu_{\mathrm{phys}}(Y_e)$, and the readouts g_e, Δ_e^{abs} follow

Family	Particle	Theorem scope	OPH value or strongest derived benchmark	Premise boundary
Charged leptons	muon	continuation gap	n/a; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\widehat{C}_e^{\mathrm{cand}}$ → branch-generator splitting → lift → $\mu_{\mathrm{phys}}(Y_e)$ → A_{ch} → g_e closure chain, with the physical identity-mode equalizer beneath the descended scalar
Charged leptons	tau lepton	continuation gap	n/a; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\widehat{C}_e^{\mathrm{cand}}$ → branch-generator splitting → $\mu_{\mathrm{phys}}(Y_e)$ → A_{ch} → g_e closure chain, with the physical identity-mode equalizer beneath the descended scalar
Neutrino comparison	declared f-basis weighted-cycle matrix	rejected target-informed candidate	no flavor-particle mass row; the frozen declared-basis coordinate gives $\theta_{12} = 34.2259^\circ$, $\theta_{23} = 49.7228^\circ$, $\theta_{13} = 8.68636^\circ$, $\delta = 305.581^\circ$, $J = -0.02753$, and, under its declared normal-ordering labels, $\Delta m^2_{21}/\Delta m^2_{32} = 0.03072111$	the upstream flavor kernel is a hand-written template; the physical charged basis and mass-label rule are premises the template does not supply; the target-informed selector precludes a blind-prediction claim; the NuFIT 6.1 correlated profile rejects the candidate; Majorana and absolute-attachment values are compare-only
Hadrons	proton	strong-binding construction absent; empirical comparison contract emitted	no first-principles emitted prediction	first-principles prediction requires a working source-derived hadronic backend, such as GLORB/Echosahedron, plus Ward-projected two-current, higher-point, and transition spectral exports, same-scheme remainder, and production systematics; empirical closure rows use separate e^+e^- → hadrons input
Hadrons	neutron	strong-binding construction absent; empirical comparison contract emitted	no first-principles emitted prediction	same OPH construction premise, with isospin-resolved hadron systematics in the construction support class
Hadrons	neutral pion proxy	strong-binding construction absent; empirical comparison contract emitted	no first-principles emitted prediction	same OPH construction premise; local stable-channel surrogate output is not a paper prediction
Hadrons	$\rho(770)^0$ proxy	strong-binding construction absent; empirical comparison contract emitted	no first-principles emitted prediction	same OPH construction premise, plus finite-volume resonance extraction in the construction support class

4 Source Values, Screen Architecture, and Theorem Premises

The particle derivation uses the premises A1–A3, the local pixel closure, the regulated screen realization, and the imported OPH theorems stated below. The conditional screen-capacity branch is used only in results that state it as a premise.

4.1 The premises A1–A3 and the closure hypothesis

The particle-spectrum derivation uses the premises A1–A3 shared with the spacetime and gauge derivation papers [9, 10]. They are restated here for local readability because every particle-family chapter depends on them either directly or through the structural OPH chain imported from those papers.

A1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators

generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

A2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, A2 supplies a closed path γ_a and one projective implementer $[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered

source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

A3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise reference.

The closure hypothesis. The premises A1–A3 describe the observer screen. The closure hypothesis states that the Universe is a timeless, self-referential structure whose observers recover the specification of their own world from their records and realize it, so that the description and the described system are one object. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The closed object is the complete inhabited history, whose internal events and responses to interventions are its proposed intrinsic computation. This reading requires no external computer or playback time. It is distinct from fixedness of a repair state, a public record, or a probability law, and does not itself construct the history or prove a final cosmic endpoint.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point per declared map on its analytic domain. Interpreting that root as the laboratory

coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at 3.0×10^{-4} relative for the displayed map and at 2.5×10^{-6} for the gauge-width variant, as diagnostics of the declared source maps. The latter absolute residual is 3.3904×10^{-4}, compared with experimental standard uncertainty 2.1×10^{-8}; this is not agreement at measured precision. The gauge width supplies 99.2 percent of the gap between the bare root and the measured value; the residual 3.3904×10^{-4} is addressed to same-scheme hadronic transport to the Thomson endpoint.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct N, and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the A1–A3 source contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\!\left(\frac{6\pi}{P\,\alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\mathrm{pres}} = N_0 \left(1 - \frac{P}{24}\right), \qquad \ln \frac{N_{\mathrm{pres}}}{\pi} = \frac{6\pi}{P\,\alpha_U(P)} + \ln\left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\mathrm{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is 3.2921×10^{122} and the exponential candidate is 3.3001×10^{122}. The full weighted Planck base-ΛCDM chain gives the comparison coordinate 3.3129×10^{122}. The respective residuals are -0.63 percent and -0.39 percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

The spherical support in A1 is the Standard Model gauge paper's screen-first global-support branch. On the producer branch, S^2 becomes available only after spherical-incidence, mesh, cross-ratio, normalization, refinement, and carrier-to-support receipts establish it. The resulting support screen is distinct from both the local twelve-port carrier boundary and the federation overlap nerve.

Two families of premise interfaces enter beside A1–A3, each named at the results that consume it. The gravity-adjacent chapters consume the edge-center, collar-recovery, generalized-entropy, and modular-normalization interfaces of the spacetime paper; these are classified derivation targets and physical identifications with their own countermodels, outside A1–A3. The particle chapters consume declared sector completions: the generation count inside the conditional window, scalar attachment and multiplicity, and the completeness statement for extra light sectors are physical attachment premises, and the charged-lepton and quantitative selector branches state their declared premises where they appear. The transportable-sector/Tannaka reconstruction is a separate conditional classification route.

4.2 Regulated screen architecture and patch language

The SM/GR derivation in Ref. [9, 10] states the premises in algebraic language. Ref. [6] supplies one concrete regulated realization of that language: a federation of finite patch carriers with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap packets, record algebras, and local repair instruments. The formal observer patch is the bounded access-and-record structure. One echosahedral carrier is a candidate primitive physical realization on the homogeneous branch. A bare carrier body does not satisfy the observer definition; with declared preparations, retained records, record-controlled feedback and checkpoint replay, one twelve-port carrier supports overlapping observer regions, native repair and local event clocks, and joined carriers serve larger regions, distant interventions and refinement [11]. This regulated architecture is important for the particle derivation for three reasons.

First, it makes the screen picture operational. A patch is a finite local algebra in a bounded carrier, and an overlap is a boundary-visible algebra with declared shared observables. Second, it makes the edge-sector language concrete. The particle derivation uses edge sectors, transport, refinement, fusion, and overlap data to build the gauge and flavor branches; Ref. [6] shows how those objects can be realized at fixed cutoff by finite gauge-aware patch carriers. Third, it clarifies the measurement interface. The observer-facing measurement package is a theorem-bearing fixed-cutoff statement about a central record algebra embedded in a declared finite algebra-state representation, Born probabilities for its event projectors, and a declared Lüders instrument on that representation. The effect table does not select that instrument. The quantum representation is not derived from the commuting record indicators.

Three geometric objects govern this reference implementation. The local carrier boundary has the twelve-port icosahedral incidence certified on the echosahedral lineage. The federation screen is the finite federation together with its overlap nerve. The support screen is the observer-visible S^2 chart obtained only when global incidence, mesh, cross-ratio, normalization, and refinement receipts hold. A federation of local icosahedral carriers can have a nonspherical nerve, so local A_5 symmetry does not force a global S^2 support.

The framework is presentation-invariant and carrier-sensitive. A different hidden implementation is physically equivalent when its observer-visible quotient agrees. A different visible port graph, topology, record process, or response law may select a different branch. Hardware evidence enters only through a public OPH evidence bundle with stable hashes and verifier receipts.

Physical phase locking is a candidate producer for coherent overlap comparison. To support that interpretation it must produce the accepted repair relation, transactional confluence, public records, and controlled noise bounds. No current theorem identifies phase locking with consensus confluence, modular flow, or an observer clock. Separate support-visible continuum theorems supply the geometric and gauge limits.

4.3 Quantitative inputs and where they enter the particle derivation

The quantitative derivation developed here uses one incomplete declared-map coordinate and one conditional capacity coordinate together with the selected no-G scale certificate:

$$P \equiv a_{\mathrm{cell}}/\ell_\star^2, \tag{1}$$

$$D \equiv \dim \mathcal{H}_{\mathrm{cap},r,D}, \qquad N_{\mathrm{CRC}} \equiv \log D_{\mathrm{CRC}}, \tag{2}$$

$$\gamma_\star \equiv \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star \equiv \frac{3\pi}{\ell_\star^2}. \tag{3}$$

These closure values have distinct downstream dependencies.

Pixel area P. The pixel area is the declared local UV-area ratio. In this derivation it feeds the heat-kernel / edge-law input stage and, through that route, the forward electroweak surface. It is the common upstream numerical variable for the electroweak, Higgs/top, flavor, quark, and hadron-facing branches. The particle paper therefore treats $P = P_\star$ as the inherited root of the incomplete outer/inner declared map

$$P_\star = \varphi + \frac{\sqrt{\pi}}{A_T(P_\star)}$$

from the flagship synthesis [3], not as a quantity derived again inside the particle-spectrum argument itself.

Conditional screen capacity N_{CRC}. The screen capacity is not part of the local recovered-core gravity/gauge derivation. It enters the separate screen-capacity branch through the stable cosmic record-closure target

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star, \qquad \Lambda_{\mathrm{CRC}}\ell_\star^2 = \frac{3\pi}{N_{\mathrm{CRC}}}, \qquad \Lambda_{\mathrm{CRC}} = \frac{3\pi}{G N_{\mathrm{CRC}}}.$$

The observers paper [4] specifies the physical active-readback target directly. At finite cutoff, compatible reachable public record atoms and the globally coupled checkpoint family define a confusability graph G_q and

$$M_0(q) = \alpha(G_q), \qquad \mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

Only a whole-fiber singleton is scalar. A faithful capacity-carrier representation makes that scalar exact map deflationary; a confusability-reflecting capacity embedding makes it monotone, and fixed-D refinement eventually stabilizes. A source-derived fixed-cutoff public checkpoint packet, carrier representation, and whole-fiber scalarization exist at $D = 24$. A bounded target-clean all-rung counterfamily has different exact zero sets under shared base agreement, positivity, the carrier bound, and executable finite controls. The exact result is nonidentifiability on that bounded completion class. No universal all-rung membership in the complete A1–A3 capacity-source contract or executable-to-Lean bridge is supplied. Direct N is not evaluable on its incomplete source antecedent. A positive direct result requires completion of that antecedent, a unique physical zero, and a universe-level carrier attachment. Identifying that capacity with de Sitter entropy requires the horizon–record identification; identifying it with the electroweak bridge requires the common screen/electroweak load-carrier identification. The source-forward bridge value $3.5321315434\ldots \times 10^{122}$ is about 6.6 percent above the weighted Planck base-ΛCDM comparison

coordinate $3.3126\ldots \times 10^{122}$. On the declared one-class branch, physically selecting that class and attaching its scalar-weighted finite presence-survival factor gives $3.2920978773\ldots \times 10^{122}$; the exponential mean-count branch gives $3.3000722254\ldots \times 10^{122}$. Exact neutral and multiplicative completions of the same local datum satisfy the declared positive composition and regrouping laws, then disagree globally. The finite source selects no action or blocked-event semantics, so the named-law branch is not evaluable and its horizon branch has no capacity object. Both comparisons are retrospective. The last equality is therefore a conditional scale-certified display; it does not determine an SI curvature scale from $P_\star$ and N_{CRC} alone. An independently frozen discrimination scale ρ_{op} is a test of the direct code capacity through $\log M_0 - \pi/\rho_{\mathrm{op}}^2$, not its definition. A diagonal terminal-state count or its argmax does not construct the off-diagonal map. In the particle context, this matters only for the cosmological-capacity discussion that frames why local null data do not determine the cosmological constant. The implemented branch uses the static-patch normalization

$$N_{\mathrm{patch}} = \left(\frac{r_{\mathrm{dS}}}{\ell_P}\right)^2 \simeq 1.05 \times 10^{122}, \qquad N_{\mathrm{scr}} = \pi N_{\mathrm{patch}} \simeq 3.313 \times 10^{122}.$$

This capacity bookkeeping supplies no neutrino result; the weighted-cycle neutrino derivation is a separate branch.

Selected scale certificate. The gravity scale is a separate observation-located certificate. On the declared branch,

$$\gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star = \frac{3\pi}{\ell_\star^2}, \qquad G_{\mathrm{SI}} = \frac{c^3 \ell_\star^2}{\hbar}.$$

If the declared-map root acquires its missing physical identification, the local pixel $P_\star$ identifies $a_{\mathrm{cell}} = P_\star \ell_\star^2$ and $\bar{\ell}_{\mathrm{shared}} = P_\star/4$. In the Newton area law $P_\star$ cancels, so this paper does not treat G as a particle-sector consequence of $P_\star$. $\bar{\ell}_{\mathrm{shared}}$ is a shared-cut density, not the logarithm of an autonomous cell Hilbert factor. The particle branch uses it for cell/edge consistency; it does not select a fixed primitive observer capacity.

Why these are the declared closure/readback values. One of the discipline constraints of the OPH derivation is that particle-sector freedom should not be hidden in a large uncontrolled parameter set. This paper exposes the incomplete declared-map root $P_\star$, the proposed N_{CRC} correctable-public-record closure coordinate, and the selected scale certificate. The same shared-coordinate discipline applies to every conditional branch. For the particle-spectrum branch, the nontrivial common quantitative burden sits on $P_\star$: N_{scr} enters the separate capacity/cosmology side and $\gamma_\star$, equivalently $B_\star$, enters the separate gravity-scale side, whereas the reverse-engineering claim for masses and couplings is that one shared declared-map coordinate should drive the spectrum instead of a long sector-by-sector input list.

4.4 Technical theorem data carried into the particle-spectrum derivation

This paper uses the same theorem premises as *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [9, 10], together with the regulator-language clarification integrated into the consensus and microphysics appendices. This separation matters only insofar as it separates branch-internal statements from declared inputs.

On the finite identity-reference branch, premise A3 and the declared local constraint grammar give the regulator-scale local Gibbs form, the quasi-local propagation bound, and the endpoint-control estimate for bounded intervals. Refinement stability is supplied by the separate optimizer-pushforward and refinement-tail interface. In regulator language, one works with finite local Hilbert spaces and a boundary-fixed overlap action on cut data. Beyond those regulator premises, the Standard Model gauge paper supplies the support-visible Bisognano–Wichmann (BW) scaling theorem for the Lorentz/null-modular/Einstein branch. Transportability and the fixed-cutoff bosonic sector category are theorem-produced in The Standard Model gauge paper; the refinement/fiber ladder and cofinal admissible compact-gauge witness additionally require its explicit compact-gauge refinement receipt. On the central branch the matching gluing obstruction is the combined zero-obstruction transport criterion: vanishing triangle defect plus trivial represented holonomy of the strictified edge 1-cocycle. It introduces no second theorem-side input. Any local-Gibbs or collar-mixing language on that branch is fixed-cutoff recoverability/support control; it is not the source of the compact-gauge witness. For the BW/geometric side, the target is the support-visible extracted geometric subnet. The Standard Model gauge paper proves the needed scaling theorem by combining regularized support-visible modular transport, weak-$*$ extraction of the cap pair in the Gelfand–Naimark–Segal representation, support-readable modular covariance, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric 2π-Kubo–Martin–Schwinger (KMS) convergence with wrong-scale controls. The unregularized full-algebra common-floor route is deliberately not claimed, because off-support directions may collapse without affecting observer-facing matrix elements. Bare finite consensus does not by itself produce the finite cap-normal BW certificate used by that theorem. The theorem also consumes an independent mixed Gelfand–Naimark–Segal common-comparison premise on the same tower. Local maximum entropy and refinement do not produce that premise.

For the particle-spectrum derivation, this has a practical consequence. Structural statements such as compact-gauge reconstruction, the contract-conditional Standard Model quotient, the exact hypercharge lattice, the color count $N_c = 3$, and the declared generation completion are not floating independently of the declared input and theorem premises. Nor do they imply a physical family attachment. The additional classical carrier modes live on explicit quadratic action/background/phase receipts, and particle poles require the stronger quantum receipt. These statements live on a controlled chain whose scope conditions need to be visible in this paper, especially for quantitative closure, continuation, and nonperturbative sectors.

4.5 Imported core theorem packages

This paper uses the following OPH theorem packages without re-proving them in full.

1. **Patch-net and overlap language.** Ref. [5] organizes overlap repair, fixed-point language, cycle holonomy, and gauge-as-gluing in a form that is especially useful for flavor transport and observer-centric interpretation.

2. **Screen-side regulated architecture.** Ref. [6] gives the federated echosahedral patch-carrier model, the regulated patch-net embedding theorem, and the fixed-cutoff edge heat-kernel / Casimir theorem.

3. **Measurement interface.** Ref. [6], together with the integrated measurement appendices and the flagship synthesis [3], supplies the fixed-cutoff central-record package and, under a separately declared algebra-state and two-wing representation, the Born-rule, Lüders-conditioning, and Bell/CHSH theorems used when the measurement sections explain how observations pick

out definite observer-accessible outcomes and how declared two-wing quantum laws obey the Tsirelson upper bound. No Bell violation is derived from the commuting record indicators or from the repair law.

4. **Forced local Standard Model Lie algebra, conditional matter image, and conditional compact reconstruction.** *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [9, 10] and the dedicated gauge-group fragment prove that complete reversible response and endogenous proper-carrier transport force

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

The separately declared matrix current and rank-15 matter table, together with anomaly balance and tensor descent, give the maximal faithful image

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3.$$

The charge lattice and $N_c = 3$ come from the stated matrix and matter premises. The quotient is exact for the declared representation, but the construction does not select it as the physical global gauge group. Separate intrinsic-CP and weak-sector premises give $3 \leq N_g \leq 5$. The same Standard Model gauge paper gives the transportable-sector/Tannaka construction as a separate conditional compact-group route. No physical source producer, intertwiner, or laboratory attachment binds the finite current or identifies it with the Tannaka current.

5. **Conditional topic notes.** The integrated branch appendices carry useful material on supersymmetry, the finite-quotient baryogenesis source theorem without a derived record generator, proton stability, generation structure, and Yukawa hierarchy, but those sections have to be read with their stated claim boundaries intact. They are sources for conditional analyses, not recovered-core theorems.

This reading rule governs the whole paper. Structural gauge and carrier results can be used as structural results. The electroweak branch is a declared quantitative calibration branch. The Higgs/top critical stage is a conditional downstream calculation on its declared running, matching, and threshold surface. The quark, charged-lepton, neutrino, and hadron chapters state conditional or comparison results because their physical premises are incomplete.

5 Matter and interaction readout from the finite response

The finite response determines more than the abstract gauge Lie type. Its matrix generators support an exact, conditional matter decoder and local interaction tensors. The distinction between a representation and a physical particle is essential: a particle identification also consumes its spin, statistics, propagator, interaction and observer-clock interpretation.

5.1 From response operators to a fifteen-state matter module

Let a response generator on the six-dimensional carrier have blocks $D = \mathrm{diag}(A, B)$, with $A \in \mathfrak{u}(3)$ and $B \in \mathfrak{so}(3)$. Write $B = \widehat{w}$. The declared spin lift and central trace redistribution give

$$L(D) = \mathrm{diag}\left(A, -\frac{i}{2} w \cdot \sigma - \frac{\mathrm{tr}\, A}{2} I_2\right).$$

This is a Lie-algebra homomorphism. The complete twelve-direction response and all sixty-six bracket pairs are checked exactly. The central direction, after clearing denominators and choosing the weak-positive orientation, has primitive integer character

$$q = \operatorname{diag}(-2, -2, -2, 3, 3).$$

The lift and trace redistribution are hypotheses. A native weak rotation through 2π is the identity on its vector carrier and acts as minus the identity on the lifted doublet. Thus the lifted module is not a subrepresentation of the original six-dimensional response. Global descent and a physical current interpretation require the corresponding additional identifications.

Proposition 5.1 (Conditional matter decoder)**.** *For the lifted five-dimensional one-body action, the representation*

$$M = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$$

has dimension fifteen. Its charge and quadratic Casimir operators identify the following irreducible components without a field-name input table.

Component	Color	Weak dimension	$Y = q/6$	Dimension
Q	3	2	$1/6$	6
u^c	$\overline{3}$	1	$-2/3$	3
d^c	$\overline{3}$	1	$1/3$	3
L	1	2	$-1/2$	2
e^c	1	1	1	1

Proof. Split $\mathbb{C}^5 = V_c \oplus V_w$, of dimensions three and two and charges -2 and 3. Then

$$\Lambda^2(V_c \oplus V_w) = \Lambda^2 V_c \oplus (V_c \otimes V_w) \oplus \Lambda^2 V_w,$$

with integer charges $-4, 1, 6$. The degree-four summands are $\Lambda^3 V_c \otimes V_w$ and $\Lambda^2 V_c \otimes \Lambda^2 V_w$, of charges $-3, 2$. Color and weak Casimirs separate the representations, and Cartan weights distinguish the conjugate color character. Exact projectors give the same answer after a rational change of basis mixing unlike charge sectors. □

The auxiliary exterior construction has two additional invariant lines, the vacuum and top exterior degree. Removing them is part of the declared fifteen-state module. Charge traces alone do not remove neutral invariant states. Its even auxiliary parity also does not establish physical fermion statistics. Physical two-component Weyl fields require an independent CAR algebra and spin interpretation. The anomaly-free charge table and common central kernel are conditional representation results; they do not identify laboratory species or select a generation count.

5.2 One Higgs doublet and the exact interaction tensors

For two same-chirality physical Weyl fields, the antisymmetric spinor contraction makes the internal coefficient symmetric. The Lorentz-scalar internal pair space is consequently $\operatorname{Sym}^2 M$, of dimension 120. A color-Casimir kernel and weak-Casimir projector identify one doublet of hypercharge $-1/2$ and two of hypercharge $+1/2$. Their constituent integer-charge pairs are $(-4, 1)$, $(-3, 6)$, and $(1, 2)$. These are interpolating channels; their existence supplies no bound-state pole, condensate or elementary-boson commutation relation.

Proposition 5.2 (Conditional one-doublet interaction)**.** *Declare one scalar doublet H of hypercharge $1/2$. Each of the three channels above has a one-dimensional invariant contraction space. They give*

$$Qu^cH, \qquad Qd^cH^*, \qquad Le^cH^*.$$

The same scalar doublet and its conjugate occur in all three expressions.

Proof. Restrict the exact pair generators to each color-singlet doublet. The tensor sum of that doublet generator and the generator of H, or its conjugate, has a one-dimensional common kernel. Equivalently, its coefficient matrices C_α satisfy, for each of the twelve source currents,

$$T^tC_\alpha + C_\alpha T + \sum_\beta (T_H)_{\beta\alpha} C_\beta = 0.$$

Here the conjugate action is $-T_H^t$ in the skew-Hermitian convention. The physical pairing kernel $\epsilon \otimes C_\alpha$ is antisymmetric. The identity $S^t\epsilon S = \epsilon$ for $S \in \mathrm{SL}(2,\mathbb{C})$ establishes its Lorentz scalar character. The executable checks the rotation and boost generators as well as all current identities. □

For three declared families the coefficients are three complex 3×3 matrices, with 27 complex entries before flavor-basis redefinitions. The pairing convention is

$$\mathcal{L}_{\mathrm{pair}} = \frac{1}{2}\epsilon_{AB}\psi_A^t K(H)\psi_B + \mathrm{h.c.}$$

up to an overall sign. With canonical kinetic normalization and $H = (0, v)$, its singular values are tree-level mass parameters; $v = v_{\mathrm{SM}}/\sqrt{2}$ in the conventional electroweak normalization. Exact matrix-unit tests cover arbitrary complex family coefficients. Their values, the Higgs potential and its vacuum are not selected by these identities. The optional invariant $(LH)(LH)/\Lambda$ has six complex symmetric family coefficients and gives $M_\nu = \kappa v^2/\Lambda$. It is an additional dimension-five effective interaction, with no selected coefficient, scale or ultraviolet completion.

If both family factors are identified with the same unbroken irreducible rank-three source representation and the Higgs is family-neutral, the exact commutant has dimension one. In that branch each Yukawa intertwiner is yI_3, giving equal singular values. A nonscalar background can split the band; choosing that background is an additional physical hypothesis. The source family-band theorem therefore does not imply the observed hierarchy.

5.3 Evidence and mass interpretation

The self-contained calculations in `model/observer_particles/` reconstruct the response, representation projectors and interaction tensors from repository-local source operators. Their complete receipts distinguish exact rational or algebraic identities from numerical residuals. They use no measured particle masses. They supply neither a physical chiral field net nor a source-selected Higgs vacuum, Yukawa matrix or mass pole. Calibrated mass fits and rejected mass formulas retain their separate scientific status. A relaxation gap, finite cavity frequency or commuting internal modular offset requires its own mass-shell identification before it can be called a physical rest mass.

6 Gauge Architecture, Standard Model Structure, and Structural Carriers

The particle-spectrum derivation starts with the forced local gauge Lie algebra

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

The declared matrix current and rank-15 matter table, determinant balance, Spin lift, and tensor descent then give the conditional maximal faithful matter image, charge lattice, and carrier roles before any mass readout. The overlap/transport obstruction calculus and transportable-sector/Tannaka reconstruction form a separate conditional classification route. The target-free matter reduct leaves the generation count unselected. The strengthened finite screen interface selects rank three under its single-object, faithful-action, and operational-cost premises. The No physical family interpretation or extra-sector completeness theorem is supplied. This logical split matters for this paper. The gauge branch does not fix the carriers' quantum-particle masses; the action-level and quantum spectral gates below decide whether a classical mode or particle row exists.

On the declared packet, the gauge/content result used throughout this paper is

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad 3\le N_g\le 5, \qquad N_c=3.$$

Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and *Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [9, 10], the longer main derivation surface, and the dedicated gauge-group proof fragment agree on the split. The conditional finite receipts fix the maximal faithful image, hypercharge lattice, and $N_c = 3$ from their stated matrix-current and matter premises. They do not select the physical current, matter action, or global quotient. Intrinsic CP capability together with weak-sector UV completeness gives $3 \le N_g \le 5$. The exact screen-band theorem selects rank three on the strengthened finite interface, and the declared Laplacian simulator realizes the corresponding lowest-positive-frequency residue. Witten parity is a consistency check. The finite local-domain receipt checks the separately declared operator $D_\sigma \otimes I_{45}$ and conditional gap inheritance. It does not construct a source-selected local matter carrier or bridge the twelve-port Spin packet to that domain. These steps do not identify physical matter poles or exclude extra light sectors.

The exterior substrate is an explicit algebra. The ten-row table is its finite component census. For the supplied carrier $V = \mathbb{C}^3 \oplus \mathbb{C}^2$, the exterior basis is indexed by all 32 subsets of the five coordinate modes. Their color and weak degrees have multiplicities $\binom{3}{c}\binom{2}{w}$. Removing the vacuum degree $(0, 0)$ and the top degree $(3, 2)$ leaves exactly the ten component bidegrees used in the finite matter scan. On every row, the integer charge is $-2c + 3w$, fermionic parity is $c + w \bmod 2$, and charge conjugation is complement in the three color and two weak modes. Exterior creation squares to zero and distinct creation generators anticommute. These are algebraic exclusion statements on the supplied exterior carrier. They do not select that carrier as physical matter or establish continuum spin–statistics.

A supplied projective sector partition gives a precise operational composition with this exterior table. Partition pinching identifies two finite matrices exactly when every matrix in the sector-preserving commutant has the same trace statistic on them. Partition averaging is the distinct commutative public-record readout; both readouts erase cross-sector corners. If an explicit map assigns each exterior row to a nonzero partition sector and supplies the corresponding central weight, substituting those labels preserves the same diagonal $\mathbb{Z}_6$ weight kernel. Partition averaging defines a surjective adaptor into the supplied partition's commutative public projector-span algebra and factors through the block readout. Every adapted value commutes with every element of the partition commutant. The supplied central weights define a sixth-root character action on the component-labelled finite product of the ten mapped projector ranges. This action is not ambient-projector conjugation and does not bind the projector ranks to exterior multiplicities. The kernel of this central-parameter action has exactly six elements and agrees with the tensor-character kernel. A separate supplied chiral anomaly-free mask is forced to one of the conjugate exterior-degree parity sectors; its support is nonzero, invariant under the same action, and detects the same kernel. No

central parameter acts as a universal minus sign on that support. This is a nonconflation control and does not construct or identify physical fermion parity. The partition, maps, weights, and mask are premises. The theorem derives no source-selected physical matter action, parity representative, global form, or laboratory charge.

The A1/A2 finite-current route, its conditional matrix and matter continuation, and the transportable-sector/Tannaka route are separate chains. No theorem identifies them as one physical gauge-current object, and no physical source binding of the first route is supplied. The rank-three response band is therefore only a candidate family fiber. Its abstract rank-45 tensor product with the generation table does not attach that band to three physical families.

For the particle zoo this structural branch fixes three carrier roles, not three particle masses. The Standard Model gauge paper's carrier-mode theorem then supplies the following conditional action-level statements. The Maxwell action on the ordinary unbroken electromagnetic vacuum has two transverse classical modes. The pure Yang–Mills action about a flat connection has $2 \dim G$ perturbative transverse modes, but the confined color phase has no asserted gauge-invariant asymptotic gluon. The pure two-derivative Einstein–Hilbert action about Minkowski space has two classical transverse-traceless modes. In every case a quantum particle additionally requires a positive- energy physical Hilbert space, a positive-residue physical two-point pole, and the appropriate asymptotic or deconfinement receipt.

A separate screen-side construction sharpens the boundary of the electromagnetic row [6]. The complete thirty-seam source maps onto the record lattice

$$D_6 = \left\{ z \in \mathbb{Z}^6 : \sum_i z_i \equiv 0 \pmod 2 \right\},$$

whose response-metric completion is three-dimensional. Under proper-carrier naturality and a unique strictly positive normalized minimizer, the sixty directed seam moves carry equal weight and define an exact homogeneous Dirichlet operator. Its plane-wave symbol has a basis-free momentum-orthogonal fiber of real dimension two at every nonzero momentum. For a supplied nonzero coordinate dilation, if that spatial symbol is declared to be an oscillator stiffness, the resulting conditional mode obeys

$$A_T'' + \Lambda_a(k) A_T = 0, \qquad \omega_a(k)^2 = \Lambda_a(k).$$

The spatial symbol vanishes at the momentum origin, so the nonnegative square-root branch satisfies $\omega_a(0) = 0$. The projector, rank-two fiber, scalar degeneracy, oscillator equation, and zero mode are exact mathematical statements. Gauge redundancy, Gauss law, the Maxwell action, the physical electromagnetic group, a physical clock, and a photon pole are absent from this theorem.

The physical electromagnetic diagonal $\mathrm{U}(1)_Q$, generated by $Q = T_3 + Y$, also requires the certified $(\mathbf{1}, \mathbf{2})_{1/2}$ carrier and a selected nonzero neutral lower-component vacuum. The projective weak-doublet calculation above identifies the stabilizer of that supplied vacuum vector. A source-selected scalar action and its physical vacuum are separate requirements. Only after that selection, together with physical position, clock, frame, propagation, readout, quantization, and pole data, can the rank-two seam oscillator be tested for identity with the electromagnetic carrier.

The possible phenomenological routes use different inputs. A direct time-of-flight calculation is a one-particle propagation test. Electron and positron dispersion and pair-production cross sections are outside that calculation. It needs a physical frequency map with a controlled remainder, position and clock attachment, frame and boost law, cosmological transport, emission model, detector timing, nuisance treatment, and exclusive readout. A pair-production threshold or opacity calculation additionally needs the physical electron and positron actions, a common energy-momentum

conservation law, pair kinematics and interactions, background-radiation transport, cross sections, and source composition. The finite seam operator supplies neither physical comparison on its own.

Before physical identification, the complete positive cosine symbol has an exact all-momentum bound. Its nonnegative auxiliary root Ω obeys

$$|\Omega(k) - \Omega(p)| \le |k - p|$$

in the selected Euclidean carrier chart. This is a carrier-coordinate contraction. It supplies no physical position, scale, or detector map. If the seam symbol is physical photon frequency squared, its complete cosine form is bounded above by $|k|^2$. Positive-mass charged leptons with Lorentz-invariant positive-energy dispersion then exclude photon decay into an electron-positron pair under additive conservation. With the soft background photon Lorentz invariant at leading order, use $E_i^2 = p_i^2 + m_i^2 + \delta_{i,2}E_i^4$, with $i \in \{\gamma, +, -\}$, $m_\gamma = 0$, and $m_+ = m_- = m_e$. A leading head-on, collinear threshold with independent hard-photon, positron, and electron dimension-six coefficients depends at fixed positron energy share $0 < x < 1$ on

$$\delta_{\gamma,2} - x^3\delta_{+,2} - (1-x)^3\delta_{-,2}.$$

The equal-share readout has rank one and a two-dimensional coefficient fiber. The photon-only threshold translation therefore depends on the separate Lorentz-invariant-lepton premise; the screen theorem does not select it. On that branch, equal sharing is the unique global optimizer of the leading head-on, collinear residual. No general independent-lepton or full anisotropic optimization is supplied.

One nontrivial universal subcase can nevertheless be closed exactly at the same leading kinematic order. Assume the hard photon, electron, and positron all carry the common negative coefficient $\delta = -d = -a^2/20$, while the soft photon is Lorentz invariant at leading order. Writing $u = x(1-x) \in (0, 1/4]$, the required soft energy is

$$\epsilon_{\mathrm{req}}(k, u) = \frac{m_e^2}{4ku} + \frac{3dk^3}{4}u.$$

The share identity, positive-domain bound, exact square-factor AM–GM envelope, equal-share endpoint regime, transition condition $3dk^4 \le 16m_e^2$, and the absence of a solution below the envelope are machine-checked in `SeamCurrentPhotonLeptonThreshold` [1]. The same file proves that at and above the transition the balanced equality witness is realized by an open physical share, closing the constrained-minimum step. Above $k_t = 2\sqrt{m_e}(3d)^{-1/4}$, the interior minimizer $u_* = m_e/(\sqrt{3d}\,k^2)$ gives $\epsilon_{\mathrm{env}} = (\sqrt{3}/2)m_e k\sqrt{d}$. A deterministic companion calculator verifies the analytic and bracketed roots and keeps the Lorentz-invariant-lepton photon-only comparator separate. For $a = \ell_P$, representative CMB energies 6.34×10^{-4} and 3.0×10^{-3} eV have upper leading roots 7.82×10^{19} and 3.70×10^{20} eV, respectively. These are conditional kinematic windows, not an opacity or propagated-flux prediction. No published photon-only air-shower exclusion transfers to this common-coefficient branch without the interaction, background, source, cascade, shower, and detector layers.

These conditional classical modes should not be confused with quantum-particle mass rows. The group and gravity branches identify the carrier roles; the displayed actions and phases supply their classical propagation; and only a separate quantum spectral receipt can establish them as photon, gluon, or graviton particles. The boson sections address the W, Z, and Higgs rows on their own electroweak and Higgs/top quantitative stages.

7 Electroweak Calibration and the Higgs/Top Critical Stage

The bosonic calculation separates source outputs from comparison coordinates. The electroweak calibration supplies the W boson and Z boson chart, and the criticality calculation supplies conditional Higgs and top coordinates. Neither calculation emits an OPH-native particle pole.

7.1 Electroweak chart on the quantitative branch

The electroweak derivation consists of a single-P running family followed by a reduced two-scalar carrier, a selector on that carrier, an exact carrier mass chart, and a conditional quotient-transport theorem beyond the selected carrier. In practical terms, the construction starts from the declared pixel input P, builds the running electroweak family, reduces it to the selected carrier, reads the W boson/Z boson pair from the selected point on that carrier, and then evaluates the mass pair together with the Ward-projected electromagnetic transport family. On this branch, a source-only prediction must respect that ordering: first certify the shared pixel input P on the declared running/matching/threshold/scheme surface, thereby fixing the source basis

$$(\alpha_U, \alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\text{source}}, v),$$

with $\alpha_U(P)$, equivalently $t_U(P)$ and $t_{\text{tr}}(P)$, fixed by the same forward pixel-closure solve, and only then emit the electroweak transport family from that source basis. The stated theorem does not contain this full chain. Its W/Z rows are data-comparison adapter coordinates rather than OPH-native poles.

The calibration couplings are read at the source scale 91.5883371732 GeV, a coordinate of the source solve, rather than at the measured Z target. Evaluation of the conditional pole map in the raw pre-carrier basis omits the compact-carrier hypercharge step. The selector-consistent conditional evaluation lands the neutral coordinate near 90.66 GeV, about half a GeV below the converted reference target. The raw-basis and selector-consistent numbers are conditional diagnostics of the imported prescription. Neither is an OPH-native pole.

The same quantitative branch has one exact golden-ratio benchmark. If one writes

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)}$$

for the total/bulk/edge hierarchy and imposes exact self-similar balance

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle},$$

then x obeys $x^2 - x - 1 = 0$, so the unique positive equilibrium point is

$$x = \varphi := \frac{1+\sqrt{5}}{2}.$$

Equivalently the order parameter $A_\varphi(x) = x - 1 - \frac{1}{x}$ vanishes there. The flagship synthesis records the unique numerical root of the incomplete outer/inner declared map. The point of the equilibrium theorem here is that the proximity to φ has a structural reason. Exact balance is too symmetric to support durable records, structure, and dynamics, so the declared map assigns a small equilibrium-breaking detuning away from that balance point.

The compare-only W/Z adapter rows in the final bundle are

$$m_W = 80.3625 \text{ GeV}, \qquad m_Z = 91.1879 \text{ GeV}.$$

These numbers sit on a target-conditioned validation surface and do not form a target-free prediction theorem. The exact selected-carrier chart gives the local pair

$$m_W^{\mathrm{carrier}} = 80.38629169244275\ \mathrm{GeV}, \qquad m_Z^{\mathrm{carrier}} = 91.18290444674243\ \mathrm{GeV},$$

so the mathematics distinguishes the selected-carrier chart from the candidate value-law pair

$$(80.3770000154, 91.1879780779)\,\mathrm{GeV}.$$

It also distinguishes the conditional quotient-transport implication from the measured-reference inverse repair surface. The rounded pair $(80.377, 91.1879780919)\,\mathrm{GeV}$ is a boundary alias of the candidate, not the inverse adapter. None is a physical pole theorem.

Theorem 7.1 (Measured-pair reparametrization (calibration): coherent electroweak inverse fit)**.** *Let the emitted running/core electroweak basis be*

$$Q_{\mathrm{run}}(P) = \big(\alpha_{Y,m_Z}(P), \alpha_{2,m_Z}(P), v_{\mathrm{chart}}(P), \eta_{\mathrm{source}}(P)\big),$$

and write

$$\tau_Y(\tau_2) = -\frac{\tau_2 + 2\eta_{\mathrm{source}}}{1 + 4\tau_2^2}, \qquad n_{\mathrm{fiber}}(\tau_2) = 1 + \frac{\alpha_{Y,m_Z}\tau_Y(\tau_2) + \alpha_{2,m_Z}\tau_2}{\alpha_{Y,m_Z} + \alpha_{2,m_Z}}.$$

Freeze one authoritative target pair

$$\mathcal{T}^\dagger = (M_W^\dagger, M_Z^\dagger), \qquad 0 < M_W^\dagger < M_Z^\dagger,$$

and define the charged anchor

$$\tau_{2,W}^\dagger = \frac{M_W^{\dagger\,2}}{\pi v^2 \alpha_{2,m_Z}} - 1, \qquad \delta\alpha_2^\dagger = \frac{M_W^{\dagger\,2}}{\pi v^2} - \alpha_{2,m_Z}.$$

Then the fiber-parallel hypercharge leg is

$$\tau_Y^\dagger = \tau_Y(\tau_{2,W}^\dagger), \qquad n_{\mathrm{fiber}}^\dagger = n_{\mathrm{fiber}}(\tau_{2,W}^\dagger),$$

$$M_{Z,\mathrm{fiber}}^\dagger = v\sqrt{\pi(\alpha_{Y,m_Z} + \alpha_{2,m_Z})\, n_{\mathrm{fiber}}^\dagger},$$

and the orthogonal neutral-shear scalar is

$$\delta M_Z^\perp = M_Z^\dagger - M_{Z,\mathrm{fiber}}^\dagger,$$

equivalently

$$\delta n^\dagger = \frac{(M_Z^\dagger + M_{Z,\mathrm{fiber}}^\dagger)\,\delta M_Z^\perp}{\pi v^2(\alpha_{Y,m_Z} + \alpha_{2,m_Z})}, \qquad \delta\alpha_Y^\perp = \frac{(M_Z^\dagger + M_{Z,\mathrm{fiber}}^\dagger)\,\delta M_Z^\perp}{\pi v^2}.$$

The fiber-parallel hypercharge motion is

$$\delta\alpha_Y^\parallel = \alpha_{Y,m_Z}\,\frac{8\eta_{\mathrm{source}}(\tau_{2,W}^\dagger)^2 - \tau_{2,W}^\dagger}{1 + 4(\tau_{2,W}^\dagger)^2}.$$

Therefore the unique coherent repair package is

$$\Sigma_{EW}^\dagger = (\delta\alpha_2^\dagger, \delta\alpha_Y^\parallel, \delta\alpha_Y^\perp),$$

equivalently $(\tau_{2,W}^{\dagger}, \delta n^{\dagger})$*. If the selected carrier anchor is the* $\tau_2 = 0$ *fiber point, so that*

$$\alpha_{2,*} = \alpha_{2,m_Z}, \qquad \alpha_{Y,*} = \alpha_{Y,m_Z}(1 - 2\eta_{\text{source}}),$$

then the coherent validation couplings are

$$\alpha_{2,\dagger} = \alpha_{2,*} + \delta\alpha_2^{\dagger}, \qquad \alpha_{Y,\dagger} = \alpha_{Y,*} + \delta\alpha_Y^{\parallel} + \delta\alpha_Y^{\perp},$$

and they emit one coherent quintet

$$M_W^{\dagger} = v\sqrt{\pi\alpha_{2,\dagger}}, \qquad M_Z^{\dagger} = v\sqrt{\pi(\alpha_{Y,\dagger} + \alpha_{2,\dagger})},$$

$$\alpha_{\text{em},\dagger}^{-1} = \frac{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}{\alpha_{Y,\dagger}\alpha_{2,\dagger}}, \qquad \sin^2\theta_{W,\dagger} = \frac{\alpha_{Y,\dagger}}{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}.$$

The target-conditioned electroweak validation law is one unique coherent value-emission law on top of the closed fiber map.

For the reference-fitted inverse-adapter comparison,

$$\begin{aligned}
\alpha_{2,\star} &= 0.03377843630219015,\\
\alpha_{Y,\star} &= 0.009682831911900495,\\
\eta_{\text{source}} &= 0.022147000871961295, v \quad = 246.76711732749683 \text{ GeV},\\
M_W^{\dagger} &= 80.3625 \text{ GeV},\\
M_Z^{\dagger} &= 91.1879 \text{ GeV},
\end{aligned}$$

and the law emits

$$\begin{aligned}
\tau_{2,W}^{\dagger} &= -0.0005918464744071317,\\
\delta\alpha_2^{\dagger} &= -1.9991648436440412 \times 10^{-5},\\
M_{Z,\text{fiber}}^{\dagger} &= 91.16822266259587 \text{ GeV},\\
\delta M_Z^{\perp} &= 19.6773374041328 \text{ MeV},\\
\delta\alpha_Y^{\parallel} &= 5.9969727526906094 \times 10^{-6},\\
\delta\alpha_Y^{\perp} &= 1.8756950557734103 \times 10^{-5},\\
\delta n^{\dagger} &= 4.2716772021678714 \times 10^{-4}.
\end{aligned}$$

So the coherent validation couplings are

$$\begin{aligned}
\alpha_{2,\dagger} &= 0.03375844465375371,\\
\alpha_{Y,\dagger} &= 0.00970758583521092,
\end{aligned}$$

and the same coherent branch emits

$$M_W^{\dagger} = 80.3625 \text{ GeV}, \qquad M_Z^{\dagger} = 91.1879 \text{ GeV},$$

$$\begin{aligned}
\alpha_{\text{em},\dagger}^{-1} &= 132.6344411210187,\\
\sin^2\theta_{W,\dagger} &= 0.22333729871365943.
\end{aligned}$$

These two carrier-side scalars belong to the target-conditioned validation surface. They are not the physical electromagnetic readout on the Ward-projected transport branch. That inverse-fit law

provides an exact coherent calibration surface: it is a measured-pair reparametrization. On that target-conditioned surface the reference W/Z pair is hit exactly because the basis is solved from that pair; the hit is the defining condition, and the branch is calibration. It carries no predictive force. The target-free running-chart solve gives (80.330, 91.119) GeV. That running chart and the target-conditioned closure chart described just above are numerically disjoint objects under the same W/Z labels: the running chart admits no measured mass into its arithmetic, and the closure chart solves its basis from the measured pair. The Particle Data Group (PDG) comparison values are mass-dependent-width Breit–Wigner parameters, while complex-pole masses define another convention. As a stated convention diagnostic under the complex-pole conversion of the PDG 2026 reference values, the energy-pole coordinates are $(M_W, M_Z) = (80.3411410, 91.1623040)$ GeV, whereas the older $\sqrt{\mathrm{Re}\, s}$ coordinates are $(80.3340218, 91.1537725)$ GeV. These two definitions must not be mixed. The target-free running-chart coordinates carry no OPH theory covariance, and no physical readout contract is supplied. No comparison to physical resonance parameters is defined for them. The candidate forward value law has a stronger conditional theorem: it follows uniquely from a finite quotient-path certificate. The certificate is not emitted by A1–A3 or by the existing carrier artifacts.

Definition 7.2 (Finite quotient-path certificate)**.** *After quotienting gauge representatives, port labels, scheduler data, and other hidden presentation coordinates, require:*

1. *the real, CP-even, color-singlet, charge-preserving response through two returns is exactly* $\mathbb{R}q_2 \oplus \mathbb{R}q_n$*, with no third scalar or charged-neutral off-block term;*
2. *the primitive response is bilinear in* $(\eta_{\mathrm{source}}, \alpha_U)$*, one primitive event has fixed unit response normalization, and the quotient probability measure gives each of the four transmutation slots weight* $1/4$*, yielding* $\lambda_{\mathrm{EW}} = \eta_{\mathrm{source}}\alpha_U/4$*;*
3. *explicit carrier path lists, carrying one factor of* η_{source} *per return and uniform color-orbit measure* $1/3$*, have primitive, one-return, and two-return incidences* $(1, 2/3, 1)$ *for* q_2 *and* $(1, 4/3, 2)$ *for* q_n*, while the diagonal* $\mathbb{Z}_6$ *projector in the declared response representation has normalized trace* $1/6$*, source amplitude* ρ_{EW}*, and subtracts* $(\rho_{\mathrm{EW}}/6)\eta_{\mathrm{source}}^2$ *from both degree-two characters;*
4. *mismatch descent fixes the charged sign as contracting and the neutral sign as uplifting, and the parallel hypercharge fibre is the least-norm minimizer with Gram factor* $1+4\tau_2^2$ *and residual pairing* $\tau_2 + 2\eta_{\mathrm{source}}$*;*
5. *the listed paths exhaust the admissible class: deeper paths, alternative central weights, and mixed terms are absent, quotient-trivial, or separated by a positive target-independent selector gap.*

Theorem 7.3 (Conditional quotient-transport value law)**.** *Let the source-only electroweak basis be*

$$Q_{\mathrm{src}}(P) = \big(\alpha_U(P), \alpha_{2,m_Z}(P), \alpha_{Y,m_Z}(P), \eta_{\mathrm{source}}(P), v_{\mathrm{chart}}(P)\big),$$

and define

$$\rho_{\mathrm{EW}} := \frac{\alpha_{2,m_Z} - \alpha_{Y,m_Z}}{\alpha_{2,m_Z} + \alpha_{Y,m_Z}}, \lambda_{\mathrm{EW}} := \frac{\eta_{\mathrm{source}}\alpha_U}{4}.$$

Assume that every entry is emitted on the same source branch, that $\eta_{\mathrm{source}} = \rho_{\mathrm{EW}}\alpha_U$*, and that no measured* W/Z*, measured* v*, fitted electroweak parameter, or calibrated proxy is an ancestor of the tuple. Hence* $\lambda_{\mathrm{EW}} = \eta_{\mathrm{source}}^2/(4\rho_{\mathrm{EW}})$*. Assume also the finite certificate of Definition 7.2 and the*

physical-domain inequalities $\alpha_2' > 0$, $\alpha_Y' > 0$. Then the beyond-selected-carrier transport map is uniquely

$$\tau_2^{\mathrm{exact}} = -\lambda_{\mathrm{EW}}\left(1 + \frac{2}{3}\eta_{\mathrm{source}} + \left(1 - \frac{\rho_{\mathrm{EW}}}{6}\right)\eta_{\mathrm{source}}^2\right),$$
$$\delta n^{\mathrm{exact}} = \lambda_{\mathrm{EW}}\left(1 + \frac{4}{3}\eta_{\mathrm{source}} + \left(2 - \frac{\rho_{\mathrm{EW}}}{6}\right)\eta_{\mathrm{source}}^2\right),$$

with fiber hypercharge transport

$$\tau_Y^{\mathrm{fiber}} = -\frac{\tau_2^{\mathrm{exact}} + 2\eta_{\mathrm{source}}}{1 + 4(\tau_2^{\mathrm{exact}})^2}.$$

The coherent transport shifts are

$$\delta\alpha_2 = \alpha_{2,m_Z}\tau_2^{\mathrm{exact}}, \qquad \delta\alpha_Y^{\parallel} = \alpha_{Y,m_Z}\frac{8\eta_{\mathrm{source}}(\tau_2^{\mathrm{exact}})^2 - \tau_2^{\mathrm{exact}}}{1 + 4(\tau_2^{\mathrm{exact}})^2},$$

$$\delta\alpha_Y^{\perp} = (\alpha_{2,m_Z} + \alpha_{Y,m_Z})\,\delta n^{\mathrm{exact}}.$$

With

$$\alpha_{2,*} = \alpha_{2,m_Z}, \qquad \alpha_{Y,*} = \alpha_{Y,m_Z}(1 - 2\eta_{\mathrm{source}}),$$

the coherent transport couplings are

$$\alpha_2' = \alpha_{2,*} + \delta\alpha_2, \qquad \alpha_Y' = \alpha_{Y,*} + \delta\alpha_Y^{\parallel} + \delta\alpha_Y^{\perp},$$

and they emit one coherent mass chart

$$M_W^{\mathrm{chart}} = v\sqrt{\pi\alpha_2'}, \qquad M_Z^{\mathrm{chart}} = v\sqrt{\pi(\alpha_2' + \alpha_Y')}, \qquad v = v_{\mathrm{chart}}(P).$$

$$a_0(P) := \frac{\alpha_{2,m_Z}(P) + \alpha_{Y,m_Z}(P)}{\alpha_{2,m_Z}(P)\alpha_{Y,m_Z}(P)}, \qquad s_0(P) := \frac{\alpha_{Y,m_Z}(P)}{\alpha_{2,m_Z}(P) + \alpha_{Y,m_Z}(P)}.$$

Thus the certificate fixes the candidate mass pair together with the running-family anchor (a_0, s_0, v) from the source basis. The physical electromagnetic row is read from the Ward-projected unbroken $\mathrm{U}(1)_Q$ channel in Theorem 7.5.

Proof. The first condition reduces every admissible response to the two displayed coordinates. The second fixes their common primitive activity. The third gives the charged and neutral path characters, including the common $\mathbb{Z}_6$ subtraction, and the fourth fixes their signs. The fibre functional

$$\mathcal{J}_Y(t) = \frac{1}{2}(1 + 4(\tau_2^{\mathrm{exact}})^2)t^2 + (\tau_2^{\mathrm{exact}} + 2\eta_{\mathrm{source}})t$$

is strictly convex, so its unique minimizer is $-(\tau_2^{\mathrm{exact}} + 2\eta_{\mathrm{source}})/(1 + 4(\tau_2^{\mathrm{exact}})^2)$. Substitution gives the repaired couplings and the one-Higgs mass chart. The exhaustiveness condition excludes every output-changing admissible deformation, which proves uniqueness on the certified class. □

Using the same declared electroweak validation basis

$$\begin{aligned}
\alpha_{2,m_Z} &= 0.03377843630219015,\\
\alpha_{Y,m_Z} &= 0.010131601067241624,\\
\alpha_U &= 0.04112498041477454,\\
\eta_{\mathrm{source}} &= 0.022147000871961295,\\
v &= 246.76711732749683\ \mathrm{GeV},
\end{aligned}$$

the conditional quotient-transport law evaluates to

$$M_W^{\mathrm{chart}} = 80.37700001539531\ \mathrm{GeV}, \qquad M_Z^{\mathrm{chart}} = 91.18797807794321\ \mathrm{GeV},$$

$$a_0 = 128.30576920234813,$$
$$s_0 = 0.23073542347506173.$$

This specialization verifies the candidate value-law algebra. It does not derive the finite quotient-path certificate, establish that this calibration tuple belongs to the source-only pixel branch, or identify W, Z as complex propagator poles. A proof requires the finite path lists, quotient canonicalizer, exact incidence and central-trace checks, fibre Gram calculation, deformation enumeration or positive selector gap, and a same-branch target-independence proof.

Proposition 7.4 (Local nondegeneracy of the two-output mass chart)**.** *Define*

$$n_{\mathrm{fib}}(\tau_2) := 1 + \frac{\alpha_2\tau_2 + \alpha_Y\tau_Y(\tau_2)}{\alpha_2 + \alpha_Y}.$$

On the physical domain $1 + \tau_2 > 0$ *and* $n_{\mathrm{fib}}(\tau_2) + \delta n > 0$,

$$\det \frac{\partial(M_W^{\mathrm{chart}}, M_Z^{\mathrm{chart}})}{\partial(\tau_2, \delta n)} = \frac{M_W^{\mathrm{chart}} M_Z^{\mathrm{chart}}}{4(1+\tau_2)\,[n_{\mathrm{fib}}(\tau_2) + \delta n]} > 0.$$

Holding $(\alpha_2, \alpha_Y, v, \eta)$ *fixed, the two-output chart is locally nondegenerate. This determinant neither proves the two-coordinate response condition nor excludes an additional physical response channel. Within the displayed two-coordinate chart, the missing content is a source selector rather than another fit coordinate.*

Proof. M_W^{chart} is independent of δn, while

$$\frac{\partial M_W^{\mathrm{chart}}}{\partial \tau_2} = \frac{M_W^{\mathrm{chart}}}{2(1+\tau_2)}, \qquad \frac{\partial M_Z^{\mathrm{chart}}}{\partial \delta n} = \frac{M_Z^{\mathrm{chart}}}{2[n_{\mathrm{fib}}(\tau_2) + \delta n]}.$$

The determinant is the product of these diagonal entries. □

Theorem 7.5 (Ward-projected $\mathrm{U}(1)_Q$ transport law and Thomson-limit electromagnetic readout)**.** *Let the source-only running family be fixed by the forward pixel solve on the declared running/matching/threshold/scheme surface, with source basis*

$$Q_{\mathrm{src}}(P) = \big(\alpha_U(P), \alpha_{2,m_Z}(P), \alpha_{Y,m_Z}(P), \eta_{\mathrm{source}}(P), v_{\mathrm{chart}}(P)\big).$$

Assume the realized electric-charge branch, so that

$$Q = T_3 + Y$$

on the realized low-energy branch and color-singlet physical states carry integer Q-charge. Assume further that the post-selector electroweak transport object is the populated quotient-local kernel

$$K_{\mathrm{src}}^{\mathrm{EW}}(q^2; P) = \{\Pi_{AA}(q^2; P), \Pi_{AZ}(q^2; P), \Pi_{ZZ}(q^2; P), \Pi_{WW}(q^2; P)\}$$

on the declared physical observable algebra, together with a Ward projector

$$\mathcal{W}_Q : K_{\mathrm{src}}^{\mathrm{EW}}(q^2; P) \longrightarrow \Pi_Q(q^2; P)$$

onto the unbroken electromagnetic channel satisfying at $q^2 = 0$

$$\mathcal{W}_Q[\Pi_{AZ}(0;P)] = 0, \qquad \mathcal{W}_Q[\Pi_{AA}(0;P)] \neq 0.$$

Assume the abelian projected edge-sector probabilities obey the U(1) *heat-kernel law*

$$p_n(q^2;P) \propto e^{-t_Q(q^2;P)\lambda_n}, \qquad \lambda_n = n^2,$$

equivalently

$$g_Q^2(q^2;P) = \frac{t_Q(q^2;P)}{2\pi},$$

and that the scalar readout uses the same source family, declared renormalization scheme, and originating electroweak kernel throughout. Assume also that $t_Q(q^2;P) > 0$ *on the readout domain and that it has a finite, strictly positive Thomson limit*

$$t_Q(0;P) := \lim_{q^2\to 0} t_Q(q^2;P) > 0.$$

Then the unique electromagnetic coupling readout on the Ward-projected branch is

$$\alpha_{\mathrm{em}}^{-1}(q^2;P) = \frac{8\pi^2}{t_Q(q^2;P)}.$$

If

$$a_0(P) := \alpha_{\mathrm{em}}^{-1}(m_Z^2;P),$$

then the unique zero-normalized affine scalar on the same source-locked family is

$$\delta_\alpha(q^2;m_Z^2;P) := \frac{t_Q(m_Z^2;P)}{t_Q(q^2;P)} - 1,$$

so that

$$\alpha_{\mathrm{em}}^{-1}(q^2;P) = a_0(P)(1 + \delta_\alpha(q^2;m_Z^2;P)).$$

Hence the Thomson-limit electromagnetic readout on that same Ward-projected electromagnetic kernel:

$$\alpha_{\mathrm{Th}}^{-1}(P) := \lim_{q^2\to 0} \alpha_{\mathrm{em}}^{-1}(q^2;P) = a_0(P)\,\frac{t_Q(m_Z^2;P)}{t_Q(0;P)}.$$

Consequently, the low-energy electromagnetic row on this branch is read as the Thomson endpoint of the electroweak transport family instead of as a mass-chart byproduct.

Proof. The realized electric-charge branch fixes the charge operator by $Q = T_3 + Y$, and the quotient-protected charge theorem fixes the physical color-singlet Q-lattice to be integral. The abelian edge-sector theorem supplies the heat-kernel extraction rule

$$p_n \propto e^{-t_Q\lambda_n}, \qquad g_Q^2 = \frac{t_Q}{2\pi},$$

so the Ward-projected Q-channel determines a unique positive transport time $t_Q(q^2;P)$ on the assumed readout domain. The separately assumed finite positive limit makes the Thomson ratio defined. Because the Ward projector kills A - Z mixing at $q^2 = 0$, the selected channel is the physical unbroken electromagnetic channel. Then

$$\alpha_{\mathrm{em}} = \frac{g_Q^2}{4\pi} = \frac{t_Q}{8\pi^2},$$

hence

$$\alpha_{\mathrm{em}}^{-1} = \frac{8\pi^2}{t_Q}.$$

Evaluating at $q^2 = m_Z^2$ defines the source-locked anchor $a_0(P)$, and dividing by the same expression at general q^2 yields

$$\delta_\alpha = \frac{t_Q(m_Z^2)}{t_Q(q^2)} - 1.$$

The provenance-equality clause forces one running-family source, one frozen scheme, and one origin kernel for all scalar readouts, so no mixed-family or Z-only surrogate is promotable. The Thomson formula is the $q^2 \to 0$ limit of the same identity. □

Corollary 7.6 (Conditional Ward-projected Maxwell normalization)**.** *On the same Ward-projected* $\mathrm{U}(1)_Q$ *branch, assume in addition the low-energy Maxwell action with the displayed nonzero kinetic normalization. Then the electromagnetic field strength and current obey*

$$F_Q = dA_Q, \qquad dF_Q = 0, \qquad d{*}F_Q = g_Q^2(q^2;P){*}J_Q,$$

with

$$g_Q^2(q^2;P) = \frac{t_Q(q^2;P)}{2\pi}, \qquad \alpha_{\mathrm{em}}^{-1}(q^2;P) = \frac{8\pi^2}{t_Q(q^2;P)}.$$

At the Thomson endpoint, this is the Maxwell normalization used by the public fine-structure row.

Proof. The compact reconstruction supplies the unbroken abelian electromagnetic connection direction $\mathrm{U}(1)_Q$ with charge generator $Q = T_3 + Y$, but does not supply the Maxwell kinetic action. On an abelian factor the compact-gauge curvature restricts to $F_Q = dA_Q$, so $d^2 = 0$ gives $dF_Q = 0$. Varying the quadratic electromagnetic action in the OPH coupling convention gives $d{*}F_Q = g_Q^2{*}J_Q$. The theorem above identifies the same branch coupling as $g_Q^2 = t_Q/(2\pi)$, hence $\alpha_{\mathrm{em}}^{-1} = 8\pi^2/t_Q$. □

Remark 7.7 (Promotion contract for the declared pixel–Ward pipeline)**.** *Promotion of this particular pixel–Ward construction to a source-only fine-structure row requires that no endpoint target feed the same-source Ward kernel. Its provenance packet must contain a source-derived Ward-current spectral measure and source kernel, the electromagnetic normalization, a same-scheme finite remainder, a no-target-leak dependency graph, the full pixel-map contraction, a target-independent selector for one admitted map, and a typed same-quantity bridge between its input and output. The equality*

$$\lim_{q^2 \to 0} \alpha_{\mathrm{em}}^{-1}(q^2;P) = \alpha_{\mathrm{Th}}^{-1}(P)$$

is definitional in this construction and supplies no independent validation. Empirical decision readiness additionally requires a prospectively frozen uncertainty model and decision rule. A predictive interval narrower than a comparison experiment can improve competitive power, but it is not a condition for source-only provenance. Without the listed provenance packet, the displayed endpoint retains its empirical hadron-closure classification. This contract does not rule out another analytic or computational source-only construction with a different, fully typed provenance chain.

For this pipeline, the same-source backend contract treats the scalar row as a two-current marginal of a larger source-derived hadronic spectral backend. Using that backend for both the Thomson endpoint and the HVP branch requires the source law to record the QCD quotient ensemble, source parameter map, Ward current normalization, systematics accounting, and no-target-leak

DAG. Full hadronic precision claims require higher-point and transition spectral exports from that backend instead of reusing the scalar fine-structure residual.

The endpoint table records the source-locked anchor from the runtime candidate,

$$a_0(P) = 128.307965473286248209961108741756 7\ldots,$$

and the OPH-plus-empirical hadron-closure value

$$\alpha_{\mathrm{emp}}^{-1}(0) = 136.3827548174946, \qquad \alpha_{\mathrm{emp}}^{-1}(0) \in [136.3670480603, 136.3984651934].$$

The source derivation contains no built-in reference inverse-α value. The measured endpoint 137.035999177(21) is excluded as a source-only closure-solve input and is a separate comparison coordinate. The source-side calculation emits the certified value $\alpha_{\mathrm{root}}^{-1} = 136.994835177413\ldots$, so its difference from the endpoint readout is an endpoint-and-matching comparison packet. This certificate establishes a unique root of the incomplete declared numerical map. It does not derive a relation between that root and the physical Thomson endpoint.

Remark 7.8. *The empirical closure row is a Thomson-limit inverse fine-structure evaluation using external hadron data. The source anchor* $a_0(P) = 128.307965473286248209961108741756 7\ldots$ *is the electroweak-scale running-family value at* m_Z^2*, while the measured row is separate. The 2022 CODATA/NIST reference value is* $\alpha^{-1} = 137.035999177(21)$ *[13].*

The measured-reference inverse repair surface provides diagnostic validation and carries no predictive force. The companion carrier-side pair

$$\alpha_{\mathrm{em},\dagger}^{-1} = 132.6344411210187,$$
$$\sin^2\theta_{W,\dagger} = 0.22333729871365943$$

is mass-chart bookkeeping on that target-conditioned validation surface. It is not the physical electromagnetic readout on the Ward-projected $\mathrm{U}(1)_Q$ branch.

The electroweak quantitative branch contains an underdetermination theorem for the quadratic repair family, a color-balanced descent, a candidate mass emitter, and the Ward-projected electromagnetic transport theorem above the source-locked running-family anchor. All of them use one source family, one declared renormalization scheme, and one originating electroweak kernel. The forward transmutation calculation reconstructs the same $\alpha_U(P)$, $t_U(P)$, and $t_{\mathrm{tr}}(P)$ as the pixel-closure solve without reading them back from measured couplings. This separation does not establish target independence. This construction supplies neither the finite quotient-path certificate nor a physical pole.

7.2 Hierarchy interpretation on the local transmutation branch

The selected branch carries a conditional dimensionless electroweak hierarchy relation. It does not certify Higgs naturalness. The large cosmic horizon ratio and the electroweak hierarchy belong to different OPH branches. A value N_{CRC} would be the output of a positive direct correctable-public-record branch. The fixed-cutoff packet and bounded counterfamily do not emit such a value. Its de Sitter interpretation and electroweak identification would require the horizon–record identification and the common screen/electroweak load-carrier identification, respectively. The local pixel/transmutation readout is the ratio

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P_\star)}\right].$$

The integer 4 has an exact representation-theoretic witness on the selected exterior package: three color copies of the weak doublet Q plus one lepton doublet L. Identifying this multiplicity with the transmutation coefficient is conditional on an intertwiner between the port and weak carriers and on equality of their normalized load traces. It is not the $SU(2)$ beta-function coefficient, and its physical load interpretation cannot be inferred from the dimension count alone. The finite-iteration hierarchy certificate records the interval

$$I_U = [0.041123336195630494,\ 0.041125336195630496],$$

the Krawczyk inclusion

$$K(I_U) \subset [0.041124335718554983,\ 0.041124336672706462] \subset \mathrm{int}(I_U),$$

and a strictly negative derivative enclosure $[-10.995768, -10.985284]$ for its declared finite map. The exact implicit-scale attachment of Theorem A.3 separately proves a unique hierarchy zero in I_U at the stored comparison decimal, with residual derivative in $[-12, -10]$ and scale in $[7, 8]\times 10^{-18}$. It retains the finite representation cutoffs. On the public endpoint branch, at the CODATA-located comparison pixel P_C,

$$\alpha_U(P_C) = 0.041124336195630495,$$
$$\frac{v}{E_\star} = 2.0199803239725553 \times 10^{-17}.$$

The source-only branch excludes the public Thomson endpoint upstream and has certified forward pixel $P_{\mathrm{fwd}} = 1.630972095858897\ldots$. These two named branches must not be merged. The dimensionless ratio is generated by the ordinary exponential transmutation factor controlled by the unified diffusion coupling; a weak scale in GeV additionally requires an independently source-closed $E_\star$.

The current packet supplies no OPH result about Higgs naturalness. On the declared electroweak/criticality surface, the observer-visible scalar mass is the normal-form readout

$$m_H = H_{\mathrm{OPH}}(P_\star)$$

on the selected branch. The split into a bare Higgs mass plus a cutoff correction is a regulator coordinate split, but declaring that split unobservable does not prove that all relevant deformations are absent or fixed. The current comparison maps are chosen so that their square commutes and their declared defect is zero. Exact countermodels on the same source antecedents also have nonzero defect. The live route is to derive the comparison maps from the source and then evaluate both residuals independently.

7.3 Local/global resonance continuation for the hierarchy

The local/global hierarchy construction compares the electroweak transmutation exponent with a proposed global capacity branch at the same joint pair $(P_\star, N_{\mathrm{CRC}})$. Conditional on an executed direct public-record fixed point and the common screen/electroweak load-carrier identification, the target relation is

$$t_{\mathrm{tr}}(P_\star) = \frac{P_\star}{12}\log\left(\frac{N_{\mathrm{CRC}}}{\pi}\right),$$

or equivalently

$$\frac{v}{E_{\mathrm{cell}}} = \left(\frac{N_{\mathrm{CRC}}}{\pi}\right)^{-P_\star/12}.$$

This packages the mathematical electroweak hierarchy bridge as a resonance between the local declared-map root and a conditional global screen-capacity coordinate. That screen branch supplies twelve curvature ports. The exterior package supplies four weak-doublet copies, and the common screen/electroweak load-carrier map is the physical identification that makes their additive load the screen load. The oriented 24-slot register and product-adjoint count are independent bookkeeping facts and carry no load or clock implication.

For comparison, the observer-visible product adjoint has the independent count. On the realized product branch,

$$m_{\mathrm{rep}} = 2\dim(\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)) = 2(8+3+1) = 24.$$

The factor 2 orients each product-adjoint channel and does not follow from the screen register. This count does not enter the direct capacity or common-load proof. The $\mathfrak{su}(5)$ adjoint has the same single-orientation integer for a different support: its X/Y mixed gauge channels are absent from the OPH product branch. This channel exclusion is narrower than proton stability. Conditional on the declared one-generation matter table and baryon and lepton labels, the exact dimension-six gauge and Lorentz census admits $QQQL$, $QQ\,u_Re_R$, QLu_Rd_R, and $u_Ru_Rd_Re_R$. Their exterior-algebra representatives are nonzero. The finite gauge and diagonal $\mathbb{Z}_6$ data supply no Wilson coefficient, physical decay amplitude, QCD matrix element, or lifetime.

After whole-fiber scalarization and faithful capacity-carrier representation, the direct active-capacity map is deflationary. It therefore has no positive two-sided Banach attractor on a domain containing smaller capacities. Under confusability-reflecting capacity extension, however, iteration from the declared maximum boundary dimension stabilizes finitely at the greatest fixed point. This order-theoretic result selects the greatest fixed point only after a scalar continuation map and its domain have been supplied. It does not choose the physical continuation map. A bounded counterfamily with shared base, positivity, and carrier controls does not entail a finite-size slack law with one physical zero. No universal all-rung membership of its countermodels in the complete A1–A3 capacity-source contract or executable-to-Lean bridge is supplied. Direct N is not evaluable on that incomplete antecedent. An independently produced ρ_{op} is available as a commuting-square test against $\log M_0$; no derivative is intrinsic at finite integer dimension.

Theorem 7.9 (Conditional common-load electroweak bridge)**.** *Assume $N_\star = \log M_0(\mathfrak{U}_{N_\star})$ is supplied by the independent correctable-public-record closure. Assume the screen-sieve branch independently supplies*

$$\Gamma_{\mathrm{scr}} = \frac{P}{12}\log\!\left(\frac{N_\star}{\pi}\right),$$

the common screen/electroweak load-carrier map identifies $\Gamma_{\mathrm{scr}} = \log(E_{\mathrm{cell}}/v)$, and the independent electroweak source relation gives $\log(E_{\mathrm{cell}}/v) = \pi/[2\alpha_U(P)]$. Then the bridge residual

$$\mathcal{B}_{\mathrm{EW}}(P, N_\star) := \alpha_U(P)\log(N_\star/\pi) - \frac{6\pi}{P} = 0.$$

Equivalently, the bridge coordinate is

$$N_{\mathrm{EW}}(P) = \pi\exp\!\left[\frac{6\pi}{P\alpha_U(P)}\right].$$

For the measured-endpoint comparison coordinate P_C,

$$N_{\mathrm{EW}}(P_C) = 3.532354622692990651118751296233054760046 2\times 10^{122}.$$

For the independently evaluated source-forward coordinate P_{fwd},

$$N_0 := N_{\mathrm{EW}}(P_{\mathrm{fwd}}) = 3.532131543418936\ldots\times 10^{122}.$$

For either declared input P, the bridge-map coordinate is conventionally labeled

$$N_{\mathrm{CRC}}^{\mathrm{EW}}(P) = \pi \exp\left[\frac{6\pi}{P\alpha_U(P)}\right],$$

with $\mathcal{B}_{\mathrm{EW}}(P, N_{\mathrm{CRC}}^{\mathrm{EW}}(P)) = 0$, and hence

$$\frac{v}{E_{\mathrm{cell}}} = \left(\frac{N_{\mathrm{CRC}}^{\mathrm{EW}}(P)}{\pi}\right)^{-P/12}.$$

The exterior representation witness gives the exact weak-doublet multiplicity $3 + 1 = 4$, and every additive isomorphism-invariant load normalized by $L_P(\mathbb{C}) = P$ gives $L_P(\mathbb{C}^4) = 4P$. Identifying this abstract weak load with the public screen load is exactly the common-carrier receipt. The notation $N_{\mathrm{CRC}}^{\mathrm{EW}}(P)$ does not identify this mathematical coordinate with a cosmic readback fixed point. That identification requires the source-derived direct public-record producer and the common screen/electroweak load-carrier map. The weighted Planck base-ΛCDM comparison coordinate $N_\Lambda = 3.31292709806038\ldots \times 10^{122}$ is about 6.6 percent lower than N_0.

Proof. Equating the screen and electroweak load expressions gives $P \log(N_\star/\pi)/12 = \pi/[2\alpha_U(P)]$. Rearrangement gives $\mathcal{B}_{\mathrm{EW}}(P, N_\star) = 0$. Substitution gives the displayed $N_{\mathrm{EW}}(P)$ and the hierarchy identity. □

Remark 7.10 (Finite-presence and mean-count reserve branches)**.** *On the protected-collar branch in Ref. [6], declared total reserve expectation $P/4$ and six-class equidistribution give finite presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and a separately proved global attachment maps its normalized collar-survival factor to the capacity coordinate, the result is*

$$N_{\mathrm{pres}} = N_0\left(1 - \frac{P_{\mathrm{fwd}}}{24}\right).$$

On the source-forward numerical branch this is $3.292097877326465\ldots \times 10^{122}$, about 0.63 percent below the weighted Planck base-ΛCDM comparison coordinate. The exponential expression

$$N_{\mathrm{Pois}} = N_0 e^{-P_{\mathrm{fwd}}/24}$$

is a distinct mean-count or continuum branch and needs a carrier beyond the finite projection trace. It gives $3.300072225377652\ldots \times 10^{122}$, about 0.39 percent below the same coordinate. The inherited screen/electroweak bridge premises, common-load identity, physical seam action, one-class selection, and scalar-weighted receipt do not provide a global action. Exact neutral and multiplicative completions of the same local survival datum are positive and obey disconnected-cut composition and finite cut-count regrouping, yet they disagree after one cut. The finite source also selects no blocked-event semantics. The declared branch therefore supplies no global-capacity object for a horizon identification. A positive construction requires a stronger source-derived global action. The target was known before these candidates were examined, so neither comparison has predictive standing.

Theorem 7.11 (Icosahedral screen sieve and gated capacity composition)**.** *On the certified echosahedral carrier lineage of Ref. [6], assume twelve primitive equal-trace port-center atoms, a declared integer atom-counting grammar on the total-twelve fiber, the normalized central-readback Hilbert–Schmidt cost, oriented $(12, 30, 20)$ incidence, and refinement-cocycle data. Then the counting theorem gives twelve unit ports. The incidence theorem gives their fixed-point-free inverse pairing, the*

proper A_5 action, and the regular rank-three icosahedral frame, naturally under refinement and relabeling. An equal-weight invariant load X is therefore read locally as $X/12$. For $X = \log(N/\pi)$, local cell entropy $P/4$, and electroweak multiplicity $\beta_{\mathrm{EW}} = 4$, define the screen composition coordinate

$$\Gamma_{\mathrm{scr}} = \beta_{\mathrm{EW}} \frac{P}{4} \frac{1}{12} \log(N/\pi) = \frac{P}{12} \log(N/\pi).$$

If, in addition, the common screen/electroweak load-carrier map identifies $\Gamma_{\mathrm{scr}} = \log(E_{\mathrm{cell}}/v)$ and matches it to the source transmutation exponent $\pi/(2\alpha_U(P))$, then

$$\log(E_{\mathrm{cell}}/v) = \frac{P}{12} \log(N/\pi), \qquad \alpha_U(P)^{-1} = \frac{P}{6\pi} \log(N/\pi),$$

which is equivalent to the electroweak bridge residual $\mathcal{B}_{\mathrm{EW}}(P, N) = 0$.

Proof. The declared counting-realization identity

$$H(q) = 12 + \sum_p (q_p - 1)^2$$

gives the unique all-one allocation with exact gap two. The oriented incidence has distance profile $(1, 5, 5, 1)$, positive automorphism group A_5, and exact Gram matrix $G^2 = 4G$ of rank three, as proved and checked in Ref. [6]. Edge-center collars expose the twelve primitive atoms as central ports. Equal-weight additivity gives $X/12$. Multiplying by $P/4$ and $\beta_{\mathrm{EW}} = 4$ gives $\Gamma_{\mathrm{scr}} = (P/12) \log(N/\pi)$. The additional identification receipt and the source transmutation equation give the remaining displays. □

The all-one split and gap are exact within the declared integer counting and normalized readback-cost realization. The integer loads also have an operational finite realization. Append-only signed atomic events generate the register values, and an admissible repair transfers one whole unit across a seam with load difference $d \geq 2$. For

$$V(q) = \sum_p q_p^2$$

each such move gives the exact decrement $\Delta V = -2(d - 1) < 0$. Hence V proves termination. The minimum number of admissible moves is a derived settling cost, natural under every declared carrier rotation and refinement. A half-unit display rescales the readback units of the same event graph; it does not implement half of an atomic event.

The load-square Lyapunov function V, the seam quadratic, the normalized readback cost H, and the Hessian associated with premise A3 are distinct objects. The positive readback scale is a unit convention. The proper-A_5 result follows independently from oriented incidence on the declared echosahedral lineage. The port/weak-carrier intertwiner, normalized port-load trace equality, and common screen/electroweak load-carrier map are physical branch premises. The exact exterior multiplicity alone does not discharge the physical load identity.

This theorem is local to one certified carrier lineage. It does not identify that carrier with a primitive observer, construct the federation nerve, or derive a global S^2 support screen. Those claims require the independent access-and-record, overlap, carrier-to-screen, and refinement receipts stated upstream.

Theorem 7.12 (EW bridge-map fixed point)**.** *Define the log-capacity map*

$$\mathcal{C}_{\mathrm{EW}}(P, x) = (1 - \lambda)x + \lambda \frac{6\pi}{P\alpha_U(P)}, \qquad 0 < \lambda \leq 1.$$

For fixed declared P and $\lambda = 1/2$, this is a contraction with Lipschitz constant $1/2$. Its unique fixed point is

$$x_\star(P) = \frac{6\pi}{P\alpha_U(P)}, \qquad N_{\mathrm{CRC}}^{\mathrm{EW}}(P) = \pi e^{x_\star(P)}.$$

Therefore

$$\mathcal{B}_{\mathrm{EW}}(P, N_{\mathrm{CRC}}^{\mathrm{EW}}(P)) = 0.$$

This theorem concerns the deliberately defined map $\mathcal{C}_{\mathrm{EW}}$, not the physical direct public-record map. Identifying the output with cosmic capacity requires an additional source law selecting one physical fixed point, a universe-level carrier attachment, and the common screen/electroweak load-carrier map. At P_{fwd} its value is $N_0 = 3.532131543418936\ldots \times 10^{122}$, about 6.6 percent above the weighted Planck base-ΛCDM coordinate. Applying the finite presence or Poisson reserve factor requires the distinct premises of Remark 7.10; neither factor follows from this contraction.

Proof. The map has derivative $1-\lambda$ in x, hence is a contraction for $0 < \lambda \leq 1$. Solving $x = \mathcal{C}_{\mathrm{EW}}(P, x)$ gives the displayed $x_\star(P)$. Substitution into $\mathcal{B}_{\mathrm{EW}}$ gives zero. □

Theorem 7.13 (Declared-map RG/Higgs commuting-square identity)**.** *On a declared hierarchy branch satisfying the selected-surface premises, let Q_s be the source hierarchy quotient and Q_H the Higgs/electroweak quotient. Define*

$$\rho_{sH}([x]_s) = [P(x), N(x), \Theta(P(x), N(x)), \Pi_{HT} F_{\mathrm{crit}} F_{\mathrm{EW}}(P(x), N(x)), 0]_H\,,$$

with

$$\Theta(P, N) = \left(\frac{N}{\pi}\right)^{-P/12}.$$

Here F_{EW} denotes the declared electroweak map and F_{crit} denotes the criticality map. For the corresponding normal-form maps and obstruction maps,

$$\rho_{sH} n_s = n_H \rho_{sH}, \qquad \chi_{sH} h_s = h_H \rho_{sH}.$$

Consequently, the defect of these declared maps is

$$\varepsilon_H^{\mathrm{decl}} = \max\{\varepsilon_{sH}^n, \varepsilon_{sH}^h\} = 0, \qquad \varepsilon_H^{\mathrm{decl}} \in [0, 0].$$

Proof. The two routes through the source and Higgs quotients preserve the same tuple

$$(P, N, \Theta(P, N), \Pi_{HT} F_{\mathrm{crit}} F_{\mathrm{EW}}(P, N)).$$

Both controlled squares preserve this tuple by their declared definitions, so their product defect is zero. This proof establishes no source selection of the maps and no statement about physical relevant deformations. □

The declared-map identity does not itself use the measured weak scale. Measured W/Z, Higgs/top, G, Planck-area, and Λ data are excluded by its declared input contract. The local/global package records the direct-capacity/common-load electroweak projection bridge, mathematical EW bridge-map fixed point, conditional operational-readback implications, and one declared RG/Higgs square. Exact finite completions sharing the current source antecedents attain defects zero and one, so the antecedents do not determine a naturality defect. A source-derived definition of all comparison maps followed by independent evaluation of both residuals is the constructive continuation. The present package does not supply that continuation, the additional capacity-selecting

source law, the universe-level carrier attachment, or the common screen/electroweak load-carrier identification. No strict source-root certificate is supplied, and the result does not independently attach $E_\star$ to a physical GeV scale. The finite producer imported from the observers paper [4] is

$$\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}, \qquad M_0(q) = \alpha(G_q),$$

with the public checkpoint packet, capacity-carrier representation, whole-fiber scalarization, confusability-reflecting extension/refinement, sewing, and fixed-point receipts kept explicit. The compatible all-rung counterfamily completions have different zero sets under shared base, positivity, carrier, and executable finite controls. The complete A1–A3 source contract has not been shown to contain those completions at every rung, and no executable-to-Lean bridge establishes that membership. The bounded completion class is nonidentifiable; the stronger source-class verdict does not follow. Direct N is not evaluable on the incomplete capacity source antecedent. A positive result requires a complete source antecedent and one physical zero. The operational resolution is an independent test. The conditional local carrier has twelve ports and an oriented 24-slot register; the latter supplies no four-load conclusion. The mathematical local/global hierarchy-resonance package is a conditional consistency surface. It does not construct the physical cosmic readback producer. The hierarchy result does not use the local/global resonance as an input.

7.4 Higgs/top critical stage

The Higgs/top critical stage is conditional on the electroweak gauge core. It is described by a downstream split law together with a Jacobian readout map on the declared running, matching, and threshold surface. Write the coupling asymmetry as ρ_{EW}, distinct from the transmutation coefficient $b_{\mathrm{tr}} = 4$. The candidate electroweak tuple

$$(\eta_{\mathrm{source}}, \rho_{\mathrm{EW}}, \lambda_{EW}, \tau_{2,\mathrm{tree}}^{\mathrm{exact}}, \delta n_{\mathrm{tree}}^{\mathrm{exact}})$$

emits the shared scalar

$$\rho_{HT} = \log(1 + \tau_{2,\mathrm{tree}}^{\mathrm{exact}})$$

and the declared residual formulas

$$R_T = -\tau_{2,\mathrm{tree}}^{\mathrm{exact}} \eta_{\mathrm{source}}^2 + \Big(1 + \frac{\rho_{\mathrm{EW}}}{28}\Big)\eta_{\mathrm{source}}^6 + \frac{\eta_{\mathrm{source}}^8}{14} + \frac{\eta_{\mathrm{source}}^9}{27},$$

$$R_H = \eta_{\mathrm{source}}^5 - \frac{3}{25}\eta_{\mathrm{source}}^6 + \frac{\lambda_{EW}\eta_{\mathrm{source}}^6}{18} + \frac{\eta_{\mathrm{source}}^8}{2\rho_{\mathrm{EW}}}.$$

The split coordinates are

$$\pi_y = \frac{\eta_{\mathrm{source}} + \left(\frac{3}{2} + \frac{\rho_{\mathrm{EW}}}{4}\right)\rho_{HT} + R_T}{\sqrt{\pi}}, \qquad \pi_\lambda = \frac{\eta_{\mathrm{source}} - \left(\frac{4}{3} - \frac{\rho_{\mathrm{EW}}}{54}\right)\rho_{HT} + R_H}{\sqrt{\pi}},$$

so the declared criticality Jacobian reads out

$$\delta y_t(\mu_t) = \pi_y\, y_t^{\mathrm{core}}(\mu_t), \qquad \delta\lambda(\mu_t) = -\frac{16}{9}\pi_\lambda\, \lambda^{\mathrm{core}}(\mu_t).$$

These equations define a physical readout only if a finite split-character certificate supplies one strict source branch and a fixed pre-split carrier, an exact two-coordinate quotient response, explicit characters deriving $\rho_{HT}, R_T, R_H, \pi_y, \pi_\lambda$ and $-16/9$, oriented mismatch descent with canonical

normalization and positivity, and an exhaustive deformation quotient proving rigidity or a target-free positive-gap selector. The displayed formulas do not supply that certificate. This gives

$$m_H = 125.1995304097179 \text{ GeV}, \qquad m_t^{\text{crit}} = 172.3523553288312 \text{ GeV}.$$

This is a coordinate on the declared electroweak/criticality Jacobian surface, obtained by back-solving from the measured pair through the synchronization-scale scan, so it is an exact interpolation of the PDG pair, a target-anchored fit; it carries no certified Higgs or top complex pole. The target-free headline is the double-criticality family evaluated below. The same surface emits a companion top coordinate. An exact negative result bounds this branch from below: at fixed P, with $u = 1 - P/24$, the exposed target-free reduct admits both the linear ($y_t = u$) and Born ($y_t = \sqrt{u}$) probability-to-amplitude completions, which agree on every reduct field and differ in the leading Higgs and top pole ratios by $2u(1-u)$ and $u(1-u)/2$. The reduct is therefore not P-complete for the Higgs/top readout. A physical conclusion requires an explicit source object that selects the amplitude lift. The Standard Model gauge paper states and proves this two-completion theorem. The selected-class quark support witness carries a target-anchored running-top comparison row using the PDG 2025 cross-section entry; it is not a separate public source-only top prediction. The bridge to the auxiliary direct-top PDG row is closed as a source-domain codomain no-go; the auxiliary row is compare-only. The one-scalar companion seed

$$\sigma_{\text{crit}} = \frac{\alpha_U \cos(2\theta_{W0})}{\sqrt{\pi}}$$

defines the fixed-ray companion branch with $\pi_y = \pi_\lambda = \sigma_{\text{crit}}$. Separately, the same criticality Jacobian admits a compare-only inverse slice that hits the canonical Higgs/top reference pair exactly: it is an exact interpolation of the PDG pair, and the landing is the defining condition of the slice rather than a derived result. That exact comparison is useful for bookkeeping and does not define the forward branch used for the public rows. The Higgs boson belongs naturally in the boson discussion. The quark-family chapter treats the top as the third-generation up-type quark. The top coordinate reported here is the conditional criticality coordinate on this surface, not a separate public top-mass prediction row. The selected-class quark support wrapper carries only a target-anchored comparison witness under the declared source-only rule.

The local implication from the candidate electroweak tuple through the displayed criticality formulas is executable and single-valued, and this arithmetic grade is certified from raw interval data: the input box contains the eleven declared branch inputs (the electroweak tuple and the criticality core/Jacobian constants, each with units or dimensionless normalization and provenance), the outward-rounded interval extension of every displayed node, the Jacobian interval enclosure over the full box, and a non-singular diagonal readout block with determinant bounded away from zero; the certified statement is scoped to the declared surface and makes no criticality-system existence or uniqueness claim. A physical Higgs mass requires a source root, independent scale, finite quotient-path selection, same-scheme running, split rigidity, top and threshold transport, a complex pole, an uncertainty enclosure, and target-independent provenance. The exact inverse slice is a compare-only validation surface.

7.5 Source-closure and physical-pole envelope

The preceding formulas end at running or declared-surface coordinates. A physical $W/Z/H$ theorem requires the following additional composition.

Proposition 7.14 (Conditional physical clock attachment)**.** *If the same strict source branch emits the dimensionless energy $\varepsilon_{\mathrm{clk}} > 0$ of a declared clock transition whose physical frequency ν_{clk} defines the operational unit, then, with $h_{\mathrm{P}} = 2\pi\hbar$ Planck's constant,*

$$E_\star = \frac{h_{\mathrm{P}}\nu_{\mathrm{clk}}}{\varepsilon_{\mathrm{clk}}}.$$

Equivalently, $\gamma_\star = \ell_\star\nu_{\mathrm{clk}}/c$ gives $E_\star = \hbar\nu_{\mathrm{clk}}/\gamma_\star$. An operator-norm perturbation of at most ϵ changes an isolated selected clock gap by at most 2ϵ. This is a conversion and stability theorem; the required physical reference transition and SI binding are absent.

Proposition 7.15 (Exact finite source-Hamiltonian gap)**.** *Let D_σ be the sign-twisted seam derivative on the connected observer-visible finite local domain, and let $H_\sigma = D_\sigma^* D_\sigma$ use the declared unit-counting measure. The signed-incidence formula makes H_σ integer symmetric and positive semidefinite. A kernel vector would supply a sign-consistent coloring on each connected component. The declared reversing seam transport frustrates every triangle, so the connected complex has no such coloring and the signed incidence rank is* 8,662*. Therefore* $\ker H_\sigma = 0$*. With maximum degree* 12*, the determinant bound gives*

$$\lambda_{\min}(H_\sigma) \geq 24^{-8661} > 0.$$

A sparse numerical refinement gives $\lambda_{\min} = 0.11753670336118684$ with relative residual 1.49×10^{-14}. The decimal is measured rather than certified. The exact statement is strict positivity and the displayed floor.

This source-derived dimensionless gap does not select a physical field fiber or laboratory transition. It therefore supplies no ν_{clk}, SI energy, particle mass, or determinant-line attachment. A sign-consistent four-cycle control has an exact zero mode, so positivity is not hardwired into the instrument [7].

The atomic component of that packet has an exact scalar reduction. Let $P = P_3 + P_4$ project onto the isolated zero-field $6S_{1/2}$, $I = 7/2$ cesium ground manifold, $\dim V_3 = 7$, $\dim V_4 = 9$, and let $Q = 1 - P$. On every real interval where $Q(\widehat{H} - z)Q$ is invertible, define the Feshbach–Schur operator $\mathcal{F}_P(z) = P(\widehat{H} - z)P - P\widehat{H}Q\,[Q(\widehat{H} - z)Q]^{-1}Q\widehat{H}P$.

Proposition 7.16 (Cesium channel scalarization and monotone clock roots)**.** *$z \in \operatorname{spec}\widehat{H}$ exactly when $0 \in \operatorname{spec}\mathcal{F}_P(z)$, with matching multiplicities. Rotational invariance and the multiplicity-free decomposition $V_3 \oplus V_4$ force $\mathcal{F}_P(z) = f_3(z)P_3 + f_4(z)P_4$, and $\mathrm{d}\mathcal{F}_P/\mathrm{d}z \preceq -P$ gives $f_F'(z) \leq -1$. One sign bracket per channel therefore isolates exactly one level, and the clock normal form is $H^{\mathrm{eff}}_{\mathrm{clock}} = E_0 P + \widehat{a}_{\mathrm{Cs}}\,\mathbf{I}\cdot\mathbf{J}$ with $E_0 = (9E_4 + 7E_3)/16$ and $\varepsilon_{\mathrm{Cs}} = E_4 - E_3 = 4\widehat{a}_{\mathrm{Cs}}$.*

The public atomic packet may therefore export two monotone scalar interval evaluators with resolvent certificates and sign brackets in place of a correlated 55-electron matrix. The unsupplied clock parents are the atomic-scheme electromagnetic convention, the absolute electron ratio $m_e c^2/E_\star$, the cesium nuclear mass/spin/current/Compton packet, the two scalar evaluators, summable refinement tails, and the no-target provenance freeze. A non-entailment theorem fixes the direction of effort: two source extensions that agree on every named object and differ in one unfixed parent produce different unique gaps, so no unique $\varepsilon_{\mathrm{Cs}}$ is a theorem emitted by the source, and further evaluation of the existing formulas cannot close the clock branch.

Proposition 7.17 (Operational scale metrology)**.** *In SI units $E/\mathrm{GeV} = [E/(h\nu_{\mathrm{Cs}})]\,[h\nu_{\mathrm{Cs}}/\mathrm{GeV}]$, and the second factor is an exact unit conversion. A source-only numerical GeV mass is therefore equivalent to a source-only dimensionless operational clock ratio. Replacing cesium by another clock species adds the burden of predicting that clock's frequency ratio to the defining cesium transition.*

The stored clock-gap candidate reproduces the displayed gravitational coupling to a relative agreement of order 10^{-49} through the clock–gravity identity, so the decimal is a calibration checksum rather than a prediction. The selection mechanism for source laws themselves is closed separately by a source-action rigidity theorem: on a finite quotient with a faithful reference measure, an affinely independent feature basis, and a source-emitted moment vector, the maximum-entropy law is unique, every feasible competitor pays a positive relative-entropy gap with the Pinsker lower bound, and the surviving action freedom consists of additive constants, exact feature redundancies, and BRST-exact terms. The physical Standard-Model operator basis and its moment vector $c_r(P_\star, N_\star)$ are required inputs not supplied by that mechanism.

Proposition 7.18 (Unique frozen RG and threshold transport)**.** *Let the canonically normalized coefficient vector $X(\mu)$ obey a locally Lipschitz beta system on each interval of a frozen finite threshold list. Assume that the maximal solution from each incoming matched value exists and remains finite on the full closed interval, and that each matching output lies in the domain of the next beta system. Also assume deterministic matching maps, fixed loop order, field content, renormalization scheme, threshold locations, and decoupling order. Then one source initial condition determines one low-energy coefficient vector. If L_j bounds the beta-system Lipschitz constant and K_j bounds threshold map j, perturbations obey the corresponding product of $e^{L_j \Delta t_j}$ and K_j, plus the declared truncation and matching remainders.*

Proof. Picard–Lindelöf gives local existence and uniqueness; the stated full-interval non-blowup and domain hypotheses extend the unique solution to every declared threshold. Grönwall bounds perturbations on each interval; applying each deterministic Lipschitz threshold map and inducting through the ordered list gives the stated composition. □

The printed running/matching packet is a declared-convention contract. It does not satisfy this source theorem because it supplies no beta provenance, threshold origins, matching interval composition, or truncation enclosure.

Theorem 7.19 (Conditional complex-pole theorem)**.** *For $B = W, Z, H$, let $\Gamma_B^{\mathrm{phys}}(s, \xi)$ be the BRST-complete physical inverse two-point block, including every declared mixing field after the Ward/Slavnov–Taylor projection, analytically continued to a declared Riemann sheet, and set*

$$D_B(s, \xi) = \det \Gamma_B^{\mathrm{phys}}(s, \xi).$$

Assume a reference determinant has exactly one simple physical zero inside a certified contour, no zero lies on the contour, and the full determinant obeys $|D_B - D_{B,0}| < |D_{B,0}|$ there. In the neutral sector also assume the Ward-protected photon line has been separated from the massive eigenvalue. Then the enclosed pole is unique and stable. With the exact convention

$$s_B = \left(M_B - \frac{i}{2}\Gamma_B\right)^2, \qquad M_B > 0, \quad \Gamma_B \geq 0,$$

it determines one mass and width. If a Nielsen identity has the form $\partial_\xi D_B = C_B^{(\xi)} D_B$, the simple pole is gauge-parameter independent [27, 36]. For W and Z, interpretation as a physical resonance additionally requires a dressed BRST-invariant current amplitude whose Laurent coefficient at that pole is nonzero; for H, the analogous scalar-amplitude coupling is a separate condition. A simple zero of the inverse-propagator determinant alone does not establish either coupling.

Proof. Rouché's theorem preserves the number of enclosed zeros under the certified perturbation. Simplicity gives the implicit-function stability bound and a locally simple determinant zero. Differentiating $D_B(s_B(\xi), \xi) = 0$ and using the Nielsen identity gives $ds_B/d\xi = 0$. An invertible analytic field redefinition multiplies D_B by a nonzero analytic factor and therefore preserves the pole set. □

No source-emitted $W/Z/H$ self-energy kernels, analytic-sheet receipt, contour enclosures, widths, residues, or pole-convention uncertainty map are present on the displayed mass-chart surface. Schema fields named for the W, Z, and top pole masses are labels only and do not supply this theorem.

Physical scope of the icosahedral Standard Model and W/Z results

The finite icosahedral package contains an axiom-forced Lie-type theorem and an exact conditional recognition result. Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence determines the antipode J. Under the explicit contract that an admissible response is a signed central involutive graph automorphism implementing inverse-port readback, the responses are exactly $\pm J$; their common sign is conventional. Conditional also on the matrix current and rank-15 matter contract with its unique charge-conjugate projector pair, the anomaly and tensor-descent certificates fix the $3+2$ block structure, hypercharge lattice, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. The kernel is computed on every declared tensor and is insensitive to the projector representative. The source does not select the matrix current, matter action, or physical global quotient. This finite calculation supplies no laboratory identification of its current or flux sectors and no physical seam action. Equality with the independently reconstructed Tannaka current, attachment of the band to three physical chiral families, exclusion of extra light sectors, four-dimensional topological attachment, scalar attachment and dynamics, and construction of a chiral quantum field theory are not supplied.

The conditional field-theory implications are explicit. A finite local action gives an exact finite gauge-invariance and locality theorem, and the familiar electroweak tree kernel is conditional on a separate canonical continuum and action-normalization bridge. An exact finite measure criterion and an exact finite Hamiltonian criterion are two parallel branches over that action. A separate formal perturbative branch carries the strict finite-order W/Z pole theorem. A nonperturbative continuum completion gives an observable-sector reconstruction implication and a distinct continued-sheet resonance-stability implication. The measure and perturbative branches are parallel descendants of the finite local action. Neither implies the other, and a perturbative pole does not imply the continuum completion.

These quantum field theory (QFT) theorems state what follows from typed action and quantization packets. They do not show that the target-free source emits those packets. The source-selected action and normalization, complete measure, target-clean perturbative matching, physical current amplitudes, source law and covariance, uncertainty enclosure, operational clock, and continuum tower with its continued-sheet packet are not supplied. The declared external Standard Model action and effective-field-theory interval provide bounded validation inputs for the perturbative protocol. Interval receipts exclude scalar zeros in the declared principal-sheet boxes and isolate, for each of W and Z, one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. They identify neither chart with the physical resonance sheet and prove no unique continuation or self-energy sign bridge. Proof-bearing physical validation also requires two genuinely independent raw loop engines, counterterm and Becchi–Rouet–Stora–Tyutin generation from the complete action, Nielsen identities, an artifact-resolving third verifier, full-matrix contours and Laurent data, dressed-current amplitudes, and complete independent replay. The external packet is a validation input rather than an OPH source producer. No source-native dimensionless or physical-unit W/Z pole follows from it.

Exact source-side frontier

The finite fifteen-state matter representation fixes the one-copy quadratic indices

$$\sum_{\text{Weyl}} T_{\mathrm{SU}(3)} = 2, \qquad \sum_{\text{Weyl}} T_{\mathrm{SU}(2)} = 2, \qquad \sum_{\text{Weyl}} d_3 d_2 Y^2 = \frac{10}{3}.$$

After importing the standard four-dimensional one-loop functional in the convention

$$\frac{\mathrm{d}g_i}{\mathrm{d}\ln\mu} = \frac{b_i g_i^3}{16\pi^2},$$

these indices give

$$b_Y = \frac{20}{9}N_g + \frac{1}{6}N_H, \qquad b_2 = -\frac{22}{3} + \frac{4}{3}N_g + \frac{1}{6}N_H, \qquad b_3 = -11 + \frac{4}{3}N_g.$$

Here N_g counts matter families and N_H counts scalar doublets. The declared choice $(N_g, N_H) = (3, 1)$ gives $(b_Y, b_2, b_3) = (41/6, -19/6, -7)$. The finite source model does not select that physical multiplicity pair.

On the declared one-doublet, rank-three family branch, four-dimensional power counting classifies the renormalizable scalar and Yukawa coefficient space as

$$\mathbb{R}_{>0} \times \mathbb{R}^2 \times \mathrm{Mat}_3(\mathbb{C})^3.$$

The three Yukawa matrices have complex dimension 27. This is an exact operator-basis theorem, with no selected coefficient, vacuum coordinate, or map from the source chart to a renormalized vacuum expectation value. A bounded check of the selected charged-response and pole-residue simulator pair finds no explicit scalar-completion input in those two artifacts. That check is not an exhaustive producer classification and does not prove physical scalar non-identifiability.

The invariant screen order-unit line and the central order-unit line of the four-copy weak multiplicity algebra are each one-dimensional. There is a unique positive unital normalized-trace isomorphism between them, natural under the declared refinements. This abstract line isomorphism does not identify a physical common load, choose the scalar sector, or normalize a laboratory current.

A source clock would convert a dimensionless gap $\epsilon_{\mathrm{clk}} = \Delta E_{\mathrm{clk}}/E_\star$ through

$$\Delta E_{\mathrm{clk}} = h\nu_{\mathrm{clk}}, \qquad E_\star = \frac{h\nu_{\mathrm{clk}}}{\epsilon_{\mathrm{clk}}}.$$

For $0 < \epsilon_{\mathrm{lo}} \le \epsilon_{\mathrm{clk}} \le \epsilon_{\mathrm{hi}}$, the exact interval is

$$\frac{h\nu_{\mathrm{clk}}}{\epsilon_{\mathrm{hi}}} \le E_\star \le \frac{h\nu_{\mathrm{clk}}}{\epsilon_{\mathrm{lo}}}.$$

The interval inversion is exact. Proposition 7.15 supplies a source-derived dimensionless spectral gap on the finite local domain. It supplies neither a physical reference transition nor its frequency, so the SI interval has no source value.

Quantization steps and the W/Z landing

The Standard Model field-theory implications split into a perturbative branch and an observable-reconstruction branch. Their dependency structure is:

$$\begin{aligned}
\text{source matter and declared action} &\longrightarrow \text{finite local classical action},\\
\text{finite local classical action} &\longrightarrow \text{finite chiral quantum object},\\
\text{finite local classical action} &\longrightarrow \text{formal BV/ST restoration} \longrightarrow \text{strict finite}-\text{order } W/Z,\\
\text{nonperturbative observable tower} &\longrightarrow \text{observable reconstruction} \longrightarrow \text{continued}-\text{sheet resonance}.
\end{aligned}$$

Source provenance for the classical action additionally requires the finite matter result and a source/action-identity receipt. A finite chiral measure or Hamiltonian does not follow from perturbative restoration, and perturbative restoration does not follow from either finite quantum construction. Strict finite-order poles do not supply a nonperturbative observable tower or a continued-sheet resonance theorem.

Construction	Mathematical implication	Required input	Conclusion
Finite matter packet	finite representation, charge, anomaly, lattice, and selector consequences on the declared packet	physical source selection and attachment	exact finite result; physical attachment not supplied
Local classical action	finite local classical G_6 action is gauge invariant and local; the standard tree kernel also needs canonical continuum normalization	source-selected action, coefficients, regulator, normalization, and ancestry	conditional implication; source production not supplied
Finite quantum object	an equivariant determinant-line section or a noncollapsing constrained Hamiltonian supplies an exact finite quantum object	full operator, measure/current or Hamiltonian/nonvacuum packet, and refinement controls	criteria proved; constructions not supplied
Perturbative renormalization and poles	stable anomaly-free BV/ST theory is formally restorable order by order; strict W/Z poles follow from the complete finite-order packet	counterterm basis, matching and Fleischer–Jegerlehner (FJ) engines, identities, currents, and numerical freeze	implications proved; external validation and OPH source production not supplied
Observable reconstruction	Osterwalder–Schrader reconstruction and a separate continued-sheet resonance theorem	reflection-positive tower and analytic-continuation packet	implications proved; constructions not supplied

Theorem 7.20 (Finite local classical G_6 action)**.** *Let K_r be a finite oriented spin four-complex with bounded incidence, positive cell weights, declared boundary conditions, paired edge orientations, and declared spin transports. Put $U_e \in G_6 = S(U(3) \times U(2))$ on oriented edges and the declared Higgs and left-handed matter variables on vertices. Require every matter action to descend to a well-defined representation of G_6, every plaquette term to be a declared class function of its G_6 holonomy, and every finite difference to be gauge covariant with defined endpoint data. Then a finite sum of these plaquette, Higgs, fermion, and invariant Yukawa terms is local and exactly gauge invariant.*

Proof. Plaquette holonomies transform by conjugation, covariant edge differences transform at their endpoints, and class functions and invariant contractions remove those transformations. In integer hypercharge normalization the three Yukawa sums are

$$1 + 3 - 4 = 0, \qquad 1 - 3 + 2 = 0, \qquad -3 - 3 + 6 = 0.$$

Every term has bounded cell support. □

Corollary 7.21 (Canonically normalized electroweak tree kernel)**.** *If, in addition, the long-wavelength map sends the finite Higgs kinetic form to $(D_\mu H)^\dagger D^\mu H$ with canonical generator*

normalization and broken background $H_0 = 2^{-1/2}(0, v)^T$*, then*

$$w = \frac{g^2 v^2}{4}, \qquad \mathcal{M}_N^2 = \frac{v^2}{4}\begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}, \qquad z = \frac{(g^2 + g'^2)v^2}{4},$$

and the other neutral eigenvalue is zero.

Remark 7.22 (Classical-action boundary)**.** *This is a classical existence template after the complex, fields, coefficients, boundary data, and normalization bridge are supplied. It does not show that OPH selects them, construct a chiral measure, remove mirrors or doublers, or produce a complex pole.*

Theorem 7.23 (Equivariant determinant-line criterion)**.** *At a fixed finite stage, let* $D_r(U)$ *be a local gauge-covariant,* γ_5*-Hermitian Ginsparg–Wilson operator on a connected admissible gauge-field component on which the chiral-projector rank is constant, with the declared spectral gap and locality bounds. A basis-independent, gauge-invariant finite chiral measure exists precisely when the determinant line admits a nowhere-zero gauge-equivariant section in the required locality and smoothness class. For a nonfree gauge action this is an equivariant statement over the action groupoid, including stabilizer actions. In local connection form the measure current must reproduce the projector curvature and obey global loop integrability. Flatness with trivial holonomy is only a sufficient special case.*

Proof. Changes of Weyl basis are transition functions of the determinant line. A nowhere-zero equivariant section cancels them and descends to the gauge quotient; conversely a basis-independent gauge-invariant phase supplies compatible nonzero fiber vectors. The curvature equation and loop condition are the local and global integrability conditions for that section. □

Theorem 7.24 (Finite Hamiltonian soundness)**.** *Let* $\mathcal{H}_{\mathrm{kin}}$ *be finite dimensional and let local Gauss generators exponentiate to the complete local* G_6 *action. Define*

$$C_G = \sum_{x,a} (G_x^a)^\dagger G_x^a, \qquad P_{\mathrm{phys}} = \mathbf{1}_{\{0\}}(C_G), \qquad 1 < \operatorname{rank} P_{\mathrm{phys}} < \dim \mathcal{H}_{\mathrm{kin}}.$$

Suppose a bounded-range self-adjoint H_r *commutes with this action and has a unique certified physical ground state* Ω_r *with a positive gap. If a bounded self-adjoint gauge-invariant observable* O_r *has strictly positive variance in* Ω_r*, and the packet also supplies its claimed chiral index, positive mirror gap, primitive completeness, and complement-complete refinement controls, then* $P_{\mathrm{phys}}\mathcal{H}_{\mathrm{kin}}$ *is a nonvacuous finite unitary gauge theory with a positive-energy nonvacuum physical excitation and the declared chiral/mirror properties.*

Proof. The vector

$$(O_r - \langle \Omega_r, O_r \Omega_r \rangle)\Omega_r$$

is physical, orthogonal to the ground state, and nonzero by the variance condition. The remaining conclusions are exactly the separately supplied index, gap, completeness, and refinement clauses. Thus neither an identity projector nor a vacuum-only physical sector passes. □

Remark 7.25 (Finite-quantum boundary)**.** *The determinant-line and Hamiltonian results are criteria and soundness theorems. Neither complete* G_6 *construction is supplied. Anomaly arithmetic alone instantiates neither theorem.*

Theorem 7.26 (Formal Slavnov–Taylor restoration)**.** *Let S_0 be the complete canonically normalized Standard-Model BV action, including the gauge-fixing and antifield sectors, and suppose $(S_0, S_0) = 0$. Fix a regulator/scheme satisfying the quantum action principle, locality, and the declared power counting. Assume stability under renormalization in a complete permitted counterterm basis with fixed normalization conditions, classification of the applicable local ghost-number-one BRST cohomology, vanishing of the declared perturbative anomaly class, and a separate check of all applicable global anomalies. Then finite local counterterms may be chosen recursively so that the formal series*

$$\Gamma = S_0 + \sum_{n\geq 1} \kappa^n \Gamma_n, \qquad \kappa = (16\pi^2)^{-1},$$

satisfies the renormalized Slavnov–Taylor identity order by order.

Proof. At order n, the quantum action principle makes the breaking a local ghost-number-one functional Δ_n. Wess–Zumino consistency makes it closed under the linearized Slavnov–Taylor operator. The cohomology classification splits it into an anomaly representative plus an exact term. Anomaly clearance removes the first, and an allowed finite counterterm cancels the second. Stability and the normalization conditions fix the remaining invariant freedom, so induction proves the formal statement. □

Remark 7.27 (Perturbative boundary)**.** *This is a formal power-series theorem. It supplies no convergence theorem, finite chiral measure, finite constrained Hamiltonian, or Wightman construction. A numerical implementation must produce its regulator-specific restoration transcript and verify the Ward, Slavnov–Taylor, and Nielsen identities.*

Lemma 7.28 (Conditional first-order FJ coordinate change)**.** *Freeze the potential normalization, bare VEV shift, tadpole prescription, field and parameter counterterms, mass arguments, mixing coordinates, and gauge-fixing convention. If the complete finite change is*

$$p_L = p_F + \kappa\, \delta p^{(1)} + O(\kappa^2)$$

and $s(p) = s_0(p) + \kappa s_1(p) + O(\kappa^2)$, equality of the exact pole in the two coordinates implies

$$s_{1,F} = s_{1,L} + \delta p^{a(1)} \partial_a s_0.$$

Proof. Substitute $p_L(p_F)$ and Taylor expand through first order. The sum must include every transformed parameter, normalization, mass argument, counterterm, and mixing coordinate; a VEV-only substitution is insufficient. □

Theorem 7.29 (Strict charged and neutral pole coefficients)**.** *On one scheme, contribution mask, and resonance sheet, write*

$$\Gamma_W^T = s - w + \kappa \Pi_{WW}^{(1)} + \kappa^2 \Pi_{WW}^{(2)} + O(\kappa^3).$$

For $s_W = w + \kappa s_{W,1} + \kappa^2 s_{W,2} + O(\kappa^3)$,

$$s_{W,1} = -\Pi_{WW}^{(1)}(w), \qquad s_{W,2} = \Pi_{WW}^{(1)}(w)\Pi_{WW}^{(1)\prime}(w) - \Pi_{WW}^{(2)}(w).$$

For the massive root of the full photon–Z matrix with tree value $z \neq 0$,

$$s_{Z,1} = -\Pi_{ZZ}^{(1)}(z),$$

$$s_{Z,2} = \Pi^{(1)}_{ZZ}(z)\Pi^{(1)\prime}_{ZZ}(z) - \Pi^{(2)}_{ZZ}(z) + \frac{\Pi^{(1)}_{ZA}(z)\Pi^{(1)}_{AZ}(z)}{z}.$$

Thus the one-loop-squared neutral mixing product is excluded at strict one loop and is one mandatory term of the complete strict-two-loop mask.

Proof. Insert the root series and compare powers of κ. In the neutral sector use the Schur complement of the photon block; both off-diagonal entries begin at order κ. □

Theorem 7.30 (Nielsen control and physical current pole)**.** *Suppose the inverse matrix and Nielsen insertions are holomorphic near a simple massive root on the frozen sheet and, through retained order N,*

$$\partial_\eta \Gamma^T = \Lambda_\eta \Gamma^T + \Gamma^T \widetilde{\Lambda}_\eta + O(\kappa^{N+1}).$$

Then

$$\partial_\eta \det \Gamma^T = \operatorname{tr}(\Lambda_\eta + \widetilde{\Lambda}_\eta) \det \Gamma^T + O(\kappa^{N+1}), \qquad \partial_\eta s_p = O(\kappa^{N+1}).$$

If the simple left/right kernel vectors ℓ, r have $\ell^\dagger \Gamma^{T\prime}(s_p) r \neq 0$, and the dressed renormalized BRST-invariant current vertices obey $J_L^\dagger r \neq 0$ and $\ell^\dagger J_R \neq 0$, then the gauge-invariant current amplitude has Laurent residue

$$\frac{(J_L^\dagger r)(\ell^\dagger J_R)}{\ell^\dagger \Gamma^{T\prime}(s_p) r}.$$

Proof. Use $\partial_\eta \det \Gamma = \operatorname{tr}(\operatorname{adj}\Gamma\, \partial_\eta \Gamma)$; the adjugate identity remains valid at a singular matrix and avoids dividing by the determinant. The simple-root implicit equation gives the omitted-order gauge variation. The rank-one Laurent expansion of Γ^{-1}, contracted with the dressed current vertices, gives the amplitude pole. This is not a positivity claim for an unstable elementary field. □

Theorem 7.31 (Gauge-invariant observable reconstruction)**.** *A compatible cofinal Schwinger family for a complete declared gauge-invariant observable algebra that satisfies distributional convergence, Euclidean covariance, graded symmetry/locality, reflection positivity, clustering, growth/regularity, noncollapse, and refinement Cauchy control reconstructs a positive Hilbert space, cyclic vacuum, positive-energy translations, and the corresponding observable-sector Wightman distributions and local graded net.*

Remark 7.32. *This observable-sector implication does not by itself construct colored local fields, charged infrared sectors, confinement, asymptotic completeness, an S-matrix, or a second-sheet resonance.*

Theorem 7.33 (Resonance and residue stability)**.** *On one common continued sheet, write*

$$G_r(s) = \frac{N_r(s)}{D_r(s)} + G_{r,\mathrm{reg}}(s).$$

Let a Jordan contour and its interior lie in a common domain on which $D_r, N_r, G_{r,\mathrm{reg}}$ and their limits are holomorphic. Assume locally uniform convergence of these data and D_r', one simple enclosed zero s_r, uniform nonzero contour and derivative bounds, and the Rouché inequality

$$\sup_C |D - D_r| < \inf_C |D_r|.$$

Then the limiting denominator has one simple zero s_*, $s_r \to s_*$, and, if $N(s_*) \neq 0$,

$$\operatorname*{Res}_{s=s_r} G_r(s) = \frac{N_r(s_r)}{D_r'(s_r)} \longrightarrow \frac{N(s_*)}{D'(s_*)} \neq 0.$$

Proof. Rouché preserves the zero count for the holomorphic denominators. Compactness and uniqueness give root convergence. Uniform numerator and derivative convergence with the lower derivative bound gives residue convergence. Ordinary uniform convergence of the meromorphic quotient through its own poles is neither assumed nor valid. □

Remark 7.34 (Producer boundary)**.** *These results specify conditional implications. A target-free shared-ξ external Standard Model harness supplies a declared action and active census, a frozen effective-field-theory interval, dual rule tables, direct and converted FJ blocks, and payload-replayed Ward and Slavnov–Taylor controls. Separate interval receipts exclude scalar zeros on declared principal-sheet boxes and isolate, for each of W and Z, one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. They identify neither chart with the physical resonance sheet and supply no unique continuation identity or sign bridge, no full-matrix Laurent packet or dressed-current amplitude, and no independent numerical replay. The independent-ξ cross-check and complete identity-control transcript are also not supplied. The external fixture is not composed with the OPH electroweak chart. The source-selected classical action, either complete finite quantum object, the OPH matching and current packet, and the observable tower with continued-sheet data are not supplied.*

Strict-one-loop W/Z pole-map kernel

The finite-order theory map is explicit and machine checked. Let the renormalized one-doublet electroweak input at scale Q be $\theta(Q) = (g, g', v_F, \dots)$, with canonical Higgs kinetic term and $v_F > 0$, and set

$$w = \frac{g^2 v_F^2}{4}, \qquad z = \frac{(g^2 + g'^2) v_F^2}{4}.$$

Use the inverse-propagator convention

$$\Gamma^T(s) = s - m_0^2 - \Delta^T(s) = s - m_0^2 + \Pi^T(s), \qquad \Delta^T = -\Pi^T,$$

where each $\Delta^{(1)}$ includes its one-loop factor, counterterms, tadpoles, and the complete declared strict-one-loop mask. Relative to the coefficient convention of Theorem 7.29,

$$\Delta_{ij}^{(1)}(s) = -\kappa \Pi_{ij}^{(1)}(s), \qquad \kappa = (16\pi^2)^{-1}.$$

The two notations are translations of one pole equation, not independent results.

Proposition 7.35 (Strict-one-loop charged and neutral pole map)**.** *Assume the tree roots $w, z > 0$ are simple and the declared one-loop entries are holomorphic near them on the frozen analytic sheet. Then*

$$s_W^{[1]} = w + \Delta_{WW}^{(1)}(w), \qquad s_Z^{[1]} = z + \Delta_{ZZ}^{(1)}(z).$$

In the neutral tree-level photon–Z basis,

$$\Gamma_N^T(s) = \begin{pmatrix} s - \Delta_{AA}^{(1)}(s) & -\Delta_{AZ}^{(1)}(s) \\ -\Delta_{ZA}^{(1)}(s) & s - z - \Delta_{ZZ}^{(1)}(s) \end{pmatrix} + O(\epsilon^2).$$

The product $\Delta_{ZA}^{(1)} \Delta_{AZ}^{(1)}$ has loop power two and is excluded from a strict-one-loop root. Its leading Schur-complement contribution is

$$-\frac{\Delta_{ZA}^{(1)}(s) \Delta_{AZ}^{(1)}(s)}{s},$$

which belongs only in a separately complete two-loop map together with the genuine two-loop entries and pole-iteration derivatives.

Proof. Write $s_W = w + \epsilon\sigma_W + O(\epsilon^2)$ in the charged inverse entry. Its order-ϵ coefficient is $\sigma_W - \delta^{(1)}_{WW}(w)$. For the neutral determinant, write $s = z + \epsilon\sigma_Z + O(\epsilon^2)$. The order-$\epsilon$ coefficient is $z[\sigma_Z - \delta^{(1)}_{ZZ}(z)]$; both off-diagonal entries start at order ϵ, so their product starts at order ϵ^2. □

Put $S = gv_F/2$, $t = g'/g$, $d_W = \Delta^{(1)}_{WW}(w)/w$, and $d_Z = \Delta^{(1)}_{ZZ}(z)/z$. The strict scalar consumer then obeys

$$s_W = S^2(1 + d_W), \qquad s_Z = S^2(1 + t^2)(1 + d_Z).$$

Consequently, whenever the denominator is nonzero,

$$\frac{s_W}{s_Z} = \frac{1 + d_W}{(1 + t^2)(1 + d_Z)}.$$

This is the exact quotient of the two one-loop-truncated pole coordinates. Re-expansion to strict one-loop order gives

$$\frac{s_W}{s_Z} = \frac{1}{1 + t^2}(1 + d_W - d_Z) + O(\epsilon^2).$$

The explicit common scale cancels under a passive rescaling at fixed normalized corrections. An active change of v_F can move thresholds and therefore d_W, d_Z. The identity leaves t, d_W, d_Z unselected and gives no numerical pole ratio by itself. Exact counterfamilies in the factored consumer coordinates show that these normalized inputs can change the dimensionless mass and width readouts while the common scale remains irrelevant.

For $s_V = m^2_{V,0} + \Delta^{(1)}_V$, the strict energy-pole coefficients are

$$\delta M^{(1)}_V = \frac{\operatorname{Re}\Delta^{(1)}_V}{2m_{V,0}}, \qquad \Gamma^{(1)}_V = -\frac{\operatorname{Im}\Delta^{(1)}_V}{m_{V,0}}.$$

They are distinct from the exact coordinate transform of the truncated complex number. On the lower-half-plane branch,

$$M_V = \sqrt{\frac{|s_V| + \operatorname{Re} s_V}{2}}, \qquad \Gamma_V = \sqrt{2(|s_V| - \operatorname{Re} s_V)}, \qquad s_V = (M_V - i\Gamma_V/2)^2.$$

Applying this nonlinear square root exactly resums kinematic powers of the one-loop coefficient. It is a useful display coordinate, not a strict two-loop calculation and not the object to compare in a finite-order Nielsen test.

Proposition 7.36 (Evidence cannot self-attest)**.** *An untrusted input boolean asserting an external Fleischer–Jegerlehner, matching, source-law, gauge/BRST, clock, or ancestry property cannot certify that property. A physical-claim verifier must resolve an independent hash-bound witness, validate it, and bind it to the exact numerical subject, order, mask, scheme, and analytic sheet.*

Proof. Choose a subject for which the external property is false and set the untrusted boolean to true while preserving every relation recomputed by the verifier. If the verifier resolves no independent witness, it follows the same accepting path. Hence acceptance would admit a false instance. □

The fail-closed consumer receipt implements these rules and rejects self-attested physical flags, unrelated-but-self-consistent poles, substituted empty fixtures, inflated tolerances, corrupted redundant fields, and altered neutral diagnostics. Its archived Standard Model dimensional-regularization

order-one fixture at $Q = 160$ GeV evaluates to

$$s_W^{[1]} = (6459.842027569383 - 160.532752773045i)\ \mathrm{GeV}^2,$$
$$s_Z^{[1]} = (8222.835212344102 - 218.292761806439i)\ \mathrm{GeV}^2.$$

The corresponding strict readouts are $(M_W, \Gamma_W) = (80.374161202712, 2.007425074735)$ GeV and $(M_Z, \Gamma_Z) = (90.680036075608, 2.402420059845)$ GeV. These numbers reconstruct an archived back-end row. They carry target ancestry and supply no independent self-energy evaluation, no source-selected vacuum normalization, no complete neutral matrix, no source covariance, no independent gauge or Becchi–Rouet–Stora–Tyutin receipt, and no source clock. The strict one-loop pole map is therefore conditional, and it is not source-native physical, so no physical conclusion follows. What is proved is the implication from a complete declared renormalized strict-one-loop packet to the separated pole and mass and width readouts. That antecedent is not constructed from the source.

A separate target-free shared-ξ Standard Model harness binds a declared action and active census to a frozen effective-field-theory interval. It contains two independent rule tables, direct and converted Fleischer–Jegerlehner blocks, and payload-replayed Ward and Slavnov–Taylor controls. The rule tables are coordinate variants inside one harness rather than two independently implemented raw loop engines. Separate interval receipts exclude scalar zeros on declared principal-sheet boxes and isolate, for each of W and Z, one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. They identify neither chart with the physical resonance sheet, prove no unique continuation from the principal chart, and do not supply the sign bridge, full-matrix Laurent data, dressed-current amplitude, or independent numerical replay. Proof-bearing physical validation also requires the second engine, counterterms generated from the complete action, BRST-generated ghost vertices, finite symmetry restoration, symbolic Nielsen identities, an artifact-resolving third verifier, a certified continuation identity, full-matrix contour and Laurent data, dressed-current residues, and complete independent replay. The external fixture is not composed with the OPH electroweak chart. The harness therefore establishes no physical W/Z pole and supplies no OPH-native source packet.

Theorem 7.37 (Rigidity, dependency, and W/Z/H composition criterion)**.** *Let $\mathcal{F}_{\mathrm{EW}}$ be the fully enumerated class of source maps that preserve the declared locality, symmetry, dimensional, refinement, and provenance constraints, modulo gauge and trivial reparameterization. Suppose either the pole readout is constant on this class or a target-independent selector has one minimizer with a positive winner gap. Suppose further that a hash-bound acyclic dependency graph records the source objects, selectors, conventions, uncertainty model, and outputs, and measured $W/Z/H$ data and calibrated proxies are absent from every source ancestor set.*

If, in addition, a unique source root and independently source-closed physical $E_\star$ emit the canonical electroweak and criticality coefficients, the finite quotient-path and split-character certificates hold, and the declared renormalization-group and complex-pole hypotheses hold, then OPH determines one source-separated pole triple (s_W, s_Z, s_H) with a certified uncertainty enclosure.

Proof. Rigidity or the positive-gap selector removes alternative source maps. A deterministic leaf depends only on its ancestors, so the dependency graph gives formal source separation. The unique root and scale emit one coefficient packet, declared running and matching transport it uniquely, and the pole theorem gives one stable pole per particle. Composition proves the claim. □

This criterion is not satisfied by the displayed numbers. The finite quotient-path and split-character certificates are assumptions, the public and source-only pixel branches differ, and no

physical scale, renormalization-group transport, complex pole, or target-independence proof is supplied.

The missing object is a source-emitted certificate. The structural theory admits target-free analytic counterfamilies $\tau_2 = -c\eta^2$, $\delta n = d(1-\rho_{\mathrm{EW}})\eta^2$ and $(\pi_y, \pi_\lambda) \mapsto (\pi_y + a\eta^N, \pi_\lambda + b\eta^N)$, all on the same open physical domain but with different mass-chart outputs. Likewise, a fixed running mass is compatible with both $s - m_R^2$ and $s - m_R^2 - \epsilon$, which have different poles. These countermodels prove that two-channel exhaustion and analyticity do not emit the electroweak character, criticality split, or physical kernel. Nor is a formal path realization sufficient: any finite polynomial can be encoded by assigning one weighted path to each monomial. A non-vacuous certificate must independently fix its transitions, path measure, signs, quotient action, exhaustive census, rigidity gap, and target-independence proof.

If the source selector is proved deterministic and globally unique, the source law is a delta measure and no Monte Carlo source propagation is required. If the source emits a nondegenerate law, the simulator must propagate that law through the pole map. Scale variation and truncation envelopes are not source covariance, and neither a point estimate nor componentwise intervals determine the covariance of $(\Re s_W, \Im s_W, \Re s_Z, \Im s_Z)$. Summable clock-Hamiltonian refinements give a unique limiting gap; fixed Lipschitz RG segments compose uniquely; Rouché, Nielsen, and analytic conjugacy isolate gauge-independent poles. If a determinant defect is bounded by ϵ, $a = D'(s_0) \neq 0$, and $|D''| \leq K$, every r with $\epsilon < |a|r - Kr^2/2$ encloses the unique displaced pole and propagates to mass/width bounds through the selected square root. The four absent source objects are therefore the factorized clock packet, independently weighted electroweak carrier, criticality split-character carrier, and Becchi–Rouet–Stora–Tyutin (BRST)-complete pole-kernel packet. A runtime dependency graph proves computational separation only. A source-only claim also requires a target-independent specification.

The electroweak chapter therefore presents four mathematically distinct surfaces: the selected-carrier chart, the candidate value law, the conditional finite quotient-path theorem, and the measured-reference inverse surface used for diagnostic W/Z values. A physical pole surface requires the additional source-root, scale, RG/scheme, two-point-kernel, rigidity, uncertainty, and provenance receipts.

7.6 Repair-tuple selection and the color amplitude/loop split

The candidate electroweak tuple sits inside a two-parameter freedom that the stated premises do not remove. With $\rho_{\mathrm{EW}} = (\alpha_2 - \alpha_Y)/(\alpha_2 + \alpha_Y)$, the coherent quadratic family is

$$\tau_{2,\mathrm{tree}}^{\mathrm{exact}} = -c\,\eta_{\mathrm{source}}^2, \qquad \delta n_{\mathrm{tree}}^{\mathrm{exact}} = d\,(1-\rho_{\mathrm{EW}})\,\eta_{\mathrm{source}}^2,$$

and an open neighborhood of distinct (c, d) values preserves the positive mass-chart domain and defines consistent electroweak repairs. The charged leg is the $\mathrm{SU}(2)_L$ coupling correction $\delta\alpha_2 = \alpha_2\tau_2$; the neutral leg is the hypercharge screening correction carried by δn.

The narrower current-carrier slice with $\delta n = 0$ and the fixed fiber law has one exact central-value obstruction conditional on treating the displayed finite-decimal inputs as the exact rationals they serialize. The charged mass is strictly increasing for $\tau_2 > -1$, so matching the displayed central W value fixes the unique coordinate $\tau_2 = -0.0005918464744070\ldots$. Substitution in the exact nonlinear neutral formula gives $M_Z = 91.1682226626\ldots$ GeV, a residual of $-0.0196773374\ldots$ GeV from the displayed central Z value; an exact rational interval evaluation of the squared residual excludes zero. Thus that closed one-scalar slice cannot hit both displayed central values exactly. This is not an uncertainty-region exclusion. The opposite directions requested by the two first-order germs are only a local diagnostic and are not a global no-go. Fits to uncertainty regions, an additional neutral

scalar, a source-derived change of the fiber law, and a different carrier or selector remain open (`code/particles/runs/calibration/d10_ew_tau2_current_carrier_obstruction.json`).

The compact-gauge reconstruction fixes the color count but does not fix the channel assignment or normalization that would close this freedom. Doplicher–Roberts/Tannaka reconstruction returns the color triplet sector with statistical dimension $d(\rho_3) = N_c = 3$, and its standard conjugate intertwiner satisfies $R^*R = d(\rho_3) = N_c$, so R has norm $\sqrt{N_c}$ while $R/\sqrt{N_c}$ is the isometry; a closed color loop carries the full dimension N_c. The two repair legs sit at different levels. The charged leg is the broken, mass-generating $\mathrm{SU}(2)_L$ channel, driven by the color-singlet condensate amplitude whose large-N_c scaling is the decay-constant scaling $\sqrt{N_c}$ assigned to one standard conjugate intertwiner. The neutral leg would be the unbroken hypercharge screening channel, a vacuum-polarization loop that closes the color line and carries raw trace weight N_c. Under these extra hypotheses the alternative quadratic model has

$$c = \frac{\sqrt{N_c}}{2} = \frac{\sqrt{3}}{2}, \qquad d = \frac{N_c}{2} = \frac{3}{2},$$

where the chart coefficient $d = N_c/2$ represents the raw neutral trace because $(\alpha_2+\alpha_Y)(1-\rho_{\mathrm{EW}}) = 2\alpha_Y$.

The resulting comparison is a target-informed, fixed-slice diagnostic of that alternative model. Profiling the charged coefficient against the running-tree W coordinate is independent of the criticality Jacobian core and inherits the declared α_2, v, running, and scheme surface. It selects $c = 0.8670$, while profiling the neutral coefficient returns $d = 1.461$. Both values are target-conditioned fit outputs. Their proximity to $\sqrt{3}/2$ and $3/2$ is an internal calibration observation, not physical W/Z evidence, and no experimental confidence band is assigned across the noncommensurate chart and mass conventions. That slice is not a test of the complete candidate value law, which changes both coordinates and includes higher path characters; it therefore neither excludes that law nor proves or excludes a separate running-tree companion.

The color-balanced rule does not equal the complete candidate value law in Theorem 7.3; it gives a different W coordinate. A proof of its amplitude and trace hypotheses would establish that alternative quadratic model, not the finite quotient-path law. A source derivation requires the two-channel quotient, exhaustive charged and neutral path lists, the $(1,2,1)$ and $(1,4,2)$ incidences with their $1/3$ color measure and $1/6$ central trace, the fibre Gram form and residual pairing, an exhaustive deformation class, and a positive selector gap. Measured W/Z agreement cannot replace that certificate. The pre-repair carrier must be specified independently of the desired outputs as an observer-like self-reading patch with bounded local state, typed ports, readback records, and admissible feedback or repair moves. It may contain neither the target masses nor the desired incidence and central-trace coefficients. Independent verification must recompute path legality, exhaustion, exact response rank and nullspace, central action, oriented mismatch first variation, the fibre Gram form, and the deformation census. The order of $\mathbb{Z}_6$ supplies the averaging coefficient $1/6$, not automatically a normalized trace $1/6$: the checker must separate the trivial physical matter-kernel action from the declared center-label/transport trace space and derive the trace, source amplitude, and subtraction sign in the latter.

7.7 Selector rigidity and the discrete repair-law boundary

The candidate electroweak selector closes the continuous (c, d) freedom of the preceding subsection. Writing $x = \tau_{2,\mathrm{tree}}^{\mathrm{exact}}$, $y = \delta n_{\mathrm{tree}}^{\mathrm{exact}}$, and $\kappa = (\alpha_Y + \alpha_2)/\alpha_Y$, the selector

$$J_{\mathrm{EW}}(x,y) = \frac{x^2\left[1 + (2\eta_{\mathrm{source}} + x)^2 + 4x^2\right]}{1+4x^2} + \frac{\kappa^2}{4}(1+4x^2)y^2$$

admits the exact factorization

$$J_{\mathrm{EW}} - \frac{3}{4}x^2 - \frac{\kappa^2}{4}y^2 = \frac{x^2}{4(1+4x^2)}\left[8(x+\eta_{\mathrm{source}})^2 + 8\eta_{\mathrm{source}}^2 + 1 + 4\kappa^2(1+4x^2)y^2\right],$$

so $J_{\mathrm{EW}} \geq \frac{3}{4}x^2 + \frac{\kappa^2}{4}y^2$ with equality exactly at the origin, and $x^2 + y^2 \geq r^2$ forces $J_{\mathrm{EW}} \geq \min\{3/4, \kappa^2/4\}r^2$. The selector picks the zero deformation with a quantitative gap. The candidate value-law tuple has $J_{\mathrm{EW}} \approx 3.12 \times 10^{-7} > 0$, so it is incompatible with the selector, and the residual electroweak freedom is one discrete choice between two incompatible laws rather than a continuous chart family.

On the target-free declared-map branch the two branch points give the target-free running-chart coordinates

$$\begin{aligned}
&\text{zero-selector law:} \quad M_W/E_\star = 6.579631 \times 10^{-18}, \quad M_Z/E_\star = 7.463335 \times 10^{-18},\\
&\text{carrier value law:} \quad M_W/E_\star = 6.578870 \times 10^{-18}, \quad M_Z/E_\star = 7.463750 \times 10^{-18},
\end{aligned}$$

with unclosed-clock displays (80.3301, 91.1191) and (80.3208, 91.1242) GeV and discrete ambiguity widths of 9.3 MeV on M_W and 5.1 MeV on M_Z. These are target-free running-chart coordinates. The PDG mass targets use a mass-dependent-width Breit–Wigner convention, and complex-pole masses use another convention. The exact convention map distinguishes $M = \mathrm{Re}\,\sqrt{s}$ from the legacy $\sqrt{\mathrm{Re}\,s}$ coordinate. The target-free running-chart coordinates are not physical observables and carry no theory covariance. They cannot decide between the laws; the decision object is a source-law selection principle, and the quantitative mechanism of the source-action rigidity theorem applies once the electroweak feature basis and moment vector are emitted.

The two-loop comparison gives $(79.115335, 89.802735)$ GeV. It combines a minimal supersymmetric Standard Model one-loop baseline with a Standard Model two-loop-minus-one-loop increment, an inconsistent hybrid prescription. A partial Feynman-gauge pole calculation gives approximately $(79.53284, 89.71232)$ GeV. It supplies no definition of v, tadpole treatment, field-content or threshold matching, scale choice, or higher-order corrections. The outputs establish neither a unique scheme conversion nor a unique 1–2% defect, and they do not exhaust the physical prescription family.

A conditional finite electroweak root/Cartan carrier emits the value-law coefficients exactly: the transitive C_3 color action has invariant measure $1/3$, the regular $\mathbb{Z}_6$ register has rank-one trace $1/6$, the transitive four-slot register has slot measure $1/4$, the fixed independent-product source law gives $\lambda_{\mathrm{EW}} = \eta_{\mathrm{source}}^2/(4\rho_{\mathrm{EW}})$, and the depth-two path census reproduces the candidate quadratic response polynomials. A companion SU(3)-Casimir carrier emits the declared criticality normalizations, $\kappa_\lambda = C_F^2 = 16/9$, the response coefficients $3/2 + \rho_{\mathrm{EW}}/4$ and $4/3 - \rho_{\mathrm{EW}}/54$, and the logarithmic character $\rho_{HT} = \log(1+\tau_2)$ from multiplicativity with unit derivative. Both carriers are target-exposed candidates without a source-independent quotient-path selection or census certificate.

The declared criticality Jacobian entries follow exactly from the core, $\partial m_t/\partial y_t = m_{t,0}/y_0$ and $\partial m_H/\partial\lambda = m_{H,0}/(2\lambda_0)$, and the exact readout is $m_t = m_{t,0}(1+r_y)$ and $m_H = m_{H,0}\sqrt{1-r_\lambda}$ with the exact linearization error $m_{H,0}r^2/(2(1+\sqrt{1-r})^2)$. The synchronized criticality core is target-conditioned: its synchronization scale minimizes an objective containing the Higgs and top comparison values. Without that switch, the literal candidate one-loop source equations give the target-free tree coordinates $m_t^{\overline{\mathrm{MS}}}/E_\star = 1.267758 \times 10^{-17}$ and $m_H^{\mathrm{tree}}/E_\star = 9.427653 \times 10^{-18}$, with unclosed-clock displays 154.78 GeV and 115.10 GeV and a QCD-converted top display of 164.13 GeV. These are one-loop tree coordinates on the declared surface; they are not physical poles and do not satisfy the pole conditions of §7.5.

The candidate one-loop coordinates form a zero-continuous-parameter family. The literal core derives its top boundary from the gauge sector through the double-criticality condition $\lambda(\mu_b) = 0$, $\beta_\lambda(\mu_b) = 0$, which fixes $y_t(\mu_b) = [(2g_2^4 + (g_2^2 + g_Y^2)^2)/16]^{1/4}$. The candidate branch imposes this at the gauge-unification scale μ_U, while the electroweak transmutation in the same model anchors v at the pixel energy $E_{\text{cell}} = E_\star/\sqrt{P}$, so the candidate model carries two different high-scale anchors. Evaluating the same law at the named source scales gives, at one loop, $(m_t, m_H) =$ (164.1, 115.1) GeV at μ_U, (170.4, 127.8) at E_{cell}, and (170.7, 128.3) at $E_\star$; a benchmark-validated two-loop upgrade shifts these to (169.4, 119.4), (175.7, 131.8), and (175.9, 132.3). The family brackets the measured pair in both channels at both loop orders, so the candidate deficit decomposes into the boundary-scale choice plus loop truncation, with no continuous freedom anywhere. Along the fit-free curve the Higgs coordinate at the measured top mass is 125.7 GeV at two loops, within 0.5 percent of measurement and inside the declared tree-to-pole matching band; the implied boundary scale is 4.8×10^{17} GeV, between μ_U and E_{cell}. No boundary-scale selection theorem is supplied; numerical agreement cannot select the scale, and the declared-surface values obtained by back-solving from the measured pair stay classified as target-anchored fits.

The triple-criticality equation $\mathrm{d}\beta_\lambda/\mathrm{d}\ln\mu = 0$ has no solution along the flow: the family derivative is dominated by $-24y_t^3\beta_{y_t} > 0$ and stays bounded away from zero across the whole window, so every family point is a clean λ minimum and the boundary scale requires source structure. A conditional variational selection principle uses three premises. The criticality boundary is a record that reconciles the model's two anchor records, the gauge-unification record at μ_U and the transmutation record at E_{cell}. The reconciliation cost is quadratic in renormalization time, and the two anchor records carry equal capacity. These premises give a strictly convex cost with a unique minimizer at the log-midpoint $\sqrt{\mu_U E_{\text{cell}}} = E_\star e^{-\pi} P^{-1/6}$, hence $(m_t, m_H) = (172.63, 125.77)$ GeV at two loops; the implication is proved exactly on rational sample points.

The quadratic-cost premise is exact under the declared Gaussian record model: the one-loop chart is inverse-affine, so $1/\alpha(t)$ is exactly linear in renormalization time, and the Kullback–Leibler divergence between fixed-resolution Gaussian records with an affine stored coordinate is exactly $(s^2/2\sigma^2)(t-t_i)^2$, a quadratic cost with no leading-order approximation. Premise A3 supplies relative-entropy minimization after the reference family, affine observable, trace convention, variance, and resolution have been declared. The Gaussian model and its normalization are separately declared, and no derivation from the complete A1–A3 schema is supplied. The reconciliation placement is exact within a separately declared port-additive mismatch and cost-minimizing repair model. In that model, a record with two scale-parents settles at the capacity-weighted log-mean. Equal capacity follows if the anchor registers have equal slope and readback resolution, which is another declared condition. These record and repair choices are declared inputs, and no derivation from A1–A3 is supplied. The finite criticality carrier must establish that the boundary record carries exactly two parent ports, one to the gauge-unification register and one to the transmutation register, and that those anchor registers belong to the same register class at equal refinement depth. Given the declared Gaussian record model and the stated repair premises, these two census facts select the boundary scale, and with it $(m_t, m_H) = (172.63, 125.77)$ GeV, with no remaining continuous parameter. The W/Z value law requires the same finite carrier census. The equal-capacity assumption is measurable: a capacity asymmetry moves the m_H coordinate by about 2.1 GeV per e-fold. A three-loop implied scale therefore provides a discriminating measurement of that assumption.

A conditional selection theorem follows. If the boundary record reconciles the two anchor records with port-additive quadratic cost in renormalization time and equal anchor capacities, the boundary scale is exactly the log-midpoint. The inverse-affine one-loop chart and the declared Gaussian maximum-entropy record model give the exact quadratic reconciliation cost. The declared port-additive repair model places a two-parent record at the capacity-weighted log-mean. The theorem

therefore consumes those record and repair models together with two finite carrier facts: the two-parent port structure of the boundary record and the equal-class equal-depth relation of the anchor registers. A carrier certificate of the census class supplies those two facts.

7.8 Target-conditioned closure-chart envelope

Evaluating two target-conditioned repair candidates at the endpoint pixel and across the empirical-closure interval gives the following comparison envelope. It is not a source interval: its candidate set is not an exhaustive source-derived deformation class, and it mixes the public endpoint branch with the source-only question.

$$\begin{aligned} m_H &\in [125.183,\, 125.232]\ \text{GeV}, && \text{measured } 125.13 \pm 0.11, \\ m_t &\in [172.278,\, 172.352]\ \text{GeV}, && \text{measured } 172.1 \pm 0.6, \\ c_W &\in [80.3692,\, 80.3774]\ \text{GeV}, && \text{PDG BW coordinate } 80.3692 \pm 0.0133, \\ c_Z &\in [91.1880,\, 91.1983]\ \text{GeV}, && \text{PDG BW coordinate } 91.1880 \pm 0.0020. \end{aligned}$$

The recorded conditional central coordinates of this target-conditioned closure chart are 80.373315 and 91.193124 GeV, each inside its displayed interval above. The Higgs and top entries are target-conditioned comparisons on their declared mass scheme. The W/Z entries are closure-chart coordinates. The PDG targets use mass-dependent-width Breit–Wigner parameters, and complex-pole masses use another convention, so no physical coverage or pull is assigned. At the endpoint pixel the two repair selections give closure-chart Z coordinates 91.18798 and 91.18801 GeV. The selection spread between the two displayed laws is roughly eight MeV in the W chart, thirty keV in the Z chart, sixteen MeV in m_H, and thirty MeV in m_t. These figures compare two ansätze; they do not bound the full admissible deformation class.

7.9 Residual attribution across the electroweak rows

On this comparison envelope, the Higgs and top residuals are target-conditioned mass-scheme diagnostics. No W/Z standard-deviation assignment is valid because the closure-chart coordinates, PDG mass-dependent-width parameters, and complex poles use different conventions. The W closure-chart spread is repair-selection freedom: the two candidate repair laws differ by about eight MeV. The m_H and m_t offsets track the same repair-selection width folded through the declared Higgs and top Jacobian cores.

The Z closure-chart row is the most sensitive adapter case. The empirical-closure pixel inherits the frozen electromagnetic anchor deficit through the fine-structure endpoint, and c_Z tracks the pixel with $\mathrm{d}c_Z/\mathrm{d}P \approx 123$ GeV per unit P. Two propagation branches separate the mechanism. The direct branch injects the certified anchor shift into the running couplings and re-evaluates the candidate W/Z closure chart. That branch shifts c_Z by -66 to -87 MeV on the hypercharge line and by roughly -230 to -302 MeV on the proportional line, exceeding the adapter displacement by an order of magnitude, so it is excluded. The pixel branch moves the endpoint to the measured fine-structure value and propagates the induced pixel shift through $\mathrm{d}P/\mathrm{d}A_{\mathrm{Th}}$. That branch moves the empirical pixel onto the calibration pixel to within 4×10^{-7} in P, equivalently -0.05 MeV in the Z chart. This is an internal adapter closure check; it supplies no physical mass or pole.

For this comparison surface, a closed fine-structure anchor bridge would collapse the endpoint pixel interval onto the calibration pixel and remove this particular chart displacement. It would not supply the finite quotient-path certificate, the independent scale, the RG/scheme certificate, or the complex-pole gate. The anchor bridge is the object developed in the fine-structure companion, and its source branch is the subject of Section 7.10.

7.10 Source branch of the fine-structure anchor bridge

The certified anchor gap [0.620, 0.651] in inverse fine-structure units names the shift the source side supplies at the electroweak anchor for the empirical-closure endpoint to reach the measured fine-structure value. The gap is defined by that measured requirement, so inserting it back is circular. The non-circular source route continues the declared running convention to the next order and tests whether the induced anchor shift lands inside the certified gap.

That test is executed. The one-loop anchor is reproduced from the declared coefficients and the transmutation-certificate boundary data to a residual of 10^{-11}. The standard two-loop running system, integrated down the same 33.2 e-folds with a top-Yukawa convention scan, shifts the anchor by [+1.62, +2.14] in inverse fine-structure units for every convention. The shift has the correct sign and overshoots the certified gap by a factor of two and a half to three and a half. Restoring the balance requires the threshold and scheme-conversion data, which the matching contract classifies as hidden fit parameters when they are undeclared. The balance requirement is quantitative: the threshold map removes about 2.14 inverse-alpha, equivalent under naive coefficient accounting to a single effective threshold near 458 GeV. The anchor bridge has no source scheme-lock or threshold map; its endpoint is empirical.

8 Flavor Transport, Generation Structure, and the Yukawa Dictionary

The gauge branch supplies a canonical rank-three response band under its separate premises. It does not prove physical family attachment or produce the full flavor dictionary. This paper therefore needs a separate flavor chapter. The imported input here is the color carrier $N_c = 3$ together with the declared physical identification of the rank-three candidate. This chapter is the bridge between that conditional input and the matter-family chapters for quarks, charged leptons, and neutrinos.

The flavor derivation is deliberately constructive. It does not claim that the full OPH flavor observable is theorem-level. Instead, it builds one explicit sufficient proof-obligation chain that the present construction would have to discharge; other carriers, direct source actions, dynamics, and observables are not excluded. This chain has the following main architecture: start from a refinement-indexed family transport kernel, derive a centered generation-bundle branch generator, lift that to same-label transport data, derive the induced overlap-edge transport cocycle, reduce the cocycle to a persistent flavor observable, and then push that observable into common sector-response objects for the downstream quark, charged-lepton, and neutrino derivations.

Mathematically, this derivation is where "why this family exists" becomes a concrete technical question. The finite Standard Model packet gives a rank-three candidate; the flavor derivation is where one tries to turn that candidate and the declared count into a physical attachment, transport, splitting, suppression, phase, and excitation data. The relevant objects are the intermediate transport and spectral structures, not the final fermion masses themselves, that the mass readouts consume.

The active chain, in slightly compressed form, is:

1. normalize a refinement-indexed family transport kernel;
2. derive a centered generation-bundle branch generator on the conditionally declared three-generation charged bundle;
3. lift this to same-label edge-line transport and then to an overlap-edge cocycle with explicit defect and gap bookkeeping;

4. reduce those data to projectors, spectral gaps, pair suppressions, and cycle phases;

5. push the resulting family object into sector-response objects and then into suppression/phase tensors for the downstream matter derivations.

Three branch-specific claim boundaries are attached to this region of the derivation. The shared excitation dictionary is the common proof-facing base. Above that base, the charged branch carries exact centered readback, a closed common-shift no-go, the declared same-label q_e readback, a source-side determinant character for any fixed formal exponent vector

$$S_M = \sum_e M_e^{\mathrm{ch}} \log q_e,$$

a determinant-line lift on theorem-grade physical charged data, and an algebraic mass readout from theorem-grade $A_{\mathrm{ch}}(P)$. The theorem branch does not emit a theorem-grade sector-isolated charged determinant exponent vector, and it does not identify a source-side determinant character with the physical charged determinant line. The charged-lepton theorem gap is the determinant trace-lift attachment $3\mu(r) = S_M(r)$ on the charged determinant channel. The charged determinant channel carries a current-corpus gap certificate: the uncentered trace lift is not emitted, the attachment $3\mu(r) = S_M(r)$ is the named route, and the only no-go on it is the common-shift theorem of the charged absolute-scale section. The neutrino branch carries a target-informed weighted-cycle candidate above a template family kernel, with compare-only bridge and absolute-attachment diagnostics. The quark branch carries an exact restricted source-spread obstruction after the candidate ordered shape law is granted: its ordered profile shapes then leave a free $(\mathbb{R}_{>0})^2$ fiber. Selected-class descent proves representative independence but does not choose either positive modulus. Target-anchored mixed-convention mass coordinates and GeV-valued mass textures are comparison data; they are neither a source-only running sextet nor physical dimensionless Yukawa matrices. A separate compare-only microphysical bridge records edge-statistics transport on a diagnostic surface. Together these boundaries mark where the flavor derivation leaves the common transport backbone and passes to branch-specific closure contracts.

9 Quark Family Derivation

Conditional on the current registered source signature containing no typed datum that separates or excludes the displayed centered-spread orbit, the quark branch carries a non-identifiability argument on a declared generic fixed-frame quark package. Separately, granting the candidate-only ordered three-point shape law supplied by the template-descended branch generator, the target-free source packet fixes the ordered up- and down-sector profile rays and leaves their endpoint spans free as two independent positive moduli. That shape law is unproven and is rejected on the common-scale physical surface below, so the generic interface carries six unemitted scalars and the two-modulus statement is a restricted lower bound inside the granted subfamily. Selected-class descent does not remove this freedom, so it emits no source-only mass row. Sections 9.6 and 9.7 supply separate conditional numerical constructions. Same-family and restricted common-refinement artifacts reproduce their chosen targets only after those moduli are obtained by target inversion. Their rows also mix renormalization conventions, while their GeV-valued matrices are mass textures rather than physical dimensionless Yukawa matrices. Neither result classifies all public quark frame classes.

9.1 What quarks are in this derivation

Quarks are the color-charged elementary constituents of hadronic matter. In nature, the up quark and down quark dominate protons and neutrons, while the strange quark, charm quark, bottom quark, and top quark appear in progressively heavier and more unstable sectors. In the OPH particle derivation, the quark branch has a closed downstream algebraic readout conditional on a source-only physical spread datum. The target-free source equations do not emit that datum. Granting the candidate-only ordered shape law, the source packet fixes two ordered profile rays and leaves their positive endpoint spans independent; without that grant the shape law, the sector means, and the spans are all unemitted. The compatible fiber inside the granted subfamily is $(\mathbb{R}_{>0})^2$, so the six numeric rows are not source-only predictions.

The quark route starts from the shared flavor excitation dictionary, internalizes the target-free mass bridge on the quark mass-profile ray, and proves selected-bridge-fiber representative independence for the attached physical spread datum on f_P, and applies the affine mean law once both positive moduli are supplied. The stored target-anchored matrices have GeV-valued singular values drawn from several comparison conventions. They are mass textures, not physical dimensionless Yukawa matrices. The same-label left-handed selector surface closes to the singleton σ_{ref}. That lower object is a negative sheet-selector statement on the selected quark sheet. It does not break the independent positive-rescaling action on the two sector profiles.

9.2 Source-selected quark rows

The following table concerns source-selected physical masses. A separate conditional numerical construction, with its flavor law and mass-chart attachments assumed explicitly, is given in Section 9.6.

Running quark masses are coordinates on a renormalization-group trajectory. Finite renormalizations and scale changes alter those coordinates while preserving physical amplitudes, so OPH can emit an RG-covariant trajectory or invariant but cannot derive the human choice of an $\overline{\mathrm{MS}}$ chart. The chart must be declared after source emission. The top row is a pole extraction coordinate rather than a sixth member of one common running-mass packet. All six rows are withheld because, under the stated registry-completeness premise, the current registered source signature does not identify the spread pair. A physical Yukawa construction would additionally transport every running coordinate to one common scale, convert the top coordinate, emit the running Higgs expectation value in the same scheme, and apply $y_q(\mu) = \sqrt{2}\, m_q(\mu)/v(\mu)$.

9.3 Target-free mass bridge and selected-class theorem

The same-label left-handed solver surface closes to the singleton σ_{ref}. This is a negative same-sheet selector statement: same-sheet rephasing preserves CKM moduli, so it does not move that selected sheet to the physical CKM shell. Same-sheet overlap scans, chirality-swapped basis diagnostics, and other non-sector-attached orbit improvements are compare-only and do not change the selected-class theorem.

A separate target-free mass bridge is internalized on the emitted quark mass-profile ray. On the minimal light branch

$$y_u = c_u\, \varepsilon^6, \qquad y_d = c_d\, \varepsilon^6, \qquad \varepsilon = \frac{1}{6},$$

the light-quark overlap-defect theorem emits

$$\Delta_{ud}^{\mathrm{overlap}} = \frac{1}{6} \log \frac{c_d}{c_u}.$$

The emitted same-family mass object is the ray

$$\mathcal{R}_{ud}^{\mathrm{mass}} := \{(c_u, c_d) \in \mathbb{R}^2_{>0} : c_d/c_u \text{ fixed}\},$$

and on that ray the same scalar is equivalently the one-scalar law

$$\Theta_{ud}^{\mathrm{mass}} := \frac{1}{6} \log \frac{c_d}{c_u}.$$

The exact scalar identities are

$$\Delta_{ud}^{\mathrm{overlap}} = \frac{t_1}{5}, \qquad \log \frac{c_d}{c_u} = \frac{6}{5}\, t_1, \qquad t_1 = 5\, \Delta_{ud}^{\mathrm{overlap}} = \frac{5}{6} \log \frac{c_d}{c_u}.$$

The mass-profile package is therefore functorial:

$$\begin{aligned} \eta_Q^{\mathrm{centered}} &= -\frac{1 - x_2^2}{27}\, t_1, \\ \kappa_Q &= -\frac{t_1}{54}, \\ x_2 &= -0.5175863354681689. \end{aligned}$$

The odd source package is also forced:

$$\begin{aligned} \beta_{u,\mathrm{diag},B}^{\mathrm{source}} &= \frac{t_1}{10}, \\ \beta_{d,\mathrm{diag},B}^{\mathrm{source}} &= -\frac{t_1}{10}, \end{aligned}$$

$$\begin{aligned} \mathbf{r}_u &= \left(-\frac{t_1}{10},\, 0,\, +\frac{t_1}{10}\right), \\ \mathbf{r}_d &= \left(+\frac{t_1}{10},\, 0,\, -\frac{t_1}{10}\right), \end{aligned}$$

For any supplied positive spread pair one obtains

$$\tau_u = \frac{\sigma_d}{10(\sigma_u + \sigma_d)}\, t_1, \qquad \tau_d = \frac{\sigma_u}{10(\sigma_u + \sigma_d)}\, t_1.$$

These identities do not select σ_u or σ_d. The mass bridge lies outside the selected-class theorem boundary. The pure-B payload pair is a lower realization object beneath the exact theorem, and $\Theta_{ud}^{\mathrm{mass}}$ is its derived scalar wrapper.

Write the same-label left-handed physical carrier as

$$\Sigma_{ud}^{\mathrm{phys}} := \left\{(\sigma_{\mathrm{id}}, \tau, U_{u,L}, U_{d,L}, V_{\mathrm{CKM}}, I_{\mathrm{CKM}}) : V_{\mathrm{CKM}} = U_{u,L}^\dagger U_{d,L}\right\} / \sim,$$

where

$$(U_{u,L}, U_{d,L}, V) \sim (U_{u,L} D_u,\ U_{d,L} D_d,\ D_u^\dagger V D_d)$$

for diagonal $D_u, D_d \in U(1)^3$. On the selected public quark frame class f_P, represented on the realized branch by the explicit common-refinement transport-frame class $[F_0^\dagger F_1]$, the exact $\Sigma_{ud}^{\mathrm{phys}}$ element and the attached exact sigma datum are independent of the declared representative inside the selected bridge fiber. Therefore the exact sigma readout descends uniquely to the selected comparison domain on f_P. It is not a public source-only mass prediction because the physical sigma datum is target-derived.

Sigma descent non-selection. Let $R_{\mathrm{decl}}(f_P)$ be the declared selected bridge fiber over the public quark frame class. If a map

$$\Sigma : R_{\mathrm{decl}}(f_P) \to \mathbb{R}^4$$

is constant on that fiber, then it descends to a well-defined public datum $\overline{\Sigma}(f_P)$. This proves representative independence only. It does not select the value of $\overline{\Sigma}(f_P)$, because every constant vector in $\mathbb{R}^4$ would also descend. Therefore a value obtained from the declared running-quark target surface is target-derived after descent.

Common-scale physical rejection of the reciprocal-ray candidate. A physical comparison first converts all six quark eigenvalues to dimensionless Yukawa singular values in one effective theory, scheme, and scale. Using the Standard Model $\overline{\mathrm{MS}}$ values at M_Z tabulated by Antusch, Hinze, and Saad [17], the two ordered log-gap ratios are

$$\rho_u = 1.1108888543, \qquad \rho_d = 0.7519410008, \qquad \rho_u\rho_d = 0.8353228768.$$

The reciprocal-ray law requires the last product to equal one. It is therefore falsified on the properly typed target surface. Even after granting the four endpoint Yukawas (y_u, y_t, y_d, y_b), the minimax reciprocal shape misses the held-out y_c and y_s by 21.556% at M_Z. The result persists under running: the products are 0.836092, 0.837752, 0.845227, and 0.843611 at 10^3, 10^5, 10^{12}, and 10^{16} GeV, with best held-out errors between 19.93% and 21.42%. The stored $\rho = 1.294285$ formula misses the charm coordinate by 56.5% at M_Z, even under the endpoint grant.

The generic physical interface is therefore

$$(\mu_u, \sigma_u, \rho_u, \mu_d, \sigma_d, \rho_d) \in \mathbb{R}^6,$$

with an exact inverse to the six ordered positive Yukawas. The three-scalar theorem is exact only inside the nonsingular domain of its imposed reciprocal-ray, affine-mean, fixed-shared-scale subfamily. The two-spread counterfamily below is a valid restricted non-identifiability lower bound, but it is not a parameterization of the whole physical interface.

A separate selector analysis sharpens the missing source object. Generation-blind flavor-singlet scalars cannot emit a fixed nonzero bifundamental Yukawa matrix. Simultaneous S_3 invariance restricts a fixed matrix to the span of I and J, which has a degenerate eigenspace and aligned up/down frames. A physical construction therefore needs a source-derived flavor-orbit selector, such as a spontaneous invariant functional or genuine bifundamental boundary data, together with a quark–Higgs carrier. The calibration simulator contains no quark, Higgs, or Yukawa coupling and its enumerated observables have zero Fisher information for these coordinates. Its 64k runs are at the calibration null: the two fusion-tower readings are 1.495245 and 1.484301, while the dense first/full readings are 1.498275 and 1.497823. Those values are not quark-mass evidence. The public receipt binds the four principal run inputs by hash; their upstream arrays are not mirrored in that compact receipt archive [8].

Restricted source-spread non-identifiability theorem. Remove every running-mass target, exact-witness, fitted-spread, and compare-only ancestor. Grant the remaining source packet its strongest ordered three-point shape law. For $\rho_{\mathrm{ord}} > 0$, define

$$v_u = \frac{1}{3(1+\rho_{\mathrm{ord}})}(-(2\rho_{\mathrm{ord}}+1), \rho_{\mathrm{ord}}-1, \rho_{\mathrm{ord}}+2),$$

$$v_d = \frac{1}{3(1+\rho_{\mathrm{ord}})}(-(\rho_{\mathrm{ord}}+2), 1-\rho_{\mathrm{ord}}, 2\rho_{\mathrm{ord}}+1).$$

Both vectors have zero trace and unit endpoint span. Their adjacent-gap ratios are ρ_{ord} and ρ_{ord}^{-1}, respectively. Conversely, those three conditions determine each profile up to one positive endpoint span. Hence every compatible pair is

$$E_u = \sigma_u v_u, \qquad E_d = \sigma_d v_d, \qquad (\sigma_u, \sigma_d) \in (\mathbb{R}_{>0})^2.$$

The group $(\mathbb{R}_{>0})^2$ acts freely and transitively by independent rescaling of the two spans while fixing the source shape data. The requested four-tuple

$$\left(\frac{\sigma_u + \sigma_d}{2}, \frac{\sigma_u - \sigma_d}{2}, \sigma_u, \sigma_d\right)$$

is not invariant under this action. No unique source-only spread package follows from the stated source data.

Conditional non-definability from the current registered signature. This ambiguity is a mathematical non-definability result. Fix any generic admissible one-Higgs three-generation package and write

$$Y_q = U_{q,L}\ \mathrm{diag}\Big(e^{\mu_q+\sigma_q v_{q,1}}, e^{\mu_q+\sigma_q v_{q,2}}, e^{\mu_q+\sigma_q v_{q,3}}\Big)\, U_{q,R}^{\dagger}, \qquad q \in \{u, d\},$$

with $\sum_i v_{q,i} = 0$. For every $\lambda_u, \lambda_d > 0$, replace σ_q by $\lambda_q \sigma_q$ while holding the frames fixed. Gauge representations, anomaly cancellation, hypercharges, one-Higgs Yukawa completability, the CKM matrix, intrinsic CP capability, and the weak-sector counting clause are unchanged. In addition

$$\det \exp(\lambda_q \sigma_q v_q) = \exp\left(\lambda_q \sigma_q \sum_i v_{q,i}\right) = 1,$$

so determinant normalization does not select either modulus.

Every member of the family carries the same gauge group, matter content, color count, and generation count. The members are nevertheless physically inequivalent because the declared physical-equivalence relation preserves Yukawa invariants. Premise A3 also does not remove the family through any registered typed output: its gauge-invariant local constraint values and associated multipliers are not numerically specified and no registered map from P to quark Yukawa multipliers is emitted. The load-bearing registry-audit premise is that the current registered A1–A3 signature and declared structural package contain no other typed constraint, dynamics, action, carrier, or output map that separates or excludes this orbit. Under that premise, the current antecedents plus fixed P admit a free $(\mathbb{R}_{>0})^2$ family of equally admissible quark spectra. There is no smallest positive rescaling; its infimum is zero, which would give the massless or degenerate boundary rather than the observed spectrum. Therefore, conditional on the registry-audit premise, the current registered signature does not define a unique quark mass spectrum. Any completion retaining this counterfamily must add or derive data that separate or exclude this explicit counterfamily down to a unique physical-equivalence class. A source functional nonconstant on the family is one possible route, but so are a derived constraint, a source-selected dynamics or action, a stronger carrier or representation, or an independently justified new physical premise with the required separating power.

The kernel interface: the exact content of the missing derivation. An exact spectral construction and finite numerical witnesses distinguish admissibility from selection. Given positive

(r, s), gaps $sr/(1+r)$ and $s/(1+r)$ determine a trace-free spectrum; shifting it positive and setting $T = Q\,\mathrm{diag}(\sqrt{\mu})\,Q^\dagger$ gives $TT^\dagger = Q\,\mathrm{diag}(\mu)\,Q^\dagger$ with the prescribed gaps. The numerical battery additionally imposes fixed gap and overlap floors. If both gaps must exceed $\delta > 0$ (nominally 10^{-12}), then necessarily $s > 2\delta$; its passing kernels cannot cover every positive span. Shrinking a frame difference may improve the defect-to-gap margin while violating the overlap floor. Twelve computed witnesses at distinct ratios and spans pass all recorded checks, with invariant errors below 10^{-9}. They establish numerical non-uniqueness on that grid, not universal surjectivity or a physical source obstruction. The rounded template-context point is not an exact operating-point witness.

The three-scalar interface is exact on the nonsingular domain of its imposed reciprocal-ray subfamily. At fixed shared scale g_{ch}, it factors through

$$(r, \sigma_u, \sigma_d) \in (\mathbb{R}_{>0})^3, \qquad \rho_{\mathrm{ord}} = \frac{3}{2+r}, \qquad x_2 = \frac{r-1}{r+1},$$

with the rays, the affine mean coefficients A_{ud}, B_{ud}, the sector means, and the six coordinates closed-form in the triple, and with the explicit left inverse $\sigma_u = \frac{1}{2}\ln(m_t/m_u)$, $\sigma_d = \frac{1}{2}\ln(m_b/m_d)$, and $r = 3/\rho - 2$ at $\rho = \ln(m_c/m_u)/\ln(m_t/m_c)$. Its domain further requires $p(r) = -r^3+4r^2+23r-2 \neq 0$, since $B_{ud} = (r+1)^2(r+5)/(2p(r))$. The polynomial has positive roots in $(2/25, 9/100)$ and $(71/10, 36/5)$, by its endpoint signs and continuity. The exact domain identity and both root-existence statements are machine-checked in `QuarkMeanDomain`. The forward map is injective on this domain, and the executed round trip closes at machine precision inside that subfamily. It is not the general interface of two ordered three-point spectra.

Indeed, for any $x \neq \pm 1$, let

$$L(x) = \mathrm{ctr}(-1, x, 1), \qquad Q(x) = \mathrm{ctr}(1, x^2, 1).$$

Their Gram determinant is

$$\det \mathrm{Gram}(L, Q) = \frac{4}{3}(1-x^2)^2.$$

Thus L, Q form a basis of the centered three-vector plane, and every centered sector spectrum has unique coordinates $E_q = a_q L + b_q Q$. Once both a_q and b_q are free, x, and hence r, is a basis choice rather than an additional invariant of the spectrum. The general common-scale eigenvalue interface has six scalar coordinates: two centered coordinates and one mean for each sector. A proposed r plus four centered coordinates plus two means is a redundant seven-coordinate chart. The reciprocal-ray law removes one centered coordinate per sector and thereby recovers its special three-scalar chart.

Under the non-definability theorem's registry premise above, the stated source data select neither the ray triple nor the six-scalar physical interface. The executable acceptance harness is only a fail-closed score of the reciprocal-ray subfamily.

The edge data do not remove the ambiguity. Even after granting source values $S_{13}, S_{23}, \delta_{21} > 0$, every positive pair can be written as

$$\sigma_u = S_{13} + c_u\delta_{21}, \qquad \sigma_d = S_{23} + c_d\delta_{21}$$

for suitable (c_u, c_d). The source equations emit no rule fixing these two coefficients. The specific coefficients used by the numerical candidate are therefore a declared ansatz from a hand-written family-transport template, not an OPH-derived kernel.

Define

$$\sigma^{ud}_{\mathrm{seed}} := \frac{\sigma_u + \sigma_d}{2}, \qquad \eta_{ud} := \frac{\sigma_u - \sigma_d}{2},$$

$$A_{ud} := \frac{1}{2(1+\rho_{\mathrm{ord}}-x_2^2)},$$
$$B_{ud} := \frac{1}{2\Big(1-x_2^2-\frac{x_2^2}{1+\rho_{\mathrm{ord}}}\Big)},$$
$$\rho_{\mathrm{ord}} = 1.294284936377706.$$

Then

$$g_u = g_{\mathrm{ch}}\exp\!\big(-(A_{ud}\sigma^{ud}_{\mathrm{seed}} - B_{ud}\eta_{ud})\big), \qquad g_d = g_{\mathrm{ch}}\exp\!\big(-(A_{ud}\sigma^{ud}_{\mathrm{seed}} + B_{ud}\eta_{ud})\big).$$

The Jacobian from (σ_u, σ_d) to $(\log(g_u/g_{\mathrm{ch}}), \log(g_d/g_{\mathrm{ch}}))$ has determinant $-A_{ud}B_{ud} \neq 0$. The two free moduli therefore change the mass readout; they are not a gauge redundancy. Given a source-only physical spread datum, the affine mean law emits (g_u, g_d) algebraically on f_P. The target-derived packet is retained as a mixed-convention comparison witness. Its dimensionful matrices are mass textures. The conditional selected-class route can be written as

$$f_P + \Sigma^{\mathrm{source}}_{ud}(P) \Longrightarrow (\sigma_u, \sigma_d, \sigma^{ud}_{\mathrm{seed}}, \eta_{ud})_{\mathrm{phys}} \Longrightarrow (g_u, g_d) \Longrightarrow (m_u, m_d, m_s, m_c, m_b, m_t),$$

followed, only after common-scale RG transport and division by the running Higgs expectation value, by dimensionless Yukawa matrices. The first arrow is obstructed by the free $(\mathbb{R}_{>0})^2$ action on the stated source domain.

Target-anchored S_3 two-mode witness. An ansatz built with visible target coordinates reproduces the six mixed-convention comparison coordinates numerically. Its exact mathematical component starts from the transposition Cayley graph of S_3. The adjacency spectrum is $3, 0, -3$ with multiplicities $1, 4, 1$, so the Laplacian spectrum is $0, 3, 6$ and

$$\frac{e^{-3\tau} - e^{-6\tau}}{1 - e^{-3\tau}} = e^{-3\tau}.$$

This identity is a finite-group theorem. No OPH theorem identifies the three distinct isotypic heat values with three generations or supplies the proposed heat time

$$\tau_f = \frac{P}{4} - \frac{\pi\alpha_U}{5}.$$

The ansatz sets $w = \pi\alpha_U$, $r = e^{-3\tau_f}$, $\rho = 3/(2+r)$, and $x = (r-1)/(r+1)$, and then uses

$$e_u = S_{13} + \frac{\rho\delta_{21}}{1+\rho}, \quad e_d = S_{23} + \frac{\delta_{21}}{2(1+\rho-x^2)},$$
$$\bar\sigma_u = e_u - w\Delta S_{13}, \quad \bar\sigma_d = e_d - w(1-\Delta S_{13}),$$
$$a_u = e_u + \frac{w}{2}, \qquad a_d = e_d - \frac{w}{5},$$
$$b_u = b_u^{\mathrm{ray}} - \frac{w\rho}{10}, \qquad b_d = b_d^{\mathrm{ray}} - \frac{w}{4}.$$

The ray coordinates have the symbolic form

$$b_u^{\mathrm{ray}} = a_u\frac{-\rho x + \rho - x - 1}{(1+\rho)(x^2-1)}, \qquad b_d^{\mathrm{ray}} = a_d\frac{-\rho x - \rho - x + 1}{(1+\rho)(x^2-1)}.$$

The ansatz feeds $\bar\sigma_u, \bar\sigma_d$ into the candidate affine mean law above and sets $E_q = a_q L + b_q Q$. With the declared comparison inputs, this gives

$$r = 0.3179975211, \qquad \rho = 1.2942205385, \qquad x = -0.5174535369,$$
$$(a_u, b_u) = (5.6430129216, -4.9927828217), \qquad (a_d, b_d) = (3.3952170977, -1.8369623926).$$

Conditional normalized-trace lemma. Let $\ell : M_d(\mathbb{C}) \to \mathbb{C}$ be complex linear, invariant under all unitary conjugations, and normalized by $\ell(I) = 1$. Unitary conjugacy of rank-one projectors and additivity over an orthonormal resolution of I give $\ell(A) = \mathrm{Tr}(A)/d$, so every rank-one channel has weight $1/d$. Likewise, a width operator on $M_d(\mathbb{C})$ invariant under the full $U(d) \times U(d)$ left-right action is scalar by Schur's lemma; total Hilbert–Schmidt width t therefore assigns t/d^2 to each unit slot. Consequently, *if* the physical heat, up-odd, down-odd, up-even, and down-even responses are respectively identified with one isotropic slot in $M_5(\mathbb{C})$ of total width $5w$ and rank-one modules of dimensions $2, 5, 10, 4$, the coefficients $w/5, w/2, w/5, \rho w/10, w/4$ follow. The full invariance and normalization of the response law, those five module assignments, their signs and orientation, and exclusion of competing assignments are hypotheses outside OPH. Thus this lemma closes the denominator arithmetic conditional on the physical channel functor; it does not remove target dependence from the formula.

The per-particle comparison table is not a source-only prediction. It is a target-anchored diagnostic: the evaluator emits dimensionless coordinates, inserts an unproved one-GeV unit, and compares them to a table that mixes light-quark $\overline{\mathrm{MS}}$ values at 2 GeV, heavy-quark self-scale values, and a separate top extraction.

The raw diagonal residual sum against the supplied central values and nominal standard deviations is 1.1653, with a maximum relative residual of 0.2946%. It is not a likelihood or goodness-of-fit statistic: discovery used these targets, there is no source-theory covariance, and the rows are not one common-scale spectrum. In the target-conditioned 219,615-member denominator grammar, the selected tuple $(5, 2, 5, 10, 4)$ is the unique minimum of that raw residual sum, but eight formulas tie its best maximum error. The grammar contains the target-exposed inputs, graph, entries, assignments, signs, and functional forms, so it is not a global look-elsewhere correction.

The evaluator is runtime-target-separated but target-dependent. $S_{13}, S_{23}, \delta_{21}, \Delta S_{13}$, and g_{ch} descend from the explicitly hand-written family-transport template. The last quantity is the dimensionless template-eigenvalue mean plus its minimum gap; treating it as GeV supplies the missing unit by hand. The selected pixel branch depends on an internal quark-spectrum continuation. The edge laws add log-overlap suppression to a linearly scaling $TT^\dagger$ gap, and ΔS_{13} is a selected basis-dependent matrix entry. The ansatz is target-informed and supplies no physical mass law. It does not contradict the non-identifiability theorem.

Sufficient source conditions for flavor. The proof analysis isolates a sufficient conditional completion without instantiating it. A physical theorem would require all of the following on one acyclic, target-free source dependency graph:

1. a unique interval-certified source root for $P, \alpha_U(P)$, with no dependency on an internal quark continuation;

2. a physical S_3 family-carrier attachment, the heat-time law $\tau_f = P/4 - \pi\alpha_U/5$, the ρ, x dictionary, and the two edge-response laws;

3. a source-labeled simple-spectrum family generator and charged seed with exact common-refinement intertwiners, monomial transport of the label rays, positive common normalization, and selected edge magnitudes bounded away from zero;

4. a physical response-channel functor that proves the normalized-trace hypotheses, module assignments, signs, orientation, competing-channel exclusion, and the affine mean law on that carrier, on a domain where the A, B denominators do not vanish;

5. a common-scale physical readout with $m_q(\mu) = y_q(\mu)v(\mu)/\sqrt{2}$, including field normalization, units, and the running Higgs expectation value in the same scheme;

6. a source-only RG packet with existence and non-blowup across the required intervals, a locally Lipschitz beta system, fixed threshold ordering and matching maps, top conversion, and the declared comparison charts.

Given these hypotheses, composition of the single-valued maps makes the final sextet unique. This is a sufficient proof-obligation contract, not a proved-minimal OPH theorem. The numerical table is not its corollary: the evaluator's coordinates are dimensionless and no instance of the final two receipts exists. What the counterfamily proves as a necessary condition is narrower and decisive: any successful completion retaining this fixed counterfamily must separate or exclude its independent (λ_u, λ_d) centered-spread rescaling orbit. In particular, a source functional acting on this same family must be nonconstant on that orbit.

Cumulant specification. The representation-slot cumulant ansatz declares

$$F = \mathbb{C}^3_{\mathrm{perm}} \oplus V_{\mathrm{std}} \cong \mathbf{1} \oplus 2V_{\mathrm{std}}, \qquad \dim F = 5, \qquad \dim F_0 = 4,$$

and the composite response dimensions

$$(29, 432, 22, 32, 840, 1008, 432, 1584).$$

These integers are arithmetically correct for the declared direct sums and tensor products. If a continuous Gaussian source space, full unitary isotropy, reversible independent sheets, the listed effect ranks, and the listed signs are additionally assumed, the second-cumulant identity gives

$$\log \mathbb{E}[e^X] = \frac{1}{2}\operatorname{Var} X = \frac{r}{d}w^2.$$

This conditional identity does not construct the physical effects or select the modules. In particular, $\dim \mathrm{End}_{S_3}(\mathbf{1} \oplus 2V_{\mathrm{std}}) = 5$, so S_3-invariance does not force one scalar covariance on the two-copy multiplicity space. Pooling heterogeneous sums such as $\mathrm{End}(F) \oplus F_0$ into a single $1/29$ response requires an additional symmetry that mixes physically distinct blocks. The proposed F also has heat spectrum $0^{(1)}, 3^{(4)}$, not the regular representation's additional sign-sector value $6^{(1)}$; no physical regular-heat-to-family attachment is supplied. Finally, a genuine Gaussian covariance term is nonnegative. The negative w^2 entries require a separate signed response or subtraction law, not orientation reversal alone.

With $w = \pi\alpha_U$, the declared cumulant formulas are

$$\bar{\sigma}_u = 3P + \left(5 + \frac{4}{15}\right)w + \frac{w^2}{29}, \quad \bar{\sigma}_d = 2P + \frac{4}{15}w - \frac{w^2}{432},$$
$$a_u = 3P + \left(5 + \frac{4}{5}\right)w + \frac{w^2}{22}, \quad a_d = 2P + \left(1 + \frac{1}{32}\right)w + \frac{w^2}{420},$$

and

$$\frac{g_{\mathrm{ch}}}{v} = 2\exp\left[-2\pi + \frac{P}{1008} + \frac{w^2}{432} - \frac{w^2}{1584}\right].$$

Conditional on these formulas and the inherited heat-time, ray, even-response, affine A, B, and exponentiation laws, the downstream map is deterministic. Using the electroweak v-display gives a maximum residual of 0.2943587% and a raw diagonal nominal-residual sum of 1.1613966 against

the same mixed-convention comparison table. Neither number is a likelihood: the calculation uses visible target-derived effective coordinates, there is no common-scale renormalization-group packet or theory covariance, and the supplied cumulant comparison changes the nominal u-row uncertainty relative to the preceding bundle.

Target dependence and ablation are decisive. The target-inferred denominators for four visible effective coordinates are approximately

$$29.19665, \qquad 428.18579, \qquad 21.90604, \qquad 836.47734,$$

close to the selected $29, 432, 22, 840$, while no target-independent grammar of alternative module expressions is specified. The zero-w^2, zero-δ_g ablation has the lower maximum residual 0.2142293% and raw residual sum 0.6694007. This does not make the ablation a physical theory; it shows that the detailed covariance data is not selected by the numerical agreement. The pixel input depends on an internal quark model, and the selector uses mixed sources.

The ansatz is an explicit, falsifiable target-conditioned specification. The claimed rescaling-orbit statement must also be scoped correctly:

$$\|\lambda E_q - E_q\|^2 = (\lambda - 1)^2 \|E_q\|^2$$

has its minimum at one only on the orbit through the declared cumulant vector. For a generic base vector C_q, the minimizer of $\|\lambda C_q - E_q\|^2$ is $\langle C_q, E_q\rangle / \|C_q\|^2$, and no OPH dynamics requires minimizing this postulated residual. The target-conditioned module ansatz does not satisfy the source conditions above or define a physical quark-mass law.

Finite maximum-entropy boundary and conditional register rigidity. The finite-MaxEnt premise is false. On a finite spectrum, maximizing entropy under mean and covariance constraints gives a discrete exponential-quadratic Gibbs law, not a Gaussian density. Explicitly, on support $\{-2, -1, 0, 1, 2\}$,

$$p_A = (1/12, 1/6, 1/2, 1/6, 1/12), \qquad p_B = (1/16, 1/4, 3/8, 1/4, 1/16)$$

both have mean zero and variance one, while $\kappa_4(p_A) = 0$ and $\kappa_4(p_B) = -1/2$. The actual finite-support MaxEnt law at variance one has a nonzero fourth cumulant (approximately -0.46815). Hence neither finite maximum entropy nor the first two moments entails the Gaussian two-cumulant truncation. Gaussianization can be recovered only conditionally from an exported triangular array with a Lindeberg or adequate mixing/bounded-dependency condition, normalization, and covariance convergence; no such array is emitted by the source.

The quark-register proposal instead states an exact primitive-path certificate. Conditional on its nine declared path hypotheses, normalized trace fixes each path weight as rank over the declared register dimension, including

$$-\frac{1}{5}, \frac{4}{15}, \frac{1}{29}, -\frac{1}{432}, \frac{4}{5}, \frac{1}{22}, \frac{1}{32}, \frac{1}{420}, -\frac{1}{10}, -\frac{1}{4}, \frac{1}{1008}, \frac{1}{432}, -\frac{1}{1584}.$$

Exact record projectors also allow only the zero or unit scalar multiplier, and inert tensor refinement preserves normalized trace weights. This is a useful rigidity theorem, but its hypotheses contain the required physical content: the typed register construction, exhaustive primitive-path catalogue, effect ranks, structural multiplicities, signs, winding character, refinement intertwiners, implementation invariance, and positive-gap branch selector. A neutral finite register can be changed while preserving the broad structural signature and changing the normalized response, so A1–A3 alone do

not select this register. That countermodel is scoped to the broad signature; it is not a realization of every stronger carrier condition one might add.

The selector results sharpen the same boundary. An equivariant section requires a stabilizer-fixed candidate in each orbit fiber; an ambiguous transitive fiber need not have one. Conversely, a complete finite invariant candidate class with a unique positive-gap minimizer has a stable selector when refinement error is below half the gap. The proposal supplies neither a physically complete candidate class nor a source-derived cost and gap. The absolute-scale rescaling no-go, finite-renormalization scheme ambiguity, interval-root schema, and piecewise-RG uniqueness theorem are likewise exact obstruction or conditional well-posedness statements. No actual target-free source map with strict interval inclusion, operational clock, or fixed beta/threshold/matching packet accompanies the proposal. The register composes algebraically with the target-conditioned cumulant ansatz and authorizes no numerical quark-mass claim.

Oriented family-transport carrier test. The primitive oriented family-transport proposal supplies explicit complex 3×3 matrices T_0, T_1, but the simulator test must not manufacture them through a fitted readout. The fixed direct observable is therefore the mean of the natural three-label permutation matrices carried by the saved oriented S_3 edges. Its singular-value ratios are invariant under family permutations, unitary row/column phases, and an overall scale, so they give a necessary comparison before any entrywise match is considered. The proposal requires

$$s(T_0)/s_1 = (1, 0.55147\ldots, 0.25517\ldots), \qquad s(T_1)/s_1 = (1, 0.54625\ldots, 0.27451\ldots).$$

A 4,096-patch BW run, an independent 4,096-patch fusion run, a dense 65,536-patch population run, and a 65,536-patch BW run contain 830,066 edges in total. Their corresponding triples are

$$(1, 0.00522, 0.00134), \quad (1, 0.00552, 0.00361), \quad (1, 0.00179, 0.00002), \quad (1, 0.00071, 0.00028).$$

Source-node block bootstraps retain the same near-rank-one result. Every state lies within the fixed 0.02 Haar-null tolerance and more than 0.5407 from both proposed targets under the fixed 0.05 tolerance. The recorded state contains only edge endpoints, integer S_3 labels, and geometry: it exports no complex oriented family amplitude and no paired coarse/fine edge intertwiner. Hence all direct T_0, refined T_1, and joint physical-emission receipts are false. The conclusion is scoped to the natural direct carrier. A source-derived complex lift could define a different carrier. Choosing its phases or paths to reconstruct the proposed matrices would make the simulator test circular. The archived machine-readable report binds the four source-state arrays by hash but does not include those arrays, so it provides report custody rather than an independent raw-state replay [8].

Maximal source-derived quark package. Let $\mathcal{P}_q$ denote A1–A3 together with the declared finite matter, transport, and quark-source premises used in this section. The quark-side result has four layers:

1. the emitted mass-profile ray

$$\mathcal{R}_{ud}^{\mathrm{mass}} = \left\{ \lambda \left(\frac{1}{5}, -\frac{1 - x_2^2}{27} \right) : \lambda \geq 0 \right\}, \qquad \lambda = t_1;$$

2. the same-label left-handed selector value

$$\sigma_{ud} = \sigma_{\mathrm{ref}},$$

which is a negative sheet-selector statement;

3. the separate target-free mass bridge

$$\mathcal{P}_q \vdash \Delta_{ud}^{\text{overlap}} = \frac{1}{6}\log\frac{c_d}{c_u}, \qquad \mathcal{P}_q \vdash \Theta_{ud}^{\text{mass}} = \frac{1}{6}\log\frac{c_d}{c_u};$$

4. the selected-class support wrapper on f_P, which proves representative independence for the attached physical spread datum. It does not select either modulus. Conditional mass readout would also require an RG-covariant trajectory and a declared comparison chart; physical Yukawa matrices require the additional common-scale dimensionless conversion.

Separate same-family and common-refinement artifacts reproduce their chosen target rows exactly. Their sigma datum is obtained by inversion of those rows. The associated GeV-valued matrices are target-anchored mass textures. A separate calculation also backreads a mass-side scalar after identifying coefficient ratios with target mass ratios. None of these surfaces breaks the free source-spread action.

9.4 Quark Theorem Boundary

The conditional downstream closure route is

$$f_P \quad + \quad \Sigma_{ud}^{\text{source}}(P) \Longrightarrow \left(\sigma_u, \sigma_d, \sigma_{\text{seed}}^{ud}, \eta_{ud}\right)_{\text{phys}} \Longrightarrow (g_u, g_d) \Longrightarrow (m_u, m_d, m_s, m_c, m_b, m_t),$$

The first arrow is not identified by the source theory. The last arrow denotes scheme-labelled mass coordinates only after an RG trajectory and comparison chart are specified. Physical dimensionless Yukawa matrices lie one common-scale normalization beyond it. Selected-fiber descent and global frame classification are logically separate from both obstructions.

The conditional matter lift and the conditional port-current algebra do not cut this fiber. Both certificates are coefficient-blind: the lift fixes the gauge representations, the chirality certificate, the anomaly traces, and exactly one Yukawa invariant line per declared channel, the port-current receipt carries no Yukawa-adjacent datum, and independent positive rescaling of the coefficients along the two hadronic invariant lines fixes every certified conclusion of both. The spread fiber survives that certified structure set, the two receipts and the twelve frozen orbit-selector candidates with input hashes pinned at emission, with its coordinates realized as the free scalar coefficients along those lines. A machine-checked transport certificate records the survival together with the exclusion of the twelve candidates, and flips fail-closed if any scanned datum moves under the rescaling; the scan is a key-name heuristic over the receipt leaves and covers that set only. A cut of this fiber requires a physical binding of the response representation, the attachment of the screen action to three physical families, a selector admitted under the frozen single-comparison discipline, or the conditional Higgs/top criticality coordinate granted together with a source-side scheme conversion.

The same certificates permit and type a possible register relation at the unification scale; they supply neither a physical equality between independent Yukawa coefficients nor a physical generation order. The declared one-scalar package has one invariant line on each of the down-conjugate and charged-lepton-conjugate channels through the same scalar, together with a certified zero line on the channel that would use the up scalar for the down-type field. Thus any separately supplied relation of this form uses the down-type and charged-lepton channel pair rather than the forbidden up-scalar channel. Conditional on the declared alphabet and the two transitive-color-orbit constraints of measure balance and register faithfulness, exhaustive enumeration selects the unordered weight multiset $\{1/3,\ 1,\ 3\}$ with no measured mass or angle in its solve path. A separate comparison-only calculation of all six assignments ranks

$$(y_b/y_\tau,\ y_s/y_\mu,\ y_d/y_e) = (1,\ 1/3,\ 3)$$

as uniquely least discrepant. That target-informed empirical ranking does not derive the generation order from A1–A3. Conditional on the adopted assignment and the charged-lepton triple, common down-sector running cancels exactly on the declared one-loop branch, so the sharp RG-protected result is

$$\frac{m_s}{m_d} = \frac{1}{9}\,\frac{m_\mu}{m_e}\bigg|_{\mu_U} = 22.9743.$$

FLAG 2024 Tables 11–12 give $m_s/m_{ud} = 27.227(81)$ and $m_u/m_d = 0.465(24)$ for $N_f = 2+1+1$, and 27.42(12) and 0.485(19) for $N_f = 2+1$ [18]. Their derived central values $m_s/m_d = 19.9438$ and 20.3594 make the conditional route value 15.2% and 12.8% high. Across all six assignments, the distinct light-family coefficient-ratio menu is $\{1/9,\ 1/3,\ 3,\ 9\}$, and both FLAG rows reject every assignment under the conservative experimental-only comparison gate. The unavailable input covariance and absent OPH theory uncertainty preclude a covariance-aware significance or a theory-wide falsification conclusion. This result applies only to the declared common-transport assignment family. Different coefficient relations, coefficient alphabets, charged-family attachments, or generation-dependent threshold transport define other hypothesis classes. The retained conclusions are the conditional pairing theorem, the target-free unordered multiset under the declared measure-balance and register-faithfulness constraints, and the exact positive-chamber Koide identity. The stipulated charged-lepton response model is neither blind nor source-derived. The same conditional chart gives $m_b(m_b) = 6.03\,\mathrm{GeV}$, $m_s(2\,\mathrm{GeV}) = 140\,\mathrm{MeV}$, and $m_d(2\,\mathrm{GeV}) = 6.1\,\mathrm{MeV}$, respectively 44.2%, 50.3%, and 30.1% above the recorded comparison coordinates. These absolute values are not source-only results, and no third-generation normalization or threshold packet is supplied.

Scoped residual-axis no-go. Hermitian circulants on the same C_3 face fiber commute and have the same discrete-Fourier eigenbasis. A direct same-fiber residual construction therefore gives identity mixing. For a direct construction from two distinct real three-dimensional residual axes, the six fivefold, ten threefold, and fifteen twofold unoriented icosahedral axes give the exhaustive acute-angle spectrum

axis families	acute angles in degrees
5×5	63.4349
5×3	37.3774, 79.1877
5×2	31.7175, 58.2825, 90
3×3	41.8103, 70.5288
3×2	20.9052, 54.7356, 69.0948, 90
2×2	36, 60, 72, 90

The smallest nonzero entry is 20.9052°, with sine 0.356822. The compare-only PDG 2024 $K_{\mu2}$ coordinate $|V_{us}| = 0.2250(4)$ [19] corresponds to $\arcsin|V_{us}| = 13.0029°$ at its central value, which is absent. The enumeration therefore excludes direct equality between the Cabibbo angle and an acute angle between two of these 31 real axes. The same exact table lacks the two eigenvector-overlap readings $|V_{us}| = \cos\theta_{\mathrm{axis}}$ and $|V_{us}|^2 = \sin^2\theta_{\mathrm{axis}}/2$ as well. The coordinate construction of the axes is certified by the exact receipt; the Lean companion certifies the finite comparison on the emitted table. The no-go does not exclude spinorial representations, higher-order symmetry breaking, additional dynamical corrections, or overlap models beyond those two readings. The $31.7175° = \arctan(1/\varphi)$ entry provides an exact geometry self-check.

The register-Clebsch branch's $\sqrt{m_d/m_s} = 0.2086$ display is the Gatto–Sartori–Tonin texture estimate formed from the same rejected mass ratio. The branch supplies no up/down Yukawa-matrix

pair or relative left-handed eigenbasis, so the display is not an independent Cabibbo prediction.

9.5 Source constraints on a conditional flavor response

The icosahedral source constrains the carrier and repair dynamics before a mass readout is chosen. Let N_{i+}, N_{i-} be the record counts at the two ports of antipodal pair i, and center their totals $b_i = N_{i+} + N_{i-}$ to obtain

$$W_5 = \{b \in \mathbb{R}^6 : \textstyle\sum_i b_i = 0\}.$$

For unit vectors n_i along the six axes, define

$$T(b) = \sum_{i=1}^{6} b_i (n_i n_i^{\mathsf{T}} - I_3/3).$$

The icosahedral inner products give $T^*T = (4/5)P_5$, where $P_5 = I_6 - \mathbf{1}\mathbf{1}^{\mathsf{T}}/6$. Thus W_5 is the geometric quadrupole carrier. Its A_5 character has norm one, so it is irreducible; the invariant normalized linear functional on its endomorphisms is $\mathrm{Tr}\,/5$.

A geometric response carrier and its heat channels. Supply one unoriented face axis. Its normalizer is D_3, and exact restriction gives

$$W_5|_{D_3} = \mathbf{1} \oplus \mathbf{2} \oplus \mathbf{2}.$$

The group average is a canonical rank-one projector P_0 with matrix entries $\pm 1/6$ in pair coordinates. Its normalized trace is $1/5$; the complementary response has dimension four and trace weight $4/5$. This realizes the five-dimensional relabeling carrier geometrically. Selecting the face-axis background and identifying this quadrupole response with physical flavor remain additional steps. In particular, the physical geometric triplet restricts as $\mathbf{3}|_{D_3} = \mathrm{sign} \oplus \mathbf{2}$, a different representation.

There is no need to insert an arbitrary regular submodule to represent classical D_3 heat probabilities as channels on W_5. Conjugation gives

$$\mathrm{End}(W_5)|_{D_3} = 5\mathbf{1} \oplus 4\,\mathrm{sign} \oplus 8\mathbf{2}.$$

The Hilbert–Schmidt Gram matrix of its six group-conjugation superoperators is $24I_6 + J_6$, which is positive definite. Their mixture coefficients are therefore identifiable. If each of the three involutions has jump rate $c_2 \geq 0$ and each of the two three-cycles has rate $c_3 \geq 0$, the nontrivial eigenvalues of the positive heat generator are $6c_2$ and $3(c_2 + c_3)$. At time $t \geq 0$, put

$$r = e^{-3(c_2+c_3)t}, \qquad s = e^{-6c_2 t}; \qquad r^2 \leq s \leq 1.$$

The transposition-only relation $s = r^2$ requires $c_3 = 0$ for $t > 0$. The endomorphism heat trace is $5+16r+4s$; the regular-carrier trace $1+4r+s$ is a different observable. The full native 30-seam port Laplacian acts on W_5 as $6I_5$. A D_3 jump process is consequently a further dynamical hypothesis, not the native seam process expressed in another basis.

A typed color coefficient. Normalize fundamental color generators by $\mathrm{Tr}(T_a T_b) = \delta_{ab}/2$. Then $\sum_a T_a^2 = (4/3)I_3$, and the quadratic color observable

$$A = \Big(\sum_a T_a^2\Big) \otimes P_0 \quad \text{on } \mathbb{C}^3 \otimes W_5$$

has

$$\frac{\operatorname{Tr} A}{15} = \frac{4}{15}, \qquad \frac{\operatorname{Tr} A^2}{15} = \frac{16}{45}, \qquad \frac{\operatorname{Tr} e^{wA}}{15} = \frac{4 + e^{4w/3}}{5}.$$

This supplies a source realization of the linear coefficient $4/15$. The first four scalar cumulants are $4/15, 64/225, 256/1125, 1024/50625$; higher response terms are constrained once this particular observable and trace response are chosen. The observable is not a projector or a POVM effect: its nonzero eigenvalue is $4/3$. Its identification as a mass response is a physical hypothesis, and these cumulants do not derive a Gaussian law on arbitrary composite modules.

The contraction order matters. A color-invariant sharp projector on $\mathbb{C}^3 \otimes M$ has rank divisible by three. Twirling a rank-one component projector instead produces the unsharp invariant effect $(I_3/3) \otimes |m\rangle\langle m|$. It preserves the trace but divides the Hilbert–Schmidt square by three. Applying an isotropic Gaussian quadratic response after this twirl changes its coefficient accordingly; averaging nonlinear component responses afterwards need not do so. Neither source symmetry nor dimension counting fixes that choice. The exact character, channel, Casimir and moment identities are independently reconstructed by `code/particles/flavor/verify_source_w5_response_constraints.py`.

Exact constraints from strict-descent seam repairs. On a seam (i, j), the strict-descent settlement rule transfers one record from i to j only if $N_i - N_j \geq 2$. It preserves $S = \sum_i N_i$ and decreases $V = \sum_i N_i^2$ by $2(N_i - N_j - 1) > 0$. Every signed-integer fiber $\{N : \sum_i N_i = S,\ V \leq B\}$ is finite and forward invariant. With the additional assumption of state-independent uniform proposals among the 30 seams, order the states by V. The transition matrix is triangular with diagonal

$$K_{NN} = 1 - \frac{a(N)}{30},$$

where $a(N)$ is the number of admissible seams. Its eigenvalues are exactly these diagonal entries, and every stationary measure is supported on settled states. The positive eigenvalues of $I - K$ have ratios at most 30. This is a discrete proposal-count statement; neither a physical clock nor equality between these rates and particle masses is assumed. The settlement rule differs from the native seam rule of the executable observer construction, which sets the two endpoint loads to the floor and ceiling of their mean and assigns an odd remainder by a fair coin. At load difference one the native rule waits or swaps, so its balanced class keeps moving; the triangular spectrum and the matching calculation below apply to the settlement rule only [11, 12].

A complete finite calculation is possible in the invariant sector

$$N_i \in \{0, 1, 2\}, \qquad \sum_{i=1}^{12} N_i = 12.$$

An active seam joins a port with count two to a port with count zero. Repair sets both counts to one and deletes both endpoints from the active bipartite graph; it creates no new active edge. Complete repair schedules are therefore precisely orderings of maximal matchings of that graph. This proves that the minimum settling cost is the smallest maximal-matching size, and the longest accepted schedule has the maximum-matching size. Uniform fresh seam proposals give a random-greedy-matching absorption law, without a Gaussian or diffusion approximation.

Exhaustive enumeration gives 73,789 states, 1,275 rotational orbits, and 303 settled states in nine orbits. Independent enumeration of 13,133 maximal matchings over orbit representatives verifies the settling calculation; the largest minimum cost is four, and the longest accepted schedule

has six moves. The nine settled orbits leave an eight-dimensional simplex of A_5-invariant stationary mixtures. Symmetry and settling alone do not choose its weights.

The endpoint classification has a larger domain than that transient sector. Settledness implies $|N_i - N_j| \le d(i,j)$ for graph distance d. Every icosahedral vertex has distance profile $(1,5,5,1)$, hence $\sum_j d(i,j) = 18$ and

$$\frac{S}{12} - \frac{3}{2} \le N_i \le \frac{S}{12} + \frac{3}{2}.$$

If $S = 12k$ with integer k, integrality forces $N_i \in \{k-1, k, k+1\}$. Uniform shifts do not change the quadrupole. Consequently the 303 settled states and their four spectra exhaust *all signed-integer endpoints* at such a total, without an initial three-level or positivity assumption. Strict descent makes every maximal accepted repair sequence finite; fresh uniform proposals reach an endpoint almost surely, whereas an unfair proposal schedule can stall. The settled quadrupole eigenvalue multisets are exactly

$$\{0,0,0\}, \quad \{-2/\sqrt{5}, 0, 2/\sqrt{5}\}, \quad \{-1-1/\sqrt{5},\, 1-1/\sqrt{5},\, 2/\sqrt{5}\},$$

and the negative of the last multiset. Their multiplicities among settled states are respectively $63, 180, 30, 30$. Thus nonzero quadrupoles satisfy

$$\operatorname{Tr} Q^2 \in \{8/5, 16/5\}, \qquad \frac{(\operatorname{Tr} Q^3)^2}{(\operatorname{Tr} Q^2)^3} \in \{0, 9/64\}.$$

Under the additional affine log-mass readout $\log m_j = a + bq_j$, a nondegenerate ordered spectrum must consequently obey

$$\frac{\log(m_2/m_1)}{\log(m_3/m_2)} \in \{1, R, R^{-1}\}, \qquad R = \frac{3/\sqrt{5} - 1}{2}.$$

This constrains a specified pointwise readout at every total divisible by twelve, but does not select the physical total, endpoint or vacuum. It does not apply the same finite menu to an arbitrary averaged response operator of a stationary mixture.

The unrestricted total also admits a finite classification. Quotient settled counts by uniform integer shifts and set their minimum to zero. The graph diameter bounds their range by three. Range-three extrema must be antipodal; the identity $d(a,j) + d(-a,j) = 3$ forces the unique height profile $N_j = 3 - d(a,j)$. Its antipodal pair totals are constant, so its quadrupole vanishes. There are twelve such profiles. Every other representative has entries in $\{0,1,2\}$ and minimum zero. Exhausting those assignments gives 6,065 settled representatives. Thus *all* signed-integer settled source states, at arbitrary total, reduce to 6,077 classes modulo uniform shift and exactly 31 quadrupole spectra. Their full characteristic polynomials and total residues modulo twelve are recorded in the executable certificate.

There is also a sharp norm bound on every integer settled state. The two seams between any two antipodal pairs form a perfect matching. The settled seam bound therefore gives $|b_i - b_j| \le 2$ for pair totals. The variance bound for six numbers in an interval of width two yields $\sum_i (b_i - \bar{b})^2 \le 6$. Hence

$$\operatorname{Tr} Q^2 \le \frac{24}{5}.$$

Equality is attained by setting all ports of three antipodal pairs to one and all ports of the other three pairs to zero. The bound extends to real assignments satisfying the same unit edge-Lipschitz condition; the integer repair threshold alone does not impose that condition on arbitrary real counts.

For every nonzero endpoint quadrupole, the shape invariant $\chi = (\operatorname{Tr} Q^3)^2/(\operatorname{Tr} Q^2)^3$ belongs to the following exact set of eleven values:

$$0,\quad \frac{5}{192},\quad \frac{45}{686},\quad \frac{9}{64},\quad \frac{1}{6},$$
$$\frac{487 \pm 189\sqrt{5}}{6000},\quad \frac{809 \pm 216\sqrt{5}}{29478},\quad \frac{12509 \pm 2808\sqrt{5}}{146334}.$$

The value $1/6$ means a repeated eigenvalue and cannot yield three distinct masses under the affine log readout. For its ordered adjacent log-gap ratio $g > 0$, centering a triple with gaps $1, g$ gives

$$\chi(g) = \frac{(g-1)^2(g+2)^2(2g+1)^2}{24(g^2+g+1)^3}.$$

This function is invariant under $g \mapsto 1/g$ and strictly decreases from $1/6$ to zero on $0 < g \leq 1$. The ten admissible values therefore give exactly nineteen possible log-gap ratios: one at $g = 1$ and nine reciprocal pairs. This finite restriction requires neither a chosen source total nor a noise approximation. It continues to require the stated individual-state mass readout; additive log scale, overall response strength, up/down attachment and physical mass scheme are not selected by the stationary-shape theorem.

For the actual accepted pair-total increment, rejection or deletion of an event can only decrease the sum of its outer products. Since each edge of the six-pair complete graph has two native seam representatives, uniform proposals give the statewise bound

$$0 \preceq \operatorname{Cov}(\Delta b \mid N) \preceq \mathbb{E}[\Delta b \Delta b^{\mathsf{T}} \mid N] \preceq \frac{2}{5} P_5.$$

On quadrupoles the covariance bound is $(8/25)I_5$ in Frobenius coordinates. At every settled state the accepted covariance vanishes. A symmetric proposal alphabet, Gaussian noise or a physical Poisson clock must not be inferred from this bound. The finite enumeration, matching identities and exact moment checks are reproduced by `code/particles/flavor/verify_native_repair_flavor_constraints.py`.

A conditional driven record state. A definite probability law for the relative record state follows from an additional preparation protocol using the source's existing atomic events. After each complete settlement, choose one of the twenty-four signed port events (i, s), $s = \pm 1$, uniformly, append that event, and settle again using fresh uniform seam proposals. The choice of this renewal protocol is a hypothesis; it is not a consequence of the source grammar merely admitting the events. No mass target, fitted orbit weight, or physical time scale enters.

If N is settled, the state during repair after this event has the form $N + se_j$. A repair is possible exactly along a seam from j to k with $s(N_j - N_k) = 1$, and moves the single defect from j to k. Thus a particle walks strictly downhill in the original integer heights, and a hole strictly uphill. The graph diameter is three, so at most three repairs occur. Every accepted move has actual source–sink difference two. Rejected proposals change waiting counts but not the uniform choice among admissible neighbors.

Normalize heights by subtracting their minimum. On the resulting $6,077$ settled relative configurations the protocol defines a finite Markov kernel with $66,192$ nonzero entries. Removing a record at a global maximum leaves the state settled. Repeating reaches the zero representative, and reversing these direct events gives positive-probability paths back. The kernel is therefore irreducible. Every renewal flips the parity of the total residue, and each direct edge has a two-cycle, so

its period is exactly two. There is a unique positive stationary distribution and Cesàro convergence; ordinary stepwise convergence from one parity class is not claimed.

Proper rotations give 136 orbits and a strongly lumped kernel with $1,000$ nonzero entries. The unique orbit weights are computed as exact rational numbers and independently certified on every one of the $6,077$ states. Their total-residue marginal is $1/12$ on each residue modulo twelve. The law also respects sign reversal. For the centered count quadrupole,

$$Q(N) = \sum_i N_i \left(n_i n_i^{\mathsf{T}} - \tfrac{1}{3} I \right), \qquad \mathbb{E}_\pi Q = 0, \qquad \mathbb{E}_\pi[Q \otimes Q] = \frac{\mathbb{E}_\pi \operatorname{tr} Q^2}{5} I_5.$$

The independently checked moments are

$$\mathbb{E}_\pi \operatorname{tr} Q^2 \simeq 1.4647638612, \qquad \mathbb{E}_\pi (\operatorname{tr} Q^2)^2 \simeq 2.8795832702.$$

All odd centered-count moments vanish. In particular, $\mathbb{E}_\pi \operatorname{tr} Q^3 = 0$. The radial fourth-moment difference from a centered isotropic five-dimensional Gaussian is

$$\mathbb{E}_\pi (\operatorname{tr} Q^2)^2 - \frac{7}{5} \big(\mathbb{E}_\pi \operatorname{tr} Q^2 \big)^2 \simeq -0.1241631665 \neq 0.$$

These are moments of the declared record law; they cannot be transferred to an independently postulated physical response without its identification.

There is also an exact cost identity. Write $W(N) = \sum_i (N_i - S/12)^2$. A uniform signed preparation increases W by $11/12$ on average. Each accepted one-defect repair decreases it by two. Since W is bounded on the relative state space, stationarity gives

$$\boxed{\mathbb{E}_\pi[\text{accepted repairs per preparation}] = \frac{11}{24}.}$$

The stationary process is not reversible. For example, the transition from e_{11} to $e_{10} + e_{11}$ has probability $1/20$, its reverse has probability $1/6$, and the exact stationary edge flows are unequal. This is a driven stationary record law, not an identified thermal equilibrium or quantum vacuum.

Only shift-invariant observables have this stationary law. The unquotiented protected total performs an unbiased walk on $\mathbb{Z}$ and has no normalizable stationary distribution. Subtracting a common height is a readout projection, not a physical gauge identification of different total records. Any Higgs or mass normalization depending on the mean count requires that additional physical input. Likewise, averaging an A_5-equivariant $\operatorname{End}(P_3)$ readout gives a scalar by irreducibility, even for a nonlinear readout. The distribution of individual sampled spectra is not the spectrum of that averaged operator. The renewal law fixes the orbit weights conditionally; it supplies no quark–Higgs map or mass scheme by itself.

The full exterior Higgs response and its kinetic normalization. Let the declared matter carrier be $C \oplus W = \mathbb{C}^3 \oplus \mathbb{C}^2$, with orthonormal exterior monomials, and write $H = h_0 e_3 + h_1 e_4$. The normalized top-form contractions

$$B_u(Q, H, u^c) = [Q \wedge H \wedge u^c]_5, \qquad B_d(Q, \overline{H}, d^c) = [(\iota_{\overline{H}} Q) \wedge d^c]_5$$

give, in the ordered color and weak basis,

$$U(H) = I_C \otimes \begin{pmatrix} h_1 \\ -h_0 \end{pmatrix}, \qquad D(H) = I_C \otimes \begin{pmatrix} -\overline{h}_0 \\ -\overline{h}_1 \end{pmatrix}.$$

Here the sign convention for d^c can be absorbed into its independently supplied channel coefficient. Exact exterior and independent creation-operator calculations give

$$U^\dagger U = D^\dagger D = \|H\|^2 I_C, \qquad U^\dagger D = 0.$$

Thus the full doublet, including its charged direction, has unit-magnitude color Clebsches. The channel coefficients g_u, g_d are not determined by normalizing these invariant tensors. Color covariance forces the color intertwiner to be a scalar identity; the traceless three- and five-dimensional parts of the internal color current cannot be read as family masses.

On the conditional family attachment $F \otimes M$, restricting every supplied operator to $I_F \otimes A$ gives $Y_u = g_u I_F$ and $Y_d = g_d I_F$. This restriction is preserved by products, adjoints, inverses, positive functional calculus, and elimination of heavy modes with a family-independent block decomposition. It is a precisely bounded direct-contraction model, not a classification of every source interaction. It produces no hierarchy. After converting the all-left Weyl coefficients to the four-component Dirac Yukawa convention, a local chiral action with positive, Higgs-independent kinetic matrices has normalized vertex

$$Y_{q,\mathrm{can}} = Z_Q^{-1/2} Y_q Z_q^{-1/2} / \sqrt{Z_H}.$$

Here Z_q is the right-handed kinetic metric. For a literal all-left bilinear the left normalization factor instead carries a transpose; the two coefficient conventions must not be mixed. Consequently a proposed record matrix $R > 0$ with $Z_Q = Z_q = R$ and $Y_q = g_q R$ cancels from the vertex: $Y_{q,\mathrm{can}} = g_q I_F / \sqrt{Z_H}$. Equivalently, $\det(pR - h g_q R) = \det(R)(p - h g_q)^3$. This is a conditional identity; no source selection of $Y_q = g_q R$ is assumed. An interacting source completion must therefore emit a Higgs derivative that is distinguishable from its kinetic metric, together with the common state and normalization. For a differentiable quadratic kernel with blocks A, B, C, D, that derivative after eliminating an invertible heavy block is

$$\partial_H (A - BD^{-1}C) = A_H - B_H D^{-1} C - B D^{-1} C_H + B D^{-1} D_H D^{-1} C.$$

The finite carrier construction does not supply this Higgs-dependent source kernel or a physical mass prescription.

The independent finite replays are `code/particles/flavor/verify_native_record_renewal.py` and `code/particles/flavor/verify_exterior_higgs_response.py`.

9.6 Conditional six-mass construction at independently fixed P

A fixed flavor-response ansatz gives six numerical mass coordinates once its physical interpretation is assumed. Its usefulness is that the assumptions can be separated from the forward arithmetic and tested individually. It is not a source-only derivation: the rational coefficient and module assignments were chosen with knowledge of the observed spectrum. Absence of quark reference values in the forward evaluator does not make that discovery independent of the targets.

Here the pixel input is fixed solely by the measured electromagnetic coupling and the golden ratio,

$$P = \varphi + \sqrt{\pi}\,\alpha = 1.630968209403959\ldots, \qquad \alpha^{-1} = 137.035999177.$$

The electromagnetic value is an external calibration [13]; this equation does not independently predict it. No quark exponent pattern or mass-dependent electromagnetic closure is used to set P. The declared finite-heat gauge branch is recomputed at this same P, giving

$$\alpha_U = 0.04112433619563048\ldots, \qquad \frac{v}{E_\star} = P^{-1/2} e^{-\pi/(2\alpha_U)} = 2.019980323972526\ldots \times 10^{-17}.$$

An external calibration supplies the absolute unit $E_\star = 1.220890 \times 10^{19}\,\mathrm{GeV}$. The resulting coordinate is $v = 246.6173778\,\mathrm{GeV}$. Ratios to v and $E_\star$ are distinguished from the conversion to GeV.

For an explicit statement of the conditional law, put

$$w = \pi\alpha_U, \quad \tau = P/4 - w/5, \quad r = e^{-3\tau}, \quad \rho = \frac{3}{2+r}, \quad x = \frac{r-1}{r+1},$$

and let ctr subtract a vector's arithmetic mean. Define $L = \mathrm{ctr}(-1, x, 1)$ and $Q = \mathrm{ctr}(1, x^2, 1)$. The two mean parameters and two linear response coefficients are

$$\mu_u = 3P + \frac{79w}{15} + \frac{w^2}{29}, \quad \mu_d = 2P + \frac{4w}{15} - \frac{w^2}{432},$$
$$a_u = 3P + \frac{29w}{5} + \frac{w^2}{22}, \quad a_d = 2P + \frac{33w}{32} + \frac{w^2}{420}.$$

The remaining coefficients are

$$b_u = \frac{a_u(-\rho x + \rho - x - 1)}{(1+\rho)(x^2-1)} - \frac{w\rho}{10},$$
$$b_d = \frac{a_d(-\rho x - \rho - x + 1)}{(1+\rho)(x^2-1)} - \frac{w}{4},$$
$$A = \frac{1}{2(1+\rho-x^2)}, \qquad B = \frac{1}{2(1-x^2-x^2/(1+\rho))},$$
$$\sigma = (\mu_u + \mu_d)/2, \qquad \eta = (\mu_u - \mu_d)/2,$$
$$\delta_g = \frac{P}{1008} + \frac{w^2}{432} - \frac{w^2}{1584}, \qquad \frac{g_{\mathrm{ch}}}{v} = 2e^{-2\pi+\delta_g}.$$

Componentwise exponentiation then defines the ordered triples

$$\frac{(m_u, m_c, m_t)}{v} = \frac{g_{\mathrm{ch}}}{v} e^{-A\sigma + B\eta + a_u L + b_u Q}, \qquad \frac{(m_d, m_s, m_b)}{v} = \frac{g_{\mathrm{ch}}}{v} e^{-A\sigma - B\eta + a_d L + b_d Q}.$$

The geometric calculation in Section 9.5 provides a source realization of the response carrier $F = W_5|_{D_3}$ and of the coefficients $1/5, 4/5, 4/15$. The evaluator consumes these exact source coefficients: $79/15 = 5 + 4/15$ and $29/5 = 5 + 4/5$. Their placement in this physical mass law is a hypothesis. The eight composite module incidences of dimensions $29, 432, 22, 32, 840, 1008, 432, 1584$, and their projectors, ranks, orientations and signs must be the physical ones. Full unitary isotropy, the Gaussian two-cumulant response, its signed terms, the displayed heat and affine readout laws, and minimization of the declared residual functional are assumed. A count-only 24-slot register additionally needs a physical nonsinglet family attachment. The response relabeling action $F|_{S_3} = \mathbf{1} \oplus \mathbf{2} \oplus \mathbf{2}$ is not the geometric restriction $\mathbf{3}|_{D_3} = \mathrm{sign} \oplus \mathbf{2}$; classical group heat probabilities have a faithful channel representation on W_5, but choosing that heat process, its time, and the physical family attachment remains additional input. The transmutation normalization and each mass scheme below are also hypotheses. None follows merely from the existence of these finite modules or from P and the icosahedron.

Quark	Conditional coordinate	Assumed chart	PDG comparison
u	$2.15896\,\mathrm{MeV}$	$\overline{\mathrm{MS}}(2\,\mathrm{GeV})$	$2.16 \pm 0.07\,\mathrm{MeV}$
d	$4.71104\,\mathrm{MeV}$	$\overline{\mathrm{MS}}(2\,\mathrm{GeV})$	$4.70 \pm 0.07\,\mathrm{MeV}$
s	$93.0701\,\mathrm{MeV}$	$\overline{\mathrm{MS}}(2\,\mathrm{GeV})$	$93.5 \pm 0.8\,\mathrm{MeV}$
c	$1.272260\,\mathrm{GeV}$	$\overline{\mathrm{MS}}(m_c)$	$1.2730 \pm 0.0046\,\mathrm{GeV}$
b	$4.189458\,\mathrm{GeV}$	$\overline{\mathrm{MS}}(m_b)$	$4.183 \pm 0.007\,\mathrm{GeV}$
t	$172.0645\,\mathrm{GeV}$	pole-extraction coordinate	$172.4 \pm 0.7\,\mathrm{GeV}$

All six coordinates fall within these individual quoted ranges [20]. The five running-mass ranges are quoted at 90% confidence; the top summary row does not state that same confidence convention. The top attachment assumes a pole-extraction chart, without deriving its physical pole or extraction matching. The table is a set of mixed native charts, not one common-scale Yukawa matrix. Its printed digits describe arithmetic, not a quantified theory uncertainty.

The same fixed law gives

$$\frac{m_u}{m_d} = 0.45827670, \qquad \frac{m_s}{(m_u + m_d)/2} = 27.09465908.$$

The second ratio is outside the PDG range $27.33^{+0.18}_{-0.14}$. Its marginal differences from FLAG's $N_f = 2+1+1$ value 27.227(81) and $N_f = 2+1$ value 27.42(12) are respectively about -1.63 and -2.71 quoted standard uncertainties [18]. These comparisons overlap in experimental and lattice information. They are retained alongside the individual mass agreements, with no joint likelihood or theory error supplied. QCD/QED separation and flavor-threshold matching of the assumed chart to these lattice conventions are additional physical requirements. A fixed lower-order control omits the four w^2 terms in μ_u, μ_d, a_u, a_d and all of δ_g; it is evaluated and compared without selecting between branches by agreement. The executable specification, independent heat-equation and log-spectrum checks, and complete comparison are in `code/particles/flavor/conditional_quark_mass_replay.py` and `code/particles/runs/flavor/conditional_quark_mass_comparison.json`. The result is a reproducible conditional numerical construction informed by known masses, rather than six independently predicted observations.

A separate conditional top calculation. A criticality ansatz provides a second route that consumes neither the six-mass flavor law nor quark targets in its forward calculation. With the same P and gauge boundary, assume

$$\mu_b = E_\star e^{-\pi} P^{-1/6}, \qquad \lambda(\mu_b) = \beta_\lambda(\mu_b) = 0,$$

choose the weak perturbative Yukawa root, and run to $\mu = y_t(\mu)v/\sqrt{2}$. The two-loop criticality equation also has a large positive root; weak-root selection is an explicit hypothesis. In the Standard Model five-coupling truncation $(g_1, g_2, g_3, y_t, \lambda)$, with all other Yukawas zero, the coupled two-loop equations [21] give approximately 162.0595 GeV in this running-mass coordinate, compared with the PDG cross-section extraction $m_t(m_t) = 162.5^{+2.1}_{-1.5}$ GeV [20]. An independently implemented integrator and hypercharge convention check the forward result.

The three-loop QCD conversion gives a pole proxy of 171.7516 GeV. The one-loop route gives 157.8983 GeV and 167.4083 GeV, respectively; two-loop Yukawa/quartic running with one-loop gauges gives 162.8813 GeV and 172.6439 GeV. These shifts are truncation diagnostics, not a confidence interval. The finite-heat gauge branch uses the coefficients $(33/5, 1, -3)$ as a declared low-boundary recipe; the subsequent Standard Model transport does not establish a common threshold-complete unification history. Identifying its fixed v coordinate with a renormalized electroweak mass convention requires a further matching hypothesis. This constant-VEV route supplies no finite electroweak, tadpole or complex-pole matching, so its QCD proxy is not a certified physical pole prediction. The separately attached Landau-VEV calculation in Section 9.7 supplies those finite matching functions under additional assumptions. Boundary-scale selection and criticality also remain physical assumptions. The three routes and numerical residuals are retained in `code/particles/runs/calibration/conditional_top_fixed_p.json`.

A top-coordinate band without equal anchor capacities. The scale hypothesis can be weakened. The same input packet supplies two anchors,

$$\mu_U = E_\star e^{-2\pi} P^{1/6} \simeq 2.47361105 \times 10^{16}\,\mathrm{GeV}, \qquad E_{\mathrm{cell}} = E_\star P^{-1/2} \simeq 9.55990967 \times 10^{18}\,\mathrm{GeV}.$$

Write $t = \log \mu$, $t_U = \log \mu_U$, and $t_E = \log E_{\mathrm{cell}}$. For positive capacities w_U, w_E, the declared additive mismatch

$$C_{\mathrm{rec}}(t) = w_U (t - t_U)^2 + w_E (t - t_E)^2$$

has the unique minimum

$$t_b = \frac{w_U t_U + w_E t_E}{w_U + w_E} \in (t_U, t_E).$$

Every point of this open interval occurs for a positive capacity ratio. Equal capacities recover the earlier logarithmic midpoint, but are not needed for a bounded scale. The weaker assumption of two differentiable strictly convex costs, each minimized at its own anchor, also places the minimum between the anchors by the signs of their derivatives there.

Gaussian records are unnecessary within a common regular exponential family $p_\theta = \exp(\theta T - \psi(\theta))h$, with $\psi'' > 0$ and $\theta(t) = a_0 + b_0 t$, $b_0 \neq 0$. Direct differentiation gives

$$\frac{d}{dt} \sum_i w_i D(p_{\theta(t)} \| p_{\theta(t_i)}) = b_0^2 \psi''(\theta(t)) \sum_i w_i (t - t_i).$$

The same weighted log mean follows. Reversing the relative entropy instead averages $\psi'(\theta_i)$; its inverse lies between the anchors. Source realization of the two records and physical criticality remain hypotheses. For independent Gaussian ports the capacities are $w_i = \sum_p s_{ip}^2/(2\sigma_{ip}^2)$; equal formula depth or equal functional form does not establish their equality.

At one loop the entire resulting top-coordinate interval follows from monotonicity, rather than a grid scan. Put $k = (16\pi^2)^{-1}$ and $(a, b, c) = (g_Y^2, g_2^2, g_3^2)$. The gauge equations are

$$a' = \tfrac{41}{3} k a^2, \qquad b' = -\tfrac{19}{3} k b^2, \qquad c' = -14 k c^2.$$

Define $X = a^2 + 2ab + 3b^2$, $F = (X/16)^{1/4}$, $C = 17a/12 + 9b/4 + 8c$, and $f(t, y) = ky(9y^2/2 - C)$. Criticality gives $y(t_b) = F(t_b)$. Differentiating the solution with respect to its boundary time yields

$$\frac{\partial y(t; t_b)}{\partial t_b} = \exp\!\left(\int_{t_b}^{t} f_y(s, y(s; t_b))\, ds \right) [F'(t_b) - f(t_b, F(t_b))].$$

Since $F'/F \geq -(19/6)kb$ and $\sqrt{X} \leq a + \sqrt{3}b$,

$$\frac{F' - f(t, F)}{F} \geq k \left[\frac{7a}{24} + 8c - \left(\frac{11}{12} + \frac{9\sqrt{3}}{8} \right) b \right] > 0$$

whenever $c \geq b/2$. The inverse gauge couplings are affine in t, so their ratios are fractional linear and endpoint checks suffice. For the declared input packet, $c/b > 0.947$ throughout the anchor interval.

The self-scale crossing is unique and ordered. If $R_y = 9y^2/2 - C$, then $R_y < 0$ at criticality. Also

$$C' = k[(697/36)a^2 - (57/4)b^2 - 112c^2] < 0$$

when $c \geq a/2$. At a putative $R_y = 0$ crossing, $R_y' = -C' > 0$, so $R_y < 0$ is preserved backwards in time. The declared packet has $c/a > 1.076$ between 50 GeV and E_{cell}, hence $y' < 0$ on each backwards solution down to 50 GeV. Positivity and existence follow independently by setting $z = 1/y^2$,

which solves the linear equation $z' = 2kCz - 9k$ with a positive backwards integral representation. The boundary bounds give $y(50) > 0.332$ while $50\sqrt{2}/v < 0.287$, and $y(\mu_U) < \sqrt{2C(t_U)/9} < 0.831$. Consequently $vy(\log\mu)/\sqrt{2} - \mu$ decreases through zero exactly once below μ_U. Its zero increases strictly with t_b.

Let m_- and m_+ denote the endpoint-defined one-loop self-scale coordinates. The exact conditional range for strictly positive capacities is (m_-, m_+), with numerical values

$$m_- \simeq 154.79231\,\mathrm{GeV}, \qquad m_+ \simeq 160.80576\,\mathrm{GeV}.$$

These decimals are approximations, not outward-certified enclosures. Coupled two-loop controls give the following coordinates:

$w_E/(w_U + w_E)$	0	1/4	1/2	3/4	1
m_{chart} (GeV)	158.99069	160.55379	162.05945	163.51339	164.92084

These five controls do not prove a two-loop continuum band. Neither calculation quantifies physical matching or truncation uncertainty. Independent quadrature, differential-equation transport and symbolic checks are in `code/particles/calibration/verify_top_positive_capacity_band.py`.

Normalization constraints that survive canonical fields. For positive kinetic matrices Z_Q, Z_u, Z_d, Higgs normalization $Z_H > 0$, raw vacuum v_0 and raw Yukawas Y_q, canonical normalization gives

$$v = \sqrt{Z_H}v_0, \qquad M_q = \frac{v_0}{\sqrt{2}} Z_Q^{-1/2} Y_q Z_q^{-1/2}.$$

The Higgs rescaling cancels. If the down determinant is nonzero,

$$\frac{\prod_j m_{u,j}^2}{\prod_j m_{d,j}^2} = \frac{\det(Y_u^\dagger Y_u)}{\det(Y_d^\dagger Y_d)} \frac{\det Z_d}{\det Z_u}.$$

The common vacuum and left metric cancel, leaving a relative right-sector normalization to determine. Gauge compatibility alone does not equate the right-sector metrics. Similarly, an assumed vacuum matching factor r_v changes the top readout to $m = r_v vy(\log m)/\sqrt{2}$, with

$$\frac{d\log m}{d\log r_v} = \frac{1}{1 - \beta_y/y}.$$

This lies between zero and one on the proved one-loop branch, but no source-derived finite range of r_v is supplied. The capacity band therefore constrains the stated mass coordinate, not an unconditionally matched physical top mass.

9.7 Explicit electroweak matching of a conditional source boundary

A definite finite matching calculation becomes possible after making the vacuum convention explicit. Retain the independently fixed P, the finite-heat gauge packet and the equal-capacity criticality boundary above. At the pre-existing source running-Z coordinate, impose the additional identification

$$Q_0 = 91.59609090\,\mathrm{GeV}, \qquad v_{\mathrm{Landau},\overline{\mathrm{MS}}}(Q_0) = v_{\mathrm{chart}} = 246.61737777\,\mathrm{GeV}.$$

The source gauge coordinates are identified with the full Standard Model $\overline{\text{MS}}$ couplings at Q_0. Neither identification is a source-selection theorem. In particular, Q_0 is not replaced by a measured Z mass or by a top-dependent scale. The two-loop criticality trajectory gives

$$(g_Y, g_2, g_3, y_t, \lambda)_{Q_0} \simeq (0.35681261,\ 0.65151015,\ 1.21944839,\ 0.96238089,\ 0.14002413).$$

All other Yukawas vanish, as in the original top-only calculation. This is a declared chiral truncation, not a complete physical flavor model.

Use the Landau-gauge minimum of the Goldstone-resummed effective potential to define the running vacuum, with $V = m^2 H^\dagger H + \lambda (H^\dagger H)^2$. The finite matching and renormalization-group functions are imported from SMDR [37, 38]. The minimum fixes $m^2(Q_0)$; at tree, one-loop and two-loop orders its values are approximately -8516.29, -8271.04 and $-8373.75\,\text{GeV}^2$. The same boundary is transported to $Q_0, 2Q_0, 4Q_0$ with two-loop running, including the vacuum anomalous dimension. Thus

$$v(Q_0), v(2Q_0), v(4Q_0) \simeq 246.61738,\ 244.72892,\ 243.08648\,\text{GeV}.$$

Reimposing $v = v_{\text{chart}}$ at each evaluation scale would change the model. The intermediate $y_t v/\sqrt{2}$ is a full-Standard-Model running parameter in this convention, not the pure-QCD short-distance top mass.

The pole convention is $s = (M - i\Gamma/2)^2$. For comparison with the running-width Breit–Wigner convention used for W and Z, write $\delta = \Gamma^2/(4M^2)$; then

$$M_{\text{BW}} = \frac{M(1+\delta)}{\sqrt{1-\delta}}, \qquad \Gamma_{\text{BW}} = \frac{\Gamma(1+\delta)}{(1-\delta)^{3/2}}.$$

The top calculation retains both expansion methods and all zero-, one- and two-loop controls. Its designated value includes four-loop pure QCD and two-loop remaining contributions; the boson and charged-current values use two-loop matching. No measured top mass, boson mass or Fermi constant is used to set a boundary or choose an evaluation scale.

Quantity	Q_0	$2Q_0$	$4Q_0$
M_t (GeV)	170.62553	170.59650	170.60124
M_H (GeV)	124.88031	124.78766	124.68169
$M_{W,\text{BW}}$ (GeV)	79.86215	79.87044	79.86794
$M_{Z,\text{BW}}$ (GeV)	90.59047	90.59651	90.59763
$10^5 G_F^{\text{local}}$ (GeV^{-2})	1.1827834	1.1825618	1.1826107

Here G_F^{local} denotes the local charged-current coefficient in the zero-muon truncation. The matching implementation's fixed empirical finite-muon correction is removed algebraically; a massless-muon model does not predict an operational muon lifetime. Experimental-reference globals are disabled in the forward calculation. Independent Python calculations reproduce the running vacuum, one-loop charged-current coefficient, one-loop top complex pole and one-loop potential minimum. The full multiloop table is separately reproduced with the pinned C implementation; a stored numerical digest supplies integrity, not an independent derivation of its multiloop integrals.

This calculation tests the additional identification. At Q_0 the W and Z Breit–Wigner masses lie approximately 0.51 and 0.60 GeV below their measured values, while the local charged-current coefficient is about 1.41 percent above the conventional Fermi constant. These differences are retained

along with the top and Higgs comparisons. The transported scale controls are not confidence intervals, and the chiral truncation, matching uncertainty and physical source normalization do not have a complete error budget. No joint acceptance or statistical significance is assigned. The result therefore supplies a conditional finite matching map and a quantitative constraint on the vacuum/gauge attachment, rather than confirming that attachment. Since zero Yukawas remain zero under the stated perturbative transport, this top-only branch cannot generate the five lighter quark masses. It also does not establish the mixed-chart attachment of the separate six-mass ansatz.

9.8 Strong-CP branch

The selected-class quark wrapper carries target-anchored mass textures on the public quark frame class f_P. Strong CP is a separate phase-side invariant. The stated source domain does not derive the bare QCD angle θ_{QCD}, does not emit the physical anomaly-invariant combination $\bar{\theta}$, and does not prove that the physical strong-CP phase vanishes.

This phase problem is independent of the source-spread and common-scale Yukawa obstructions. A closure would require a source-emitted quark mass matrix at one declared scale, its physical determinant-line phase contribution, and a theorem fixing the topological-angle contribution on the realized branch. The present GeV mass textures do not supply that input.

10 Charged-Lepton Family Derivation

The charged-lepton derivation differs from the quark derivation in a precise way. It does not emit public charged masses from P. It emits an exact same-family witness, a closed common-shift no-go, the declared same-label q_e readback, a determinant-line lift on theorem-grade physical charged data, and a downstream algebraic mass readout from theorem-grade $A_{\mathrm{ch}}(P)$. For any fixed formal source exponent vector $M_{\bullet}^{\mathrm{ch}}$, the same-label readback defines a source-side determinant character

$$S_M = \sum_e M_e^{\mathrm{ch}} \log q_e.$$

The theorem branch does not emit a theorem-grade sector-isolated charged determinant exponent vector, and it does not identify a source-side determinant character with the physical charged determinant line. The source-only theorem therefore supplies no electron, muon, or tau mass.

The mathematical split is equally precise. The charged-lepton derivation starts from the ordered charged package, proves that the realized support is a one-dimensional linear subray, exposes the canonical quadratic support-extension direction, maps that into the charged excitation gaps, closes a two-scalar support-extension law shell, isolates the smaller eta source-readback primitive on that same carrier, and then builds the log-spectrum and forward shape/scale surface. Those centered objects are common-shift invariant. The determinant line fixes the physical affine scalar once theorem-grade physical charged data are present, and the mass readout is then algebraic. The charged theorem boundary has two distinct pieces: physical selection of the latent charged sector-response candidate as $\widehat{C}_e$, and source-to-physical determinant attachment. For a fixed formal source exponent vector $M_{\bullet}^{\mathrm{ch}}$, that attachment is the identity

$$3\mu(r) = \sum_e M_e^{\mathrm{ch}} \log q_e(r),$$

equivalently zero determinant-normalization defect

$$N_{\det}(P) = s_{\det}(P) - \sum_e M_e^{\mathrm{ch}} \log q_e(P),$$

on the charged determinant channel.

Physically, the electron is the light charged lepton that makes atoms and chemistry possible, the muon is its heavier unstable cousin, and the tau lepton is the heaviest charged lepton. A physical charged-lepton derivation would have to explain how those three rows emerge from the shared flavor dictionary without Koide-assisted fitting. The theorem-grade branch must select the latent charged sector-response candidate $\widehat{C}_e^{\mathrm{cand}}$ as $\widehat{C}_e$ through the branch-generator splitting theorem. On theorem-grade physical Y_e, a refinement-stable uncentered lift collapses the determinant-line section and affine anchor to one descended physical affine scalar $\mu_{\mathrm{phys}}(Y_e)$, with

$$\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\mathrm{phys}}(Y_e)\,\mathbf{1}, \qquad s_{\det}(Y_e) = 3\mu_{\mathrm{phys}}(Y_e), \qquad A_{\mathrm{ch}}(Y_e) = \mu_{\mathrm{phys}}(Y_e).$$

Theorem 10.1 (Charged same-carrier source-pair readback)**.** *Fix the ordered charged carrier*

$$(-1, x_2, 1),$$
$$x_2 = -0.5175863354681689.$$

If the charged source pair

$$(\eta_{\mathrm{ext}}, \sigma_{\mathrm{ext}}) = (\eta_{\mathrm{source_support_extension_log_per_side}}, \sigma_{\mathrm{source_support_extension_total_log_per_side}})$$

is emitted on that carrier, then the centered charged logs are

$$e_{\mathrm{log,centered}} = -\frac{(3+x_2)\sigma_{\mathrm{ext}} - \eta_{\mathrm{ext}}}{6},$$
$$\mu_{\mathrm{log,centered}} = \frac{x_2\sigma_{\mathrm{ext}} - \eta_{\mathrm{ext}}}{3},$$
$$\tau_{\mathrm{log,centered}} = \frac{(3-x_2)\sigma_{\mathrm{ext}} + \eta_{\mathrm{ext}}}{6},$$

and therefore the charged masses are

$$m_e = g_e\, e^{e_{\mathrm{log,centered}}}, \qquad m_\mu = g_e\, e^{\mu_{\mathrm{log,centered}}}, \qquad m_\tau = g_e\, e^{\tau_{\mathrm{log,centered}}}.$$

This theorem is exact on the same-carrier shell, but the absolute values are not emitted. The visible scalar order is

$$\eta_{\mathrm{ext}} \quad \text{then} \quad \sigma_{\mathrm{ext}},$$

and the charged absolute scale g_e is not determined. The scalars η_{ext} and σ_{ext} are the first same-carrier residuals. If the latent candidate $\widehat{C}_e^{\mathrm{cand}}$ is selected as the physical operator, then η_{ext} and σ_{ext} become charged spectral invariants instead of separate primitive goals, and the absolute-scale burden is pushed to one affine-covariant absolute charged anchor A_{ch}. In the local chain, $\widehat{C}_e$ itself is undeclared. The available object is only the centered compressed generation-bundle candidate $\widehat{C}_e^{\mathrm{cand}}$. Its missing theorem is the compression-descendant commutator condition after the central split. The centered common-shift quotient is closed negatively, so centered data alone do not emit the affine anchor A_{ch}.

Theorem 10.2 (Charged absolute-scale underdetermination)**.** *Let*

$$E_e^{\mathrm{centered}} = \big(e_{\mathrm{log,centered}}, \mu_{\mathrm{log,centered}}, \tau_{\mathrm{log,centered}}\big)$$

be the centered charged log triple emitted from the charged source pair $(\eta_{\mathrm{ext}}, \sigma_{\mathrm{ext}})$*. Then for every* $c \in \mathbb{R}$*,*

$$Y_e(c) := \exp(c)\,\mathrm{diag}(e^{E_e^{\mathrm{centered}}})$$

has the same charged spectral invariants

$$\eta_{\mathrm{ext}}, \qquad \sigma_{\mathrm{ext}}, \qquad \gamma_{21}, \qquad \gamma_{32},$$

the same centered log vector, and the same ratio data. Only the absolute masses scale:

$$(m_e, m_\mu, m_\tau) \longmapsto e^c\,(m_e, m_\mu, m_\tau).$$

Hence every readout that factors through the centered readback and these four invariants determines only the quotient class

$$E_e^{\mathrm{centered}} \in \mathbb{R}^3/\langle(1,1,1)\rangle,$$

and leaves the absolute scalar $g_e = e^c$ free. The same-label q_e readback and the determinant character S_M are chain outputs outside this list; the theorem makes no statement about them.

The proof is immediate from the centered sum rule

$$e_{\log,\mathrm{centered}} + \mu_{\log,\mathrm{centered}} + \tau_{\log,\mathrm{centered}} = 0,$$

which implies

$$\det Y_e^{\mathrm{shape}} = 1, \qquad \det Y_e = g_e^3.$$

All emitted charged invariants depend only on differences of log entries, so a common shift in the $(1,1,1)$ direction leaves them unchanged. The stated source domain therefore gives a no-go for a P-driven charged mass read from centered data alone: the theorem fixes the centered shape, while the absolute normalization sits in the determinant-line landing, an underived bridge on the corpus.

The charged absolute-scale branch is explicitly typed. The charged scale is a linear quantity

$$g_e = e^{\mu_e^{\mathrm{abs}}},$$

so the log-coordinate μ_e^{abs} must not be mixed directly with centered log gaps. The same-family writeback therefore records the type-consistent shell

$$\begin{aligned}\mu_{e,\mathrm{seed}}^{\mathrm{abs}} &= \log(0.9231656602589082)\\ &= -0.07994658034676537,\end{aligned}$$

$$\mu_{e,\mathrm{cand}}^{\mathrm{abs}} = \mu_{e,\mathrm{seed}}^{\mathrm{abs}} - \gamma_{\min} = -0.38231224060567365,$$

$$g_e^{\mathrm{cand}} = e^{\mu_{e,\mathrm{cand}}^{\mathrm{abs}}} = 0.6822819838027987.$$

This is a representation-consistency shell only, not a charged-mass theorem. It fixes the linear-vs-log coordinate discipline. This surface does not emit a theorem-grade value law for g_e. The checked shortcut

$$\begin{aligned}\Delta_e^{\mathrm{abs}} &= 0.30236566025890826,\\ g_e &= 0.6822819838027987\end{aligned}$$

is not a theorem-grade closure. It merely chooses one representative on the common-shift orbit and does not land on the physical charged masses.

A physical completion requires a theorem-grade affine-covariant section A_{ch} of the quotient map, satisfying

$$A_{\mathrm{ch}}(\ell + c\,\mathbf{1}) = A_{\mathrm{ch}}(\ell) + c.$$

The clean realization of that section is an uncentered charged response lift carrying a determinant line, in which

$$A_{\mathrm{ch}} = \tfrac{1}{3}\log\det(Y_e) = \tfrac{1}{3}\operatorname{tr}(\log Y_e).$$

Once such an anchor exists,

$$g_e = e^{A_{\mathrm{ch}}(\ell)}, \qquad \Delta_e^{\mathrm{abs}} = \log g_{\mathrm{ch}}^{\mathrm{shared}} - A_{\mathrm{ch}}(\ell).$$

Proposition 10.3 (Twenty-four-register rate no-go)**.** *An oriented register with twenty-four slots, a twenty-four-stage automaton, or a decomposition into twenty-four formal update steps determines no physical frequency, Hamiltonian gap, or mass scale. Indeed, rescaling every physical generator by $H \mapsto \lambda H$, $\lambda > 0$, preserves the register, transition graph, grading, and settled quotient while multiplying all generator gaps and rates by λ. A repair iteration counter is likewise not an operational clock. A physical rate requires a source-derived clock instrument, calibrated comparison line, and residual bound.*

Theorem 10.4 (Conditional charged determinant-clock attachment)**.** *Let L_{24} be a one-dimensional source-derived operational clock line with a distinguished calibrated positive gap vector ℓ_{24} of norm Δ_{24}. Suppose there is a refinement-natural, quotient-visible norm-preserving line isomorphism*

$$\Theta_e : L_{24}^{\otimes 3} \longrightarrow \det M_e$$

to the physical charged-lepton determinant line, together with the load-bearing element compatibility

$$\Theta_e(\ell_{24}^{\otimes 3}) = \det(M_e).$$

Then the common charged mass scale is fixed by

$$g_e := |\det M_e|^{1/3} = \Delta_{24}.$$

More generally, if the element-compatible map rescales norms by a separately derived factor $\kappa > 0$, then $g_e = \kappa^{1/3}\Delta_{24}$. Thus the implication from a calibrated clock-line attachment to the determinant scale is exact.

Proof. The distinguished tensor-cube vector has norm $\|\ell_{24}^{\otimes 3}\| = \Delta_{24}^3$. Element compatibility and norm preservation give $|\det M_e| = \Delta_{24}^3$; take the positive cube root. □

The register count does not construct L_{24}, calibrate Δ_{24}, or produce Θ_e. Those are the removable physical clock, determinant-descent, and normalization gates. The determinant scale is not derived here. Koide balance fixes a shape condition and is independent of this affine scale attachment.

Positive-chamber face-circulant identity and conditional finite-GNS balance. The regular C_3 face fiber has the general Hermitian carrier

$$C = aI + \rho\left(e^{i\delta}R + e^{-i\delta}R^2\right), \qquad a, \rho \geq 0,$$

where R is the regular cyclic shift, $R^3 = I$. The commutant of the regular C_3 action is spanned by I, R, R^2, which makes C the general Hermitian C_3-equivariant carrier. Fourier diagonalization gives

$$r_k = a + 2\rho\cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2.$$

In a positive spectral chamber, $m_k = s r_k^2$ and $\sqrt{m_k} = \sqrt{s}\, r_k$. The roots-of-unity identities

$$\sum_{k=0}^{2} \cos\!\left(\delta + \frac{2\pi k}{3}\right) = 0, \qquad \sum_{k=0}^{2} \cos^2\!\left(\delta + \frac{2\pi k}{3}\right) = \frac{3}{2}$$

give the exact Koide identity

$$Q = \frac{\sum_k m_k}{(\sum_k \sqrt{m_k})^2} = \frac{1 + 2(\rho/a)^2}{3},$$

so $\rho/a = 1/\sqrt{2}$ is exactly equivalent to $Q = 2/3$ in the positive-eigenvalue chamber. At balance, positivity holds for $|\delta| \le \pi/12 \pmod{2\pi/3}$, and throughout that chamber Koide is independent of δ. Outside it, physical square roots are absolute eigenvalues and the physical Koide ratio differs from the signed trace expression. Equivalently, if $E_0 = 3a^2$ and $E_c = 6\rho^2$ are the singlet and charged-plane Hilbert–Schmidt powers, then

$$Q = \frac{1 + E_c/E_0}{3}, \qquad Q = \frac{2}{3} \iff E_c = E_0.$$

The positive square-root-mass vector lies on the 45° Koide cone about the democratic direction. Circulant symmetry leaves the relative norm of the singlet and charged plane free.

The finite response packet sets that norm under its declared hypotheses. Let

$$\mathcal{V}_K = \mathbf{1} \oplus \chi \oplus \bar{\chi}, \qquad \mathcal{H}_{\mathrm{or}} = \mathbb{C}^2, \qquad \mathcal{A}_K = B(\mathcal{V}_K \otimes \mathcal{H}_{\mathrm{or}}) \simeq M_6(\mathbb{C}).$$

A one-state record cannot distinguish orientation reversal, so $\mathbb{C}^2$ is the minimal faithful orientation record. Minimality of this response-local register is an explicit declared hypothesis. With q_+ one oriented outcome, define

$$E_+ = P_0 \otimes I_{\mathrm{or}} + P_c \otimes q_+, \qquad Z_0 = P_0 \otimes I_{\mathrm{or}}, \qquad Z_c = P_c \otimes q_+.$$

The blocks Z_0, Z_c both have rank two. Under the declared Lüders instrument, Born–Lüders conditioning of the unique MaxEnt state $I_6/6$ on E_+ gives

$$p_0 = p_c = \frac{1}{2}.$$

The orientation-blind singlet has event action zero, while the selected charged orientation has action $\ln 2$. Charged multiplicity two times its weight $1/2$ equals the singlet weight. The unique normalized trace on $M_6(\mathbb{C})$ fixes these probabilities without the free trace parameter present on $\mathbb{C} \oplus M_2(\mathbb{C})$.

In $L^2(B(\mathbb{C}^3), \tau_3)$, the operators I, R, R^2 are orthonormal, and

$$\mathcal{J}(e_0) = I, \qquad \mathcal{J}(e_+) = R, \qquad \mathcal{J}(e_-) = R^2$$

is unitary and intertwines the C_3 action, orientation reversal, and the two block projections. The canonical GNS square-root amplitude is

$$\xi = \sqrt{p_0}\, e_0 + \sqrt{\frac{p_c}{2}} \left(e^{i\delta} e_+ + e^{-i\delta} e_-\right),$$

so its response has $a = \sqrt{p_0}$ and $|b| = \sqrt{p_c/2}$. Hence

$$\frac{|b|^2}{a^2} = \frac{p_c}{2p_0} = \frac{1}{2}, \qquad \frac{|b|}{a} = \frac{1}{\sqrt{2}}, \qquad Q = \frac{2}{3}.$$

An orientation action gap ΔS gives the general form

$$\frac{p_c}{p_0} = 2e^{-\Delta S}, \qquad \frac{|b|^2}{a^2} = e^{-\Delta S}, \qquad Q(\Delta S) = \frac{1 + 2e^{-\Delta S}}{3}.$$

For comparison, the target-informed minimal complete public response coordinate has

$$Q_{\mathrm{MCPR}} = 0.6666644634090389, \qquad \frac{|b|}{a} = 0.7071044442750720,$$

whose modulus is 3.30 ppm below $1/\sqrt{2}$. The PDG 2026 central charged-lepton masses give

$$Q_{\mathrm{PDG}} = 0.6666644634026367.$$

The numerical proximity is diagnostic. The response architecture and phase were stipulated with target knowledge, so this comparison is not a source-only prediction [22].

A physical Koide implication is conditional on faithful trace-preserving u.c.p. maps Φ and Ψ from the source process to a physical response process and back, with $\Psi\Phi = \mathrm{id}$. Kadison–Schwarz applied to both maps puts every source element in the multiplicative domain of Φ, so Φ is a $*$-monomorphism and an exact L^2 isometry. If it also intertwines Z_0, Z_c with physical response records and sends the conditioned source state to the normalized positive square-root-mass response $C = (Y_e^\dagger Y_e)^{1/4}$, then it transports $p_0 = p_c$ to $E_0 = E_c$ and proves physical Koide. For an accepted finite chiral checkpoint with three left and three right family modes, positive kinetic metrics, and a committed neutral Higgs direction, the reversible-checkpoint conditions force $\Phi = \mathrm{Ad}_{J_L \oplus J_E}$, preserving chirality and the regular C_3 action. If $M_F = X_F^2$ is the positive source response, then

$$\widehat{Y}_e = \frac{\sqrt{2}}{v} J_L M_F J_E^\dagger, \qquad \mathcal{M}_L = J_L M_F J_L^\dagger.$$

The source and canonical physical square-root responses have equal block powers. This closes the finite balanced shape implication on supplied physical charged data, not the determinant scale. The extended branch $\mathrm{OPH}^+_{\mathrm{ch}}$ adds graded physical completion and quotient source-law selection. Given an exhaustive declared carrier class with a positive winner gap, a source-closed BV/BRST continuum, and an interval or contraction certificate for the charged QFT self-map, it selects one accepted charged sector and one dressed mass readout. The stable electron mass is a dressed spectral lower edge; the muon and tau masses are real parts of specified dressed resonance roots. A balanced C_3 fixed point gives $Q = 2/3$. A C_3-symmetric fixed point with charged attenuation $\chi_\star$ gives

$$Q = \frac{1 + e^{-2\chi_\star}}{3}.$$

An off-plane response requires the full response operator. These extended-branch principles and the QFT certificate are additional conditions beyond A1–A3. The relation $\mathrm{Hom}_{C_3}(\mathbf{1}, \chi \oplus \bar{\chi}) = 0$ prevents the neutral source grammar from producing a labeled charged-family vector through symmetry. A charged source tensor, connection, or selected orbit must supply the phase and individual ratios.

The count $(N_c + 1)/(2N_cN_g) = 2/9$ is arithmetic, and the value $2/9$ is the empirical Brannen–Koide fitted phase [14, 15]; the integer factorization is conditioned on that value. A physical phase theorem requires a regular-C_3 family bundle, an oriented family connection whose declared loop has holonomy $\exp[i\beta_{\mathrm{EW}}/(2N_cN_g)]$, and an attachment identifying that holonomy with the phase of the charged Fourier coefficient. Hypercharge acts as a common scalar on the generation copies, while the shift R permutes them. Pure finite A_5/C_3 holonomy yields cube-root phases. The

required continuous family connection and its source selector are not constructed. Under the stated connection and attachment hypotheses, $\delta = 2/9$. The phase of one equal link differs by a factor of three from the total triangular holonomy. With the scale-free normalization $a = 1$, the ordered roots

$$r_k = a + 2\rho\cos\left(\delta + \frac{2\pi k}{3}\right)$$

become

$$\begin{aligned} r_e &= 0.040349908219207475, \\ r_\mu &= 0.5802119201475368, \\ r_\tau &= 2.3794381716332555. \end{aligned}$$

Under those assumptions the bridge selects the same-carrier pair

$$\begin{aligned} \eta_{\mathrm{ext}} &= -6.729586682888832, \\ \sigma_{\mathrm{ext}} &= 8.154061112725994, \end{aligned}$$

with ordered gap values

$$\begin{aligned} \gamma_{21} &= 5.33160859254774, \\ \gamma_{32} &= 2.822452520178255, \\ \kappa_{\mathrm{ext}} &= -4.59605680397234, \end{aligned}$$

and centered logs

$$E_{\log,\mathrm{centered}} = [-4.495223235091244,\ 0.836385357456495,\ 3.6588378776347503].$$

Against the charged references, this continuation-centered shape has residual norm

$$\left\| E^{\mathrm{cont}}_{\log,\mathrm{centered}} - E^{\mathrm{ref}}_{\log,\mathrm{centered}} \right\| \approx 2.13 \times 10^{-5},$$

so this compare-only branch is a near-exact centered-shape closure up to one common absolute scale. The ppm-scale shape agreement carries a large pull in measurement units: the same $\delta = 2/9$ ansatz misses the measured m_μ/m_e ratio by 9.83 ppm, approximately 440σ of the measurement precision, so the row is excluded as a prediction and stands as an approximate fit. The theorem branch is unpromoted because the public affine absolute anchor is external to this near-exact centered charged-shape branch. The compare-only common scale required for exact absolute masses is

$$g_e^\star = 0.04577885783568762.$$

Equivalently, relative to the stored shared-budget seed

$$g_{\mathrm{ch}}^{\mathrm{shared}} = 0.9231656602589082,$$

the missing affine absolute anchor on the determinant-line route would have to take the target value

$$\Delta_e^{\mathrm{abs},\star} = \log\frac{g_{\mathrm{ch}}^{\mathrm{shared}}}{g_e^\star} = 3.003986333402356.$$

This identifies the charged theorem boundary. The compare-only branch nearly solves the centered charged shape. The derivation does not select the latent candidate $\widehat{C}_e^{\mathrm{cand}}$ as the physical $\widehat{C}_e$, and it lacks one affine-covariant absolute anchor A_{ch} that would turn that centered readback into public charged masses on the theorem branch.

Icosahedral face-incidence carrier. The screen geometry supplies a sharper candidate carrier than a scalar twelve-port moment. The twenty outward-oriented icosahedral faces form A_5/C_3, and the face stabilizer cyclically permutes the three corners. Thus each face has a canonical local regular-C_3 fiber, and an equivariant Hermitian circulant has a face-representative-independent unordered spectrum. This is an exact geometric lemma. It is not a physical generation attachment: the sixty face-corner flags form the regular A_5 torsor, not one canonical three-dimensional matter-family space.

A conditional theorem package declares the diagonal affine repair map

$$T(\kappa, \chi_\rho, \zeta_\delta) = (s_\kappa - q_\kappa \kappa,\ s_\chi + q_\chi \chi_\rho,\ s_\zeta + q_\zeta \zeta_\delta).$$

For that declared map it proves $\|DT\|_\infty = 0.0015356510519\ldots < 1$, so Banach's theorem closes existence, uniqueness, and iterative stability of its fixed point. This is a conditional closure, not a derivation of the repair dynamics. A source-multiplier family $T_{\boldsymbol{\lambda}}$ with the same displayed symmetry, analytic degree, Jacobian, and contraction but different masses exists. It proves scoped selector non-identifiability under those properties, not non-entailment from A1–A3. Exact zeroth-order block balance gives $|b|/a = 1/\sqrt{2}$, whereas the endpoint operator uses $|b|/a = e^{-\chi_\rho}/\sqrt{2}$; no source-derived bare-to-endpoint repair bridge is supplied.

Proposition 10.5 (Finite charged-register realization)**.** *The stipulated charged register-and-path class is nonempty. There is an explicit finite algebraic model with register dimensions*

$$(50, 31, 10, 512, 77, 21, 27, 5),$$

connected transition graphs, normalized tracial states, rank-one event projectors, and the eight declared path weights

$$\left(\frac{1}{50}, -\frac{1}{31}, -\frac{1}{310}, \frac{1}{512}, \frac{1}{77}, \frac{1}{21}, \frac{1}{27}, \frac{1}{135}\right).$$

Each noncentral event in $B(H_r)$ admits an accepted/rejected central record dilation in $B(H_r) \otimes D_2$. The graph family is covariant under the sixty proper icosahedral rotations, and normalized path weights are unchanged under inert ancillary stabilization $X \mapsto X \otimes I_k$.

Proof. For each declared connected graph, the diagonal matrix units and both directed units on every edge generate E_{ij} along graph paths and hence generate the full matrix algebra. The normalized trace of a rank-one event is the reciprocal register dimension, and tensor products multiply the two depth-two weights. For an event P, the pinching map

$$\mathcal{E}_P(X) = PXP + (I - P)X(I - P)$$

is a unital trace-preserving conditional expectation. Its instrument writes $P\rho P$ and $(I-P)\rho(I-P)$ into orthogonal accepted/rejected pointers in the central D_2 factor. Explicit permutation intertwiners preserve graph incidence and ranks for all sixty proper rotations. Finally, $\tau_{dk}(P \otimes I_k) = \tau_d(P)$, and the partial-trace coarse map retracts the inert embedding. These facts construct one model of the declared schema. □

Proposition 10.5 proves formal nonemptiness. It supplies no derivation of the charged source. The eight dimensions, path automaton, coupling degrees, grading, signs, clock, and scalar response are model inputs. The verifier exhausts that automaton, not the physically admissible OPH path space. The construction resolves the abstract event-versus-central-record issue at fixed cutoff. Physical response selection, cofinal screen refinement, and charged-sector attachment are not supplied.

Proposition 10.6 (Conditional charged nature and singularity transport)**.** *Let M_F be the positive face endpoint. Suppose a physical chiral three-family carrier supplies a natural unitary J_L, canonical kinetic metrics, and*

$$\frac{v^2}{2}\widehat{Y}_e\widehat{Y}_e^\dagger = J_L M_F^2 J_L^\dagger.$$

Then its positive left charged response is $J_L M_F J_L^\dagger$. Suppose in addition that the exact renormalized charged kernel is supplied, its singular lines satisfy the declared infrared and regularity conditions, and a finite-register/Dyson map has singularity readout $J_L M_F J_L^\dagger$. Regular invertible field changes preserve the zero set, and a regular Nielsen factorization $\partial_\xi K = A_\xi K + K B_\xi$ makes each simple zero gauge independent.

Proof. The first statement follows from uniqueness of the positive square root. Multiplication of K on the left and right by analytic invertible matrices multiplies its determinant by nonvanishing factors. Jacobi's identity gives

$$\partial_\xi \det K = (\operatorname{tr} A_\xi + \operatorname{tr} B_\xi) \det K,$$

so a simple zero cannot move while the Nielsen factors are regular. These are the standard mixed-fermion and gauge-independence implications used in the pole literature [26, 27]; the stable charged-line infrared qualification follows the infraparticle boundary [28, 29]. □

Proposition 10.6 closes the downstream logic after its physical premises are supplied. It does not supply them. The displayed square identity is the desired operator attachment, and the singularity premise sets the Dyson readout equal to the face response. The explicit kernel $K_0(s) = sI - M_F^2$ has zero self-energy; it is a free algebraic existence witness rather than the interacting charged-lepton one-particle-irreducible kernel. The physical carrier attachment, interacting kernel, and singularity transport are not supplied. The verifier reconstructs the fixed point and response shape but does not check $M_F = g_{\mathrm{end}} S_F$: a coherent change of the ordered spectrum by factors $(2, 1/2, 1)$, together with the matching mass matrix, Yukawa matrix, and pole roots, preserves the determinant and satisfies the tested mass relations. The package is therefore a conditional conditional theorem. It does not identify physical charged matter, construct an interacting pole, or predict a mass.

These results do not yield physical charged-lepton masses. Such a conclusion requires a quotient-visible face-to-charged-family intertwiner, the $\ln 2$ MaxEnt and ensemble-to-Yukawa bridge, a charged connection deriving both the $2/9$ base phase and its correction, a normalized $\mathbb{Z}_6$ determinant character, an OPH dynamical theorem selecting the physical registers, state, exhaustive path category, grading, clock, and response rather than declaring them, cofinal refinement naturality, one target-independent coherent source tuple, and a mass-scheme map. The icosahedral face carrier, face-incidence theorem, and conditional finite-quotient weights supply only the stated finite ingredients.

10.1 Absolute charged-lepton mass intervals from electromagnetic transport

The missing absolute anchor is one real scale. Fixing A_{ch} is equivalent to fixing the single family rescaling $Y_e \mapsto e^\kappa Y_e$, under which every charged mass scales as $m_i \mapsto e^\kappa m_i$ while the centered shape and every mass ratio are invariant. The declared charged antecedents are invariant under that rescaling, which is the content of the common-shift no-go: the gauge representations, the anomaly structure, the complexity vector, the determinant algebra, and the same-label readback take the same value on the whole family $\{e^\kappa Y_e\}$, so no source object built from those antecedents alone selects κ.

The rescaling is not a symmetry of the declared electromagnetic transport. The charged leptons enter the photon vacuum polarization, and the leptonic packet in the fine-structure endpoint map carries

$$P_{\rm lep}(\kappa) = \frac{1}{3\pi}\sum_i \left[2\log\frac{M_Z}{m_i} - \frac{5}{3}\right], \qquad \frac{{\rm d}P_{\rm lep}}{{\rm d}\kappa} = -\frac{2}{\pi}.$$

The declared endpoint pixel map $P = \varphi + \sqrt{\pi}/A_{\rm Th}(P)$ consumes $P_{\rm lep}$, so the emitted endpoint and the pixel move with κ. The common-shift no-go omits this transport from its antecedent list. Source-only, the system is the stiff curve $(P(\kappa), \kappa)$ with $|{\rm d}P/{\rm d}\kappa| \approx 6 \times 10^{-5}$, and the absolute masses are not derived from the source branch.

On the empirical-closure branch the transport identifies κ. Inverting the on-shell decomposition

$$a_0 + g = A^{-1}(0)\,(1 - P_{\rm lep}(\kappa) - \Delta_{\rm had} - \Delta_{\rm top})$$

with the frozen anchor $a_0 = 128.308$, the certified anchor-gap interval $g \in [0.620, 0.651]$, the hadronic payload $\Delta_{\rm had} = 0.027609 \pm 0.000112$ pinned to the published data-driven compilation of Keshavarzi, Nomura and Teubner (Phys. Rev. D 101, 014029), the target-anchored mass ratios, and the measured $A^{-1}(0)$ supplied as a compare-only exclusion inside the interval solve path gives

$$\kappa \in [-0.070,\, +0.061], \qquad \kappa_{\rm central} = -0.004.$$

An independent 100-digit decimal calculation brackets π, pads the transcendental evaluation, and rounds every endpoint outward. The resulting target-anchored diagnostic intervals are

$$\begin{aligned} m_e &= 0.5089\ [0.4765,\, 0.5434]\ \text{MeV}, && \text{measured } 0.5110\ \text{MeV},\\ m_\mu &= 105.22\ [98.5,\, 112.4]\ \text{MeV}, && \text{measured } 105.66\ \text{MeV},\\ m_\tau &= 1.7695\ [1.657,\, 1.890]\ \text{GeV}, && \text{measured } 1.7769\ \text{GeV}, \end{aligned}$$

with central values within about 0.4% of measurement and the physical triple inside every interval. The physical on-shell anchor requirement lies inside the certified gap interval. The solve also inverts exactly at the witness: the anchor-gap value $g = 0.6379$ closes the branch on the measured triple, and its distance $+0.0070$ from the standard on-shell reference deficit 0.631 is the scheme correction required by the bridge. A source-emitted bridge value is a sharp falsification target: a value at 0.6379 within the payload width closes the lepton branch on the witness, and a value outside the certified interval refutes the decomposition. The interval width reduces to the published payload uncertainty and the anchor gap. A source derivation requires the hadron backend and the scheme bridge.

The certified anchor-gap interval is the exact affine image of the hadronic payload interval, $g(X) = A^{-1}(0)(1-X) - A_{\rm lep} - a_0$, so the rectangle solve above carries the one payload uncertainty in two anti-correlated slots. On the payload-coherent reading, where the gap is evaluated at the same payload value as the hadronic term, the payload cancels from the decomposition and the surviving width is the higher-order leptonic remainder and the kernel truncation. The coherent central coincides with the rectangle central, and the certified width contracts by a further factor of 3.8:

$$\begin{aligned} m_e &= 0.5089\ [0.5001,\, 0.5178]\ \text{MeV}, && \text{measured } 0.5110\ \text{MeV},\\ m_\mu &= 105.22\ [103.40,\, 107.06]\ \text{MeV}, && \text{measured } 105.66\ \text{MeV},\\ m_\tau &= 1.7695\ [1.7390,\, 1.8004]\ \text{GeV}, && \text{measured } 1.7769\ \text{GeV}, \end{aligned}$$

with the measured triple inside every coherent interval. The coherent logarithmic half-width is 0.01732, corresponding to multiplicative one-sided widths of -1.72% and $+1.75\%$. The rectangle

interval remains the independent-gap statement for every bridge value inside its certified range; the coherent row is conditional on the declared payload-coherent anchor-gap premise. Both intervals require the same hadron backend and scheme bridge for source closure. The surviving coherent width is a premise floor rather than a budget slack: the higher-order band matches the published per-order structure of the leptonic running with negligible κ-sensitivity across the certified interval, so the certified floor is the scheme-bridge ambiguity itself, and a tighter certified width requires the source bridge rather than a smaller budget.

The determinant prescription is a consistency test on this scale. It reproduces the physical leptonic packet at $|\kappa| \approx 0.006$. Its runtime has no continuous fit parameter, while its architecture and charged ratios retain target ancestry. The construction adds no premise. The interval branch carries absolute masses only on the empirical-closure branch; the source-only charged no-go is in force.

10.2 Conditional response candidate and symmetry-breaking boundary

A conditional package defines the charged response by stipulating a typed, incidence-complete response architecture on an oriented icosahedral flag: eight finite registers of dimensions $50, 31, 10, 512, 77, 21, 27, 5$, eight primitive path classes with normalized rank-one trace weights, a public-block amplitude, an oriented process phase, and a $\mathbb{Z}_6$ determinant character. Conditional on that architecture, the three-channel response map is a Banach contraction with contraction constant 1.5×10^{-3}; its unique fixed point, the regular-C_3 shape functional, and the determinant character emit one dimensionless charged triple on the declared-architecture branch with zero runtime charged-mass input:

$$\frac{(m_e, m_\mu, m_\tau)}{E_\star} = (4.18511 \times 10^{-23},\ 8.65348 \times 10^{-21},\ 1.45532 \times 10^{-19}),$$

with mass ratios $m_\mu/m_e = 206.7683$, $m_\tau/m_e = 3477.3655$, $m_\tau/m_\mu = 16.8177$, and unclosed-clock displays (0.51096, 105.649, 1776.78) MeV. A segregated compare-only calculation places this coherent branch about 84 ppm below the comparison triple; the candidate computation has no dependency on the comparison file.

The register dimensions, path table, signs, amplitude, phase, and determinant exponent are declared model inputs, so the triple is a declared-model coordinate. Those architecture choices are target-informed and stipulated with knowledge of the charged-family target. Their executable evaluation is runtime-reference-free, and the candidate is neither blind nor source-only. The 84 ppm comparison is diagnostic. The Koide subsection derives the finite public-block amplitude $1/\sqrt{2}$ from the connected M_6 response register. The candidate supplies no recoverable attachment between that event and a physical chiral mass channel, so its mass claim is conditional. The arithmetic identity $(N_c + 1)/(2N_cN_g) = 2/9$ supplies no oriented phase; a holonomy reading requires link variables, a loop, and a nonlocal readout map. A product of fourteen normalized rank-one traces is neither a $\mathbb{Z}_6$ character nor a determinant, and the canonical determinant norm carries the kinetic factor $\det \mathcal{M}_L = (v/\sqrt{2})^3\, |\det Y_e|/\sqrt{\det Z_L \det Z_E}$; the declared 6^{-14} weight is a candidate positive path weight.

The source-side problem is a finite symmetry-breaking analysis on the icosahedral carrier. The vertex permutation module decomposes as $\mathbb{R}^{12} \cong W_1 \oplus W_3 \oplus W_3' \oplus W_5$; the traceless quadrupole map $Q(w) = \sum_i w_i(p_i p_i^T - I/3)$ satisfies $Q = QP_5$ and $Q^*Q = \frac{8}{5}P_5$, so it sees exactly W_5, and the two adjacent gaps of its spectrum span the centered family plane. A unique invariant MaxEnt state has zero expectation on every nontrivial irreducible module, and the homogeneous twelve-port branch has the unique uniform minimizer, so charged family shape requires a source-selected W_5

orbit. Because $\mathrm{Sym}^2_0(\mathbf{3}) \cong \mathrm{Sym}^2_0(\mathbf{3}') \cong W_5$, the family attachment is multiplicity-one up to one sign once the chiral family fibers carry a three-dimensional A_5 representation.

Residual symmetry alone does not select the mass ratios. Realize W_5 as the traceless symmetric real 3×3 matrices. Invariance under a threefold or fivefold rotation about n forces

$$A = \alpha(nn^T - I/3),$$

with spectrum $(2\alpha/3, -\alpha/3, -\alpha/3)$. Those fixed points have a double eigenvalue. The twofold fixed locus is

$$A = \begin{pmatrix} u & v & 0 \\ v & w & 0 \\ 0 & 0 & -u-w \end{pmatrix},$$

which is three-dimensional, hence two-dimensional after overall scaling, and admits simple spectrum. It leaves exactly enough continuous freedom to carry the two independent mass ratios without predicting them. The Klein four-group of three perpendicular twofold axes fixes the diagonal traceless forms, one projective parameter with generic simple spectrum, and every subgroup of A_5 other than the trivial group, C_2, and V_4 contains a threefold or fivefold rotation, so no residual stabilizer pins an isolated simple-spectrum orbit. The source-side family problem is therefore selection by a specific screen-derived A_5-invariant potential.

The leading cubic entropy truncation can be classified globally, rather than by the former finite-seed search. A W_5 port vector consists of six values a_i, each repeated on an antipodal pair, with $\sum_i a_i = 0$ and $2\sum_i a_i^2 = 1$. At a constrained extremum of $S_3 = 2\sum_i a_i^3$, the Lagrange equation makes every a_i a root of one quadratic. Hence there are at most two distinct values, and enumerating their multiplicity $m = 1, \ldots, 5$ gives

$$S_{3,m} = \frac{3-m}{\sqrt{3m(6-m)}}.$$

The global maximum is $2/\sqrt{15}$, uniquely up to permutation, with one antipodal pair against the other five. It is the C_5-axis orbit, and its quadrupole spectrum is exactly $(-2/\sqrt{15}, -2/\sqrt{15}, 4/\sqrt{15})$. Thus the leading cubic truncation selects a double eigenvalue and fails the required simple-spectrum shape. Interpreting that degeneracy as two equal physical charged masses requires the separately unproved W_5-to-family attachment (`code/particles/runs/leptons/charged_entropic_branch_no_go.json`).

At finite band amplitude, the global minimizers of one coefficient-free candidate—the quartic truncation of the relative-entropy expansion on W_5—can also be classified exactly. Write $b_i = 1 + ra_i = 12q_i$. Exact stationarity, second-variation, and face factorizations show that every global interior minimizer has two coordinate values and that no omitted face wins. With

$$r_c = \frac{32\sqrt{15} - 20\sqrt{6}}{27} = 2.7757656378\ldots,$$

the strict record domain $b_i > 0$ selects the degenerate multiplicity-one C_5 orbit for $0 < r < r_c$, both multiplicity-one and multiplicity-two orbits at $r = r_c$, and the simple-spectrum multiplicity-two orbit for $r_c < r < \sqrt{24}$. The latter has centered eigenvalues proportional to $(-(\varphi^4 - 1), -1, \varphi^4)$ and sorted-gap ratio φ. For $\sqrt{24} \le r < \sqrt{60}$, the strict problem has no attained minimum; its infimum lies on the boundary.

The quartic truncation also admits a simple-spectrum boundary branch. On the closed probability simplex the global minimizer above $\sqrt{24}$ has four pair weights zero and two weights $b_\pm =$

$3\pm\frac{1}{2}\sqrt{r^2-24}$. Its quadrupole spectrum is proportional to $(-2, 1-d, 1+d)$, where $d=\sqrt{(r^2-15)/5}$, so its upper-to-lower sorted-gap ratio is

$$R(r)=\frac{2d}{3-d}, \qquad r^2=15+\frac{45R^2}{(R+2)^2}.$$

It reaches the compare-only central charged-lepton ratio $R=1.889009\ldots$ at the target-attached value $r=5.061326\ldots$. This is not a prediction: eight of twelve port probabilities then vanish, the Taylor expansion about the uniform record is least controlled there, no source emits r, and the physical quadrupole-to-log-mass attachment is not supplied. The strict full-support quartic minima are conditionally excluded as the observed shape. The closed boundary remains a minimum of that truncated objective (`code/particles/runs/flavor/entropy_w5_shape_certificate.json`).

Full relative entropy at fixed pair norm. The untruncated uniform-reference objective gives a sharper answer for the same six-pair model. Set

$$p_i=\frac{1+ra_i}{6}, \qquad \sum_i p_i=1, \qquad \sum_i p_i^2=\frac{1}{6}+\frac{r^2}{72}, \qquad p_i\geq 0,$$

and minimize $D(p\|\mathbf{1}/6)=\sum_i p_i\log(6p_i)$, with $0\log 0=0$. Pairing the equal antipodal port probabilities leaves this divergence unchanged. For every $0<r\leq\sqrt{60}$, all global minima are the six permutations of

$$p_{\text{high}}=\frac{1+5r/\sqrt{60}}{6}, \qquad p_{\text{low}}=\frac{1-r/\sqrt{60}}{6} \quad \text{(five entries)}.$$

This is the standard maximum-Shannon-entropy branch at fixed index of coincidence [16]; its implication for the icosahedral quadrupole is the application here.

For completeness, compactness gives a minimum. At an interior extremum with $0<r<\sqrt{60}$, the two constraint gradients are independent, and stationarity gives $\log p_i+1=\lambda+2\mu p_i$. Strict concavity of $\log x-2\mu x$ allows at most two positive roots $x<y$. If the larger root occurs twice, the variation exchanging those two coordinates is tangent to both constraints, while its constrained Hessian coefficient is negative:

$$\frac{1}{y}-2\mu=\frac{1}{y}-\frac{\log y-\log x}{y-x}<0.$$

Thus an interior minimum has one high and five low entries, and the two constraints give the displayed values.

No nonvertex boundary point minimizes the full objective. If its positive coordinates include two unequal values, adding a new weight $\epsilon>0$ at a zero coordinate and adjusting those two values by $O(\epsilon)$ preserves both constraints, by the implicit function theorem. The change in the objective is $\epsilon\log\epsilon+O(\epsilon)<0$. If instead the support consists of $k\geq 2$ equal weights $1/k$, replace those weights by $(1-\epsilon)/k$, add the new weight ϵ, and add and subtract δ at two old coordinates, where

$$\delta^2=\frac{\epsilon}{k}-\frac{k+1}{2k}\epsilon^2.$$

The perturbed weights stay positive for sufficiently small ϵ, both constraints hold exactly, and the old-coordinate contribution changes by $O(\epsilon)$; the same negative leading term excludes this boundary point. At $r=\sqrt{60}$, purity is one and only the six vertices are feasible. At $r=0$, the distribution is uniform and does not select a normalized direction a.

For $r > 0$, every selected normalized port direction has pair values $(5, -1, -1, -1, -1, -1)/\sqrt{60}$, up to permutation. With six unit antipodal-axis representatives n_i, the forward twelve-port map is $Q(a) = 2\sum_i a_i(n_i n_i^{\mathsf{T}} - I/3)$; its normalization obeys $n_i^{\mathsf{T}} Q(a) n_i = 8a_i/5$. Its spectrum is

$$\operatorname{spec} Q(a) = \left(-\frac{2}{\sqrt{15}}, -\frac{2}{\sqrt{15}}, \frac{4}{\sqrt{15}}\right).$$

The six branches are the antipodal vertex axes; each has the order-ten dihedral stabilizer D_5 in the rotational icosahedral group (the C_5-axis terminology names its fivefold axis). The perturbation quadrupole $Q(ra) = rQ(a)$ is zero at $r = 0$ and has a repeated eigenvalue at every positive feasible amplitude. Consequently, no amplitude rule $r = r(P)$ rescues three distinct family masses through an attachment that reads only these quadrupole eigenvalues and treats equal eigenvalues equally. This excludes that uniform-reference, fixed-norm full-entropy selector with that readout, including its closed boundary; it does not exclude OPH, other references or constraints, different actions, or different physical readouts. The quartic boundary formula remains valid for its own objective; its two-positive-weight branch does not minimize the full entropy, although both objectives share the pure endpoint. Neither a physical mass attachment nor numerical quark masses follow from this result (`code/particles/runs/flavor/full_entropy_w5_selector.json`).

The finite theorem does not supply an effective action on W_5 with a unique simple-spectrum minimizing orbit. The invariant algebra has one quadratic, two cubic, and two quartic coefficients, so the W_5-restricted Hessian scalar h_5 at the homogeneous state is a necessary local discriminator. A physical charged shape additionally requires the physical A_5 family lift, normed determinant-line descent with kinetic factors, an interacting charged kernel, an operational scale, and a source packet fixed before comparison.

The affine scale has a second admissible closure through electromagnetic transport. With the exact one-loop Ward kernel $I(z) = \int_0^1 x(1-x)\log(1 + zx(1-x))\,\mathrm{d}x$ in closed form, the charged response $\mathcal{W}_Q(\mu; \ell) = \frac{2}{\pi}\sum_i I(q^2 e^{-2(\mu+\ell_i)})$ is a smooth, strictly decreasing bijection of the common log-scale μ onto $(0, \infty)$ at fixed centered shape ℓ, with high-energy slope $-2/\pi$. A source-complete spacelike Ward endpoint pair with a source-complete nonleptonic subtraction therefore fixes a unique μ_{ch}, hence the determinant line $|\det(M_e/E_\star)| = e^{3\mu_{\mathrm{ch}}}$ and the electron ratio $m_e c^2/E_\star$ consumed by the cesium clock packet. The common-shift no-go of the preceding subsections holds on the reduct without mass-dependent electromagnetic transport; a Ward endpoint response and a determinant-line basepoint are the two admissible closures of that orbit. The Ward inversion, its interval enclosure, and its fail-closed source-packet schema are encoded in the Ward determinant-line receipt. A source result requires the endpoint pair, nonleptonic subtraction, higher-order monotonicity certificate, and atomic transport factor.

11 Neutrino Family Derivation

The neutrino branch contains one exact no-go and one failed comparison candidate. The intrinsic isotropic branch is excluded by its spectral cap. The weighted-cycle construction descends from two hand-written family-transport matrices and declared label, orientation, cycle, and exponent choices selected with target information. Its bridge and absolute attachment are compare-only.

For the isotropic matrix $M = aI + \rho C$ with unit-modulus off-diagonal entries, the general Gershgorin estimate applied to $M^\dagger M$ gives

$$\max_{i,j} |\Delta m_{ij}^2| \le 8a\rho + 4\rho^2.$$

The certified bound is $1.52304 \times 10^{-6}\,\mathrm{eV}^2$, about three orders of magnitude below the atmospheric scale.

The weighted-cycle comparison candidate gives

$$\begin{aligned}\theta_{12} &= 34.225904631810025^\circ,\\ \theta_{23} &= 49.72282845058266^\circ,\\ \theta_{13} &= 8.686355527700156^\circ,\end{aligned}$$

$$\begin{aligned}\delta_{\mathrm{PMNS}} &= 305.58061231449796^\circ,\\ J &= -0.02753115613565372,\end{aligned}$$

and the dimensionless hierarchy invariant

$$\frac{\Delta m^2_{21}}{\Delta m^2_{32}} = 0.030721110097966534.$$

The official NuFIT 6.1 normal-ordering profile gives

$$\begin{aligned}\Delta\chi^2_{23,\delta} &= 20.11955 \quad \text{with the tabulated atmospheric likelihood},\\ \Delta\chi^2_{23,\delta} &= 18.43528 \quad \text{without it},\end{aligned}$$

at $(\sin^2\theta_{23}, \delta_{\mathrm{CP}}) = (0.582056, -54.419^\circ)$ [25]. Both values exceed the two-parameter 3σ contour value 11.83. Separate marginal intervals do not test this correlation. The published profiles overlap and are not summed; their maximum is a lower bound on the unavailable full fixed-candidate $\Delta\chi^2$.

The declared shared-basis construction sets

$$M_{\mathrm{shared}} = U^*_{e,\mathrm{left}}\, M_{\mathrm{wc}}\, U^\dagger_{e,\mathrm{left}}, \qquad U_{\nu,\mathrm{shared}} = U_{e,\mathrm{left}}\, U_{\mathrm{wc}},$$

the identity

$$U^\dagger_{e,\mathrm{left}} U_{\nu,\mathrm{shared}} = U_{\mathrm{wc}}.$$

This cancellation is a tautology because $U_{\nu,\mathrm{shared}}$ was defined as $U_{e,\mathrm{left}}U_{\mathrm{wc}}$. It does not independently identify the physical charged-lepton basis or validate the PMNS point. No theorem in the combined theorems places the f-labelled weighted-cycle operator in the charged-lepton mass basis.

No charged-basis construction is supplied, and the three normalized shape singular values of the template differ only at about the 10^{-12} relative level, so its left singular vectors do not define a stable physical family basis. The charged-lepton continuation is support-extension gated and does not supply a replacement labelled basis. At source level, the physical PMNS matrix is therefore unformed.

Taking the stored basis declarations literally gives a sharply different diagnostic. Reordering M_{wc} from $[f3, f1, f2]$ to the independently declared shared basis $[f1, f2, f3]$, computing its Takagi matrix $U^{(f)}_\nu$, and then forming $U^\dagger_{e,\mathrm{left}} U^{(f)}_\nu$ yields

$$\theta_{12} = 38.6812^\circ, \qquad \theta_{23} = 76.2759^\circ, \qquad \theta_{13} = 44.8283^\circ,$$

with $\delta = 324.2140^\circ$ and $J = -0.0233164$. In particular, $\sin^2\theta_{13} = 0.4970$ and $\sin^2\theta_{23} = 0.9437$, both outside the corresponding NuFIT 6.1 profile grids. This is a diagnostic conditional on the stored matrices and label order. It is not a source-closed prediction because the charged basis and family kernel also descend from template-level inputs.

The transported Takagi identity is

$$U^T_{\nu,\mathrm{shared}} M_{\mathrm{shared}} U_{\nu,\mathrm{shared}} = \mathrm{diag}(m_i) \in \mathbb{R}_{>0}.$$

Here U_{wc} denotes the canonical phase-fixed Takagi unitary, not the raw arbitrary-phase eigensystem of $M_{\mathrm{wc}}^{\dagger}M_{\mathrm{wc}}$. Equivalently, $U_{\mathrm{wc}}^{T}M_{\mathrm{wc}}U_{\mathrm{wc}}$ is positive diagonal in its declared basis. Canonical Takagi readout under the declared charged-basis assumption and the electron-row gauge $U_{e1} \in \mathbb{R}_{>0}$ gives the comparison-only Majorana pair

$$\alpha_{21}^{(\mathrm{Maj})} = 153.6185177794357^{\circ}, \qquad \alpha_{31}^{(\mathrm{Maj})} = 257.0032408220805^{\circ}.$$

The declared exponent law uses the positive transport-load segment between $\chi = 1+\epsilon$ and $1+\gamma_{1/2}$: on a one-dimensional affine segment, the balanced selector and the least-distortion selector for any positive translation-invariant quadratic form coincide at the midpoint, so

$$D_{\nu} = \frac{\chi + (1+\gamma_{1/2})}{2}, \qquad p = 1 + \gamma + \frac{\epsilon}{D_{\nu}}.$$

The midpoint fact does not derive the segment endpoints, the exponent law, the cycle topology, or their application to neutrino transport. The formula is target-informed and is not a source-side consequence of OPH. On the declared positive selector segment there is a stronger compare-only two-parameter adapter: under the declared normal-ordering hypothesis, solving τ_{ν} against the representative central ratio and then solving λ_{ν} against Δm_{32}^2 gives

$$(m_1, m_2, m_3) = (0.01745663295,\ 0.01948419960,\ 0.05308139066)\ \mathrm{eV},$$

$$\begin{aligned} \Delta m_{21}^2 &= 7.49 \times 10^{-5}\ \mathrm{eV}^2, \\ \Delta m_{31}^2 &= 2.5129 \times 10^{-3}\ \mathrm{eV}^2, \\ \Delta m_{32}^2 &= 2.438 \times 10^{-3}\ \mathrm{eV}^2. \end{aligned}$$

These exact central numbers are compare-only. The reduced invariant contains target feedback, so the following values are diagnostic coordinates:

$$\begin{aligned} C_{\nu} &= 0.9994295999075177, \\ P_{\nu} &= 6.699825740519345, \\ B_{\nu} &= P_{\nu}C_{\nu} = 6.696004159297337, \end{aligned}$$

and therefore

$$\lambda_{\nu} = \frac{m_{\star,\mathrm{eV}}}{q_{\mathrm{mean}}^{p_{\nu}}} P_{\nu}C_{\nu} = 1.7237014208357415,$$

$$m_i = \lambda_{\nu}\widehat{m}_i, \qquad \Delta m_{ij}^2 = \lambda_{\nu}^2 \widehat{\Delta}_{ij}.$$

The normalized same-label overlap-defect calculation is exact after the template and declared selectors are supplied. The positive-segment adapter, bridge corridor, and bridge-coordinate calculations are diagnostic. In particular, the following quantity is only a diagnostic coordinate:

$$C_{\nu} := \frac{B_{\nu}}{I_{\nu}^{1/2}\,\widehat{\mathrm{ratio}}^{\,1/2}\,\mathrm{sum_defect}^{-1}}$$

It is defined from the declared proxy and normalizer. The construction also factors exactly through $q_e = q_{\mathrm{mean}} qbar_e$, so a $qbar_e$-only collapse law for the bridge factor does not follow from the conditional algebraic surface stated here. The best algebraically complete conditional local object beneath that bridge is the defect-weighted same-label edge family $q_e = \sqrt{g_e d_e}$ together with the induced μ_e family. That family sits below C_{ν} and the induced amplitude B_{ν}; it does not replace them. No source-only PMNS, hierarchy, Majorana-phase, or absolute-mass prediction survives these gates.

Cosmological-neutrino pressure surface. The compare-only absolute attachment displays

$$\sum_i m_{\nu_i} = 0.09001192964464505 \text{ eV}.$$

Under the declared normal-ordering hypothesis, standard relic-neutrino inheritance, and absent an explicitly declared extra OPH relativistic coherence channel, the diagnostic cosmological-neutrino-background branch uses

$$N_{\text{eff}}^{\text{OPH}} = 3.044,$$

with the usual broad, low-amplitude normal-ordering free-streaming imprint. This is a diagnostic cosmological pressure surface for the rejected candidate. The displayed mass sum sits below the Planck+BAO 0.12 eV bound [23] but is exposed to strict DESI DR2 ΛCDM-style bounds near 0.0642 eV under their model assumptions [24]. A cosmological upper bound below the displayed compare-only value would exclude the displayed absolute-mass continuation under a model class that also respects oscillation lower limits and the OPH background assumptions.

On the same-label neutrino-only branch, the centered eta-class is exactly S_3-isotropic, so the same-label data are edge-constant and the solar $1-2$ split cannot open there by itself. Any solar mover must therefore consume data beyond this same-label S_3-isotropic eta-class. Realized flavor-side same-label gap/defect readback is one route; a neutrino-side extension with new non-isotropic source data, spontaneous breaking, another carrier or representation, or additional dynamics is not excluded.

11.1 Intrinsic neutrino eta-chain

The proof-facing input is smaller than the raw same-label matrix payload. It compresses to the same-label scalar certificate

$$(g_e,\ \omega_e)_{e\in\{12,23,31\}}, \qquad \omega_e = \text{same-label overlap}_e^2, \qquad d_e = 1-\omega_e,$$

modulo one common scale. From that certificate one forms

$$q_e = \sqrt{g_e d_e}, \qquad \eta_e = \log q_e - \frac{1}{3}\sum_f \log q_f, \qquad e \in \{12,23,31\},$$

the intrinsic selector depends only on the centered class

$$[\eta_e] \in \mathbb{R}^3/\langle(1,1,1)\rangle.$$

Equivalently one may work with any positive normalized family

$$\mu_e = \frac{e^{\eta_e}}{\frac{1}{3}\sum_f e^{\eta_f}}.$$

Common scaling cancels identically, so the intrinsic selector is determined by the centered eta-class alone.

This factorization result is conditional on the scalar inputs; it is not a source-closure result. The scalar values are numerically complete. Their gap data inherit the template family-transport kernel, and their overlap data inherit a candidate-only line lift. Neither input is source-closed, so the intrinsic spectrum is a conditional diagnostic.

On the isotropic neutrino-only branch one has

$$(\eta_{12},\eta_{23},\eta_{31}) = (0,0,0),$$

which gives the conditional intrinsic singular values

$$(s_0, s_1, s_2) = (2.3986448447627196,\ 2.3986448447627196,\ 2.590074050773907) \times 10^{-12}\ \text{GeV},$$

with ascending squared-mass gaps

$$\begin{aligned} s_1^2 - s_0^2 &= 0, \\ s_2^2 - s_0^2 &= 9.549864971855843 \times 10^{-25}\ \text{GeV}^2. \end{aligned}$$

This is the exact neutrino-only isotropy obstruction behind the solar-splitting boundary.

Theorem 11.1 (Fixed isotropic reference plus same-label scalars suffice for the intrinsic deformation)**.** *Fix the isotropic reference M_0, equivalently its parameters (a, ρ, Ω). Assume the same-label gap and overlap scalars are supplied on all realized arrows. Then the full intrinsic neutrino mass-eigenstate bundle factors through the scalar certificate*

$$(g_e, \omega_e)_{e \in \{12,23,31\}}$$

or equivalently through the centered eta-class $[\eta_e]$. No additional raw matrix payload is needed to form the intrinsic selector, the depressed-cubic spectrum, the ascending gaps, or the intrinsic spectral subspaces. When the singular spectrum is simple and strictly positive, canonical Takagi congruence fixes an ordered column basis up to column signs. If a simple spectrum contains one zero singular value, its zero column retains an arbitrary $U(1)$ phase while the nonzero columns are fixed up to signs. At a degeneracy, including the isotropic $s_0 = s_1$ point, only the degenerate spectral subspace and its projector are fixed. Ascending singular-value sorting does not select the physical normal or inverted mass-eigenstate labels.

Theorem 11.2 (Exact principal-branch selector from the centered eta-class)**.** *Fix the cycle sum*

$$\Omega = \psi_{12} + \psi_{23} + \psi_{31},$$

and a positive weight family μ_e. Define

$$\mu_{\min} = \min_e \mu_e, \qquad F_{\max} = \sum_e \arcsin\left(\frac{\mu_{\min}}{\mu_e}\right).$$

If $|\Omega| < F_{\max}$, then on the principal branch $\psi_e \in (-\pi/2, \pi/2)$, the affine energy

$$A(\psi) = \sum_e \mu_e (1 - \cos\psi_e)$$

has a unique minimizer. It is given by

$$\psi_e^\star = \arcsin(\lambda/\mu_e),$$

where λ is the unique solution of

$$\sum_e \arcsin(\lambda/\mu_e) = \Omega, \qquad \lambda \in (-\min_e \mu_e, \min_e \mu_e).$$

Proof. The Euler–Lagrange equations are $\mu_e \sin\psi_e = \lambda$. On the principal branch the scalar function

$$F(\lambda) = \sum_e \arcsin(\lambda/\mu_e)$$

is strictly increasing because

$$F'(\lambda) = \sum_e \frac{1}{\sqrt{\mu_e^2 - \lambda^2}} > 0.$$

Its range on $(-\mu_{\min}, \mu_{\min})$ is $(-F_{\max}, F_{\max})$, so $F(\lambda) = \Omega$ has a unique solution under the stated condition. Strict convexity follows because the Hessian of A is

$$\nabla^2 A = \operatorname{diag}(\mu_e \cos \psi_e),$$

which is positive definite on the principal branch and is positive definite after restriction to the affine plane $\sum_e \psi_e = \Omega$. □

For the normalized family

$$(\mu_{12}, \mu_{23}, \mu_{31}) = (1.4497003,\ 0.9968526,\ 0.5534471),$$

one has $F_{\max} = 2.5511001$, $|\Omega| = 1.7561443$, and positive domain margin 0.7949558. The numerical selector lies inside the theorem domain.

Let

$$a = m_\star = \frac{v^2}{\mu_u}, \qquad \rho = |(M_0)_{12}|.$$

Then the intrinsic Majorana matrix is

$$M(\eta) = \begin{pmatrix} a & \rho e^{i\psi_{12}} & \rho e^{i\psi_{31}} \\ \rho e^{i\psi_{12}} & a & \rho e^{i\psi_{23}} \\ \rho e^{i\psi_{31}} & \rho e^{i\psi_{23}} & a \end{pmatrix}.$$

Theorem 11.3 (Exact intrinsic spectral cubic)**.** *Let $H = M^\dagger M$. Then*

$$H = dI + T, \qquad d = a^2 + 2\rho^2,$$

where T has zero diagonal and off-diagonals

$$x_{12} = 2a\rho \cos \psi_{12} + \rho^2 e^{i(\psi_{23} - \psi_{31})},$$
$$x_{23} = 2a\rho \cos \psi_{23} + \rho^2 e^{i(\psi_{31} - \psi_{12})},$$
$$x_{13} = 2a\rho \cos \psi_{31} + \rho^2 e^{i(\psi_{23} - \psi_{12})}.$$

Define

$$P = |x_{12}|^2 + |x_{23}|^2 + |x_{13}|^2, \qquad Q = \Re(x_{12} x_{23} \overline{x_{13}}).$$

Then the eigenvalues λ_k of T are exactly the three real roots of

$$\lambda^3 - P\lambda - 2Q = 0,$$

and the intrinsic squared masses are

$$m_k^2 = d + \lambda_k.$$

If $P > 0$, equivalently,

$$\lambda_k = 2\sqrt{\frac{P}{3}} \cos\left(\frac{1}{3} \arccos\left(\frac{3\sqrt{3}\, Q}{P^{3/2}} \right) - \frac{2\pi k}{3} \right).$$

If $P = 0$, all three off-diagonal entries vanish, $T = 0$, and $\lambda_0 = \lambda_1 = \lambda_2 = 0$; the trigonometric expression is not used.

Numerical mass readout. The cubic identity is an exact algebraic statement, not a stability claim for its floating-point evaluation. Near a massless locus, subtracting $d+\lambda_k$ loses small positive squared masses; forming $M^\dagger M$ also squares the condition number. For example, at $\Omega = 0$, $\eta = 0$, $\rho = 1$, and $a = 1 + 10^{-8}$, the two small masses are $a - \rho$, although binary64 evaluation of the cubic can round both to zero. The executable mass readout uses a scaled direct singular-value decomposition of M, checks the positive Takagi congruence, and retains the cubic calculation only as a diagnostic. These are normwise floating-point residual checks, not interval enclosures or relative-error certificates for arbitrarily small masses. A degenerate block requiring additional congruence resolution is explicitly refused. This numerical repair supplies no physical mass-eigenstate labels or source closure.

Corollary 11.4 (Intrinsic eta-chain spectral closure)**.** *Once the centered eta-class is emitted at the flavor boundary, the intrinsic neutrino branch emits*

$$s_0 \le s_1 \le s_2, \qquad s_1^2 - s_0^2,\ s_2^2 - s_0^2,\ s_2^2 - s_1^2,$$

and the corresponding intrinsic spectral subspaces with no PMNS import and no flavor-label leakage. A columnwise Takagi basis up to signs is emitted only on the strictly positive simple-spectrum locus; a simple zero mode retains its phase freedom. The physical normal-ordering assignment is $(\nu_1, \nu_2, \nu_3) = (s_0, s_1, s_2)$*, while the inverted-ordering assignment is* $(\nu_3, \nu_1, \nu_2) = (s_0, s_1, s_2)$*. The source does not choose between them and therefore does not select a physical ordering.*

The matrix used in this corollary is Majorana, so its physical column matrix is the Takagi matrix U_ν defined by

$$U_\nu^T M U_\nu = \mathrm{diag}(s_0, s_1, s_2) \in \mathbb{R}_{\ge 0}.$$

The Takagi matrix, whose columns diagonalize $M^\dagger M$, gives maximum congruence off-diagonal residual $5.4 \times 10^{-28}\,\mathrm{GeV}$ on the anisotropic matrix. On the stored charged matrix, the intrinsic combination gives

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta) = (45.0137^\circ,\ 2.50489^\circ,\ 2.53597^\circ,\ 210.6251^\circ).$$

That diagnostic is strongly incompatible with the NuFIT angle surface, but it is not an OPH prediction because no charged basis is constructed and the basis used is template-derived.

11.2 Perturbative laws around the isotropic point

Write

$$\psi_e = \varphi + \delta_e, \qquad \delta_{12} + \delta_{23} + \delta_{31} = 0.$$

At first order in the centered eta-class,

$$\delta_e = -\tan\varphi\, \eta_e + O(\eta^2).$$

Define

$$\sigma^2 = \frac{2}{3} \sum_e \delta_e^2.$$

Assume $2a\cos\varphi + \rho \neq 0$ and that the collective state is separated from the isotropic doublet. Then the two ascending doublet eigenvalues satisfy

$$s_{0,1}^2 = m_d^2 \mp 2a\rho |\sin\varphi|\, \sigma + O(\delta^2),$$

so

$$s_1^2 - s_0^2 = 4a\rho|\sin\varphi|\,\sigma + O(\delta^2) = 4a\rho\frac{\sin^2\varphi}{|\cos\varphi|}\sqrt{\frac{2}{3}\sum_e \eta_e^2} + O(\eta^2).$$

Under the declared normal-ordering hypothesis, the ascending atmospheric gaps obey

$$\Delta m_{31}^2 = \Delta_{\mathrm{atm,iso}} + \frac{1}{2}\Delta m_{21}^2 + O(\delta^2), \qquad \Delta m_{32}^2 = \Delta_{\mathrm{atm,iso}} - \frac{1}{2}\Delta m_{21}^2 + O(\delta^2),$$

so the first-order invariant atmospheric object is the collective-to-doublet centroid gap

$$\Delta_{\mathrm{cent}} = s_2^2 - \frac{1}{2}(s_0^2 + s_1^2) = \Delta_{\mathrm{atm,iso}} + O(\delta^2).$$

Its quadratic shift is

$$\delta\Delta_{\mathrm{cent}} = -a\rho\,\sigma^2\frac{a(4\cos^2\varphi - 1) + 6\rho\cos\varphi}{2a\cos\varphi + \rho} + O(\delta^3).$$

The projective largest-mass right-singular direction, in the phase gauge that approaches the real democratic vector at the isotropic point, obeys the first-order deformation law

$$u_3 = u + \kappa\begin{pmatrix}\delta_{23}\\ \delta_{31}\\ \delta_{12}\end{pmatrix} + O(\delta^2), \qquad u = \frac{1}{\sqrt{3}}(1,1,1)^T, \qquad \kappa = \frac{\sqrt{3}\,(2a\sin\varphi + 3i\rho)}{9(2a\cos\varphi + \rho)}.$$

This equation fixes only the complex line. If v_3 denotes its normalized right-hand side, the canonical Takagi representative used by the executable construction is

$$u_3^{\mathrm{Tak}} = e^{-\frac{i}{2}\arg(v_3^T M_\nu v_3)}v_3, \qquad (u_3^{\mathrm{Tak}})^T M_\nu u_3^{\mathrm{Tak}} = s_3 > 0,$$

up to the unavoidable column sign. In particular, at the isotropic point the real democratic vector is generally outside positive Takagi gauge. Equivalently,

$$u_3 = u - \tan\varphi\,\kappa\begin{pmatrix}\eta_{23}\\ \eta_{31}\\ \eta_{12}\end{pmatrix} + O(\eta^2).$$

One exact demonstration uses centered eta-class

$$(\eta_{12},\eta_{23},\eta_{31}) = \begin{matrix}(0.13631072512014578,\ -0.18362987264261327,\\ 0.0473191475224675),\end{matrix}$$

and returns the ascending intrinsic singular values

$$(s_0, s_1, s_2) = (2.3929601069646055,\ 2.4048200109774875,\ 2.589606227283229)\times 10^{-12}\ \mathrm{GeV},$$

with

$$s_1^2 - s_0^2 = 5.690121167370743\times 10^{-26}\ \mathrm{GeV}^2,$$
$$s_2^2 - s_0^2 = 9.798023388600230\times 10^{-25}\ \mathrm{GeV}^2.$$

These are exact outputs of the intrinsic eta-chain once the fixed isotropic reference and centered eta-class are supplied. They are not flavor-labeled OPH rows, and ascending sorting does not pick a physical ordering. Under the declared normal-ordering hypothesis $(\nu_1,\nu_2,\nu_3) = (s_0, s_1, s_2)$; under the inverted-ordering hypothesis $(\nu_3,\nu_1,\nu_2) = (s_0, s_1, s_2)$. The source-side label rule is absent.

11.3 Weighted-cycle bridge rigidity and absolute attachment

The weighted-cycle route supplies a frozen PMNS/hierarchy candidate. The reduced invariant C_ν was selected on a compare-only correction surface, with $B_\nu = P_\nu C_\nu$ retained as a diagnostic amplitude parameterization:

$$\begin{aligned} C_\nu &= G^2 Q S^{-1/2} = 0.9994295999075177, \\ P_\nu &= 6.699825740519345, \\ B_\nu &= P_\nu C_\nu = 6.696004159297337. \end{aligned}$$

The blocked absolute attachment displays

$$\begin{aligned} \lambda_\nu &= \frac{m_{\star,\mathrm{eV}}}{q_{\mathrm{mean}}^{p_\nu}} P_\nu C_\nu = 1.7237014208357415, \\ m_i &= \lambda_\nu \hat{m}_i, \\ \Delta m^2_{ij} &= \lambda_\nu^2 \widehat{\Delta m^2_{ij}}. \end{aligned}$$

The basis permutation, holonomy orientation, and CP sign lack source-side selection. Exhaustive enumeration of both stored orientations and all row and mass-column relabelings of the stored candidate matrix finds no normal-ordering-consistent relabeling that passes both NuFIT 6.1 correlated 3σ gates. This excludes a convention-only rescue of that matrix; it neither derives nor exhausts possible source-side physical charged-basis placements. The one-parameter atmospheric-anchor slice and the two-parameter positive-segment adapter are compare-only continuations.

12 Hadrons, QCD, and the Emergence of Ordinary Matter

The hadron derivation is the most operationally demanding part of the particle construction. The source-only artifact is not a fitted hadronic residual and not a bare scalar ρ_{had}. It is a source-derived hadronic spectral backend: a QCD quotient ensemble, source QCD parameter map, Euclidean slab/vacuum-transfer construction, hadronic Hilbert quotient, Ward-normalized electromagnetic current accounting, positive two-current spectral export $d\rho_Q^{(2)}$, higher-point and transition spectral exports, same-scheme remainder Ξ_Q, and systematics accounting. A source-only hadron mass row requires a working OPH hadron backend, such as GLORB/Echosahedron, that can emit these objects with verifiable provenance and production systematics. Small local surrogates cannot provide the required quantum chromodynamics. The two-current spectral measure is the marginal needed for running-α and HVP transport; it is insufficient for HLbL and rare-decay long-distance amplitudes without the four-current and transition sectors. The source-backend boundary has an empirical comparison contract. The empirical comparison uses a separate $e^+e^- \to$ hadrons payload class for the hadronic spectral contribution and stays separate from source-only OPH rows. The source-only derivation nevertheless keeps the mathematical execution bridge, deterministic sampling law, evaluator, and surrogate validation as an execution contract for that backend.

12.1 Seeded $2+1$ family and unquenched measure

Let

$$m_l := \frac{m_u + m_d}{2}, \qquad \rho_l := \frac{m_u + m_d}{2\,\Lambda^{(3)}_{\overline{\mathrm{MS}}}}, \qquad \rho_s := \frac{m_s}{\Lambda^{(3)}_{\overline{\mathrm{MS}}}}.$$

The seeded fixed-physics family is

$$a\Lambda_n = a\Lambda_{\mathrm{seed}}\, 2^{-n}, \qquad a\Lambda_{\mathrm{seed}} = \sqrt{\rho_l \rho_s},$$

$$am_l^{(n)} = a\Lambda_n \rho_l, \qquad am_s^{(n)} = a\Lambda_n \rho_s,$$

$$\beta_n = 6 + \frac{9}{2\pi^2}\log\frac{1}{a\Lambda_n}, \qquad L_n = \left\lceil \frac{\lambda_L^{\mathrm{target}}}{a\Lambda_n} \right\rceil, \qquad T_n = \left\lceil \frac{\lambda_T^{\mathrm{target}}}{a\Lambda_n} \right\rceil.$$

The unquenched ensemble measure is

$$d\mu_n(U) = Z_n^{-1} \exp[-S_g(U;\beta_n)] \det D_l(U; am_l^{(n)})^2 \det D_s(U; am_s^{(n)})\, dU.$$

On this branch, $N_f = 2 + 1$ and QED is off.

12.2 Deterministic configuration and source contract

The emitted execution law is

$$U_{n,c} = K_n^{N_{\mathrm{therm}} + cN_{\mathrm{sep}}}(U_{\mathrm{cold}}; \mathrm{seed}_{n,c}),$$

with stop-time formula

$$t_{\mathrm{stop}}(n,c) = N_{\mathrm{therm}} + cN_{\mathrm{sep}}.$$

The deterministic cfg seed law is

$$\mathrm{seed}_{n,c} = \mathrm{Decode}\big(h_{n,c}\big),$$

with

$$h_{n,c} = \mathrm{SHA256}\Big(\mathrm{Serialize}\big(n, \beta_n, L_n, T_n, am_l^{(n)}, am_s^{(n)}, c\big)\Big).$$

The source set is fixed as

$$S_{n,c} = \left\{ (0,0,0,0), \left(\left\lfloor \frac{L_n}{2} \right\rfloor, \left\lfloor \frac{L_n}{2} \right\rfloor, \left\lfloor \frac{L_n}{2} \right\rfloor, \left\lfloor \frac{T_n}{2} \right\rfloor \right) \right\}.$$

The surrogate execution uses

$$N_{\mathrm{therm}} = 2048, \qquad N_{\mathrm{sep}} = 512.$$

These values are surrogate execution inputs only; they are not claimed as theorem outputs.

The production geometry summary makes the runtime boundary concrete. On the emitted seeded $2 + 1$ family there are three ensembles and six configurations total. If one stores all four links at every site as full double-complex 3×3 matrices, the naive raw gauge storage estimate is

$$576 \text{ bytes/site},$$

which gives total naive raw gauge storage

$$2.80071464105088 \times 10^{14} \text{ bytes}$$

over the production schedule. The normalized backend correlator dump needed by the analysis layer is tiny:

$$195264 \text{ bytes}$$

for the full π_{iso}, $N_{\mathrm{iso,dir}}$, and $N_{\mathrm{iso,ex}}$ payload on the production schedule. The hadron requirement is the backend export bundle that would feed the normalized dump from real production execution.

12.3 Operational reading of the required computation

The lightweight analysis layer is separate from the physical backend. The seeded ensemble family, deterministic configuration and source contract, jackknife evaluator, forward-window selector, and uncertainty schema are explicit. A physical result requires Ward-projected hadronic spectral data from a working OPH hadron backend.

Technically, the required production computation is the standard lattice-QCD stable-channel workflow on the emitted family: run unquenched rational or standard hybrid Monte Carlo for the three seeded ensembles; for each realized configuration and fixed source solve the clover-improved Wilson light/strange systems; construct the zero-momentum π_{iso}, $N_{\mathrm{iso,dir}}$, and $N_{\mathrm{iso,ex}}$ two-point sequences; write the normalized backend bundle; then apply the jackknife and forward-window estimators to obtain $am_{X,\mathrm{ground}}$, R_X, $m_X[\mathrm{GeV}]$, and the published σ_{stat}, δ_{cont}, δ_{vol}, and δ_χ fields. This gives a complete source-only execution contract and a separate empirical display policy.

The engineering burden is asymmetric. Local execution suffices for surrogate validation and downstream readout tests because those calculations use a tiny correlator payload. The physical branch needs backend hardware and execution semantics. This paper treats GLORB/Echosahedron-class OPH hardware, or an equivalent working OPH hadron backend, as the source-only backend requirement. Production hadron masses enter the particle construction only with a backend-emitted Ward-projected spectral measure and its uncertainty budget. This is a source-backend scope boundary.

12.4 Stable-channel correlators, effective masses, and forward windows

The pion stable channel is

$$p_\pi^{(n,c,s)}(t) = \sum_x \Re\,\mathrm{tr}_{c,\mathrm{spin}}\left[\gamma_5 S_l(x;s)\gamma_5 S_l(s;x)\right].$$

The nucleon stable channel is

$$p_{N,\mathrm{dir}}^{(n,c,s)}(t) = \sum_x G_d, \qquad p_{N,\mathrm{ex}}^{(n,c,s)}(t) = \sum_x G_x,$$

$$p_N^{(n,c,s)}(t) = p_{N,\mathrm{dir}}^{(n,c,s)}(t) - p_{N,\mathrm{ex}}^{(n,c,s)}(t).$$

Cfg/source and ensemble averaging are

$$\bar{p}_X^{(n,c)}(t) = \frac{1}{|S_{n,c}|}\sum_{s\in S_{n,c}} p_X^{(n,c,s)}(t),$$

$$C_X^{(n)}(t) = \frac{1}{|C_n|}\sum_{c\in C_n} \bar{p}_X^{(n,c)}(t).$$

The effective-mass laws are

$$am_{\mathrm{eff},\pi}(t) = \log\frac{C_\pi(t)}{C_\pi(t+1)}, \qquad am_{\mathrm{eff},N}(t) = \log\frac{|C_N(t)|}{|C_N(t+1)|}.$$

The forward window is

$$W_n = \{t : 1 \le t+1 < \lfloor T_n/2 \rfloor\}.$$

The evaluator monitors log-convexity,

$$R_{\log\mathrm{conv},\pi}(t) = C_\pi(t)^2 - C_\pi(t-1)C_\pi(t+1),$$

$$R_{\log\,\mathrm{conv},N}(t) = |C_N(t)|^2 - |C_N(t-1)|\,|C_N(t+1)|,$$

tail-drop,

$$D_X(t) = am_{\mathrm{eff},X}(t) - am_{\mathrm{eff},X}(t+1),$$

and mirror suppression,

$$M_X(t) = \exp[-am_{\mathrm{eff},X}(t)(T_n - 2t)].$$

The selected forward window is the longest contiguous run satisfying finite effective masses, nonnegative log-convexity up to tolerance, nonnegative tail-drop up to tolerance, mirror suppression below threshold, and local plateau flatness. The candidate ground-state mass is then the weighted window average

$$am_{X,\mathrm{ground}} = \frac{\sum_{t\in W_X^{\mathrm{sel}}} w_t\, am_{\mathrm{eff},X}(t)}{\sum_{t\in W_X^{\mathrm{sel}}} w_t}, \qquad w_t = \frac{1}{\max(\sigma_t^2, \varepsilon)}.$$

12.5 Statistics, systematics, and dimensional readout

Delete-1 jackknife is performed over the cfg axis after source averaging inside each cfg. With n_{cfg} configurations and integrated autocorrelation time $\tau_{\mathrm{int,cfg}}$, the effective cfg count is

$$n_{\mathrm{eff,cfg}} = \frac{n_{\mathrm{cfg}}}{2\,\tau_{\mathrm{int,cfg}}}.$$

The published statistical error is

$$\sigma_{\mathrm{stat},X} = \mathrm{JKstderr}(am_{X,\mathrm{ground}}).$$

The machine-readable systematics field uses

$$\sigma_{\mathrm{sys},X} = \sqrt{\delta^2_{\mathrm{cont},X} + \delta^2_{\mathrm{vol},X} + \delta^2_{\chi,X}}.$$

Continuum, volume, and chiral proxies are encoded as

$$R_X^{(n)} = \frac{am_X^{(n)}}{a\Lambda_n}, \qquad R_X^{(n)} \approx R_X(0) + c_X(a\Lambda_n)^2,$$

$$\delta^{(n)}_{\mathrm{cont},X} = a\Lambda_n \left| R_X^{(n)} - R_X(0) \right|,$$

$$\delta^{(n)}_{\mathrm{vol},\pi} = am_\pi^{(n)} \frac{e^{-am_\pi^{(n)} L_n}}{\max(am_\pi^{(n)} L_n, 1)},$$

$$\delta^{(n)}_{\mathrm{vol},N} = am_N^{(n)} \frac{e^{-am_\pi^{(n)} L_n}}{\max(am_\pi^{(n)} L_n, 1)},$$

$$\delta^{(n)}_{\chi,\pi} = a\Lambda_n \left| R_\pi^{(n)} - \langle R_\pi \rangle_n \right|,$$

$$Q_N^{(n)} = \frac{R_N^{(n)}}{R_\pi^{(n)}}, \qquad \delta^{(n)}_{\chi,N} = a\Lambda_n \langle R_\pi \rangle_n \left| Q_N^{(n)} - \langle Q_N \rangle_n \right|.$$

These are surrogate publication proxies, not production physical systematics. Given a ground-state candidate,

$$R_X = \frac{am_{X,\mathrm{ground}}}{a\Lambda^{(3)}_{\overline{\mathrm{MS}}}}, \qquad m_X[\mathrm{GeV}] = R_X\, \Lambda^{(3)}_{\overline{\mathrm{MS}}}[\mathrm{GeV}],$$

with

$$\Lambda^{(3)}_{\overline{\mathrm{MS}}} = 0.3344017073 \text{ GeV}.$$

12.6 Surrogate execution bundle and frontier

The surrogate execution evolves a latent state $z \in \mathbb{R}^d$ with Hamiltonian

$$H(z,p) = S(z) + \frac{1}{2}\sum_i p_i^2,$$

where

$$S(z) = \frac{1}{2}\sum_i \omega_i^2 z_i^2 + \lambda_4 \sum_i z_i^4 + \kappa \sum_i (z_{i+1} - z_i)^2 + 2\alpha_l \sum_i \log(\mu_l^2 + z_i^2) + \alpha_s \sum_i \log(\mu_s^2 + \tfrac{1}{2} z_i^2).$$

Leapfrog integration plus Metropolis accept/reject gives the surrogate HMC update

$$(z,p) \mapsto (z',p'), \qquad P_{\mathrm{acc}} = \min(1, e^{-\Delta H}).$$

This kernel is a deterministic executable surrogate honoring the emitted receipt and seed law, with no physical lattice-QCD RHMC/HMC claim.

For validation of the execution bridge, the surrogate locks the ground-state masses to the hadron comparison proxy values

$$m_{\pi,\mathrm{proxy}} = 0.13497682776768472 \text{ GeV},$$
$$m_{N,\mathrm{iso},\mathrm{proxy}} = \frac{m_p + m_n}{2} = 0.93891875434 \text{ GeV}.$$

Thus

$$am^{(n)}_{\pi,\mathrm{proxy}} = \frac{m_{\pi,\mathrm{proxy}}}{\Lambda^{(3)}_{\overline{\mathrm{MS}}}[\mathrm{GeV}]}\, a\Lambda_n,$$
$$am^{(n)}_{N,\mathrm{iso},\mathrm{proxy}} = \frac{m_{N,\mathrm{iso},\mathrm{proxy}}}{\Lambda^{(3)}_{\overline{\mathrm{MS}}}[\mathrm{GeV}]}\, a\Lambda_n.$$

The surrogate correlator is

$$C_X^{\mathrm{sur}}(t) = A_{X,0} e^{-am_X t} + A_{X,1} e^{-am_{X,\mathrm{ex}} t} + A_{X,\mathrm{mir}} e^{-am_X (T_n - t)}$$

up to a multiplicative correlated noise factor. Direct and exchange nucleon pieces are written as

$$C^{\mathrm{sur}}_{N,\mathrm{dir}}(t) = f_{\mathrm{dir}}\, C^{\mathrm{sur}}_N(t), \qquad C^{\mathrm{sur}}_{N,\mathrm{ex}}(t) = (f_{\mathrm{dir}} - 1)\, C^{\mathrm{sur}}_N(t),$$

so that

$$C^{\mathrm{sur}}_N(t) = C^{\mathrm{sur}}_{N,\mathrm{dir}}(t) - C^{\mathrm{sur}}_{N,\mathrm{ex}}(t).$$

On the finest surrogate ensemble, the resulting diagnostic candidates are

$$m^{\mathrm{sur}}_{\pi,\mathrm{iso}} = 0.135039383836 \text{ GeV}, \qquad m^{\mathrm{sur}}_{N,\mathrm{iso}} = 0.938960210578 \text{ GeV},$$

with worst absolute error approximately 6.26×10^{-5} GeV across the surrogate stable-channel family. These numbers validate the emitted execution bridge. They are not promotable production hadron predictions.

So the hadron derivation closes the full

$$\text{receipt} \to \text{execution} \to \text{writeback} \to \text{evaluation} \to \text{budgets} \to \text{forward-window summary}$$

path on executed surrogate data, while the physical closure requires production unquenched RHMC/HMC, real Dirac solves and baryon contractions, production autocorrelation studies, and production continuum / finite-volume / chiral systematics.

13 Observer-Centric Particle Ontology and Measurement

OPH is observer-centric at the level of its basic ontology, so particle physics also requires an account of what a particle *is* and how a measurement turns an excitation into an observed state. The fixed-cutoff record and measurement theorems give that account precise mathematical content.

13.1 Observer patches, overlap consistency, and particle data

Ref. [5] formulates the basic kinematic picture in patch-net language: each observer patch carries a local state, neighboring patches compare a shared interface alphabet, and a global state is physically admissible exactly when neighboring projections agree on the overlap. The algebraic OPH form of the same statement is the first and second axioms: local physical data are carried by patch algebras $\mathcal{A}(P)$, and those local states must agree on shared subalgebras whenever patches overlap.

The particle interpretation starts from patch-local algebras, overlap-visible observables, edge sectors, and transport data rather than a single absolute global particle basis. A particle row in the reported derivation is a readout claim about a stable or candidate-stable excitation structure visible to a family of observer patches and consistent on their overlaps.

For this record-algebra discussion, use the support-local algebra-state-record reduct

$$O_{\mathrm{red}} = (P, \mathcal{A}(P), \rho, R),$$

where P is the support patch, $\mathcal{A}(P)$ its local algebra, ρ the local state, and R the record algebra. The full operational observer additionally carries overlap interface algebras and restriction maps, allowed update and repair instruments, and checkpoint data used for continuation. For the particle derivation, R belongs to the quantum observer surface itself. On the exact fixed-cutoff measurement surface it is generated by central record projectors, and practical readout may use approximately commuting projectors that are close to that central reference algebra. That is what makes the observer-facing record surface shareable across overlapping observer descriptions without violating the usual no-cloning constraints on generic quantum states.

13.2 Record algebras and definite outcomes

The integrated measurement appendices and Ref. [6] make the measurement interface precise at fixed cutoff. On the declared operational surface, the completed write/verify slice carries a finite commutative central record algebra. Practical readout may instead use projectors Q_a on the same declared slots with commuting central reference projectors $\widehat{Q}_a$, where

$$\delta_{\mathrm{rec}} := \max_a \|Q_a - \widehat{Q}_a\|.$$

Then

$$\|[Q_a, Q_b]\| \le 4\,\delta_{\mathrm{rec}},$$

and, whenever $\|\widetilde{\rho} - \rho\|_1 \le \varepsilon$,

$$\left|\mathrm{Tr}(\widetilde{\rho} Q_a) - \mathrm{Tr}(\rho \widehat{Q}_a)\right| \le \varepsilon + \delta_{\mathrm{rec}}.$$

On the explicit fixed-cutoff screen architecture, the exact record algebra after a completed write/verify cycle is

$$\mathcal{Z}_{\mathrm{rec}}(t) = \mathrm{Alg}\Big(\{P_{m_I^{(J)}(t)}\}_{I\neq J}, \{P_{r_I^{\mathrm{bulk}}(t)}\}_I, \{\Pi_\alpha^{(IJ)}(t)\}_{I<J,\alpha}\Big).$$

Ref. [6] proves that, on the declared operational measurement surface, $\mathcal{Z}_{\mathrm{rec}}(t)$ is commutative and central for the readout instrument being used.

This fixed-cutoff centrality result is what supports definite outcomes without adding a second ontology. The integrated measurement construction phrases the same idea through edge-center decomposition: on a fixed collar with exact Markov structure aligned with the edge split (the Standard Model gauge paper's Markov-split alignment hypothesis), the state splits into superselection blocks,

$$\rho_{ABC} = \bigoplus_j q_j \, \rho^{(j)}_{Ab_L^{(j)}} \otimes \rho^{(j)}_{b_R^{(j)}C},$$

and the label j is classical center data. The supplement then goes one step further: in the physical algebra there are no interference observables between different sector blocks. So once a particle detection event has been recorded in the observer-accessible center/record algebra, the accessible physics is organized as a classical mixture over those blocks instead of as a superposed measurement record.

The finite matrix theorem states the exact boundary of that claim. For a supplied projective partition $\{P_j\}$, let $\mathcal{E}(X) = \sum_j P_j X P_j$. Two matrices X and Y give equal trace statistics against every observable commuting with all P_j if and only if $\mathcal{E}(X) = \mathcal{E}(Y)$. In particular, $\mathcal{E}(P_i X P_j) = 0$ for $i \neq j$, so each cross-sector corner is invisible to the complete sector-preserving commutant. The result is relative to the supplied partition and readout. It does not construct the physical edge center used by a detector.

Accordingly, a measured particle family or excitation channel should be thought of as an observer-accessible sector or record value carried by the central measurement surface, not as a mysterious absolute collapse of the full universe-state into a preferred basis chosen by hand.

13.3 Born probabilities and post-measurement states

The fixed-cutoff measurement package is also explicit about probabilities and updates under its supplied density-state valuation and declared Lüders instrument. For every event E in the sigma algebra generated by $\mathcal{Z}_{\mathrm{rec}}(t)$, the measurement probability is

$$\mathbb{P}_t(E) = \mathrm{Tr}(\rho_t P_E),$$

where $P_E \in \mathcal{Z}_{\mathrm{rec}}(t)$ is the projector for that event. Conditioning on the event then produces the post-measurement state

$$\rho_t\,|_E = \frac{P_E \rho_t P_E}{\mathrm{Tr}(\rho_t P_E)}.$$

In the supplement, the same conditional statement appears as the declared Lüders update on the record algebra. In the flagship synthesis [3], this fixed-cutoff Born/Lüders package and its Bell/CHSH extension are imported as theorem-bearing under their displayed representation and instrument premises, not non-theorem commentary.

On the declared binary-icosahedral spinor branch, an exact finite candidate has $|S_{\mathrm{CHSH}}| = 1 + 3/\sqrt{5}$ for an incidence-defined setting family. The carrier admits (960) maximizing quadruples, while the declared family contains (120) of them. No source mechanism selects those settings or supplies a completed two-wing record instrument, so the candidate is not a physical Bell prediction.

Within this declared instrument model, measurements assign particles conditioned observer-accessible states on the central record algebra. A detector click, a stable overlap-sector readout, or a pointer value does more than announce a pre-existing classical label; it defines the conditioned state for subsequent observer-accessible physics. After a nonzero-weight event is observed and the

declared Lüders instrument is applied, an immediate reread with the same projector has probability 1. An intervening repair preserves that conclusion only when its output state is explicitly supported in the same event projector. So the conditional measurement interface is well defined and operationally auditable.

13.4 Particle identity as transport-stable structure

Once one asks "which event happened?" and then "which particle family is this?", the flavor derivation becomes essential. The active flavor derivation does not start with the names electron, muon, tau lepton, or up quark built into the fundamental observable. It starts with transport kernels, generation-bundle data, same-label eigenline transport, overlap-edge cocycles, and a persistent flavor observable carrying intrinsic labels $f1, f2, f3$. This is an important discipline condition in the paper. At the deepest flavor surface used here, family identity is transport-stable intrinsic structure first and named experimental family assignment second.

The shared Yukawa/excitation dictionary carries substantial weight even though it is not itself the leading missing object. The flavor construction gives an invariant base: projectors, spectral gaps, pair suppressions, cycle phases, and common sector-response objects. No map from that intrinsic base to the named low-energy fermion families is derived. The quark calculation supports direct numerical comparison, while the charged-lepton theorem emits no physical mass. The neutrino weighted-cycle construction is a target-informed template candidate without a physical flavor-labelled attachment.

Accordingly, particle identity is a refinement-stable observer-accessible transport pattern whose named interpretation is inherited from the available dictionary and declared comparison domain. Without that dictionary, no named physical identity follows.

14 Affine Event Records and Cross-Boundary Token Stitching

The sector-identity analysis concerns which transport-stable excitation type is being read. A different question is whether two located record tokens, seen near a chart or partition boundary, are the same continuing observer-visible token. This section closes that second question at the finite-certificate level. It does not add a particle species, mass, charge, or scattering theorem. It says when a bounded patch system is allowed to stitch affine event records into nonbranching record-worldlines, and when it must report ambiguity.

14.1 Event chart, frame, and clock atlas

The compact cap-normal theorem supplies the Lorentz-natural frame hyperboloid. It identifies sky directions with $q(\Omega) = (1, \Omega)$, represents oriented round caps by $n_C = (\cot\alpha, \csc\alpha\, \mathbf{c})$, and maps each cap to a geodesic half-space in H^3. The source-causal continuum certificate separately supplies affine physical event charts only after exact two-way cone/order agreement, count–volume, manifoldlikeness, topology, and refinement convergence. This section consumes event supports, observer clocks, chart transitions, gauge transport, and worldline stitching. Frame estimation, event location, overlap descent, and worldline stitching are separate certificates.

Fix a curvature radius $R_H > 0$ and write

$$H^3_{R_H} = \{X \in \mathbb{R}^{1,3} : \langle X, X\rangle_L = -R_H^2,\ X^0 > 0\}, \qquad \langle X, Y\rangle_L = -X^0Y^0 + X^1Y^1 + X^2Y^2 + X^3Y^3.$$

The hyperbolic distance used for conditioned frame comparison is

$$d_H(X,Y) = R_H \operatorname{arcosh}\left(-\frac{\langle X,Y\rangle_L}{R_H^2}\right).$$

An event-record atlas is a tuple

$$\mathcal{A}_H = (R_H, \{U_i, \chi_i, u_i, J_i, \vartheta_i\}, \{G_{ji}, F_{ji}\}, E, \nabla, \mathsf{Prov}_{R_H}, \mathsf{Prov}_u).$$

Here χ_i are affine event charts, $u_i \in H^3_{R_H}$ is the selected local observer-frame anchor when such an anchor is claimed, J_i are event-chart domains for record support, and $\vartheta_i : J_i \to T$ are observer-clock maps into a common comparison-time line T. Prov_{R_H} records whether R_H is a unit convention, an imported physical scale, or an independently derived OPH scale. The unit-chart convention alone supplies no independent OPH scale. Prov_u records which observer, tetrad, or clock object selected the anchors u_i. The Lorentz part $G_{ji} \in \mathrm{SO}^+(1,3)$ transports frames and tetrads; the full Poincaré transition transports affine event coordinates. The map F_{ji} is the sector or gauge transport functor on the record payload. These two holonomies are deliberately separate: atlas holonomy tests geometry, and gauge holonomy tests transported sector data. When two charts describe the same transported observer frame, the atlas additionally requires

$$G_{ji}u_i = u_j.$$

Theorem 14.1 (H3 geometric naturality)**.** *For every $A \in \mathrm{SO}^+(1,3)$, the map $X \mapsto AX$ preserves $H^3_{R_H}$, d_H, geodesics, exponential and logarithm maps, parallel transport, Hausdorff distances between compact frame supports, and Fréchet means whenever the mean is unique.*

Proof. The defining equation of $H^3_{R_H}$ and the formula for d_H depend only on the Lorentz inner product and the future sheet. Elements of $\mathrm{SO}^+(1,3)$ preserve both. The Levi-Civita connection, geodesic exponential, logarithm, and parallel transport are functorial for isometries, so the listed constructions commute with A. Hausdorff distance and unique Fréchet means are metric constructions, hence are preserved by any isometry. □

14.2 Cap-response frame balls and affine event supports

The Standard Model gauge paper supplies the conditional inverse theorem that turns calibrated modular cap responses into conditioned observer-frame supports. For every descended record token i and clock slice t, the frame layer consumes a certificate

$$R_i(C,t,O) = \omega_{i,O}\left(\sigma_t^{C,O}(M_{C,0,O})\right), \qquad y_{i,j} = F_j(X_i(t)) + e_{i,j},$$

on a compact frame domain $\Omega \subset H^3_{R_H}$, with quantitative observability

$$\alpha \bar{d}(X,Y) \le |F(X) - F(Y)|_W \le L\bar{d}(X,Y), \qquad |e_i|_W \le \sigma_i(t).$$

Given an ε-net and residual tolerance $\tau_i(t)$, the imported theorem gives the conservative enclosure

$$S_i(t) \subseteq B_H(\widehat{X}_i(t), r_i(t)), \qquad r_i(t) = R_H\left[\frac{L}{\alpha}\varepsilon + \frac{2}{\alpha}\sigma_i(t) + \frac{1}{\alpha}\tau_i(t)\right].$$

Not every point in the displayed ball is asserted to fit the responses. A frame estimate $\widehat{X}_i(t)$ is a unique finite output only when $\Delta_{\mathrm{frame},i}(t) > 0$. If this gap fails, the output is a frame ambiguity

set or support ball. This ball cannot serve as an event position. Event stitching consumes an affine event support $A_i(t)$ produced by the physical source-causal continuum and its ancestry/placement receipts. Distinct event supports at a shared clock slice require a certified separation in the affine event metric. A conditioned-frame separation may also be recorded:

$$d_H(\widehat{X}_i, \widehat{X}_k) > r_i + r_k + m_{\mathrm{sep}}$$

for a declared positive frame margin m_{sep}. It does not replace event separation.

14.3 Proto-record extraction and descent

A raw local proto-record is a connected component of an invariant detector scalar above a frozen threshold, together with its certified affine event support, conditioned frame ball, local clock interval, sector payload, and uncertainty collar. The detector scalar is built from record-algebra data, not from implementation object IDs. Extraction is chart-natural when transporting the raw field by G_{ji} and F_{ji} carries extracted components in chart i to the extracted components in chart j, up to the declared support collar.

Before temporal stitching, overlapping charts are descended. The overlap-descent relation joins two proto-records only when their transported affine event supports lie in the same connected component on a real chart overlap and the same-component margin dominates the support collar. If distinct component separation does not dominate the collar, the descent emits an ambiguity certificate instead of choosing a label. On component-connected covers, this descent produces canonical global descended records by the usual sheaf gluing argument: cocycle-compatible local components with positive separation margins have a unique global component, and the construction is natural under atlas restriction.

14.4 Boundary-crossing germs and transported residuals

A cross-boundary candidate edge is admissible only when the two descended records meet a real patch interface in adjacent observer-clock intervals and share a common comparison chart. The interface contact must be oriented and transverse: the signed-distance interval brackets the interface, and the normal velocity has a positive lower bound. Tangencies, grazing contacts, and file or shard-name boundaries without a physical interface are rejected. If the local slab contains an interaction marker, the record is routed to the interaction solver instead of stitched by the free-propagation rule.

For an admissible pair, the certificate transports the left record into the right comparison chart and evaluates frozen residuals

$$r_t, \quad r_\partial, \quad r_x, \quad r_v, \quad r_{\mathrm{int}},$$

for clock adjacency, boundary contact, affine event position, transported velocity or finite difference, and interaction absence. The cost function is declared before seeing the proposed matching. Record IDs, global IDs, shard-local IDs, and stitch keys are forbidden in admissibility, costs, and tie-breaking. All affine event-position and velocity residuals are evaluated as certified interval or uncertainty-box distances. A point-estimate residual may be printed as a diagnostic; it cannot replace the event-support radius in an admissibility or matching certificate. Conditioned H^3 frame residuals are a separate tetrad-consistency check.

Lemma 14.2 (Candidate-edge naturality)**.** *The admissible candidate-edge set and its residual intervals are invariant under common* $\mathrm{SO}^+(1,3)$ *chart changes and under gauge-bundle trivialization changes.*

Proof. Every event term is computed from the affine overlap cocycle, oriented interface data, and the common clock line. Frame terms use d_H and Theorem 14.1. Sector and gauge residuals compare transported payloads after applying the declared connector, so changing a local trivialization conjugates both sides of the comparison. The residual intervals and the resulting admissibility predicate are therefore unchanged. □

14.5 Assignment gap, nonbranching paths, and ambiguity

Let L and R be the descended record sets on two adjacent time slabs. A stitch matching is a partial one-to-one assignment between L and R, augmented by declared appearance and disappearance interval costs. Its certified gap is

$$\Delta_{\mathrm{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*),$$

where M_* is the proposed matching, $\overline{C}(M_*)$ is its upper cost bound, and $\underline{C}(M)$ is the lower cost bound for every competing matching. The event-location gap Δ_{event} and stitch-assignment gap Δ_{cert} are distinct receipts: the first certifies a unique event support inside one clock slice, while the second certifies a unique temporal assignment across slices.

Theorem 14.3 (Stable ID-independent stitch assignment)**.** *If the candidate-edge set is natural, the costs are frozen and ID-independent, the proposed matching is one-to-one, and $\Delta_{\mathrm{cert}} > 0$, then M_* is the unique certified minimum-cost stitch. The certified matching is natural under chart changes, gauge trivializations, and relabeling of implementation record IDs.*

Proof. The strict gap says every competing matching has lower cost bound above the proposed matching's upper cost bound, so no competitor can tie or beat it under any realization inside the certified intervals. Lemma 14.2 makes the candidate graph and cost intervals invariant under chart and gauge presentation changes. Because IDs are excluded from admissibility and costs, relabeling them changes only bookkeeping fields, not the optimum. □

Theorem 14.4 (Unavoidable ambiguity)**.** *If a nontrivial permutation of the visible descended records preserves the atlas, clocks, interfaces, sector/gauge payloads, candidate graph, and cost intervals, and its induced action fixes no admissible stitch, then no natural ID-independent finite procedure can certify a unique stitch across that slab. Without the fixed-point-free clause, a unique natural stitch need only be invariant under the symmetry and ambiguity does not follow.*

Proof. A natural procedure must commute with the permutation. A unique certified output would therefore be a fixed admissible stitch, contrary to the fixed-point-free hypothesis. Choosing one element of a nontrivial orbit would introduce a label or presentation dependence. Thus no natural single-valued admissible-stitch output depending only on the listed preserved visible inputs exists. The declared fail-closed protocol records this situation as AMBIGUOUS; an invariant orbit/set-valued output or rejection is a different protocol and is not excluded. □

The certified stitch graph is then the graph obtained by chaining the unique matchings across adjacent slabs. Since each matching is partial one-to-one, every connected component is a nonbranching path, ray, or isolated record. These are the affine event-record worldlines of this section.

14.6 Refinement naturality and dynamical lift

For a refinement $s \to r$, require that record contraction induces a map $Q_{sr} : \mathcal{M}_s \to \mathcal{M}_r$ from every admissible fine matching to an admissible coarse matching. The coarse optimum M_r^* must have at least one admissible fine lift $\widetilde{M}_s$, with $Q_{sr}(\widetilde{M}_s) = M_r^*$. Suppose the certified cost bounds imply

$$|C_s(M) - C_r(Q_{sr}M)| \le \eta_{sr} \qquad (M \in \mathcal{M}_s)$$

for all admitted realizations, and the coarse certified gap obeys $\Delta_r > 2\eta_{sr}$. If a fine optimum M_s^* contracted to a competitor, then

$$C_s(M_s^*) \ge C_r(M_r^*) + \Delta_r - \eta_{sr} > C_r(M_r^*) + \eta_{sr} \ge C_s(\widetilde{M}_s),$$

a contradiction. Thus every fine optimum contracts to M_r^*. The lift premise is essential: coarse costs $C_r(A) = 0, C_r(B) = 1$, a single fine matching mapping to B with cost 1, and $\eta_{sr} = 0$ satisfy the gap and distortion inequalities but cannot recover A. This is the refinement-naturality gate: feasibility, the lift, and the uniform cost bound must all be certified before the coarse/fine agreement follows from the margin.

The stitch certificate lives on the conditional effective Lorentzian manifold. A geodesic or free-particle interpretation requires a separate dynamical receipt with a timelike speed margin, connection compatibility, and an action or propagation law. The stitch theorem proves token continuation in event charts; it does not prove geodesic motion.

Theorem 14.5 (Certified affine event-record stitching)**.** *Given a source-selected refinement family satisfying the spacetime paper's source-causal continuum certificate, together with a valid affine event atlas compatible with its effective limit, a common operational clock atlas, affine event supports for every descended record token and clock slice, and separately typed conditioned-frame certificates where observer frames are used, chart-natural proto-record extraction, overlap descent with positive event-support separation margins, real transverse interface-crossing germs, sector and gauge transport continuity, a frozen ID-independent one-to-one assignment with $\Delta_{\mathrm{cert}} > 0$, no interaction marker in the free-propagation slab, and a passing coarse/fine contraction gate, the emitted stitch graph is a unique natural nonbranching affine event-record graph. Under the declared fail-closed protocol, failure to certify a positive event-location gap, separation margin, or assignment gap returns* AMBIGUOUS *and certifies no unique stitch by this gap contract. That protocol output is not a proof of mathematical nonuniqueness; unavoidable ambiguity additionally requires the fixed-point-free symmetry hypothesis of Theorem 14.4. The protocol returns* INTERACTION REQUIRED *when an interaction marker is present, and rejects the stitch when its atlas, interface, event-location, or transport gates fail.*

Proof. The atlas, event-location, and extraction gates make local proto-records presentation-independent. The positive Δ_{event} gate licenses event-position outputs; otherwise the procedure carries event boxes or ambiguity sets forward. Descent turns overlap-compatible local components into descended records before any temporal decision is made. The interface and transport gates define a natural candidate-edge graph by Lemma 14.2. The strict assignment gap gives a unique ID-independent matching by Theorem 14.3. Chaining partial one-to-one matchings gives nonbranching components, and the refinement gate makes the result stable under the declared coarse/fine contraction. The other labels are the declared protocol's fail-closed responses to failed or uncertified gates; they need not be disjoint or logically exhaustive. Theorem 14.4 upgrades the ambiguity label to a mathematical obstruction only under its fixed-point-free action on admissible stitches. □

14.7 Scope boundary of the measurement chapter

The fixed-cutoff consensus theorem gives the central record algebra. Given the separately declared algebra-state, Lüders instrument, and two-wing representation, the measurement theorem gives Born probabilities, the selected Lüders conditioning, immediate repeated-read stability (and persistence under an explicit support-preserving intervening operation), and the stated Bell/CHSH bound. It does not prove global observer continuation or strange-loop closure. Those metaphysical proposals do not enter the technical argument.

15 Discussion: Matter, Antimatter, Supersymmetry, and Physical Boundaries

The OPH theorems distinguish structural results, conditional quantitative branches, and target-conditioned comparisons. Matter versus antimatter, supersymmetry, and several adjacent questions lie on that boundary. The quark calculation is an obstruction and target-conditioned surface rather than a set of source-only numeric predictions. Its S_3 table is a target-anchored diagnostic.

15.1 Matter and antimatter

The Standard Model structural branch fixes the chiral gauge architecture within which matter and antimatter are defined. Once a chiral fermion representation is present, its conjugate representation gives the corresponding antiparticle content. In that limited sense, OPH explains why matter and antimatter both exist: they are paired by the realized gauge and chiral structure of the low-energy branch.

The cosmological asymmetry requires a separate finite source object. At regulator r, write

$$\mathfrak{B}_r = (Q_r, C_r, E_r, \omega_{R,r}, L_{r,T}, p_{r,i}, \tau_r, T_r, \rho_{R,r}).$$

Here Q_r is the physical quotient, C_r is its CP involution, $\omega_{R,r}$ is a CP-odd integral winding cocycle, $L_{r,T}$ is the physical quotient repair generator, $p_{r,i}$ is the initial law, τ_r and $T_r(\tau)$ are the physical clock and temperature map, and $\rho_{R,r}$ assigns record charges r_ψ to chiral matter.

For a record phase acting as $\psi \mapsto e^{ir_\psi \Theta_R}\psi$, the finite-quotient anomaly-and-current theorem gives

$$\boxed{k_R = \sum_{\psi\,\mathrm{LH}} r_\psi \, 2T_2(R_\psi)}$$

and

$$\boxed{\dot{\Theta}_R(T) = \theta_0 \sum_{q,q' \in Q_r} p_r(q,T) L_{r,T}(q,q') \omega_{R,r}(q,q')}, \qquad \theta_0 = \frac{2\pi}{m_R}.$$

The first equation is the mixed electroweak anomaly index. The second is an oriented probability current in proper time. Quotient settlement supplies no rate matrix, and a repair iteration count supplies no physical clock.

The theorem also fixes the sign boundary. If the initial law and generator are CP symmetric,

$$p_{r,i}(C_r q) = p_{r,i}(q), \qquad L_{r,T}(C_r q, C_r q') = L_{r,T}(q,q'),$$

then every transition cancels its CP image and

$$\boxed{\dot{\Theta}_R(T) = Y_B = 0.}$$

An oriented register and a normal-form map therefore select no cosmological matter sign. A nonzero sign requires a quotient-intrinsic CP-odd boundary condition, action term, or transition affinity.

The conditional $\mathbb{Z}_6$ determinant/deck winding supplies an integral phase coordinate with inversion under orientation reversal. Its natural Standard Model attachment is the central hypercharge direction. The associated coefficient vanishes generation by generation:

$$\boxed{k_R^{YWW} = N_g(3Y_Q + Y_L) = 3\left(3 \cdot \frac{1}{6} - \frac{1}{2}\right) = 0.}$$

This is the mixed $SU(2)_L^2 U(1)_Y$ anomaly-cancellation condition. The gauge/deck phase cannot directly supply the required $\Theta_R W\widetilde{W}$ coupling. Quotient periodicity and the twelve electroweak zero modes do not select $k_R = 1$ or any other nonzero record attachment.

A distinct global record attachment can carry a nonzero index. On the three-generation branch,

$$k_R(Y) = k_R(B - L) = 0, \quad k_R(B) = k_R(L) = 3, \quad k_R(B + L) = 6.$$

A quotient-visible gauge-singlet $B+L$ record phase is therefore a viable conditional attachment. No derivation from OPH repair data is supplied. On that branch, the freeze-out transport functional requires

$$\left\langle \frac{\dot{\Theta}_R}{T} \right\rangle_{\text{fo}} = (4.463 \pm 0.028) \times 10^{-9}.$$

This is a source-generator target. It may not be consumed to select $L_{r,T}$, the attachment, or the CP-odd affinity.

The exact source theorem and gauge/deck no-go are proved at finite quotient. The physical baryon abundance is not derived. Its required source packet consists of a distinct anomalous record phase, source-only transition rates, a proper-time and temperature calibration, a CP-odd initial or boundary law, sign-domain coherence, and the resulting full source history. Transport, washout, and freeze-out must then be evaluated on that emitted history.

15.2 Why the declared branch does not require supersymmetry

The heat-kernel appendix below contains the gauge-coupling calibration. The declared branch needs no supersymmetric partner spectrum to produce its unification-like running benchmark. This does not rule out supersymmetry in a different physical completion.

At one loop, the appendix records the familiar fact that Standard Model beta-function coefficients do not produce successful naive unification, whereas minimal supersymmetric Standard Model coefficients do. The conditional calibration hypothesis is that edge-sector heat-kernel weights shift the effective beta-function coefficients in that direction through geometric or entropic sector multiplicity instead of through an actual low-energy superpartner spectrum. In that reading, "unification-like behavior" and "a supersymmetric particle zoo" come apart.

This calibration is not a derived unification theorem. The appendix records one calibration branch in which Peter–Weyl multiplicities, the printed one-loop running frame, the threshold conventions, and an additional fermionic-grading restriction together produce an supersymmetric benchmark shift. It does not prove that those effective loop multiplicities, statistics restrictions, and decoupling conventions are uniquely forced by OPH edge sectors alone.

For this paper, the correct conclusion is therefore scope-limited and modest:

1. the conditional derivation uses the declared Standard Model branch without having to introduce a supersymmetric partner for every known field;

2. the heat-kernel appendix contains a conditional calibration in which unification-style running features could arise from edge-sector structure instead of supersymmetric particle content;

3. the beta-shift calculation uses explicit calibration assumptions rather than a closed edge-sector theorem;

4. none of this is a universal theorem that supersymmetry is impossible.

So if the paper asks "why is there no supersymmetry in the derived spectrum?", the best technical answer is: because the declared branch stops at the Standard Model content, and the existing unification discussion is trying to reproduce the relevant running behavior through a separate heat-kernel calibration, without extending the particle content to a full supersymmetric multiplet structure.

15.3 Three Generations And Flavor Closure

The target-free structural matter packet leaves the three values in $3 \leq N_g \leq 5$ indistinguishable. On the strengthened screen interface, the operational Laplacian cost selects the canonical rank-three band uniquely among the single complete faithful candidates. The declared unitary simulator recovers the same band at its lowest positive generator frequency. This supplies an exact finite family candidate. The fifteen-state charge table has exact anomaly cancellation, diagonal $\mathbb{Z}_6$ action, and a nondegenerate chirality grading. Tensoring the rank-three response band with that table gives a conditional complex rank-45 candidate. On a separate finite local domain, the receipt checks the declared scalar signed operator tensored with I_{45}; its zero kernel and exact positive dimensionless gap are inherited with multiplicity 45. The source does not select this matter action. The twelve-port Spin packet and the 8,662-node operator domain have no certified source, domain, or transport bridge. Matter-pole identification, the continuum Spin/locality limit, physical seam selection, the missing persistence leg, and laboratory current identification have no receipts [7]. The full flavor dictionary, excitation map, and CKM closure are not derived. The charged source landing from P to physical charged data is an underived bridge, a corpus gap certificate whose named live route is the trace-lift attachment $3\mu(r) = S_M(r)$, and the neutrino branch has no source-closed flavor prediction. Its weighted-cycle candidate fails the NuFIT 6.1 correlated profile [25]. The flavor and family chapters therefore spend substantial space on constructive object boundaries as well as final numbers.

15.4 Ordinary matter and the hadron backend boundary

A second distinction belongs in discussion form instead of in the results sections: the difference between "elementary rows exist" and "ordinary matter is fully derived." Ordinary visible matter is dominated by hadrons, especially protons and neutrons. The OPH derivation has a hadron construction, with source-only hadron masses gated by a production backend export. The paper emits the eligible elementary rows and retains the weighted-cycle coordinates only as a rejected comparison record, while the full observed matter spectrum is not emitted from a source-only OPH backend.

15.5 Claim boundary for conditional branches

The reader should therefore interpret this whole discussion section with the same rule used by *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and Deriving Standard*

Model Gauge Structure from Observer Overlap Consistency [9, 10]. The structural gauge and gravity theorems do not depend on a quark source-spread selector, charged-lepton source landing, a source-derived flavor-labelled neutrino kernel, or source-only hadron masses. The independently supported bosonic rows have the same separation.

16 Conclusion

Complete reversible response and endogenous proper-carrier transport force the local Standard Model gauge Lie algebra

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

Under the separately declared matrix current and rank-15 matter table, anomaly balance and tensor descent give the exact hypercharge lattice, three-color carrier, common $\mathbb{Z}_6$ kernel, and maximal faithful image of that representation. The construction does not select the physical global quotient. The transportable-sector/Tannaka chain is a separate conditional classification. No physical source binding, laboratory current identification, the two routes' current identity, matter-pole and continuum identification of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, scalar multiplicity/dynamics, or continuum QFT is supplied. The strengthened finite screen interface selects its rank-three candidate under the stated premises. Exclusion of extra light sectors is a declared completion.

On the declared quantitative branch, one local coordinate P organizes the electromagnetic, electroweak, hierarchy, and Higgs readouts. The common-load premise gives a conditional dimensionless hierarchy. The separately declared comparison maps give only the packaging identity $\epsilon_H^{\mathrm{decl}} = 0$; no physical naturality value is emitted. The separate weak-boson and Higgs chart calculations give running-chart and closure-chart coordinates. Section 9.7 additionally supplies a finite Landau-VEV attachment and a target-free imported-Standard-Model matching calculation, retaining its electroweak disagreement. The conditional perturbative implication from a complete renormalized packet to strict charged and neutral complex-pole coefficients is specified. The OPH source does not produce that packet: its source-selected action, complete Yukawas, Fleischer–Jegerlehner coordinate [34, 35], source-selected running/matching, two independent complete engines, executable ST/Nielsen/current-pole receipts, source law/covariance, physical reference transition, and SI clock binding are not supplied. The finite local-domain Hamiltonian has an exact positive gap for its signed seam operator. This finite graph operator is distinct from the compact-gauge repair generator and the continuum Yang–Mills Hamiltonian. It supplies no physical clock or mass scale. A nonperturbative chiral gauge construction would additionally face the known lattice chiral-measure program [30, 31, 32] and the γ_5-scheme restoration bookkeeping [33]. A finite chiral quantum object and a nonperturbative observable reconstruction are not prerequisites for the bounded imported-Standard-Model perturbative validation. Neither is constructed for those stronger claims.

The flavor analysis produces exact algebraic statements, conditional numerical constructions and explicit negative results. Native renewal fixes an exact relative-state law under a declared drive, and exterior Higgs tensors and kinetic normalization delimit the missing family interaction. The independent-alpha six-mass replay supplies target-informed coordinates and retains the adverse light-ratio comparison; it does not close that interaction. The positive-chamber face-circulant identity reduces Koide balance to $|b|/a = 1/\sqrt{2}$, and equal rank-two event blocks give that modulus in the conditional finite tracial-GNS packet. No physical chiral-mass attachment, phase, or numerical ratios are supplied. Conditional on the current registered source signature, the analysis proves a quark counterfamily; after granting a candidate fixed-ray shape law, it gives an exact restricted two-modulus fiber. It also rejects the tested register-Clebsch, reciprocal-quark, and weighted-cycle

neutrino routes. The Clebsch rejection leaves a conditional channel-pairing boundary and a target-free unordered weight set under declared constraints; neither supplies Yukawa-coefficient equality or a source-derived family order. The exact real-axis enumeration excludes direct identification of the Cabibbo angle with an acute angle between two of the 31 real three-dimensional icosahedral residual axes, while leaving spinorial, higher-order, and dynamical routes outside its scope. The W_5 stabilizer theorem shows why residual symmetry cannot finish the charged-family problem: threefold and fivefold fixed points are degenerate, the twofold fixed locus retains two parameters after scaling, the Klein four-group fixed locus retains one, and every other subgroup of A_5 contains a threefold or fivefold rotation. Absolute charged-lepton masses require a source-derived clock and determinant map. Source-only hadron masses require a working hadronic production backend. Empirical hadronic transport is an empirical input.

The relation between the particle hierarchy coordinate and cosmic capacity is conditional on two constructions in the companion papers: the direct public-record fixed point from a complete positive source packet and the identification of its carrier with the screen/electroweak load. The particle results are stated separately as theorems, conditional implications, comparisons, rejected candidates, and requirements for physical identification.

A Heat-Kernel Calibration Supplement

This appendix gives the compact-group heat-kernel calculation used by the particle calibration. The fixed-cutoff same-overlap thermal/Casimir law is proved in Ref. [6]. The calculation below supplies the compact-group Peter–Weyl lift and beta-shift bookkeeping. It does not derive the physical one-loop beta coefficients from edge sectors alone.

A.1 $\mathcal{R}_U$ Unified-Coupling Certificate

The $\mathcal{R}_U$ certificate supplies the unified diffusion coupling used by the hierarchy discussion of the flagship synthesis and by the electroweak transmutation factor

$$\frac{v}{E_\star} = P^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P)}\right].$$

It does not certify the full clock branch

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}_{\mathrm{QCD/nuc}}^{^{133}\mathrm{Cs}} + \mathcal{R}_{\mathrm{atom}}^{^{133}\mathrm{Cs}}.$$

The complete $\mathcal{R}_\gamma$ construction requires a source-only electromagnetic endpoint, charged-lepton absolute scale, cesium nuclear data, and atomic spectral enclosures. The numerical package is a witness for $\mathcal{R}_U$. Its mathematical conclusion requires the stated outward-rounded interval arithmetic.

Declared source packet. The public-pixel branch uses

$$P_C = \frac{\texttt{1.6309682094039593248792798477826489413359828516279250606661}}{\texttt{507533907793398933432}}.$$

The source-only branch uses the certified coordinate

$$P_{\mathrm{fwd}} = 1.630972095858897\ldots.$$

It is interval-certified unique on its interval and, by the domain-global certificate, the only fixed point of its readout map on the declared analytic domain $\alpha^{-1} \in [100, 200]$ (the interval contraction

certificate of 2026-07-14, whose closure row is a certified source-root row). The finite heat-kernel cutoffs are

$$N_2 = 128, \qquad N_3 = 64.$$

The one-loop running packet is

$$(b_1, b_2, b_3) = \left(\frac{33}{5}, 1, -3\right), \qquad N_c = 3, \qquad \beta_{\mathrm{EW}} = N_c + 1 = 4.$$

This packet is an explicit premise of the certificate. This appendix does not derive it from the recovered OPH core.

Definition A.1 (No-hidden-free-variable rule for $\mathcal{R}_U$)**.** *The following objects must be fixed before comparison with electroweak or gravity data:*

$$(b_1, b_2, b_3), \quad N_2, \quad N_3, \quad M_U(P), \quad E_{\mathrm{cell}}(P),$$
$$\beta_{\mathrm{EW}}, \quad \textit{Casimir normalization}, \quad \textit{matching scheme}.$$

Changing any of them after comparison with M_W, M_Z, $\alpha_s(M_Z)$, low-energy electroweak couplings, or G_{exp} demotes the row to calibration.

Dimensionless source map. Set $E_\star = 1$ for the dimensionless calculation. For a trial coupling a, define

$$M_U(P) = e^{-2\pi} P^{1/6}, \qquad E_{\mathrm{cell}}(P) = P^{-1/2},$$
$$v(P, a) = E_{\mathrm{cell}}(P) \exp\left[-\frac{2\pi}{4a}\right].$$

For $i = 1, 2, 3$,

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \qquad \alpha_Y(\mu; P, a) = \frac{3}{5}\alpha_1(\mu; P, a).$$

The source Z-scale satisfies the implicit equation

$$\mu = \frac{v(P, a)}{2} \sqrt{4\pi\alpha_2(\mu; P, a) + 4\pi\alpha_Y(\mu; P, a)}.$$

For SU(2), with $j = n/2$ and $0 \le n \le N_2$,

$$d_j = 2j + 1, \qquad C_2(j) = j(j + 1).$$

For SU(3), with $0 \le p, q \le N_3$,

$$d_{p,q} = \frac{(p + 1)(q + 1)(p + q + 2)}{2}, \qquad C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}.$$

The group sums are

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \qquad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

Define

$$t_2(P, a) = 4\pi^2 \alpha_2(\mu_Z(P, a); P, a), \qquad t_3(P, a) = 4\pi^2 \alpha_3(\mu_Z(P, a); P, a),$$

and

$$\Phi_U(P, a) = \bar{\ell}_{\mathrm{SU}(2)}(t_2(P, a)) + \bar{\ell}_{\mathrm{SU}(3)}(t_3(P, a)) - \frac{P}{4}.$$

The certificate value is the zero $\Phi_U(P, \alpha_U(P)) = 0$.

Numerical witness. For $P = P_C$, the checker emits

$$\alpha_U(P_C) = 0.041124336195630495,$$
$$\alpha_U(P_C)^{-1} = 24.316501918546496.$$

The corresponding dimensionless hierarchy data are

$$\frac{M_U(P_C)}{E_\star} = 0.0020260720012100037,$$
$$\frac{v(P_C)}{E_\star} = 2.0199803239725553 \times 10^{-17},$$
$$\frac{\mu_Z(P_C)}{E_\star} = 7.502403238986022 \times 10^{-18}.$$

The emitted couplings at μ_Z are

$$\alpha_1 = 0.016885708038988166,$$
$$\alpha_Y = 0.010131424823392899,$$
$$\alpha_2 = 0.033777889448908055,$$
$$\alpha_3 = 0.11833602747445048.$$

The heat-kernel parameters and entropy readouts are

$$t_2 = 1.3334976254578108,$$
$$t_3 = 4.6717191102770705,$$

$$\bar{\ell}_{\mathrm{SU}(2)} = 0.39488144681077636,$$
$$\bar{\ell}_{\mathrm{SU}(3)} = 0.012860605540213493.$$

Thus

$$\bar{\ell}_{\mathrm{SU}(2)} + \bar{\ell}_{\mathrm{SU}(3)} = 0.40774205235098987,$$
$$P_C/4 = 0.4077420523509898,$$

and

$$\Phi_U(P_C, 0.041124336195630495) = 5.55 \times 10^{-17}.$$

A centered derivative check gives

$$\partial_a \Phi_U(P_C, a) \simeq -10.990524683118785.$$

With $\delta = 10^{-6}$, the bracket signs are

$$\Phi_U(P_C, \alpha_U - \delta) = 1.0990748697592423 \times 10^{-5},$$
$$\Phi_U(P_C, \alpha_U + \delta) = -1.0990300680135956 \times 10^{-5}.$$

For the certified source-only pixel,

$$P_{\mathrm{fwd}} = 1.630972095858897\ldots,$$
$$\alpha_U(P_{\mathrm{fwd}}) = 0.041124247441816685\ldots,$$
$$\frac{v(P_{\mathrm{fwd}})}{E_\star} = 2.0198114078576331 \times 10^{-17}.$$

This branch is the one to cite when avoiding the measured Thomson endpoint in the upstream pixel solve.

Theorem A.2 (Electroweak unified-coupling numerical witness)**.** *Assume the declared source map, running packet, heat-kernel cutoff policy, and no-hidden-free-variable rule above. Let*

$$I_U = [0.041123336195630494,\ 0.041125336195630496].$$

If outward-rounded interval arithmetic verifies the directed endpoint signs

$$\Phi_U(P_C, 0.041123336195630494) > 0, \qquad \Phi_U(P_C, 0.041125336195630496) < 0,$$

and verifies

$$0 \notin \partial_a \Phi_U(P_C, I_U),$$

then there is a unique $\alpha_U(P_C) \in I_U$ satisfying

$$\Phi_U(P_C, \alpha_U(P_C)) = 0.$$

The comparison pixel P_C inherits its CODATA electromagnetic input. Holding that pixel fixed, this calculation uses no additional measured low-energy coupling, measured G, Planck unit formed using measured G, measured Λ, measured M_Z, or measured M_W.

Proof. Continuity follows because the running couplings, self-consistent Z-scale solution, and finite heat-kernel sums are continuous on the declared positive interval. The endpoint sign change gives existence by the intermediate value theorem. The derivative exclusion gives uniqueness. The dependency statement follows by inspection of the declared source map: the residual is built only from P_C, the declared running packet, the self-consistent Z-scale equation, and finite compact-group representation sums. □

Theorem A.3 (Implicit-scale attachment at the comparison pixel)**.** *Let $F(a, \mu)$ be the right-hand side of the source Z-scale equation, with the exact stored decimal*

$$\widehat{P}_C = 1.630968209403959324879279847782648941, \qquad J = [7 \times 10^{-18}, 8 \times 10^{-18}].$$

For every $a \in I_U$, all running denominators are positive on J and

$$F(a, J) \subset [7.49, 7.52] \times 10^{-18} \subset \operatorname{int} J, \qquad 0 \le \partial_\mu F(a, J) \le 0.006 < 1/100.$$

There is exactly one scale $\mu(a) \in J$, it is continuously differentiable, and the finite-cutoff hierarchy residual evaluated at this exact scale obeys

$$\frac{\mathrm{d}}{\mathrm{d}a} \Phi_U(\widehat{P}_C, a) \in [-12, -10].$$

The lower and upper endpoint values lie respectively in

$$[1.09907, 1.09908] \times 10^{-5}, \qquad [-1.09904, -1.09902] \times 10^{-5}.$$

There is therefore exactly one zero in I_U. The cutoffs remain $N_2 = 128, N_3 = 64$; neither an infinite representation sum nor a scale outside J is part of this claim.

Proof. The uniform self-map and derivative bound give a contraction on J. The interior image and $1 - \partial_\mu F \ge 99/100$ give the implicit derivative

$$\mu'(a) = \frac{\partial_a F}{1 - \partial_\mu F}.$$

Every interval image of a box containing this fixed point retains it. Eight images on $I_U \times J$ enclose the scale for the derivative test; sixteen images at each endpoint enclose the endpoint residuals. These are inclusions of the exact fixed point, not assertions that an iterate equals it. Interval automatic differentiation and an independent analytic calculation verify the displayed bounds at 60-digit interval precision. For the latter, write $S = \alpha_2 + 3\alpha_1/5$ and $B = \alpha_2^2 + 99\alpha_1^2/25$. Then

$$F_\mu = \frac{FB}{4\pi\mu S}, \qquad F_a = F\left[\frac{\pi}{2a^2} + \frac{\alpha_2^2 + 3\alpha_1^2/5}{2a^2 S}\right].$$

The finite entropy derivative is $\mathrm{d}\bar{\ell}_G/\mathrm{d}t = -\operatorname{Cov}_t(C_2, \log d)$; combining it with $t_i' = 4\pi^2\alpha_i^2[a^{-2} + b_i\mu'/(2\pi\mu)]$ gives the certified residual derivative. Continuity, endpoint signs, and strict monotonicity finish the argument. The separate 60- and 64-iteration certificates retain their own finite-map scope and Krawczyk enclosures. □

Precision policy. The exponential map makes downstream gravity displays sensitive to α_U:

$$\frac{\partial \log G}{\partial \alpha_U} \simeq \frac{\pi}{\alpha_U^2} \approx 1.86 \times 10^3.$$

Therefore the number of digits printed for any downstream G or $\varepsilon_{\mathrm{Cs}}$ row may not exceed the interval precision certified by

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}_{\mathrm{QCD/nuc}}^{133\mathrm{Cs}} + \mathcal{R}_{\mathrm{atom}}^{133\mathrm{Cs}}.$$

The witness supports an electroweak hierarchy certificate. It does not support a 52-digit source-only gravity prediction.

A.2 One-Loop Calibration Frame

At one loop, if couplings unify at (M_U, α_U),

$$\alpha_i^{-1}(M_Z) = \alpha_U^{-1} + \frac{b_i}{2\pi}\ln\frac{M_U}{M_Z}.$$

On the forward calibration branch these are the values $(M_U(P), \alpha_U(P))$ emitted by the source solve. They are not inferred by inverse readback from measured low-energy couplings. Writing $A_i := \alpha_i^{-1}(M_Z)$ and $L := \ln(M_U/M_Z)$, one finds

$$L = \frac{2\pi}{b_1 - b_2}(A_1 - A_2),$$

and the corresponding consistency relation for this calibration is

$$A_3^{\mathrm{pred}} = \frac{b_3 - b_2}{b_1 - b_2}A_1 + \frac{b_1 - b_3}{b_1 - b_2}A_2.$$

This appendix keeps that algebra on the page as a calibration relation, not as a theorem that OPH derives a supersymmetric UV spectrum.

A.3 Benchmark coefficient comparison

For the standard one-loop benchmark coefficients,

Model	(b_1, b_2, b_3)	$\alpha_s(M_Z)^{\text{pred}}$
Standard Model	$(41/10, -19/6, -7)$	≈ 0.071
minimal supersymmetric Standard Model	$(33/5, 1, -3)$	≈ 0.116
Observed [22]		≈ 0.118

At one loop, the printed minimal supersymmetric Standard Model (MSSM) coefficients reproduce the observed closure far better than the plain Standard Model coefficients. The forward calibration matches this benchmark. It does not imply that OPH derives a supersymmetric ultraviolet spectrum. The observed row is the Particle Data Group world average, quoted at the precision this benchmark separates, and the one-loop comparison is the textbook unification benchmark rather than an output of the OPH source, which is why it carries no claim class and no comparison row of its own.

A.4 Heat-Kernel Sector Law

Read this subsection as the compact-group merge boundary above the fixed-cutoff microphysics theorem. MaxEnt yields the familiar $d_R e^{-tC_2(R)}$ law only after the quadratic-Casimir generator and one-sided edge algebra below are declared; the Peter–Weyl lift alone does not select either input. This is not a second, independent proof of the finite-cutoff same-overlap Casimir branch.

Theorem A.4 (Heat-kernel edge-sector weights)**.** *Let the MaxEnt edge generator be the factorwise group Laplacian $H_{\text{edge}} = \sum_i c_i \Delta_{G_i}$, with eigenvalue $C_2(R)$ in the one-factor notation, and let the probability space be the one-sided electric-center algebra $\bigoplus_R B(V_R)$, where the sector projector has trace d_R. Then the sector probabilities take heat-kernel form:*

$$p_R(t) \propto d_R\, e^{-tC_2(R)},$$

where $d_R = \dim R$ and $C_2(R)$ is the quadratic Casimir. Bi-invariance by itself permits other central energies, such as $C_2(R)^2$. A Gibbs trace on the two-sided Peter–Weyl space $L^2(G)$ instead gives the multiplicity d_R^2.

Lemma A.5 (Bi-invariant operators)**.** *If $G = \prod_i G_i$ is a compact semisimple Lie group written as a product of compact simple factors, then any bi-invariant second-order differential operator on G has the form*

$$D = c_0 \mathbf{1} - \sum_i c_i \Delta_{G_i},$$

where Δ_{G_i} is the Laplace–Beltrami operator on the factor G_i.

Remark A.6. *Bi-invariance is equivalent to $D \in Z(U(\mathfrak{g}))$. Linear terms vanish because there are no invariant vectors in the adjoint representation, and the quadratic part is proportional to the Killing form on each simple factor, giving one independent coefficient per factor. The Casimir element then acts as $-\Delta_G$ in the regular representation.*

A.5 Peter–Weyl Multiplicity and Beta Shifts

By Peter–Weyl decomposition, the effective refinement-limit edge representation space is

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*.$$

Entanglement traces over one factor give multiplicity d_R in p_R, but loops see both factors, so the effective multiplicity entering the calibration branch is

$$N_{\text{eff}}(R) = d_R\, p_R.$$

The one-loop beta shift from edge sectors is then

$$\Delta b_a = \sum_R p_R\, d_R\, T_a(R),$$

where $T_a(R)$ is the Dynkin index for gauge factor a. This is the exact mathematical point where edge-sector heat-kernel weights feed the calibration branch. Matching to MSSM-like one-loop running is a comparison benchmark, not a theorem-level MSSM spectrum claim.

The branch boundary for theorem-level unification closure is explicit. One requires a derived refinement-limit identification of which transportable sectors actually contribute to the running carrier, a derived statistics/grading rule rather than an imposed fermionic-grading restriction, controlled threshold and decoupling conventions on that same carrier, a derived representation content/truncation rule for the sectors retained in the comparison, and a proof that the effective loop multiplicities entering Δb_a are exactly the ones used in this calibration package rather than nearby alternatives. The declared runtime surface does not emit the full RG/matching/threshold/scheme packet: scheme lock, threshold map, beta provenance, and interval composition are separate certificate requirements.

At the branch unification diffusion parameter $t_U(P) \approx 1.64$ selected by the forward source solve, the benchmark shift is

$$\Delta b \approx (2.49, 4.38, 3.97) \qquad \text{vs MSSM} \qquad (2.50, 4.17, 4.00).$$

With the additional calibration assumption of a fermionic-grading restriction to half-integer SU(2) sectors, this sharpens to

$$\Delta b \approx (2.50, 4.17, 3.97),$$

which matches the MSSM-style one-loop benchmark at the percent level under that declared calibration package. Equivalently, the benchmark ratio

$$\frac{\Delta b_3}{\Delta b_2} \approx 0.91$$

sits close to the MSSM comparison value 0.96.

B H3 Worldline-Stitch Certificate

This appendix records the finite evidence-facing certificate for Theorem 14.5. It is separate from the heat-kernel calibration package above. A passing certificate is a record-continuation statement in a declared H^3 observer chart, not a particle-species, mass, charge, or scattering-amplitude theorem.

The certificate payload has the following required blocks.

1. **Atlas.** It declares the hyperboloid model $H^3_{R_H} \subset \mathbb{R}^{1,3}$, the Lorentz signature $(-+++)$, the curvature radius R_H, chart domains, and chart transitions $G_{ji} \in \mathrm{SO}^+(1,3)$. Transition residuals must certify Lorentz inner-product preservation, orientation, and future-sheet preservation.

2. **Clock map.** Each local record interval is mapped to a common comparison-time line. The observer-time adjacency margin must dominate the declared clock uncertainty.

3. **Chart-natural extraction.** Raw local records are connected components of a frozen detector scalar on the H^3 chart. The detector scalar, thresholds, support collars, and chart-naturality residuals are included. Implementation record IDs are not inputs.

4. **Overlap descent.** Records are first descended across genuine chart overlaps. The payload records same-component join margins, distinct-component separation margins, support errors, and triple-overlap cocycle residuals.

5. **Interface crossing.** A candidate stitch must cross a real interface in adjacent clock intervals. It includes oriented signed-distance margins and a positive lower bound on normal velocity. Tangential, grazing, and file-boundary-only contacts are rejected.

6. **Transport.** The candidate edge is evaluated in a common chart after sector and gauge transport. The atlas holonomy comparison is separate from the gauge-holonomy comparison.

7. **Assignment.** The matching is one-to-one with appearance/disappearance interval costs. The proposed winner has an upper cost bound, every competitor has a lower cost bound, and

$$\Delta_{\mathrm{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*) > 0.$$

   Record IDs, global IDs, stitch keys, and shard-local IDs are forbidden in admissibility, costs, and tie-breaking.

8. **Refinement.** A coarse/fine pair supplies Q_{sr}, a distortion bound η_{sr}, and a contracted-graph isomorphism check. The coarse gap must satisfy $\Delta_r > 2\eta_{sr}$.

9. **Interaction firewall.** A free-propagation slab is required. If an interaction marker is present, the terminal label is H3 interaction required and the event is handed to the interaction solver.

The allowed terminal outcomes are

H3 stitch certified, H3 stitch ambiguous, H3 stitch rejected,
H3 interaction required, H3 atlas invalid, H3 certificate incomplete.

H3 stitch certified is emitted only when every block above passes and the stable assignment gap is positive. Equal-cost matchings, preserved nontrivial permutations of the visible data, or coarse/fine gaps below margin make the fail-closed protocol emit H3 stitch ambiguous, meaning that it has certified no unique stitch. This verdict is not by itself a theorem of mathematical nonuniqueness; a symmetry obstruction requires a proved action that is fixed-point-free on the admissible stitch set. Missing fields force H3 certificate incomplete. Failed atlas, interface, metric, or transport gates force rejection or atlas invalidity. In particular, Euclidean distances on the stored coordinate triples are not certificate distances; the certificate distance is the hyperboloid geodesic d_H from Section 14.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- A5 flavor potentials and residual-stabilizer selection [54, 72].
- Burnside_orbit_counting [56, 46].
- Classical electroweak symmetry breaking and Goldstone counting in the Standard Model Higgs sector [108, 95, 63].
- Classical Koide relation, standard one-loop running and the existing OPH certified alpha and tau artifacts [73, 14, 95].
- complex_pole_to_running_width_Breit_Wigner_conversion [41, 101].
- cumulant expansions [74].
- De Broglie internal-clock and Zitterbewegung phase kinematics [52, 65, 99].
- Electroweak phenomenology, exterior weak multiplicities, hierarchy-problem literature, and commuting-square naturality diagnostics [20, 40, 103, 104, 80].
- exact_quadratic_field_linear_algebra [48].
- exact_rational_linear_algebra [105, 42].
- Exterior or Fock algebra and Standard Model exterior embeddings [40, 45, 59].
- exterior_algebra_and_CAR [45, 69].
- Finite clock models, Hamiltonian scaling, and determinant-line mass invariants [94, 66].
- Finite-group heat kernels [53, 98].
- finite_absorbing_Markov_chains_and_triangular_spectra [70, 66].
- finite_group_invariant_operator_averaging [100].
- finite_group_representation_theory_and_Schur_lemma [100, 58].
- finite_irreducible_periodic_Markov_chain_stationarity [75, 90].
- finite_random_unitary_channels_and_class_generators [107, 53].
- Flavor-basis invariants, singular-value decompositions, and parameter non-identifiability [68, 43, 96].
- Free plane-wave evolution and de Broglie phase kinematics under translation flows [52, 109].
- Georgi–Jarlskog/Clebsch flavor textures, one-loop Yukawa running, and lattice light-quark ratios [60, 79, 18].
- Harremoes and Topsoe (2001), DOI 10.1109/18.959272, Shannon entropy at fixed index of coincidence [16].
- Koide's charged-lepton relation, Hermitian C3 circulants, and finite GNS constructions [73, 50, 44].
- maximal_matching_combinatorics [76].
- moment_and_cumulant_algebra [87, 93].
- numerical_quadrature_and_coupled_ODE_shooting [102, 51].

- Pauli algebra sigma_a sigma_b = delta_ab + i epsilon_abc sigma_c and the Pauli Hamiltonian derivation of g = 2 [92, 97].
- PDG and FLAG quark summaries [20, 18].
- perturbative renormalization-group transport [78, 77].
- positive_Hermitian_kinetic_normalization [66, 71].
- Primary mathematical and physical sources [40, 89].
- QCD running-to-pole conversion [47, 88].
- Residual-symmetry flavor mixing and icosahedral axis geometry [72, 54].
- scalar_ODE_parameter_sensitivity_and_comparison_barriers [64, 106].
- Schur_complement_differentiation [111, 81].
- Signed graph Laplacians, integer positive-definite matrices, and spectral residual estimates [110, 67, 91].
- SMDR_1_3_complex_pole_and_local_charged_current_matching [86, 85].
- Standard Model Higgs near-criticality calculations [57, 21].
- Standard_Model_beta_functions_top_only_specialization [79, 77].
- Standard_Model_two_loop_beta_functions_and_effective_potential [77, 55, 82].
- standard_SU3_fundamental_Casimir [61, 95].
- strict_convexity_and_exponential_family_KL_projection [49, 39].
- SU3_and_SU2_intertwiner_identities [61, 58].
- Tree-level charged-current couplings g/sqrt(2) with weak-isospin raising and lowering operators [108, 95].
- Tree-level electroweak neutral-current couplings Q = T3 + Y and T3 - Q sin^2 theta [62, 95, 108].
- TSIL_and_3VIL_loop_integral_evaluation [83, 84].

References

[1] B. Müller et al., *Observer-Patch Holography: formal proofs and executable receipts*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2052/Lean/Screen

[2] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/extra/observable_normal_forms.pdf.

[3] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, J. Kim, D. Matscheko, D. A. Friedman, J. Hill, W. T. Glynn, M. A. Visser, K. A. Anirudha, and B. de La Fournière, *Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/flagship/from_observer_consensus_to_standard_physics.pdf

[4] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, and D. A. Friedman, *Observers Are All You Need*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/paper/observers_are_all_you_need.pdf

[5] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/paper/reality_as_consensus_protocol.pdf

[6] B. Müller, A. Osika, K. Xue, B. Cassie, B. de La Fournière, and D. A. Friedman, *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in Observer-Patch Holography*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/paper/screen_microphysics_and_observer_synchronization.pdf

[7] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Source: https://github.com/muellerberndt/oph-physics-sim. Canonical evidence archives: https://github.com/FloatingPragma/observer-patch-holography/tree/r2052/evidence. Exact local-domain archive and checker: https://github.com/FloatingPragma/observer-patch-holography/tree/r2052/evidence/local_domain.

[8] FloatingPragma, *Particle-simulation receipt archive*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2052/evidence/particle_simulation_receipts.

[9] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[10] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[11] B. Müller, M. Poneder, J. Kim, and D. A. Friedman, *An Executable Observer for Observer Patch Holography: Specification, Certified Components and Acceptance Criteria*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/paper/executable_observer_construction.pdf

[12] FloatingPragma, *Repair laws: what is fixed, what is free and how to tell*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2052/model/CANDIDATES.md

[13] National Institute of Standards and Technology, *2022 CODATA Recommended Values of the Fundamental Constants of Physics and Chemistry*, NIST SP 959, May 2024. https://physics.nist.gov/cuu/pdf/wallet_2022.pdf

[14] Y. Koide, *New view of quark and lepton mass hierarchy*, Physical Review D **28**, 252 (1983).

[15] C. A. Brannen, *The Lepton Masses*, preprint, 2006. http://brannenworks.com/MASSES2.pdf

[16] P. Harremoës and F. Topsøe, *Inequalities between entropy and index of coincidence derived from information diagrams*, IEEE Transactions on Information Theory **47**, 2944–2960 (2001). https://doi.org/10.1109/18.959272

[17] S. Antusch, K. Hinze, and S. Saad, *Updated Running Quark and Lepton Parameters at Various Scales*, arXiv:2510.01312, 2025. https://arxiv.org/abs/2510.01312

[18] Y. Aoki et al. (Flavour Lattice Averaging Group), *FLAG Review 2024*, arXiv:2411.04268, 2024, Tables 11–12. https://arxiv.org/abs/2411.04268

[19] S. Navas et al. (Particle Data Group), *CKM Quark-Mixing Matrix*, Review of Particle Physics 2024, Sec. 12.2.2, revised May 31, 2024. https://pdg.lbl.gov/2024/reviews/rpp2024-rev-ckm-matrix.pdf

[20] S. Navas et al. (Particle Data Group), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024), 2025 update, quark summary table. https://pdg.lbl.gov/2025/tables/rpp2025-sum-quarks.pdf

[21] D. Buttazzo et al., *Investigating the near-criticality of the Higgs boson*, JHEP **12** (2013) 089, Appendix B. https://arxiv.org/abs/1307.3536

[22] Particle Data Group, *Review of Particle Physics: 2026 particle listings*, 2026. https://pdg.lbl.gov/2026/listings/particle_properties.html

[23] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astronomy & Astrophysics 641, A6, 2020. https://arxiv.org/abs/1807.06209

[24] DESI Collaboration, *Constraints on Neutrino Physics from DESI DR2 BAO and DR1 Full Shape*, arXiv:2503.14744, 2025. https://arxiv.org/abs/2503.14744

[25] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, JHEP **12** (2024) 216, arXiv:2410.05380, with the NuFIT 6.1 (2025) profile-table release at https://www.nu-fit.org/?q=node/309.

[26] B. A. Kniehl, *All-order renormalization of the propagator matrix for fermionic systems with flavor mixing*, Physical Review D **89**, 096005 (2014), arXiv:1308.3140.

[27] P. Gambino and P. A. Grassi, *The Nielsen identities of the Standard Model and the definition of mass*, Physical Review D **62**, 076002 (2000), arXiv:hep-ph/9907254.

[28] D. Buchholz, *Gauss' law and the infraparticle problem*, Physics Letters B **174**, 331–334 (1986).

[29] P. Duch and W. Dybalski, *Infrared problem in quantum electrodynamics*, arXiv:2307.06114.

[30] M. Lüscher, *Abelian chiral gauge theories on the lattice with exact gauge invariance*, Nuclear Physics B **549**, 295–334 (1999), arXiv:hep-lat/9811032.

[31] M. Lüscher, *Lattice regularization of chiral gauge theories to all orders of perturbation theory*, Journal of High Energy Physics **06** (2000) 028, arXiv:hep-lat/0006014.

[32] D. Kadoh and Y. Kikukawa, *A simple construction of fermion measure term in $U(1)$ chiral lattice gauge theories with exact gauge invariance*, Journal of High Energy Physics **02** (2008) 063, arXiv:0709.3658.

[33] H. Bélusca-Maïto, A. Ilakovac, M. Mađor-Božinović, and D. Stöckinger, *Dimensional regularization and Breitenlohner–Maison/'t Hooft–Veltman scheme for γ_5 applied to chiral Yang–Mills theory*, Journal of High Energy Physics **08** (2020) 024, arXiv:2004.14398.

[34] V. Dūdėnas and M. Löschner, *Vacuum expectation value renormalization in the Standard Model and beyond*, Physical Review D **103**, 013003 (2021), arXiv:2010.15076.

[35] S. Dittmaier and H. Rzehak, *Electroweak renormalization based on gauge-invariant vacuum expectation values of non-linear Higgs representations*, Journal of High Energy Physics **05** (2022) 125, arXiv:2203.07236.

[36] P. A. Grassi, B. A. Kniehl, and A. Sirlin, *Width and partial widths of unstable particles in the light of the Nielsen identities*, Physical Review D **65**, 085001 (2002), arXiv:hep-ph/0109228.

[37] S. P. Martin and D. G. Robertson, *Standard Model parameters in the tadpole-free pure* $\overline{\text{MS}}$ *scheme*, arXiv:1907.02500.

[38] S. P. Martin, *Top-quark pole mass in the tadpole-free MS-bar scheme*, arXiv:1604.01134.

[39] Amari, S. i., and Nagaoka, H. (2000). *Methods of Information Geometry*. American Mathematical Society. https://doi.org/10.1090/mmono/191

[40] Baez, J., and Huerta, J. (2010). "The algebra of grand unified theories." *Bulletin of the American Mathematical Society* **47**, 483–552. https://doi.org/10.1090/S0273-0979-10-01294-2

[41] Bardin, D. Y., Leike, A., Riemann, T., and Sachwitz, M. (1988). "Energy-dependent width effects in e+e- annihilation near the Z-boson pole." *Physics Letters B* **206**, 539–542. https://doi.org/10.1016/0370-2693(88)91627-9

[42] Bareiss, E. H. (1968). "Sylvester's identity and multistep integer-preserving Gaussian elimination." *Mathematics of Computation* **22**, 565–578. https://doi.org/10.1090/S0025-5718-1968-0226829-0

[43] Branco, G. C., Lavoura, L., and Silva, J. P. (1999). *CP Violation*. Clarendon Press, Oxford. ISBN 0-19-850399-7.

[44] Bratteli, O., and Robinson, D. W. (1987). *Operator Algebras and Quantum Statistical Mechanics 1: C*- and W*-Algebras, Symmetry Groups, Decomposition of States*. Springer. https://doi.org/10.1007/978-3-662-02520-8

[45] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2*. Springer. https://doi.org/10.1007/978-3-662-03444-6

[46] Burnside, W. (1911). *Theory of Groups of Finite Order*. Cambridge University Press. https://doi.org/10.1017/CBO9781139237253

[47] Chetyrkin, K., and Steinhauser, M. (2000). "The relation between the $\overline{\text{MS}}$ and the on-shell quark mass at order α_s^3." *Nuclear Physics B* **573**, 617–651. https://doi.org/10.1016/s0550-3213(99)00784-1

[48] Cohen, H. (1993). *A Course in Computational Algebraic Number Theory*. Springer. https://doi.org/10.1007/978-3-662-02945-9

[49] Csiszár, I. (1975). "I-divergence geometry of probability distributions and minimization problems." *Annals of Probability* **3**, 146–158. https://doi.org/10.1214/aop/1176996454

[50] Davis, P. J. (1979). *Circulant Matrices*. Wiley, New York. ISBN 0-471-05771-1.

[51] Davis, P. J., and Rabinowitz, P. (1984). *Methods of Numerical Integration*. Academic Press. https://doi.org/10.1016/C2013-0-10566-1

[52] de Broglie, L. (1925). "Recherches sur la théorie des quanta." *Annales de Physique* **10**, 22–128. https://doi.org/10.1051/anphys/192510030022

[53] Diaconis, P. (1988). *Group Representations in Probability and Statistics*. Institute of Mathematical Statistics (Lecture Notes–Monograph Series). https://doi.org/10.1214/lnms/1215467407

[54] Everett, L. L., and Stuart, A. J. (2009). "Icosahedral (A5) family symmetry and the golden ratio prediction for solar neutrino mixing." *Physical Review D* **79**, 085005. https://doi.org/10.1103/PhysRevD.79.085005

[55] Ford, C., Jack, I., and Jones, D. (1992). "The standard model effective potential at two loops." *Nuclear Physics B* **387**, 373–390. https://doi.org/10.1016/0550-3213(92)90165-8

[56] Frobenius, G. (1887). "Ueber die Congruenz nach einem aus zwei endlichen Gruppen gebildeten Doppelmodul." *Journal für die reine und angewandte Mathematik* **101**, 273–299. https://doi.org/10.1515/crll.1887.101.273

[57] Froggatt, C., and Nielsen, H. (1996). "Standard Model criticality prediction: top mass 173 ± 5 GeV and Higgs mass 135 ± 9 GeV." *Physics Letters B* **368**, 96–102. https://doi.org/10.1016/0370-2693(95)01480-2

[58] Fulton, W., and Harris, J. (1991). *Representation Theory: A First Course*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4612-0979-9

[59] Georgi, H., and Glashow, S. L. (1974). "Unity of All Elementary-Particle Forces." *Physical Review Letters* **32**, 438–441. https://doi.org/10.1103/PhysRevLett.32.438

[60] Georgi, H., and Jarlskog, C. (1979). "A new lepton-quark mass relation in a unified theory." *Physics Letters B* **86**, 297–300. https://doi.org/10.1016/0370-2693(79)90842-6

[61] Georgi, H. (1999). *Lie Algebras in Particle Physics: From Isospin to Unified Theories*. Perseus Books. ISBN 0-7382-0233-9.

[62] Glashow, S. L. (1961). "Partial-symmetries of weak interactions." *Nuclear Physics* **22**, 579–588. https://doi.org/10.1016/0029-5582(61)90469-2

[63] Goldstone, J., Salam, A., and Weinberg, S. (1962). "Broken Symmetries." *Physical Review* **127**, 965–970. https://doi.org/10.1103/PhysRev.127.965

[64] Hartman, P. (2002). *Ordinary Differential Equations*. Society for Industrial and Applied Mathematics (Classics in Applied Mathematics). https://doi.org/10.1137/1.9780898719222

[65] Hestenes, D. (1990). "The zitterbewegung interpretation of quantum mechanics." *Foundations of Physics* **20**, 1213–1232. https://doi.org/10.1007/BF01889466

[66] Horn, R. A., and Johnson, C. R. (2012). *Matrix Analysis*. Cambridge University Press. https://doi.org/10.1017/CBO9781139020411

[67] Hou, Y., Li, J., and Pan, Y. (2003). "On the Laplacian Eigenvalues of Signed Graphs." *Linear and Multilinear Algebra* **51**, 21–30. https://doi.org/10.1080/0308108031000053611

[68] Jarlskog, C. (1985). "Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation." *Physical Review Letters* **55**, 1039–1042. https://doi.org/10.1103/PhysRevLett.55.1039

[69] Jordan, P., and Wigner, E. (1928). "Über das Paulische Äquivalenzverbot." *Zeitschrift für Physik* **47**, 631–651. https://doi.org/10.1007/BF01331938

[70] Kemeny, J. G., and Snell, J. L. (1976). *Finite Markov Chains*. Springer-Verlag, New York (Undergraduate Texts in Mathematics). ISBN 0-387-90192-2.

[71] King, S., and Peddie, I. (2004). "Canonical normalisation and Yukawa matrices." *Physics Letters B* **586**, 83–94. https://doi.org/10.1016/j.physletb.2004.01.087

[72] King, S. F., and Luhn, C. (2013). "Neutrino mass and mixing with discrete symmetry." *Reports on Progress in Physics* **76**, 056201. https://doi.org/10.1088/0034-4885/76/5/056201

[73] Koide, Y. (1983). "A fermion-boson composite model of quarks and leptons." *Physics Letters B* **120**, 161–165. https://doi.org/10.1016/0370-2693(83)90644-5

[74] Kubo, R. (1962). "Generalized cumulant expansion method." *Journal of the Physical Society of Japan* **17**, 1100–1120. https://doi.org/10.1143/JPSJ.17.1100

[75] Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*. American Mathematical Society. https://doi.org/10.1090/mbk/107

[76] Lovász, L., and Plummer, M. D. (2009). *Matching Theory*. AMS Chelsea Publishing, Providence. https://doi.org/10.1090/chel/367

[77] Luo, M., Wang, H., and Xiao, Y. (2003). "Two-loop renormalization group equations in general gauge field theories." *Physical Review D* **67**, 065019. https://doi.org/10.1103/PhysRevD.67.065019

[78] Machacek, M. E., and Vaughn, M. T. (1983). "Two-loop renormalization group equations in a general quantum field theory." *Nuclear Physics B* **222**, 83–103. https://doi.org/10.1016/0550-3213(83)90610-7

[79] Machacek, M. E., and Vaughn, M. T. (1984). "Two-loop renormalization group equations in a general quantum field theory (II). Yukawa couplings." *Nuclear Physics B* **236**, 221–232. https://doi.org/10.1016/0550-3213(84)90533-9

[80] Mac Lane, S. (1998). *Categories for the Working Mathematician*. Springer. https://doi.org/10.1007/978-1-4757-4721-8

[81] Magnus, J. R., and Neudecker, H. (2019). *Matrix Differential Calculus with Applications in Statistics and Econometrics*. Wiley. https://doi.org/10.1002/9781119541219

[82] Martin, S. P. (2002). "Two-loop effective potential for a general renormalizable theory and softly broken supersymmetry." *Physical Review D* **65**, 116003. https://doi.org/10.1103/PhysRevD.65.116003

[83] Martin, S. P., and Robertson, D. G. (2006). "TSIL: a program for the calculation of two-loop self-energy integrals." *Computer Physics Communications* **174**, 133–151. https://doi.org/10.1016/j.cpc.2005.08.005

[84] Martin, S. P., and Robertson, D. G. (2017). "Evaluation of the general three-loop vacuum Feynman integral." *Physical Review D* **95**, 016008. https://doi.org/10.1103/PhysRevD.95.016008

[85] Martin, S. P. (2019). "Matching relations for decoupling in the standard model at two loops and beyond." *Physical Review D* **99**, 033007. https://doi.org/10.1103/PhysRevD.99.033007

[86] Martin, S. P., and Robertson, D. G. (2019). "Standard Model parameters in the tadpole-free pure MS-bar scheme." *Physical Review D* **100**, 073004. https://doi.org/10.1103/PhysRevD.100.073004

[87] McCullagh, P. (1987). *Tensor Methods in Statistics*. Chapman and Hall, London. ISBN 0-412-27480-9.

[88] Melnikov, K., and van Ritbergen, T. (2000). "The three-loop relation between the MS-bar and the pole quark masses." *Physics Letters B* **482**, 99–108. https://doi.org/10.1016/S0370-2693(00)00507-4

[89] Nielsen, M. A., and Chuang, I. L. (2010). *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press. https://doi.org/10.1017/CBO9780511976667

[90] Norris, J. R. (1997). *Markov Chains*. Cambridge University Press. https://doi.org/10.1017/CBO9780511810633

[91] Parlett, B. N. (1998). *The Symmetric Eigenvalue Problem*. Society for Industrial and Applied Mathematics. https://doi.org/10.1137/1.9781611971163

[92] Pauli, W. (1927). "Zur Quantenmechanik des magnetischen Elektrons." *Zeitschrift für Physik* **43**, 601–623. https://doi.org/10.1007/BF01397326

[93] Peccati, G., and Taqqu, M. S. (2011). *Wiener Chaos: Moments, Cumulants and Diagrams*. Springer, Milan (Bocconi & Springer Series). https://doi.org/10.1007/978-88-470-1679-8

[94] Peres, A. (1980). "Measurement of time by quantum clocks." *American Journal of Physics* **48**, 552–557. https://doi.org/10.1119/1.12061

[95] Peskin, M. E., and Schroeder, D. V. (1995). *An Introduction to Quantum Field Theory*. Addison-Wesley. https://doi.org/10.1201/9780429503559

[96] Rothenberg, T. J. (1971). "Identification in Parametric Models." *Econometrica* **39**, 577–591. https://doi.org/10.2307/1913267

[97] Sakurai, J. J., and Napolitano, J. (2020). *Modern Quantum Mechanics*. Cambridge University Press. https://doi.org/10.1017/9781108587280

[98] Saloff-Coste, L. (2004). "Random Walks on Finite Groups." In *Probability on Discrete Structures*, 263–346. Springer, Berlin (Encyclopaedia of Mathematical Sciences). https://doi.org/10.1007/978-3-662-09444-0_5

[99] Schrödinger, E. (1930). "Über die kräftefreie Bewegung in der relativistischen Quantenmechanik." *Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, 418–428. https://zbmath.org/?q=an:56.0754.06

[100] Serre, J. P. (1977). *Linear Representations of Finite Groups*. Springer. https://doi.org/10.1007/978-1-4684-9458-7

[101] Sirlin, A. (1991). "Theoretical considerations concerning the Z0 mass." *Physical Review Letters* **67**, 2127–2130. https://doi.org/10.1103/PhysRevLett.67.2127

[102] Stoer, J., and Bulirsch, R. (2002). *Introduction to Numerical Analysis*. Springer, New York (Texts in Applied Mathematics). https://doi.org/10.1007/978-0-387-21738-3

[103] Susskind, L. (1979). "Dynamics of spontaneous symmetry breaking in the Weinberg-Salam theory." *Physical Review D* **20**, 2619–2625. https://doi.org/10.1103/PhysRevD.20.2619

[104] 't Hooft, G. (1980). "Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking." In *Recent Developments in Gauge Theories*, 135–157. Springer. https://doi.org/10.1007/978-1-4684-7571-5_9

[105] von zur Gathen, J., and Gerhard, J. (2013). *Modern Computer Algebra*. Cambridge University Press. https://doi.org/10.1017/CBO9781139856065

[106] Walter, W. (1970). *Differential and Integral Inequalities*. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-86405-6

[107] Watrous, J. (2018). *The Theory of Quantum Information*. Cambridge University Press. https://doi.org/10.1017/9781316848142

[108] Weinberg, S. (1967). "A model of leptons." *Physical Review Letters* **19**, 1264–1266. https://doi.org/10.1103/PhysRevLett.19.1264

[109] Weinberg, S. (1995). *The Quantum Theory of Fields, Volume 1: Foundations*. Cambridge University Press. https://doi.org/10.1017/CBO9781139644167

[110] Zaslavsky, T. (1982). "Signed graphs." *Discrete Applied Mathematics* **4**, 47–74. https://doi.org/10.1016/0166-218X(82)90033-6

[111] Zhang, F. (2005). *The Schur Complement and Its Applications*. Springer. https://doi.org/10.1007/b105056

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.

Result class	Exact output(s)	Exact derivation	Conditions
Carrier roles and classical modes	no photon/gluon/graviton particle-mass output	A1–A3 → realized electromagnetic/color/dynamical-metric roles; additional action/background/phase receipt → transverse/TT quadratic modes	quantum-particle receipt not supplied; confined color has no free asymptotic-gluon claim
Electroweak chart and analytic consumer	no nonzero source-only mass output	source tuple → exact two-coordinate chart; complete external renormalized packet → strict scalar pole consumer and bounded analytic checks; finite quotient-path certificate ⇒ candidate value law; inverse adapter separate	the external fixture is not composed with the OPH chart. The receipts do not identify a physical sheet or full matrix amplitude, and no source-only pole pair or mass comparison follows
Hierarchy bridge and declared-map square	exact conditional dimensionless bridge residual and a zero defect only for the declared comparison maps	A1–A3 → P-fixed source branch → exact exterior weak multiplicity four → mathematical N_{EW} bridge map; separately declared source-to-Higgs comparison maps → commuting-square identity; conditional screen branch: 12 ports and an oriented 24-slot register	conditional theorem for $v/E_\star$; the declared zero defect is a packaging identity, because the antecedents do not select the maps and admit both zero- and nonzero-defect completions; the register count does not construct the weak load; equality with cosmic capacity requires the direct public-record fixed point and the common screen/electroweak load-carrier map; a naturality result requires source-derived maps and independent residual evaluation; no physical $E_\star$ or mass row follows
Higgs/top inverse slice	no nonzero source-only mass output	gauge core → declared Jacobian → exact inverse slice	target-conditioned benchmark only; the conditional coordinates require every unsupplied physical premise
Charged exact witness	exact same-family charged triple withheld	A1–A3 → shared excitation dictionary → ordered charged carrier → exact centered readback → closed quadratic readout theorem → same-family exact witness	same-family target-anchored comparison witness only; theorem branch carries exact centered readback plus the closed common-shift no-go. The theorem does not provide a $\widehat{C}_e$ lift that emits the physical scalar $\mu_{\mathrm{phys}}(Y_e)$
Quark physical boundary	numeric predictions withheld; common-scale reciprocal-ray candidate rejected; mixed-convention formula residuals are target-anchored	six dimensionless Yukawas in one scheme and scale → reciprocal-ray rejection → six-scalar interface and selector no-go	at M_Z, $\rho_u = 1.110889$, $\rho_d = 0.751941$, product 0.835323; even with four endpoints granted, the held-out miss is 21.56%. The two-spread theorem is a restricted non-identifiability result
Neutrino candidate	declared scale-free comparison point; absolute mass attachments withheld	template family transport → same-label scalar certificate → target-informed weighted-cycle law → compare-only bridge and absolute attachment → shared-basis identity	rejected by the NuFIT 6.1 correlated profile; target dependence, source closure, basis orientation, and absolute normalization preclude a prediction [25]

Carrier role	Theorem scope	Action-level output	Physical degrees of freedom	Quantum-particle requirements
Electro-magnetic connection	conditional Maxwell mode	transverse $k^2 = 0$ kernel	two transverse classical modes	photon requires quantization and a positive-residue physical pole
Color connection	conditional perturbative pure-Yang–Mills mode	one transverse $k^2 = 0$ kernel per generator	$2 \dim G$ perturbative modes	deconfined BRST/LSZ receipt required; no free-gluon claim in confined QCD
Metric perturbation	conditional pure-Einstein mode	$D^{\mathrm{TT}} \propto i\Pi^{\mathrm{TT}}/(k^2 + i0)$	two classical TT modes	graviton requires metric quantization, physical Hilbert space, and pole receipt

Particle	Calculation	Theorem scope	Value
W boson	electroweak calibration	no source-only physical mass	no OPH-native pole
Z boson	electroweak calibration	no source-only physical mass	no OPH-native pole
Higgs boson	Higgs/top criticality	no source-only physical mass	no OPH-native pole
Top quark	Higgs/top criticality	companion coordinate on declared surface; direct-top auxiliary codomain no-go	not a separate source-only prediction

Quark	Theorem scope	Public value	Mass convention
Up quark	current-signature spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = 2\,\mathrm{GeV}$, comparison chart
Down quark	current-signature spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = 2\,\mathrm{GeV}$, comparison chart
Strange quark	current-signature spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = 2\,\mathrm{GeV}$, comparison chart
Charm quark	current-signature spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = m_c(\mu)$, comparison chart
Bottom quark	current-signature spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = m_b(\mu)$, comparison chart
Top quark	separate extraction coordinate	withheld	cross-section pole-mass chart